Oracle-FHIR — R4 REST API — part 3 of 3 (r4-rest-api-v32/api-observation.html – r4-rest-api-v32/op-structuredefinition-id-get.html)

Source: Oracle Health Millennium FHIR R4 REST API Guide v32, Bulk Data Access Guide v10 and API guides (oracle-fhir-docs.zip). Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

Observation

File: r4-rest-api-v32/api-observation.html

Description

Observation

This topic describes information for the Observation resource.

Overview

The Observation resource provides measurements or simple assertions about a patient that are useful for establishing baselines or trends, monitoring a patient's progress, and establishing diagnoses. Most observations are simple name and value pair assertions, but some observations, such as blood pressure, group other observations together logically. Examples of common observations are laboratory results (blood sugar, hemoglobin), vital signs (temperature, blood pressure), personal characteristics (height, weight), and social history (tobacco and alcohol use, employment status). Pathology reports, radiology reports, and other textual reports are represented by the DiagnosticReport resource.

See the following examples of common observations:

* Laboratory results (blood sugar, hemoglobin)

* Imaging (CMS quality measure around CT radiation dose and ensuring that the dosage is As Low As Reasonably Achievable [ALARA])

* Vital signs (temperature, blood pressure)

* Personal characteristics (height, weight)

* Social history (tobacco and alcohol use, employment status)

* Survey (PRAPARE, AUDIT-C)

Pathology reports, radiology reports, and other textual reports are represented by the DiagnosticReport resource.

This resource supports the following HL7 FHIR US Core Implementation Guide STU 6.1.0 profiles:

US Core Observation Clinical Result Profile

* US Core Laboratory Result Observation Profile

* US Core Observation Occupation Profile

* US Core Observation Pregnancy Intent Profile

* US Core Observation Pregnancy Status Profile

* US Core Observation Screening Assessment Profile

* US Core Observation Sexual Orientation Profile

* US Core Simple Observation Profile

* US Core Smoking Status Observation Profile

US Core Vital Signs Profile

* US Core Pediatric Head Occipital Frontal Circumference Percentile Profile

* US Core Pediatric BMI for Age Observation Profile

* US Core Pediatric Weight for Height Observation Profile

* US Core Blood Pressure Profile

* US Core BMI Profile

* US Core Body Height Profile

* US Core Body Temperature Profile

* US Core Body Weight Profile

* US Core Head Circumference Profile

* US Core Heart Rate Profile

* US Core Pulse Oximetry Profile

* US Core Respiratory Rate Profile

The examples provided here are non-normative and replaying them in the public sandbox is not guaranteed to yield the results shown on the site. The field is present only for confidential laboratory and social-history observations. Confidential social-history observations can contain confidential members, represented by the .

Note:

meta.security

inline security label extension

Fields

ID

Identifier

Status

Category (laboratory, social history, vital-signs, imaging)

Code (Observation name or text)

Subject (Patient)

Encounter

Effective date/time (collection date/time for laboratory)

Extensions (for more information, see the Extensions section of this topic)

Issued (date/time observation made available, entered, verified)

Performer

Observation value or result

For Observations with valueQuantity

Quantity comparator (<, <=, >, >=)

Quantity units

For Observations with valueCodeableConcept

Codeable concept

Data absent reason

Interpretation (abnormal flagging)

Note (comments)

* Specimen

Reference range

low

high

type

text

Member Observations (for limited use cases)

Component (for example, systolic and diastolic for blood pressure)

* derivedFrom

Terminology Bindings

Observation.category

Details: Observation Category Codes

System: http://terminology.hl7.org/CodeSystem/observation-category

Details: Observation US Core Category Codes

System: http://hl7.org/fhir/us/core/CodeSystem/us-core-category

Observation.code

Description: The codes identifying names of simple observations.

Details: LOINC

System: http://loinc.org

Details: SNOMED CT

System: http://snomed.info/sct

Details: For more information, see Code Set 72 Clinical Event Codes under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/72

Observation.performer.extension

Description: The codes identifying the function of the performer.

Details: Performer function codes

System: http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Details: Provider Roles

System: https://terminology.hl7.org/CodeSystem-v2-0443.html

Observation.dataAbsentReason

Description: The codes for reasons that data is absent.

Details: v4 data absentreason

System: http://terminology.hl7.org/CodeSystem/data-absent-reason

Observation.valueCodeableConcept

Description: The codes for the value of the observation.

Details: SNOMED CT

System: http://snomed.info/sct

Observation.interpretation

Description: The codes identifying interpretations of observations.

Details: v3 Interpretation Codes

System: http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation

Details: For more information, see Code Set 52 Interpretation Result under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/52

Observation.component.code

Description: The codes identifying names of simple observations.

Details: LOINC

System: http://loinc.org

Details: SNOMED CT

System: http://snomed.info/sct

Details: For more information, see Code Set 72 Clinical Event Codes under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/72

Observation.component.interpretation

Description: The codes identifying interpretations of observations.

Details: v3 Interpretation Codes

System: http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation

Details: For more information, see Code Set 52 Interpretation Result under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/52

Observation.component.dataAbsentReason

Description: The codes for reasons that data is absent.

Details: v4 data absentreason

System: http://terminology.hl7.org/CodeSystem/data-absent-reason

Observation.meta.security

Description: These tags connect specific resources to the overall security policy and infrastructure.

Details: ValueSet

System: http://terminology.hl7.org/CodeSystem/v3-ActCode

Extensions

convertedMeasurement: This extension returns a converted measurement of a different measurement system than the original quantity.

performerFunction

* valueAttachment:

* The URL for this extension is defined as: https://hl7.org/fhir/5.0/StructureDefinition/extension-Observation.valueAttachment

This extension is defined and referenced from the newer version of FHIR. See Extensions for converting between versions and R5 Snapshot of Observation.value for more information.

* Inline Security Label

Observation

The operations from the Observation category.

Create an observation

Method: post Path: /Observation

Get a list of observations

Method: get Path: /Observation

Get an observation by ID

Method: get Path: /Observation/{ID}

Update an observation

Method: put Path: /Observation/{ID}

Create an observation

File: r4-rest-api-v32/op-observation-post.html

Creates a new observation.

Note:

* See Understand Supported Vital Signs in the FHIR Observation Resource for a list of vital signs that are supported for the create operation.

* Components are supported only when writing observation blood pressure and pulse oximetry profiles.

* Individual systolic and diastolic components are paired on subsequent searches or reads as long as the blood pressures are paired in Millennium. See Configure Blood Pressure Event Set Pairing Hierarchy for more information.

* The category of is supported to update a limited number of observations. If the category of is provided with an other than the LOINC codes below, the transaction returns a 422 (Unprocessable Entity) status code.

imaging

imaging

Observation.code

* LOINC 96912-1 - Calculated CT global noise

* LOINC 96913-9 - Calculated CT size-adjusted dose

* LOINC 96914-7 - CT dose and image quality category

* Only the body fields mentioned below are supported. Unsupported fields are ignored.

* Do not provide modifier fields because they cause the transaction to fail.

* You can also create observations with proprietary codes with the latest version of observation.

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : ObservationCreateBody

Type:

object

Show Source

category(required): array

category

A code that classifies the general type of observation being made.

Note:

* VitalSigns, Laboratory, and Imaging categories are supported.

* Only one category may be supplied on create.

Example:

```
{
  "category": [
    {
      "coding": [
        {
```

```
          "system": 'http://terminology.hl7.org/CodeSystem/observation-category',
          "code": 'vital-signs',
          "display": 'Vital Signs'
        }
      ],
      "text": 'Vital Signs'
    }
  ]
}
```

code(required): object

code

Describes what was observed. Also called the observation name.

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8331-1'
      }
    ],
    "text": 'Temperature Oral'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8302-2'
      },
      {
        "system": 'http://loinc.org',
        "code": '3137-7'
      }
    ],
    "text": 'Height/Length Measured'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '96914-7'
      },
    "text": 'CT dose and image quality category'
  }
}
```

component: array

component

Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations. Component is required to create blood pressure.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8480-6'
      }
    ],
    "text": 'Systolic Blood Pressure Invasive'
  },
  "valueQuantity": {
    "value": 120,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
          "code": 'N'
        }
```

```
      ]
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 45.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "high": {
        "value": 75.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

component.code(required): object

component.code

Describes what was observed. Also called the observation name.

Example:

```
{
  "coding": [
    {
      "system": 'http://loinc.org',
      "code": '8480-6'
    }
  ],
  "text": 'Systolic Blood Pressure Invasive'
}
```

component.interpretation: array

component.interpretation

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
   "coding": [
     {
       "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
       "code": 'N'
     }
   ],
   "text": 'NORMAL'
 }
```

component.referenceRange: object

component.referenceRange

Guidance on how to interpret the value by comparison to a normal or recommended range.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "low": {
    "value": 45.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "high": {
    "value": 75.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
```

```
  },
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal'
      }
    ],
    "text": 'Normal Range'
  }
}
```

component.value[x]:

The information determined as a result of making the observation, if the information has a simple value. Supports writing observations with only valueQuantity.

Note:

Example:

```
{
  "value": 120,
  "unit": 'mmHg',
  "system": 'http://unitsofmeasure.org',
  "code": 'mm[Hg]'
}
```

effective[x](required):

The date and time when the observation was made. Only a type of is supported.

Note:

dateTime

Example:

```
{
  "effectiveDateTime": "2020-07-07T20:37:00.000Z"
}
```

encounter: object

encounter

The encounter during which the observation was first made.

Example:

```
{
  "encounter": {
    "reference": "Encounter/631923"
  }
}
```

interpretation: array

interpretation

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
  "interpretation": [
    {
      "coding": [
        {
          "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',
          "code": '211',
          "display": 'LOW',
          "userSelected": true
        }
      ],
      "text": 'LOW'
    }
  ]
}
```

```
{
  "interpretation": [
    {
      "coding": [
        {
          {
            "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
            "code": 'L',
            "display": 'Low'
          }
        ],
        "text": 'LOW'
      }
    ]
  }
```

issued: string

The instant the observation was made.

Example:

```
{
  "issued": "2020-07-06T21:02:47.000Z"
}
```

note: array

note

Comments about the observation

Example:

```
{
  "note": [
    {
      "authorReference": {
        "reference": 'Practitioner/62955753'
      },
      "time": '2018-08-17T09:55:42.000Z',
      "text": 'Charted at Incorrect Time'
    }
  ]
}
```

performer: array

performer

Limited to "real" people rather than equipment.

Note:

* Must be a reference to a practitioner.

* Vital signs require a performer.

Example:

```
{
"performer": [
    {
        "reference": 'Practitioner/12724064'
      }
  ]
}
```

performer.extension: object

performer.extension

Distinguishes the performer's type of involvement in the event.

Note:

* Vital signs support and codes.

PPRF

* When only one performer is provided, the code must be LA.

* When multiple performers are provided, the code must be LA for one of the performers.

* Laboratory supports LA, PPRF, PRF, OP, and AUTHEN codes.

Example:

```
{
   "extension": [
     {
       "valueCodeableConcept": {
         "coding": [
           {
             "system": 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
             "code": 'LA',
             "display": 'legal authenticator'
           }
         ],
         "text": 'legal authenticator'
       },
       "url": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'
     }
   ],
 }
{
   "extension": [
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "https://terminology.hl7.org/CodeSystem-v2-0443.html",
                  "code": "OP",
```

```
                  "display": "ordering provider"
               }
            ],
            "text": "ordering provider"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
               }
            ],
            "text": "primary performer"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"PRF",
                  "display":"performer"
               }
            ],
            "text":"performer"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"AUTHEN",
                  "display":"authenticator"
               }
            ],
            "text":"authenticator"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
   ]
}
```

referenceRange: array

referenceRange

Provides guidance for interpretating the observation.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "referenceRange": [
    {
      "low": {
        "value": 35.2,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "high": {
        "value": 36.7,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal',
            "display": 'Normal Range'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
```

```
}
```

referenceRange.high: object

referenceRange.high

The upper bound of the reference range.

Example:

```
{
  "low": {
    "value": 36.7,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```

referenceRange.low: object

referenceRange.low

The lower bound of the reference range.

Example:

```
{
  "low": {
    "value": 35.2,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```

referenceRange.text: string

The text-based reference range in an observation, which may be used when a quantitative range is not appropriate for an observation.

Example:

```
{
  "text": 'Negative'
}
```

referenceRange.type: object

referenceRange.type

Codes to indicate what part of the targeted reference population it applies to. Only a of normal is supported.

Note:

ReferenceRange.type

Example:

```
{
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal',
        "display": 'Normal Range'
      }
    ],
    "text": 'Normal Range'
  }
}
```

* resourceType(required): string Allowed Values: ["Observation"] The type of the FHIR resource.

status(required): string

Codes identifying the lifecycle stage of an event. The , , and status codes are supported.

Note:

final

amended

corrected

Example:

```
{
  "status": "final"
}
```

subject(required): object

subject

Indicates the patient who the observation record is associated with. Must only include references to a patient.

Note:

Example:

```
{
  "subject": {
    "reference": "Patient/631923"
  }
}
```

value[x](required):

The value of the observation.

Note:

* Vital signs support writing observations with valueQuantity.

* Laboratory supports valueQuantity, valueCodeableConcept, valueString, valueAttachment, and valueDateTime.

* Imaging supports valueQuantity and valueCodeableConcept.

Example:

```
{
  "valueQuantity":
  {
    "value": 162,
    "unit": 'cm',
    "system": 'http://unitsofmeasure.org',
    "code": 'cm'
  }
}
```

```
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": 'http://snomed.info/sct',
        "code": '373066001',
        "display": 'Yes (qualifier value)'
      }
    ],
    "text": 'Yes'
  }
}
```

```
{
  "valueString": '2'
}
```

```
{
  "valueDateTime": '2020-07-04T07:15:00Z'
}
```

```
{
    "type":"object",
    "required":[
        "resourceType",
        "status",
        "category",
        "code",
        "subject",
        "effective[x]",
        "value[x]",
        "component.code"
    ],
    "properties":{
        "resourceType":{
            "description":"The type of the FHIR resource.\n",
            "type":"string",
            "enum":[
                "Observation"
            ]
        },
        "status":{
            "description":"Codes identifying the lifecycle stage of an event.<br>\n\n<b>Note:</b> The
    <code>final</code>, <code>amended</code>, and <code>corrected</code> status codes are
    supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"status\": \"final\"\n}</code></pre>\n",
            "type":"string"
        },
        "category":{
            "description":"A code that classifies the general type of observation being
    made.<br>\n\n<b>Note:</b>\n<ul>\n  <li> <code>VitalSigns</code>, <code>Laboratory</code>, and <code>Imaging</code>
    categories are supported. </li>\n  <li> Only one category may be supplied on create.
    </li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"category\": [\n    {\n      \"coding\": [\n        {\n
              \"system\": 'http://terminology.hl7.org/CodeSystem/observation-category',\n          \"code\": 'vital-
    signs',\n          \"display\": 'Vital Signs'\n        }\n      ],\n      \"text\": 'Vital Signs'\n    }\n
    ]\n}</code></pre>\n",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
```

```
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
    },
    "code":{
        "description":"Describes what was observed. Also called the observation
name.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8331-1'\n      }\n    ],\n    \"text\": 'Temperature Oral'\n
}\n}</code></pre>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8302-2'\n      },\n      {\n        \"system\": 'http://loinc.org',\n
        \"code\": '3137-7'\n      }\n    ],\n    \"text\": 'Height/Length Measured'\n  }\n}</code></pre>\n\n
<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": 'http://loinc.org',\n        \"code\":
'96914-7'\n      },\n    \"text\": 'CT dose and image quality category'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "subject":{
        "description":"Indicates the patient who the observation record is associated with.<br>\n\n<b>Note:</b> Must
 only include references to a patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
\"Patient/631923\"\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "encounter":{
        "description":"The encounter during which the observation was first
made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"encounter\": {\n    \"reference\": \"Encounter/631923\"\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "effective[x]":{
        "description":"The date and time when the observation was made.<br>\n\n<b>Note:</b> Only a type of
<code>dateTime</code> is supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"effectiveDateTime\":
\"2020-07-07T20:37:00.000Z\"\n}</code></pre>\n"
    },
    "issued":{
        "description":"The instant the observation was made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"issued\":
\"2020-07-06T21:02:47.000Z\"\n}</code></pre>\n",
        "type":"string"
    },
    "performer":{
        "description":"Limited to \"real\" people rather than equipment.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Must be a
reference to a practitioner.</li>\n  <li>Vital signs require a
performer.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n\"performer\": [\n    {\n        \"reference\":
'Practitioner/12724064'\n      }\n  ]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "reference":{
```

```
                    "type":"string"
                }
            }
        }
    },
    "performer.extension":{
        "description":"Distinguishes the performer's type of involvement in the event.<br>\n\n<b>Note:</b>\n<ul>\n
   <li>Vital signs support <code>LA</code> and <code>PPRF</code> codes.\n      <ul>\n         <li>When only one
performer is provided, the code must be <code>LA</code>.</li>\n         <li>When multiple performers are provided,
the code must be <code>LA</code> for one of the performers.</li>\n      </ul>\n   </li>\n   <li>Laboratory supports
<code>LA</code>, <code>PPRF</code>, <code>PRF</code>, <code>OP</code>, and <code>AUTHEN</code>
codes.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n   \"extension\": [\n     {\n       \"valueCodeableConcept\":
 {\n         \"coding\": [\n           {\n             \"system\":
'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',\n             \"code\": 'LA',\n
\"display\": 'legal authenticator'\n           }\n         ],\n         \"text\": 'legal authenticator'\n       },\n
       \"url\": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'\n     }\n   ],\n
}</code></pre>\n\n<pre><code>{\n   \"extension\": [\n      {\n         \"valueCodeableConcept\": {\n
\"coding\": [\n             {\n                \"system\":
\"https://terminology.hl7.org/CodeSystem-v2-0443.html\",\n                  \"code\": \"OP\",\n
\"display\": \"ordering provider\"\n              }\n            ],\n            \"text\": \"ordering provider\"\n
         },\n         \"url\": \"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n      },\n
{\n         \"valueCodeableConcept\": {\n            \"coding\": [\n               {\n                  \"system\":
\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n                  \"code\": \"PPRF\",\n
                \"display\": \"primary performer\"\n               }\n            ],\n            \"text\":
\"primary performer\"\n         },\n         \"url\": \"http://hl7.org/fhir/StructureDefinition/event-
performerFunction\"\n      },\n      {\n         \"valueCodeableConcept\":{\n            \"coding\":[\n
               {\n                  \"system\":\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                  \"code\":\"PRF\",\n                  \"display\":\"performer\"\n               }\n            ],\n
            \"text\":\"performer\"\n         },\n         \"url\":\"http://hl7.org/fhir/StructureDefinition/event-
performerFunction\"\n      },\n      {\n         \"valueCodeableConcept\":{\n            \"coding\":[\n
               {\n                  \"system\":\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                  \"code\":\"AUTHEN\",\n                  \"display\":\"authenticator\"\n               }\n
            ],\n            \"text\":\"authenticator\"\n         },\n
\"url\":\"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n      }\n   ]\n}</code></pre>\n",
        "type":"object"
    },
    "value[x]":{
        "description":"The value of the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Vital signs support writing
observations with <code>valueQuantity</code>.</li>\n  <li>Laboratory supports <code>valueQuantity</code>,
<code>valueCodeableConcept</code>, <code>valueString</code>, <code>valueAttachment</code>, and
<code>valueDateTime</code>.</li>\n  <li>Imaging supports <code>valueQuantity</code> and
<code>valueCodeableConcept</code>.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"valueQuantity\":\n  {\n
\"value\": 162,\n    \"unit\": 'cm',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'cm'\n
}\n}</code></pre>\n\n<pre><code>{\n  \"valueCodeableConcept\": {\n    \"coding\": [\n      {\n        \"system\":
'http://snomed.info/sct',\n        \"code\": '373066001',\n        \"display\": 'Yes (qualifier value)'\n      }\n
    ],\n    \"text\": 'Yes'\n  }\n}\n</code></pre>\n<pre><code>{\n  \"valueString\":
'2'\n}\n</code></pre>\n<pre><code>{\n  \"valueDateTime\": '2020-07-04T07:15:00Z'\n}</code></pre>\n"
    },
    "interpretation":{
        "description":"A categorical assessment of an observation value. For example, high, low,
normal.<br>\n\n<b>Note:</b> Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
\"interpretation\": [\n    {\n      \"coding\": [\n        {\n          \"system\":
'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',\n          \"code\": '211',\n
\"display\": 'LOW',\n          \"userSelected\": true\n        }\n      ],\n      \"text\": 'LOW'\n    }\n
]\n}</code></pre>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n      \"coding\": [\n        {\n          {\n
            \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n            \"code\":
 'L',\n            \"display\": 'Low'\n          }\n        ],\n        \"text\": 'LOW'\n      }\n    ]\n
}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    },
    "note":{
        "description":"Comments about the observation<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"note\": [\n    {\n
     \"authorReference\": {\n        \"reference\": 'Practitioner/62955753'\n      },\n      \"time\":
'2018-08-17T09:55:42.000Z',\n      \"text\": 'Charted at Incorrect Time'\n    }\n  ]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
```

```
            "properties":{
                "authorReference":{
                    "type":"object",
                    "properties":{
                        "reference":{
                            "type":"string"
                        }
                    }
                },
                "time":{
                    "type":"string"
                },
                "text":{
                    "type":"string"
                }
            }
        }
    },
    "referenceRange":{
        "description":"Provides guidance for interpretating the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only
one referenceRange is supported. </li>\n  <li><code>ReferenceRange.text</code> is not supported when
<code>referenceRange.high</code>, <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type
of <code>normal</code> is supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"referenceRange\": [\n
{\n      \"low\": {\n        \"value\": 35.2,\n        \"unit\": 'degC',\n        \"system\":
'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n      \"high\": {\n        \"value\": 36.7,\n
        \"unit\": 'degC',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n
      \"type\": {\n        \"coding\": [\n          {\n            \"system\":
'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n            \"code\": 'normal',\n
\"display\": 'Normal Range'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object"
        }
    },
    "referenceRange.low":{
        "description":"The lower bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 35.2,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "value":{
                "type":"number"
            },
            "unit":{
                "type":"string"
            },
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    },
    "referenceRange.high":{
        "description":"The upper bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 36.7,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "value":{
                "type":"number"
            },
            "unit":{
                "type":"string"
            },
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    },
    "referenceRange.text":{
        "description":"The text-based reference range in an observation, which may be used when a quantitative range
 is not appropriate for an observation.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"text\":
'Negative'\n}</code></pre>\n",
        "type":"string"
    },
    "referenceRange.type":{
        "description":"Codes to indicate what part of the targeted reference population it applies
to.<br>\n\n<b>Note:</b> Only a <code>ReferenceRange.type</code> of normal is
supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": {\n    \"coding\": [\n      {\n        \"system\":
'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n        \"code\": 'normal',\n        \"display\":
'Normal Range'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
```

```
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "component":{
        "description":"Some observations have multiple component observations. These component observations are
expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic
component observations for blood pressure measurement and multiple component observations for genetics
observations.<br>\n\n<b>Note:</b> Component is required to create blood
pressure.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8480-6'\n      }\n    ],\n    \"text\": 'Systolic Blood Pressure Invasive'\n
  },\n  \"valueQuantity\": {\n    \"value\": 120,\n    \"unit\": 'mmHg',\n    \"system\":
'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"interpretation\": [\n    {\n      \"coding\": [\n
        {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
\"code\": 'N'\n        }\n      ]\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n        \"value\":
 45.0,\n        \"unit\": 'mmHg',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n
      },\n      \"high\": {\n        \"value\": 75.0,\n        \"unit\": 'mmHg',\n        \"system\":
'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n      },\n      \"type\": {\n        \"coding\": [\n
          {\n            \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
\"code\": 'normal'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object"
        }
    },
    "component.code":{
        "description":"Describes what was observed. Also called the observation
name.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"coding\": [\n    {\n      \"system\": 'http://loinc.org',\n
\"code\": '8480-6'\n    }\n  ],\n  \"text\": 'Systolic Blood Pressure Invasive'\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "component.value[x]":{
        "description":"The information determined as a result of making the observation, if the information has a
simple value.<br>\n\n<b>Note:</b> Supports writing observations with only
valueQuantity.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"value\": 120,\n  \"unit\": 'mmHg',\n  \"system\":
'http://unitsofmeasure.org',\n  \"code\": 'mm[Hg]'\n}</code></pre>\n"
    },
    "component.interpretation":{
        "description":"A categorical assessment of an observation value. For example, high, low,
normal.<br>\n\n<b>Note:</b> Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
\"coding\": [\n     {\n       \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
       \"code\": 'N'\n     }\n   ],\n   \"text\": 'NORMAL'\n }</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
```

```
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    },
    "component.referenceRange":{
        "description":"Guidance on how to interpret the value by comparison to a normal or recommended
   range.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one referenceRange is supported.</li>\n
   <li><code>ReferenceRange.text</code> is not supported when <code>referenceRange.high</code>,
   <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type of <code>normal</code> is
   supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n    \"value\": 45.0,\n    \"unit\":
   'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"high\": {\n    \"value\":
   75.0,\n    \"unit\": 'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n
   \"type\": {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-
   meaning',\n        \"code\": 'normal'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
},
"example":"{\n  \"resourceType\": \"Observation\",\n  \"status\": \"final\",\n  \"category\": [\n    {\n
\"coding\": [\n        {\n          \"system\": \"http://terminology.hl7.org/CodeSystem/observation-category\",\n
          \"code\": \"vital-signs\",\n          \"display\": \"Vital Signs\"\n        }\n      ],\n      \"text\":
\"Vital Signs\"\n    }\n  ],\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": \"http://loinc.org\",\n
        \"code\": \"8331-1\"\n      }\n    ],\n    \"text\": \"Temperature Oral\"\n  },\n  \"subject\": {\n
\"reference\": \"Patient/12457981\"\n  },\n  \"encounter\": {\n    \"reference\": \"Encounter/97845408\"\n  },\n
\"effectiveDateTime\": \"2020-04-03T19:21:00.000Z\",\n  \"issued\": \"2020-04-03T19:21:40.000Z\",\n  \"performer\":
[\n    {\n      \"extension\": [\n        {\n          \"valueCodeableConcept\": {\n            \"coding\": [\n
              {\n                \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                \"code\": \"LA\",\n                \"display\": \"legal authenticator\"\n              }\n
            ],\n            \"text\": \"legal authenticator\"\n          },\n          \"url\":
\"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n        }\n      ],\n      \"reference\":
\"Practitioner/11638321\"\n    }\n  ],\n  \"valueQuantity\": {\n    \"value\": 37.2,\n    \"unit\": \"degC\",\n
\"system\": \"http://unitsofmeasure.org\",\n    \"code\": \"Cel\"\n  },\n  \"interpretation\": [\n    {\n
\"coding\": [\n        {\n          \"system\":
\"http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation\",\n          \"code\": \"N\",\n
\"display\": \"Normal\"\n        }\n      ]\n    }\n  ],\n  \"note\": [\n    {\n      \"authorReference\": {\n
        \"reference\": \"Practitioner/11638321\"\n      },\n      \"time\": \"2020-09-03T15:03:10.000Z\",\n
\"text\": \"No additional test require\"\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n
\"value\": 35.8,\n        \"unit\": \"degC\",\n        \"system\": \"http://unitsofmeasure.org\",\n        \"code\":
 \"Cel\"\n      },\n      \"high\": {\n        \"value\": 37.3,\n        \"unit\": \"degC\",\n        \"system\":
\"http://unitsofmeasure.org\",\n        \"code\": \"Cel\"\n      },\n      \"type\": {\n        \"coding\": [\n
          {\n            \"system\": \"http://terminology.hl7.org/CodeSystem/referencerange-meaning\",\n
\"code\": \"normal\",\n            \"display\": \"Normal Range\"\n          }\n        ],\n        \"text\":
\"Normal Range\"\n      }\n    }\n  ]\n}\n"
}
```

Example:

```
{
  "resourceType": "Observation",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
```

```
      {
        "system": "http://loinc.org",
        "code": "8331-1"
      }
    ],
    "text": "Temperature Oral"
  },
  "subject": {
    "reference": "Patient/12457981"
  },
  "encounter": {
    "reference": "Encounter/97845408"
  },
  "effectiveDateTime": "2020-04-03T19:21:00.000Z",
  "issued": "2020-04-03T19:21:40.000Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/11638321"
    }
  ],
  "valueQuantity": {
    "value": 37.2,
    "unit": "degC",
    "system": "http://unitsofmeasure.org",
    "code": "Cel"
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
          "code": "N",
          "display": "Normal"
        }
      ]
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/11638321"
      },
      "time": "2020-09-03T15:03:10.000Z",
      "text": "No additional test require"
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 35.8,
        "unit": "degC",
        "system": "http://unitsofmeasure.org",
        "code": "Cel"
      },
      "high": {
        "value": 37.3,
        "unit": "degC",
        "system": "http://unitsofmeasure.org",
        "code": "Cel"
      },
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
            "code": "normal",
            "display": "Normal Range"
          }
        ],
        "text": "Normal Range"
      }
    }
  ]
}
```

Nested Schema : category

Type:

array

A code that classifies the general type of observation being made.

Note:

* VitalSigns, Laboratory, and Imaging categories are supported.

* Only one category may be supplied on create.

Example:

```
{
  "category": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/observation-category',
          "code": 'vital-signs',
          "display": 'Vital Signs'
        }
      ],
      "text": 'Vital Signs'
    }
  ]
}
```

Show Source

* Array of: object items


```
{
    "description":"A code that classifies the general type of observation being made.<br>\n\n<b>Note:</b>\n<ul>\n  <li>
    <code>VitalSigns</code>, <code>Laboratory</code>, and <code>Imaging</code> categories are supported. </li>\n  <li>
    Only one category may be supplied on create. </li>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"category\": [\n
        {\n      \"coding\": [\n        {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/observation-
    category',\n          \"code\": 'vital-signs',\n          \"display\": 'Vital Signs'\n        }\n      ],\n
    \"text\": 'Vital Signs'\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : code

Type:

object

Describes what was observed. Also called the observation name.

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8331-1'
      }
    ],
    "text": 'Temperature Oral'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8302-2'
```

```
      },
      {
        "system": 'http://loinc.org',
        "code": '3137-7'
      }
    ],
    "text": 'Height/Length Measured'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '96914-7'
      },
    "text": 'CT dose and image quality category'
  }
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"Describes what was observed. Also called the observation
    name.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8331-1'\n      }\n    ],\n    \"text\": 'Temperature Oral'\n
    }\n}</code></pre>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8302-2'\n      },\n      {\n        \"system\": 'http://loinc.org',\n
            \"code\": '3137-7'\n      }\n    ],\n    \"text\": 'Height/Length Measured'\n  }\n}</code></pre>\n\n
    <pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": 'http://loinc.org',\n        \"code\":
    '96914-7'\n      },\n    \"text\": 'CT dose and image quality category'\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : component

Type:

array

Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations. Component is required to create blood pressure.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8480-6'
      }
    ],
    "text": 'Systolic Blood Pressure Invasive'
  },
  "valueQuantity": {
    "value": 120,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "interpretation": [
    {
      "coding": [
```

```
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
          "code": 'N'
        }
      ]
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 45.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "high": {
        "value": 75.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Some observations have multiple component observations. These component observations are expressed as
     separate code value pairs that share the same attributes. Examples include systolic and diastolic component
    observations for blood pressure measurement and multiple component observations for genetics
    observations.<br>\n\n<b>Note:</b> Component is required to create blood
    pressure.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8480-6'\n      }\n    ],\n    \"text\": 'Systolic Blood Pressure Invasive'\n
      },\n  \"valueQuantity\": {\n    \"value\": 120,\n    \"unit\": 'mmHg',\n    \"system\":
    'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"interpretation\": [\n    {\n      \"coding\": [\n
            {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
    \"code\": 'N'\n        }\n      ]\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n        \"value\":
     45.0,\n        \"unit\": 'mmHg',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n
          },\n      \"high\": {\n        \"value\": 75.0,\n        \"unit\": 'mmHg',\n        \"system\":
    'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n      },\n      \"type\": {\n        \"coding\": [\n
              {\n            \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
    \"code\": 'normal'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
    ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object"
    }
}
```

Nested Schema : component.code

Type:

object

Describes what was observed. Also called the observation name.

Example:

```
{
  "coding": [
    {
      "system": 'http://loinc.org',
      "code": '8480-6'
    }
  ],
  "text": 'Systolic Blood Pressure Invasive'
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"Describes what was observed. Also called the observation name.<br>\n<b>Example:</b>\n\n<pre><code>{\n
      \"coding\": [\n    {\n      \"system\": 'http://loinc.org',\n      \"code\": '8480-6'\n    }\n  ],\n  \"text\":
    'Systolic Blood Pressure Invasive'\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
```

```
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : component.interpretation

Type:

array

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
   "coding": [
     {
       "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
       "code": 'N'
     }
   ],
   "text": 'NORMAL'
 }
```

Show Source

* Array of: object items

```
{
    "description":"A categorical assessment of an observation value. For example, high, low, normal.<br>\n\n<b>Note:</b>
     Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n   \"coding\": [\n     {\n
    \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n       \"code\": 'N'\n     }\n
    ],\n   \"text\": 'NORMAL'\n }</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : component.referenceRange

Type:

object

Guidance on how to interpret the value by comparison to a normal or recommended range.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "low": {
    "value": 45.0,
```

```
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "high": {
    "value": 75.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal'
      }
    ],
    "text": 'Normal Range'
  }
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"Guidance on how to interpret the value by comparison to a normal or recommended
    range.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one referenceRange is supported.</li>\n
    <li><code>ReferenceRange.text</code> is not supported when <code>referenceRange.high</code>,
    <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type of <code>normal</code> is
    supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n    \"value\": 45.0,\n    \"unit\":
    'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"high\": {\n    \"value\":
    75.0,\n    \"unit\": 'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n
    \"type\": {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-
    meaning',\n        \"code\": 'normal'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : encounter

Type:

object

The encounter during which the observation was first made.

Example:

```
{
  "encounter": {
    "reference": "Encounter/631923"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The encounter during which the observation was first made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
    \"encounter\": {\n    \"reference\": \"Encounter/631923\"\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : interpretation

Type:

array

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
  "interpretation": [
    {
      "coding": [
        {
          "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',
          "code": '211',
          "display": 'LOW',
          "userSelected": true
        }
      ],
      "text": 'LOW'
    }
  ]
}
```

```
{
  "interpretation": [
    {
      "coding": [
        {
          {
            "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
            "code": 'L',
            "display": 'Low'
          }
        ],
        "text": 'LOW'
      }
    ]
  }
```

Show Source

* Array of: object items

```
{
    "description":"A categorical assessment of an observation value. For example, high, low, normal.<br>\n\n<b>Note:</b>
     Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n
        \"coding\": [\n        {\n          \"system\":
    'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',\n          \"code\": '211',\n
    \"display\": 'LOW',\n          \"userSelected\": true\n        }\n      ],\n      \"text\": 'LOW'\n    }\n
    ]\n}</code></pre>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n      \"coding\": [\n        {\n          {\n
              \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n            \"code\":
     'L',\n            \"display\": 'Low'\n          }\n        ],\n        \"text\": 'LOW'\n      }\n    ]\n
    }</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : note

Type:

array

Comments about the observation

Example:

```
{
  "note": [
    {
      "authorReference": {
        "reference": 'Practitioner/62955753'
```

```
      },
      "time": '2018-08-17T09:55:42.000Z',
      "text": 'Charted at Incorrect Time'
    }
  ]
}
```

Show Source

* Array of: object items


```
{
    "description":"Comments about the observation<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"note\": [\n    {\n
    \"authorReference\": {\n        \"reference\": 'Practitioner/62955753'\n      },\n      \"time\":
    '2018-08-17T09:55:42.000Z',\n      \"text\": 'Charted at Incorrect Time'\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "authorReference":{
                "type":"object",
                "properties":{
                    "reference":{
                        "type":"string"
                    }
                }
            },
            "time":{
                "type":"string"
            },
            "text":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : performer

Type:

array

Limited to "real" people rather than equipment.

Note:

* Must be a reference to a practitioner.

* Vital signs require a performer.

Example:

```
{
"performer": [
    {
        "reference": 'Practitioner/12724064'
    }
  ]
}
```

Show Source

* Array of: object items


```
{
    "description":"Limited to \"real\" people rather than equipment.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Must be a
    reference to a practitioner.</li>\n  <li>Vital signs require a
    performer.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n\"performer\": [\n    {\n        \"reference\":
    'Practitioner/12724064'\n      }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : performer.extension

Type:

object

Distinguishes the performer's type of involvement in the event.

Note:

* Vital signs support and codes.

PPRF

* When only one performer is provided, the code must be LA.

* When multiple performers are provided, the code must be LA for one of the performers.

* Laboratory supports LA, PPRF, PRF, OP, and AUTHEN codes.

Example:

```
{
   "extension": [
     {
       "valueCodeableConcept": {
         "coding": [
           {
             "system": 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
             "code": 'LA',
             "display": 'legal authenticator'
           }
         ],
         "text": 'legal authenticator'
       },
       "url": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'
     }
   ],
 }
```

```
{
   "extension": [
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "https://terminology.hl7.org/CodeSystem-v2-0443.html",
                  "code": "OP",
                  "display": "ordering provider"
               }
            ],
            "text": "ordering provider"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
               }
            ],
            "text": "primary performer"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"PRF",
                  "display":"performer"
               }
            ],
            "text":"performer"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"AUTHEN",
                  "display":"authenticator"
               }
            ],
            "text":"authenticator"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
   ]
}
```

Nested Schema : referenceRange

Type:

array

Provides guidance for interpretating the observation.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "referenceRange": [
    {
      "low": {
        "value": 35.2,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "high": {
        "value": 36.7,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal',
            "display": 'Normal Range'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Provides guidance for interpretating the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one
    referenceRange is supported. </li>\n  <li><code>ReferenceRange.text</code> is not supported when
    <code>referenceRange.high</code>, <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type
    of <code>normal</code> is supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"referenceRange\": [\n
    {\n      \"low\": {\n        \"value\": 35.2,\n        \"unit\": 'degC',\n        \"system\":
    'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n      \"high\": {\n        \"value\": 36.7,\n
            \"unit\": 'degC',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n
          \"type\": {\n        \"coding\": [\n          {\n            \"system\":
    'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n            \"code\": 'normal',\n
    \"display\": 'Normal Range'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
    ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object"
    }
}
```

Nested Schema : referenceRange.high

Type:

object

The upper bound of the reference range.

Example:

```
{
  "low": {
    "value": 36.7,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```

Show Source

* code: string

* system: string

* unit: string

* value: number

```
{
    "description":"The upper bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 36.7,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "value":{
            "type":"number"
        },
```

```
        "unit":{
            "type":"string"
        },
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : referenceRange.low

Type:

object

The lower bound of the reference range.

Example:


```
{
  "low": {
    "value": 35.2,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```


Show Source

* code: string

* system: string

* unit: string

* value: number


```
{
    "description":"The lower bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 35.2,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "value":{
            "type":"number"
        },
        "unit":{
            "type":"string"
        },
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : referenceRange.type

Type:

object

Codes to indicate what part of the targeted reference population it applies to. Only a of normal is supported.

Note:

ReferenceRange.type

Example:


```
{
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal',
        "display": 'Normal Range'
      }
    ],
    "text": 'Normal Range'
  }
}
```


Show Source

* coding: array coding

* text: string


```
{
    "description":"Codes to indicate what part of the targeted reference population it applies to.<br>\n\n<b>Note:</b>
```

```
    Only a <code>ReferenceRange.type</code> of normal is supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"type\":
    {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
           \"code\": 'normal',\n        \"display\": 'Normal Range'\n      }\n    ],\n    \"text\": 'Normal Range'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : subject

Type:

object

Indicates the patient who the observation record is associated with. Must only include references to a patient.

Note:

Example:


```
{
  "subject": {
    "reference": "Patient/631923"
  }
}
```


Show Source

* reference: string


```
{
    "description":"Indicates the patient who the observation record is associated with.<br>\n\n<b>Note:</b> Must only
    include references to a patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
    \"Patient/631923\"\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* authorReference: object authorReference

* text: string

* time: string


```
{
    "type":"object",
    "properties":{
        "authorReference":{
            "type":"object",
            "properties":{
                "reference":{
```

```
                    "type":"string"
                }
            }
        },
        "time":{
            "type":"string"
        },
        "text":{
            "type":"string"
        }
    }
}
```

Nested Schema : authorReference

Type:

object

Show Source

* reference: string

```
{
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* reference: string

```
{
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
```

```
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider (Vital Signs and Laboratory)

* System (Vital Signs and Laboratory)

Example Request:

```
POST https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Mon, 16 Nov 2020 22:05:40 GMT
  Etag: W/"1"
  Location: https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-197392513
  Last-Modified: Mon, 16 Nov 2020 22:05:40 GMT
  Vary: Origin
  X-Request-Id: 11111111111111111111111111111111
```

x-imaging Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider (Imaging)

* System (Imaging)

Example Request:

```
POST https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation
-d{
  "resourceType": "Observation",
  "id": "f1abc4de-267e-44f9-b8aa-07c32c00b7e4",
  "identifier": [
      {
          "use": "official",
          "value": "urn:oid:1.3.6.1.4.1.14519.5.2.1.6450.2626.300453394315087810440454128499-96914-7",
          "system": "https://alaragateway.com"
      }
  ],
  "category": [
      {
          "coding": [
              {
                  "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                  "code": "imaging",
                  "display": "Imaging"
              }
          ]
      }
  ],
  "code": {
      "coding": [
          {
              "system": "http://loinc.org",
              "code": "96914-7",
              "display": "CT dose and image quality category"
          }
      ],
      "text": "CT dose and image quality category"
  },
  "subject": {
      "reference": "Patient/12461981"
  },
  "encounter": {
      "reference": "Encounter/97701407"
  },
  "effectiveDateTime": "2022-11-16T08:00:00.000+00:00",
  "status": "final",
  "valueCodeableConcept": {
```

```
      "coding": [
          {
              "system": "http://loinc.org",
              "code": "LA31771-1",
              "display": "Full Body"
          }
      ]
  }
}
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/json
  Date: Wed, 13 Nov 2024 10:30:08 GMT
  Etag: W/"1"
  Location: https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-206972543
  Last-Modified: Wed, 13 Nov 2024 10:30:07 GMT
  Vary: Origin
  X-Request-Id: 11111111111111111111111111111111
```

x-labs Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider (Vital Signs and Laboratory)

* System (Vital Signs and Laboratory)

Example Request:

```
POST https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation
-d{
  "resourceType": "Observation",
  "status": "final",
  "meta": {
    "security": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PHY"
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PRS"
      }
    ]
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ],
      "text": "Laboratory"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "718-7",
        "display": "Hemoglobin [Mass/volume] in Blood"
      }
    ],
    "text": "Hemoglobin"
  },
  "subject": {
    "reference": "Patient/2798003"
  },
  "encounter": {
    "reference": "Encounter/2673896"
  },
  "issued": "2020-07-30T20:42:00.000Z",
  "effectiveDateTime": "2020-07-30T20:42:00.000Z",
  "performer": [
    {
      "reference": "Practitioner/1477926"
    }
  ],
  "valueQuantity": {
    "value": 13.2,
    "unit": "mg/dL",
    "system": "http://unitsofmeasure.org",
    "code": "mg/dL"
  },
  "interpretation": [
```

```
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
          "code": "N",
          "display": "Normal"
        }
      ],
      "text": "Normal"
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 12.0,
        "unit": "g/dL",
        "system": "http://unitsofmeasure.org",
        "code": "g/dL"
      },
      "high": {
        "value": 15.5,
        "unit": "g/dL",
        "system": "http://unitsofmeasure.org",
        "code": "g/dL"
      }
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/1477926"
      },
      "time": "2020-07-30T20:42:15.000Z",
      "text": "Specimen slightly hemolyzed."
    }
  ]
}
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Mon, 16 Nov 2020 22:05:40 GMT
  Etag: W/"1"
  Location: https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-197392513
  Last-Modified: Mon, 16 Nov 2020 22:05:40 GMT
  Vary: Origin
  X-Request-Id: 11111111111111111111111111111111
```

x-vitals-blood-pressure Response

This resource action is still under development. This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

Note:

* Provider (Vital Signs and Laboratory)

* System (Vital Signs and Laboratory)

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation
-d '{
  "resourceType": "Observation",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "85354-9"
      }
    ],
    "text": "Blood pressure"
  },
  "subject": {
    "reference": "Patient/12457981"
  },
  "encounter": {
```

```
  "reference": "Encounter/97845408"
},
"effectiveDateTime": "2022-06-04T08:00:38.000Z",
"issued": "2022-06-04T08:00:38Z",
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8480-6"
        }
      ],
      "text": "Systolic Blood Pressure Invasive"
    },
    "valueQuantity": {
      "value": 120,
      "unit": "mmHg",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    },
    "interpretation": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
            "code": "N"
          }
        ]
      }
    ],
    "referenceRange": [
      {
        "low": {
          "value": 45.0,
          "unit": "mmHg",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        },
        "high": {
          "value": 75.0,
          "unit": "mmHg",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        },
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
              "code": "normal"
            }
          ],
          "text": "Normal Range"
        }
      }
    ]
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8462-4"
        }
      ],
      "text": "Diastolic Blood Pressure Invasive"
    },
    "valueQuantity": {
      "value": 80,
      "unit": "mmHg",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    },
    "interpretation": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
            "code": "N"
          }
        ]
      }
    ],
    "referenceRange": [
      {
        "low": {
          "value": 35.8,
          "unit": "mmHg",
          "system": "http://unitsofmeasure.org",
```

```
            "code": "mm[Hg]"
          },
          "high": {
            "value": 37.3,
            "unit": "mmHg",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          },
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
                "code": "normal"
              }
            ],
            "text": "Normal Range"
          }
        }
      ]
    }
  ]
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-vitals-pulse-oxymetry Response

This resource action is still under development. This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

Note:

* Provider (Vital Signs and Laboratory)

* System (Vital Signs and Laboratory)

Example Request:

```
POST https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation
-d '{
  "resourceType": "Observation",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "59408-5"
      }
    ],
    "text": "SpO2"
  },
  "subject": {
    "reference": "Patient/12457981"
  },
  "encounter": {
    "reference": "Encounter/97845408"
  },
  "effectiveDateTime": "2022-06-05T08:00:38.004Z",
  "issued": "2022-06-05T08:00:34Z",
  "performer": [
    {
      "extension": [
```

```
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/15441459"
    }
  ],
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "3151-8",
            "display": "Inhaled oxygen flow rate"
          }
        ],
        "text": "Oxygen Flow Rate"
      },
      "valueQuantity": {
        "value": 7,
        "unit": "L/min",
        "system": "http://unitsofmeasure.org",
        "code": "L/min"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "3150-0"
          }
        ],
        "text": "FIO2"
      },
      "valueQuantity": {
        "value": 21,
        "unit": "%",
        "system": "http://unitsofmeasure.org",
        "code": "%"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "59408-5"
          }
        ],
        "text": "SpO2"
      },
      "valueQuantity": {
        "value": 96,
        "unit": "%",
        "system": "http://unitsofmeasure.org",
        "code": "%"
      }
    }
  ]
}
```


Example Response:


```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```


Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and is used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

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Get a list of observations

File: r4-rest-api-v32/op-observation-get.html

Retrieves a list of observations that meet the supplied query parameters. Examples include laboratory, imaging, vitals, and social history observations.

Note:

* The comments field may have RTF or other formatted data rather than simple text. A resolution to this issue is planned for a future release and an alternative is being investigated.

* Searching records with the vital-signs category by code with a proprietary system returns an empty response.

* When multiple pages of observation results are returned for a single query:

* All social history observations (if any qualify for the query) are returned on the first page of results. The next item on this list does not apply to social history observations.

* Results are sorted by effective date and time in descending order. In other words, all observations on any given page of results are newer than all observations on the next page of results.

* If the query uses the _lastUpdated query parameter, results are sorted by last updated date andtime in descending order by page, not by effective date and time.

* Oracle recommends searching by code, date, or both.

Request

Query Parameters

_count: string

The maximum number of results to return per page.

* The first page is unaffected because all social history data is displayed on the first page regardless of requested count.

* The default count is 50 and the maximum supported count is 200.

* From the second page onward, the returned item count may be less than requested.

_id: string

The logical resource ID.

* This parameter is required if patient or subject is not used.

* Search by _id for laboratory observations supports multiple IDs from a single patient only.

_lastUpdated: string

The date range in which the observation was last updated. Example: or

* May be provided up to two times, and must use the eq, ge, gt, le, or lt prefixes.

* When a value is provided without a prefix, an implied eq prefix is used.

* When provided twice, the lower value must have a ge or gt prefix and the higher value must have an le or lt prefix.

* May not be provided with date.

* Only clinically significant updates are qualified, such as changes to the value or code, and other significant fields. Minor updates, such as non-clinically relevant note updates, do not qualify.

_lastUpdated=gt2014-09-24

_lastUpdated=lt2015-09-24T12:00:00.000Z

_revinclude: string

The Provenance resource entries to be returned as part of the bundle. Example:

* May be provided once with the Provenance:target value. Example: _revinclude=Provenance:target.

* May be provided with the _id/patient/subject parameter. Example: patient=12457977&category=vital-signs&_revinclude=Provenance:target.

* When _revinclude is provided in a request to the closed endpoint, the OAuth2 token must include the user/Provenance.read scope.

_revinclude=Provenance:target

* category: string The category of observations. When included through the search parameter or a granular scope, the code parameter is ignored. Example: category=laboratory

code: string

The code or component-code of the observation type. Example:

* May be a list of comma-separated values.

* Searches only Observation.code. For example, when fetching blood pressures, the resource is only returned when the search is based on 85354-9 (Systolic and Diastolic BP). Using the 8480-6(Systolic BP) or 8462-4 (Diastolic BP) component codes does not return the resource.

* This parameter is ignored when combined with the category search parameter or with granular scopes that include the category.

code=http://loinc.org|3094-0,http://loinc.org|3139-3

date: string

The date range into which the observation falls. Example: or

* May be provided up to two times, and must use the eq, ge, gt, le, or lt prefixes.

* When a value is provided without a prefix, an implied eq prefix is used.

* When provided twice, the lower value must have a ge or gt prefix and the higher value must have an le or lt prefix.

* May not be provided with _lastUpdated.

date=gt2014-09-24

date=lt2015-09-24T12:00:00.000Z

* patient: string The subject (patient) that the observation is about. This parameter is required if _id or subject is not used. Example: patient=12345

subject: string

The subject (patient) that the observation is about. Example: or

* This parameter is required if _id or patient are not used.

* The subject parameter must represent a Patient resource and may use the :Patient modifier.

subject=Patient/12345

subject:Patient=12345

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "79c71e5b-2793-4ace-bc5e-190e38784e80",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-1-73080191",
      "resource": {
        "resourceType": "Observation",
        "id": "SH-1-73080191",
        "meta": {
          "versionId": "73080191",
          "lastUpdated": "2020-06-11T04:05:33.000Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Observation</b></p><p><b>Patient Id</b>:
    12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Code</b>:
    Alcohol</p><p><b>Issued Date</b>: Jun 11, 2020  4:05 A.M. UTC</p></div>"
        },
        "status": "final",
        "category": [
```

```
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "363905002",
          "display": "Details of alcohol drinking behavior (observable entity)"
        }
      ],
      "text": "Alcohol"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/18565902"
    },
    "effectiveDateTime": "2022-06-01T10:39:02Z",
    "issued": "2020-06-11T04:05:33.000Z",
    "performer": [
      {
        "reference": "Practitioner/15441459"
      }
    ],
    "hasMember": [
      {
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      {
        "reference": "Observation/SH-73080191-q4625813"
      }
    ]
  }
},
{
  "fullUrl": "https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-73080191-q4625817",
  "resource": {
    "resourceType": "Observation",
    "id": "SH-73080191-q4625817",
    "meta": {
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      "lastUpdated": "2020-06-11T04:05:33.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Question</b>: SHX Alcohol
use</p><p><b>Response</b>: Past</p><p><b>Issued Date</b>: Jun 11, 2020  4:05 A.M. UTC</p></div>"
    },
    "status": "final",
    "category": [
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        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "363905002",
          "display": "Details of alcohol drinking behavior (observable entity)"
        }
      ],
      "text": "SHX Alcohol use"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/18565902"
    },
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    "issued": "2020-06-11T04:05:33.000Z",
    "performer": [
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        "reference": "Practitioner/15441459"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "410513005",
          "display": "In the past (qualifier value)"
        }
      ],
      "text": "Past"
    }
  }
},
{
  "fullUrl": "https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-73080191-q4625813",
  "resource": {
    "resourceType": "Observation",
    "id": "SH-73080191-q4625813",
    "meta": {
      "versionId": "73080191",
      "lastUpdated": "2020-06-11T04:05:33.000Z"
    },
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12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Question</b>: SHX Alcohol
type</p><p><b>Response</b>: Liquor</p><p><b>Issued Date</b>: Jun 11, 2020  4:05 A.M. UTC</p></div>"
    },
    "status": "final",
    "category": [
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        "coding": [
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            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "363905002",
          "display": "Details of alcohol drinking behavior (observable entity)"
        }
      ],
      "text": "SHX Alcohol type"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/18565902"
    },
    "effectiveDateTime": "2022-06-01T10:39:02Z",
    "issued": "2020-06-11T04:05:33.000Z",
    "performer": [
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        "reference": "Practitioner/15441459"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "6524003",
          "display": "Distilled spirits (substance)"
        }
      ],
      "text": "Liquor"
    }
  }
},
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-
CV-55046871-SECT-8098",
  "resource": {
    "resourceType": "Observation",
    "id": "VS-CV-55046871-SECT-8098",
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15116458</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Vital Signs</p><p><b>Code</b>:
SpO2</p><p><b>Result</b>: 95 %</p><p><b>Effective Date</b>: Jun  5, 2022  3:00 A.M. CDT</p></div>"
    },
    "status": "final",
    "category": [
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        "coding": [
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            "code": "vital-signs",
            "display": "Vital Signs"
          }
        ],
        "text": "Vital Signs"
      }
    ],
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          "code": "3623994",
          "display": "SpO2",
          "userSelected": true
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        {
          "system": "http://loinc.org",
          "code": "2708-6",
          "display": "Oxygen saturation in Arterial blood"
        },
        {
          "system": "http://loinc.org",
          "code": "59408-5",
          "display": "Oxygen saturation in Arterial blood by Pulse oximetry"
        }
      ],
      "text": "SpO2"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/18565902"
    },
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      "system": "http://unitsofmeasure.org",
      "code": "%"
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              "code": "703569",
              "display": "Oxygen Flow Rate",
              "userSelected": true
            },
            {
              "system": "http://loinc.org",
              "code": "3151-8"
            }
          ],
          "text": "Oxygen Flow Rate"
        },
        "valueQuantity": {
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          "unit": "L/min",
          "system": "http://unitsofmeasure.org",
          "code": "L/min"
        }
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        "code": {
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              "code": "2700657",
              "display": "FIO2",
              "userSelected": true
            },
            {
              "system": "http://loinc.org",
              "code": "3150-0",
              "display": "Inhaled Oxygen concentration"
            }
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          "text": "Inhaled Oxygen concentration"
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        "valueQuantity": {
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          "unit": "%",
          "system": "http://unitsofmeasure.org",
          "code": "%"
        }
      }
    ]
  }
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{
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  "resource": {
    "resourceType": "Observation",
    "id": "VS-197292857",
    "meta": {
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      "lastUpdated": "2020-07-06T21:13:24.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Vital Signs</p><p><b>Code</b>: Temperature
Axillary</p><p><b>Result</b>: 27 degC</p><p><b>Interpretation</b>: LOW</p><p><b>Effective Date</b>: Jul  6, 2020
9:11 P.M. UTC</p><p><b>Reference Range</b>: 35.2-36.7 degC</p></div>"
    },
    "identifier": [
      {
        "system": "https://fhir.cerner.com/ceuuid",
        "value": "CE87caf4b7-9397-4667-9897-702218017c9e-197292857-2020070621132500"
      }
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            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "Vital Signs"
          }
        ],
        "text": "Vital Signs"
      }
    ],
    "code": {
      "coding": [
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          "code": "703535",
          "display": "Temperature Axillary",
          "userSelected": true
        },
        {
          "system": "http://loinc.org",
          "code": "8328-7",
          "display": "Axillary temperature"
        },
        {
          "system": "http://loinc.org",
          "code": "8310-5",
          "display": "Body temperature"
        }
      ],
      "text": "Temperature Axillary"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
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    "issued": "2020-07-06T21:13:23.000Z",
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    "extension": [
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        "valueCodeableConcept": {
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              "code": "OP",
              "display": "ordering provider"
            }
          ],
          "text": "ordering provider"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/1994008"
  },
  {
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              "code": "LA",
              "display": "legal authenticator"
            }
          ],
          "text": "legal authenticator"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
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        "valueCodeableConcept": {
          "coding": [
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              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "PPRF",
              "display": "primary performer"
            }
          ],
          "text": "primary performer"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/1974008"
  }
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"valueQuantity": {
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  "unit": "degC",
  "system": "http://unitsofmeasure.org",
  "code": "Cel"
},
"interpretation": [
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    "coding": [
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        "code": "211",
        "display": "LOW",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
        "code": "L",
        "display": "Low"
      }
    ],
    "text": "LOW"
  }
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"referenceRange": [
  {
    "low": {
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      "unit": "degC",
      "system": "http://unitsofmeasure.org",
      "code": "Cel"
    },
    "high": {
      "value": 36.7,
      "unit": "degC",
      "system": "http://unitsofmeasure.org",
      "code": "Cel"
    },
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        "type": {
          "coding": [
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              "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
              "code": "normal",
              "display": "Normal Range"
            }
          ],
          "text": "Normal Range"
        }
      }
    ]
  }
},
{
  "fullUrl": "https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-12724066-o495740014",
  "resource": {
    "resourceType": "Observation",
    "id": "SH-12724066-o495740014",
    "meta": {
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      "lastUpdated": "2024-12-17T08:52:31.000Z"
    },
    "text": {
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      "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Code</b>: History of
Occupation</p><p><b>Effective Date</b>: Apr 30, 2020  8:00 P.M. CDT</p><p><b>Result</b>: Bank President [Chief
Executives]</p><p><b>Components</b>:</p><dl><dd><b>Code</b>: History of Occupation Industry</dd><dd><b>Result</b>:
Accountants' associations [Professional Organizations ]</dd></dl></div>"
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "11341-5"
        }
      ],
      "text": "History of Occupation"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "effectivePeriod": {
      "start": "2020-05-01T01:00:00.000Z",
      "end": "2021-07-01T01:00:00.000Z"
    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/PHOccupationalDataForHealthODH",
          "code": "11-1011.00.003173",
          "display": "Bank President [Chief Executives]"
        }
      ],
      "text": "Bank President [Chief Executives]"
    },
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "86188-0"
            }
          ],
          "text": "History of Occupation Industry"
        },
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/PHOccupationalDataForHealthODH",
              "code": "813920.000034",
              "display": "Accountants' associations [Professional Organizations ]"
            }
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```
          ],
          "text": "Accountants' associations [Professional Organizations ]"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-206763699",
  "resource": {
    "resourceType": "Observation",
    "id": "L-206763699",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2024-07-18T14:54:40.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Patient Id</b>:
12981145</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Code</b>: Pregnancy
Status</p><p><b>Result</b>: Not pregnant</p><p><b>Effective Date</b>: Jul 18, 2024  2:54 P.M. UTC</p></div>"
    },
    "identifier": [
      {
        "system": "https://fhir.cerner.com/ceuuid",
        "value": "CE87caf4b7-9397-4667-9897-702218017c9e-206763699-2024071814544100"
      }
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    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History",
            "userSelected": false
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
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          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
          "code": "704785",
          "display": "Pregnancy Status",
          "userSelected": true
        },
        {
          "system": "http://loinc.org",
          "code": "82810-3",
          "display": "Pregnancy status",
          "userSelected": false
        }
      ],
      "text": "Pregnancy Status"
    },
    "subject": {
      "reference": "Patient/12981145"
    },
    "encounter": {
      "reference": "Encounter/98082140"
    },
    "effectiveDateTime": "2024-07-18T14:54:00.000Z",
    "issued": "2024-07-18T14:54:22.000Z",
    "performer": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
                }
              ],
              "text": "primary performer"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
          }
        ],
        "reference": "Practitioner/12780979"
      }
```

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    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "60001007",
          "display": "Not pregnant (finding)",
          "userSelected": false
        }
      ],
      "text": "Not pregnant"
    }
  }
},
{
  "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-206763697",
  "resource": {
    "resourceType": "Observation",
    "id": "L-206763697",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2024-07-18T14:54:40.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Patient Id</b>:
12981145</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social History</p><p><b>Code</b>: Pregnancy
Intention</p><p><b>Result</b>: Wants to become pregnant</p><p><b>Effective Date</b>: Jul 18, 2024  2:54 P.M.
UTC</p></div>"
    },
    "identifier": [
      {
        "system": "https://fhir.cerner.com/ceuuid",
        "value": "CE87caf4b7-9397-4667-9897-702218017c9e-206763697-2024071814544100"
      }
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    "status": "final",
    "category": [
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        "coding": [
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            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History",
            "userSelected": false
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
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          "code": "2575443883",
          "display": "Pregnancy Intention",
          "userSelected": true
        },
        {
          "system": "http://loinc.org",
          "code": "86645-9",
          "display": "Pregnancy intention in the next year - Reported",
          "userSelected": false
        }
      ],
      "text": "Pregnancy Intention"
    },
    "subject": {
      "reference": "Patient/12981145"
    },
    "encounter": {
      "reference": "Encounter/98082140"
    },
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    "issued": "2024-07-18T14:54:22.000Z",
    "performer": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
                }
              ],
              "text": "primary performer"
            },
```

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            "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
          }
        ],
        "reference": "Practitioner/12780979"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "454411000124108",
          "display": "Wants to become pregnant (finding)",
          "userSelected": false
        }
      ],
      "text": "Wants to become pregnant"
    }
  }
},
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-197294976",
  "resource": {
    "resourceType": "Observation",
    "id": "L-197294976",
    "meta": {
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      "lastUpdated": "2020-08-04T21:56:16.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Observation</b></p><p><b>Patient Id</b>:
12742837</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Laboratory</p><p><b>Code</b>: Coronavirus
COVID-19</p><p><b>Result</b>: Detected</p><p><b>Interpretation</b>: Abnormal</p><p><b>Effective Date</b>: Aug  3,
2020  4:00 P.M. UTC</p><p><b>Reference Range</b>: Not Detected</p></div>"
    },
    "identifier": [
      {
        "system": "https://fhir.cerner.com/ceuuid",
        "value": "CE87caf4b7-9397-4667-9897-702218017c9e-197294976-2020080316283100"
      }
    ],
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      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "laboratory",
            "display": "Laboratory",
            "userSelected": false
          }
        ],
        "text": "Laboratory"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
          "code": "2572358253",
          "display": "Coronavirus COVID-19",
          "userSelected": true
        },
        {
          "system": "http://loinc.org",
          "code": "94500-6",
          "display": "SARS-CoV-2 (COVID-19) RNA [Presence] in Respiratory specimen by NAA with probe detection",
          "userSelected": false
        }
      ],
      "text": "Coronavirus COVID-19"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/97954258"
    },
    "effectiveDateTime": "2020-08-03T16:00:00.000Z",
    "issued": "2020-08-03T16:28:26.000Z",
    "performer": [
      {
        "extension": [
          {
```

```
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "LA",
              "display": "legal authenticator"
            }
          ],
          "text": "legal authenticator"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "PPRF",
              "display": "primary performer"
            }
          ],
          "text": "primary performer"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/12736052"
  },
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0912",
              "code": "OP",
              "display": "Ordering Provider"
            }
          ],
          "text": "Ordering Provider"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/4122622"
  }
],
"valueCodeableConcept": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "260373001",
      "display": "Detected (qualifier value)",
      "userSelected": false
    },
    {
      "system": "http://snomed.info/sct",
      "code": "260373001",
      "display": "Detected (qualifier value)",
      "userSelected": false
    },
    {
      "system": "http://snomed.info/sct",
      "code": "260373001",
      "display": "Detected (qualifier value)",
      "userSelected": false
    },
    {
      "system": "http://snomed.info/sct",
      "code": "260373001",
      "display": "Detected (qualifier value)",
      "userSelected": false
    },
    {
      "system": "http://snomed.info/sct",
      "code": "260373001",
      "display": "Detected (qualifier value)",
      "userSelected": false
    }
  ],
  "text": "Detected"
},
"interpretation": [
  {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52",
        "code": "201",
        "display": "ABN",
```

```
            "userSelected": true
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
            "code": "A",
            "display": "Abnormal",
            "userSelected": false
          }
        ],
        "text": "Abnormal"
      }
    ],
    "specimen": {
      "reference": "Specimen/275667708"
    },
    "referenceRange": [
      {
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
              "code": "normal",
              "display": "Normal Range"
            }
          ],
          "text": "Normal Range"
        },
        "text": "Not Detected"
      }
    ]
  }
},
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SA-123456-123456",
  "resource": {
    "resourceType": "Observation",
    "id": "SA-123456-123456",
    "meta": {
        "lastUpdated": "2020-06-11T04:05:33.000Z"
    },
    "text": {
        "status": "generated",
        "div": "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: SDOH, Survey</p><p><b>Issued Date</b>: Jun 11, 2020
4:05 A.M. UTC</p></div>"
    },
    "status": "final",
    "category": [
        {
            "coding": [
                {
                    "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
                    "code": "sdoh",
                    "display": "sdoh"
                }
            ],
            "text": "sdoh"
        },
        {
            "coding": [
                {
                    "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                    "code": "survey",
                    "display": "Survey"
                }
            ],
            "text": "Survey"
        }
    ],
    "code": {
        "coding": [
            {
                "system": "https://fhir.cerner.com/91fd1841-fb25-42f3-a48e-1264a2c6b72a/codeSet/72",
                "userSelected": true
            },
            {
                "system": "http://loinc.org",
                "code": "93025-5"
            }
        ]
    },
     "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/123123123"
    },
    "effectiveDateTime": "2022-06-01T10:39:02Z",
    "issued": "2020-06-11T04:05:33.000Z",
```

```
    "performer": [
      {
        "reference": "Practitioner/15441459"
      }
    ],
    "hasMember": [
        {
            "reference": "Observation/SA-123456-789987"
        }
    ]
  },
  "search": {
      "mode": "match"
  }
},
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SA-123456-789987",
  "resource": {
    "resourceType": "Observation",
    "id": "SA-123456-789987",
    "meta": {
      "versionId": "73080191",
      "lastUpdated": "2020-06-11T04:05:33.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: SDOH, Survey</p><p><b>Issued Date</b>: Jun 11, 2020
4:05 A.M. UTC</p></div>"
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ],
        "text": "Survey"
      },
      {
        "coding": [
          {
           "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ],
        "text": "Survey"
      }
    ],
    "code": {
      "coding": [
        {
            "system": "https://fhir.cerner.com/91fd1841-fb25-42f3-a48e-1264a2c6b72a/codeSet/72",
            "userSelected": true
        },
        {
            "system": "http://loinc.org",
            "code": "93025-5"
        }
      ],
      "text": "CM Preferred Spoken Language"
    },
    "subject": {
      "reference": "Patient/12724066"
    },
    "encounter": {
      "reference": "Encounter/123123123"
    },
    "effectiveDateTime": "2022-06-01T10:39:02Z",
    "issued": "2020-06-11T04:05:33.000Z",
    "performer": [
      {
        "reference": "Practitioner/15441459"
      }
    ],
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "373067005",
                "display": "No (qualifier value)"
            }
        ],
        "text": "No"
    }
```

```
      }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-laboratory-response Response

Example response containing confidential laboratory observations.

Example Request:

```
GET https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066&category=laboratory&_count=2
```

Example Response:


```
  {
    "resourceType": "Bundle",
    "id": "7eea4bc6-7e9b-4e06-a7d3-699926375318",
    "type": "searchset",
    "link": [
      {
        "relation": "self",
        "url": "https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066&category=laboratory&_count=2"
      },
      {
        "relation": "next",
        "url": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-
    0e701af7583d/Observation?patient=12724066&category=laboratory&_count=2&-
    pageContext=eJxljdEKwiAYRt_lv03BLdtk0E0FNbqKHkD-
    Nq1AdE2tYPjuWVeD7j4-zuFM0LloAzQlgV5pjCbvCbo4ejdCA9vjgy_37Xnzup_4cFg8d-0aCNzQS6vemQ1jVAQGvCrZORt-
    HxQCa850SStkgnJWCCpWjFEuRFkrrSrkPSQCPmSvlwYv86iNxswTGo3_a3yhlD6vGz4p&-pageDirection=NEXT"
      }
    ],
    "entry": [
      {
        "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-208846149",
        "resource": {
          "resourceType": "Observation",
          "id": "L-208846149",
          "meta": {
            "versionId": "1",
            "lastUpdated": "2025-06-06T18:06:19.000Z",
            "security":
              [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                  "code": "PRS",
                  "display": "patient requested information sensitivity",
                  "userSelected": false
                },
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                  "code": "PHY",
                  "display": "physician requested information sensitivity",
                  "userSelected": false
                }
              ]
          },
          "text": {
            "status": "generated",
            "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Meta Security</b>:
    patient requested information sensitivity, physician requested information sensitivity</p><p><b>Patient Id</b>:
    12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Laboratory</p><p><b>Code</b>: Estimated Creatinine
    Clearance</p><p><b>Result</b>: 305.03 mL/min</p><p><b>Effective Date</b>: Jun  6, 2025  6:06 P.M. UTC</p></div>"
          },
          "identifier": [
            {
              "system": "https://fhir.cerner.com/ceuuid",
              "value": "CE87caf4b7-9397-4667-9897-702218017c9e-208846149-2025060618062000"
            }
          ],
          "status": "final",
          "category": [
            {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                  "code": "laboratory",
                  "display": "Laboratory",
                  "userSelected": false
                }
              ],
```

```
          "text": "Laboratory"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
            "code": "20004425",
            "display": "Estimated Creatinine Clearance",
            "userSelected": true
          },
          {
            "system": "http://loinc.org",
            "code": "35592-5",
            "display": "Creatinine renal clearance/1.73 sq M.predicted by Cockcroft-Gault formula, BSA formula",
            "userSelected": false
          }
        ],
        "text": "Estimated Creatinine Clearance"
      },
      "subject": {
        "reference": "Patient/12724066"
      },
      "encounter": {
        "reference": "Encounter/97953477"
      },
      "effectiveDateTime": "2025-06-06T18:06:19.000Z",
      "issued": "2025-06-06T18:06:19.000Z",
      "performer": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                    "code": "LA",
                    "display": "legal authenticator"
                  }
                ],
                "text": "legal authenticator"
              },
              "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                    "code": "PPRF",
                    "display": "primary performer"
                  }
                ],
                "text": "primary performer"
              },
              "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
            }
          ],
          "reference": "Practitioner/1"
        }
      ],
      "valueQuantity": {
        "value": 305.03,
        "unit": "mL/min",
        "system": "http://unitsofmeasure.org",
        "code": "mL/min"
      }
    },
    "search": {
      "mode": "match"
    }
  },
  {
    "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-207035434",
    "resource": {
      "resourceType": "Observation",
      "id": "L-207035434",
      "meta": {
        "versionId": "1",
        "lastUpdated": "2024-12-16T14:00:43.000Z"
      },
      "text": {
        "status": "generated",
        "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Laboratory</p><p><b>Code</b>: Potassium
Level</p><p><b>Result</b>: 0.7 g/dL</p><p><b>Effective Date</b>: Dec 16, 2024  1:59 P.M. UTC</p><p><b>Reference
Range</b>: &lt;=0.6 g/dL</p></div>"
      },
      "identifier": [
```

```
            {
              "system": "https://fhir.cerner.com/ceuuid",
              "value": "CE87caf4b7-9397-4667-9897-702218017c9e-207035434-2024121614004300"
            }
          ],
          "status": "final",
          "category": [
            {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                  "code": "laboratory",
                  "display": "Laboratory",
                  "userSelected": false
                }
              ],
              "text": "Laboratory"
            }
          ],
          "code": {
            "coding": [
              {
                "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
                "code": "21704910",
                "display": "Potassium Level",
                "userSelected": true
              },
              {
                "system": "http://loinc.org",
                "code": "2823-3",
                "display": "Potassium [Moles/volume] in Serum or Plasma",
                "userSelected": false
              }
            ],
            "text": "Potassium Level"
          },
          "subject": {
            "reference": "Patient/12724066"
          },
          "encounter": {
            "reference": "Encounter/97954225"
          },
          "effectiveDateTime": "2024-12-16T13:59:38.000Z",
          "valueQuantity": {
            "value": 0.7,
            "unit": "g/dL",
            "system": "http://unitsofmeasure.org",
            "code": "g/dL"
          },
          "referenceRange": [
            {
              "high": {
                "value": 0.6,
                "unit": "g/dL",
                "system": "http://unitsofmeasure.org",
                "code": "g/dL"
              },
              "type": {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
                    "code": "normal",
                    "display": "Normal Range"
                  }
                ],
                "text": "Normal Range"
              }
            }
          ]
        },
        "search": {
          "mode": "match"
        }
      }
    ]
  }
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-social-history-response Response

Example response containing confidential social-history observations.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066&category=social-
    history
```

Example Response:


```
  {
    "resourceType": "Bundle",
    "id": "55b84239-2185-4403-a59b-e9588366bdf9",
    "type": "searchset",
    "link":
    [
      {
        "relation": "self",
        "url": "https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12724066&category=social-history"
      }
    ],
    "entry":
    [
      {
        "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-93-76551113",
        "resource":
          {
            "resourceType": "Observation",
            "id": "SH-93-76551113",
            "meta":
              {
                "versionId": "76551113",
                "lastUpdated": "2025-09-16T16:52:46Z",
                "security":
                  [
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                      "code": "PROCESSINLINELABEL",
                      "display": "process inline security label"
                    }
                  ]
              },
            "text":
              {
                "status": "generated",
                "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Meta Security</b>:
    process inline security label</p><p><b>Patient Id</b>: 12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>:
    Social History</p><p><b>Code</b>: Tobacco</p><p><b>Issued Date</b>: Sep 16, 2025  4:52 P.M. UTC</p></div>"
              },
            "status": "final",
            "category":
              [
                {
                  "coding":
                    [
                      {
                        "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                        "code": "social-history",
                        "display": "Social History"
                      }
                    ],
                  "text": "Social History"
                }
              ],
            "code":
              {
                "coding":
                  [
                    {
                      "system": "http://loinc.org",
                      "code": "88028-6",
                      "display": "Tobacco use panel"
                    }
                  ],
                "text": "Tobacco"
              },
            "subject":
              {
                "reference": "Patient/12724066"
              },
            "effectiveDateTime": "2025-09-16T16:52:46Z",
            "issued": "2025-09-16T16:52:46Z",
            "performer":
              [
                {
                  "reference": "Practitioner/1",
                  "display": "SYSTEM, SYSTEM Cerner"
                }
              ],
            "hasMember":
              [
                {
                  "extension":
                    [
```

```
                {
                  "valueCoding":
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                      "code": "PRS",
                      "display": "patient requested information sensitivity",
                      "userSelected": false
                    },
                  "url": "http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-inline-sec-
label"
                }
              ],
            "reference": "Observation/SH-76551113-q4625825"
          }
        ]
    },
  "search":
    {
      "mode": "match"
    }
},
{
  "fullUrl": "https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-76551113-q4625825",
  "resource":
    {
      "resourceType": "Observation",
      "id": "SH-76551113-q4625825",
      "meta":
        {
          "versionId": "76551113",
          "lastUpdated": "2025-09-16T16:52:46Z",
          "security":
            [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                "code": "PRS",
                "display": "patient requested information sensitivity",
                "userSelected": false
              }
            ]
        },
      "text":
        {
          "status": "generated",
          "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Meta Security</b>:
patient requested information sensitivity</p><p><b>Patient Id</b>: 12724066</p><p><b>Status</b>:
Final</p><p><b>Categories</b>: Social History</p><p><b>Question</b>: SHX Tobacco use</p><p><b>Response</b>: Never
(less than 100 in lifetime)</p><p><b>Issued Date</b>: Sep 16, 2025  4:52 P.M. UTC</p></div>"
        },
      "status": "final",
      "category":
        [
          {
            "coding":
              [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                  "code": "social-history",
                  "display": "Social History"
                }
              ],
            "text": "Social History"
          }
        ],
      "code":
        {
          "coding":
            [
              {
                "system": "http://loinc.org",
                "code": "72166-2",
                "display": "Tobacco smoking status",
                "userSelected": false
              }
            ],
          "text": "SHX Tobacco use"
        },
      "subject":
        {
          "reference": "Patient/12724066"
        },
      "effectiveDateTime": "2025-09-16T16:52:46Z",
      "issued": "2025-09-16T16:52:46Z",
      "performer":
        [
          {
            "reference": "Practitioner/1",
            "display": "SYSTEM, SYSTEM Cerner"
```

```
              }
            ],
          "valueCodeableConcept":
            {
              "coding":
                [
                  {
                    "system": "http://loinc.org",
                    "code": "LA6270-8"
                  }
                ],
              "text": "Never (less than 100 in lifetime)"
            }
        },
      "search":
        {
          "mode": "match"
        }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-revinclude Response

This operation supports the following authorization types:

* Provider

* System

Example Request:

```
https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12457977&category=vital-
    signs&_revinclude=Provenance:target
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "79c71e5b-2793-4ace-bc5e-190e38784e80",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation?patient=12457977&category=vital-
    signs&_revinclude=Provenance%3Atarget"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-196190669",
      "resource": {
        "resourceType": "Observation",
        "id": "VS-196190669",
        "meta": {
          "versionId": "2",
          "lastUpdated": "2017-02-25T02:41:03.000Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Observation</b></p><p><b>Patient
    Id</b>:12457977</p><p><b>Status</b>: Entered in Error</p><p><b>Categories</b>: Vital Signs</p><p><b>Code</b>:
    Temperature Temporal Artery</p><p><b>Result</b>: 39 degC</p><p><b>Interpretation</b>: Normal</p><p><b>Effective
    Date</b>: Feb 24, 2017  8:37 P.M. CST</p><p><b>Reference Range</b>: 36-38 degC</p></div>"
        },
        "identifier": [
          {
            "system": "https://fhir.cerner.com/ceuuid",
            "value": "CE87caf4b7-9397-4667-9897-702218017c9e-196190669-2017022502373000"
          }
        ],
        "status": "entered-in-error",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "vital-signs",
                "display": "Vital Signs"
              }
            ],
            "text": "Vital Signs"
          }
```

```
],
"code": {
  "coding": [
    {
      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
      "code": "4157752",
      "display": "Temperature Temporal Artery",
      "userSelected": true
    },
    {
      "system": "http://loinc.org",
      "code": "75539-7",
      "display": "Body temperature - Temporal artery"
    },
    {
      "system": "http://loinc.org",
      "code": "8310-5",
      "display": "Body temperature"
    }
  ],
  "text": "Temperature Temporal Artery"
},
"subject": {
  "reference": "Patient/12457977"
},
"encounter": {
  "reference": "Encounter/97697398"
},
"effectiveDateTime": "2017-02-25T02:37:00.000Z",
"issued": "2017-02-25T02:41:03.000Z",
"performer": [
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "LA",
              "display": "legal authenticator"
            }
          ],
          "text": "legal authenticator"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "PPRF",
              "display": "primary performer"
            }
          ],
          "text": "primary performer"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/11638346"
  }
],
"valueQuantity": {
  "value": 39,
  "unit": "degC",
  "system": "http://unitsofmeasure.org",
  "code": "Cel"
},
"interpretation": [
  {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52",
        "code": "214",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
        "code": "N",
        "display": "Normal"
      }
    ]
  }
],
"note": [
  {
    "authorReference": {
      "reference": "Practitioner/11638346"
```

```
        },
        "time": "2017-02-25T02:41:03.000Z",
        "text": "Other\\n"
      }
    ],
    "referenceRange": [
      {
        "low": {
          "value": 36,
          "unit": "degC",
          "system": "http://unitsofmeasure.org",
          "code": "Cel"
        },
        "high": {
          "value": 38,
          "unit": "degC",
          "system": "http://unitsofmeasure.org",
          "code": "Cel"
        },
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
              "code": "normal",
              "display": "Normal Range"
            }
          ],
          "text": "Normal Range"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/vit-4",
  "resource": {
    "resourceType": "Provenance",
    "id": "vit-4",
    "meta": {
      "versionId": "4",
      "lastUpdated": "2020-12-03T16:28:02Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>Observa
tion/VS-196190669</li></ul><p><b>Recorded</b>: Oct 25, 2019  9:57 A.M.
CDT</p><p><b>Agents</b>:</p><dl><dt>Unknown</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>:
Source</dd></dl><p><b>Entity Source</b>: DocumentReference/S-100</p></div>"
    },
    "target": [
      {
        "reference": "Observation/VS-196190669"
      }
    ],
    "recorded": "2019-10-25T14:57:43Z",
    "agent": [
      {
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
              "code": "author"
            }
          ],
          "text": "Author"
        },
        "role": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                "code": "SOURCE"
              }
            ],
            "text": "Source"
          }
        ],
        "who": {
          "reference": "Organization/101",
          "display": "Unknown"
        }
      }
    ],
    "entity": [
      {
        "role": "source",
```

```
            "what": {
              "reference": "DocumentReference/S-100"
            }
          }
        ]
      },
      "search": {
        "mode": "include"
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get an observation by ID

File: r4-rest-api-v32/op-observation-id-get.html

Retrieves an individual observation by the associated observation ID.

Note: Patient authorization read requests work similarly to patient authorization search requests.

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-197292857
```

Example Response:

```
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-197292857",
  "resource": {
    "resourceType": "Observation",
    "id": "VS-197292857",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2020-07-06T21:13:24.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Observation</b></p><p><b>Patient Id</b>:
    12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Vital Signs</p><p><b>Code</b>: Temperature
    Axillary</p><p><b>Result</b>: 27 degC</p><p><b>Interpretation</b>: LOW</p><p><b>Effective Date</b>: Jul  6, 2020
    9:11 P.M. UTC</p><p><b>Reference Range</b>: 35.2-36.7 degC</p></div>"
    },
    "identifier": [
      {
        "system": "https://fhir.cerner.com/ceuuid",
        "value": "CE87caf4b7-9397-4667-9897-702218017c9e-197292857-2020070621132500"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
```

```
        "display": "Vital Signs"
      }
    ],
    "text": "Vital Signs"
  }
],
"code": {
  "coding": [
    {
      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
      "code": "703535",
      "display": "Temperature Axillary",
      "userSelected": true
    },
    {
      "system": "http://loinc.org",
      "code": "8328-7",
      "display": "Axillary temperature"
    },
    {
      "system": "http://loinc.org",
      "code": "8310-5",
      "display": "Body temperature"
    }
  ],
  "text": "Temperature Axillary"
},
"subject": {
  "reference": "Patient/12724066"
},
"encounter": {
  "reference": "Encounter/97953477"
},
"effectiveDateTime": "2020-07-06T21:11:00.000Z",
"issued": "2020-07-06T21:13:23.000Z",
"performer": [
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://terminology.hl7.org/CodeSystem-v2-0443.html",
              "code": "OP",
              "display": "ordering provider"
            }
          ],
          "text": "ordering provider"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/1994008"
  },
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "LA",
              "display": "legal authenticator"
            }
          ],
          "text": "legal authenticator"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "PPRF",
              "display": "primary performer"
            }
          ],
          "text": "primary performer"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
    ],
    "reference": "Practitioner/1974008"
  }
],
"valueQuantity": {
  "value": 27,
  "unit": "degC",
```

```
      "system": "http://unitsofmeasure.org",
      "code": "Cel"
    },
    "interpretation": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52",
            "code": "211",
            "display": "LOW",
            "userSelected": true
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
            "code": "L",
            "display": "Low"
          }
        ],
        "text": "LOW"
      }
    ],
    "referenceRange": [
      {
        "low": {
          "value": 35.2,
          "unit": "degC",
          "system": "http://unitsofmeasure.org",
          "code": "Cel"
        },
        "high": {
          "value": 36.7,
          "unit": "degC",
          "system": "http://unitsofmeasure.org",
          "code": "Cel"
        },
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
              "code": "normal",
              "display": "Normal Range"
            }
          ],
          "text": "Normal Range"
        }
      }
    ]
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-laboratory-response Response

Reading confidential laboratory observations.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-208846149
```

Example Response:


```
  {
    "resourceType": "Observation",
    "id": "L-208846149",
    "meta":
    {
      "versionId": "1",
      "lastUpdated": "2025-06-06T18:06:19.000Z",
      "security":
        [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "PRS",
            "display": "patient requested information sensitivity",
            "userSelected": false
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "PHY",
            "display": "physician requested information sensitivity",
            "userSelected": false
          }
        ]
    },
```

```
"text":
{
  "status": "generated",
  "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Meta Security</b>: patient
requested information sensitivity, physician requested information sensitivity</p><p><b>Patient Id</b>:
12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Laboratory</p><p><b>Code</b>: Estimated Creatinine
Clearance</p><p><b>Result</b>: 305.03 mL/min</p><p><b>Effective Date</b>: Jun  6, 2025  6:06 P.M. UTC</p></div>"
},
"identifier":
[
  {
    "system": "https://fhir.cerner.com/ceuuid",
    "value": "CE87caf4b7-9397-4667-9897-702218017c9e-208846149-2025060618062000"
  }
],
"status": "final",
"category":
[
  {
    "coding":
      [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory",
          "userSelected": false
        }
      ],
    "text": "Laboratory"
  }
],
"code":
{
  "coding":
    [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
        "code": "20004425",
        "display": "Estimated Creatinine Clearance",
        "userSelected": true
      },
      {
        "system": "http://loinc.org",
        "code": "35592-5",
        "display": "Creatinine renal clearance/1.73 sq M.predicted by Cockcroft-Gault formula, BSA formula",
        "userSelected": false
      }
    ],
  "text": "Estimated Creatinine Clearance"
},
"subject":
{
  "reference": "Patient/12724066"
},
"encounter":
{
  "reference": "Encounter/97953477"
},
"effectiveDateTime": "2025-06-06T18:06:19.000Z",
"issued": "2025-06-06T18:06:19.000Z",
"performer":
[
  {
    "extension":
      [
        {
          "valueCodeableConcept":
            {
              "coding":
                [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                    "code": "LA",
                    "display": "legal authenticator"
                  }
                ],
              "text": "legal authenticator"
            },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        },
        {
          "valueCodeableConcept":
            {
              "coding":
                [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                    "code": "PPRF",
                    "display": "primary performer"
```

```
                    }
                  ],
                "text": "primary performer"
              },
            "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
          }
        ],
      "reference": "Practitioner/1"
    }
  ],
  "valueQuantity":
  {
    "value": 305.03,
    "unit": "mL/min",
    "system": "http://unitsofmeasure.org",
    "code": "mL/min"
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-socialhistory-response Response

Reading confidential social history observations.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/SH-93-76551113
```

Example Response:


```
{
  "resourceType": "Observation",
  "id": "SH-93-76551113",
  "meta":
  {
    "versionId": "76551113",
    "lastUpdated": "2025-09-16T16:52:46Z",
    "security":
      [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
          "code": "PROCESSINLINELABEL",
          "display": "process inline security label"
        }
      ]
  },
  "text":
  {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Observation</b></p><p><b>Meta Security</b>: process
  inline security label</p><p><b>Patient Id</b>: 12724066</p><p><b>Status</b>: Final</p><p><b>Categories</b>: Social
  History</p><p><b>Code</b>: Tobacco</p><p><b>Issued Date</b>: Sep 16, 2025  4:52 P.M. UTC</p></div>"
  },
  "status": "final",
  "category":
  [
    {
      "coding":
        [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
      "text": "Social History"
    }
  ],
  "code":
  {
    "coding":
      [
        {
          "system": "http://loinc.org",
          "code": "88028-6",
          "display": "Tobacco use panel"
        }
      ],
    "text": "Tobacco"
  },
  "subject":
  {
    "reference": "Patient/12724066"
  },
```

```
    "effectiveDateTime": "2025-09-16T16:52:46Z",
    "issued": "2025-09-16T16:52:46Z",
    "performer":
    [
      {
        "reference": "Practitioner/1",
        "display": "SYSTEM, SYSTEM Cerner"
      }
    ],
    "hasMember":
    [
      {
        "extension":
          [
            {
              "valueCoding":
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                  "code": "PRS",
                  "display": "patient requested information sensitivity",
                  "userSelected": false
                },
              "url": "http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-inline-sec-label"
            }
          ],
        "reference": "Observation/SH-76551113-q4625825"
      }
    ]
  }
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Update an observation

File: r4-rest-api-v32/op-observation-id-put.html

Updates an existing observation.

Note:

* Currently, only laboratory, imaging, and vital-signs are supported.

* Both read and write scopes are required.

* Updates to patient and category are not supported.

* Components are only supported for vital-sign blood pressure and pulse oximetry profiles.

* The category of is supported to update a limited number of observations. If the category of is provided with an other than the LOINC codes below, the transaction returns a 422 (Unprocessable Entity) status code.

imaging

imaging

Observation.code

* LOINC 96912-1 - Calculated CT global noise

* LOINC 96913-9 - Calculated CT size-adjusted dose

* LOINC 96914-7 - CT dose and image quality category

* See FHIR Update for additional details about update operations.

Request

Supported Media Types

* application/fhir+json

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : ObservationUpdateBody

Type:

object

Show Source

* ID(required): string The logical ID of the resource to update. The ID value must match the Observation/ value. Example: p123412

* identifier: object identifier A unique identifier assigned to the observation. Note: Identifier is required for Vital Signs and it cannot be modified. Identifier is optional for Laboratory.


```
{
    "required":[
        "ID"
    ],
    "properties":{
        "ID":{
            "type":"string",
            "description":"The logical ID of the resource to update. The ID value must match the Observation/<ID>
    value.",
            "example":"p123412"
        },
        "identifier":{
            "type":"object",
            "description":"A unique identifier assigned to the observation.<br>\n\n<b>Note:</b> Identifier is required
    for <code>Vital Signs</code> and it cannot be modified. Identifier is optional for <code>Laboratory</code>.\n",
            "example":"{\n  \"system\":\"https://fhir.cerner.com/ceuuid\",\n
    \"value\":\"CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900\"\n}\n"
        }
    },
    "allOf":[
        {
            "$ref":"#/definitions/ObservationCreateBody"
        },
        {
            "type":"object"
        }
    ],
    "example":"{\n  \"resourceType\": \"Observation\",\n  \"id\": \"VS-197356031\",\n  \"identifier\": [\n    {\n
    \"system\": \"https://fhir.cerner.com/ceuuid\",\n      \"value\":
    \"CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900\"\n    }\n  ],\n  \"status\": \"corrected\",\n
    \"category\": [\n    {\n      \"coding\": [\n        {\n          \"system\":
    \"http://terminology.hl7.org/CodeSystem/observation-category\",\n          \"code\": \"vital-signs\",\n
    \"display\": \"Vital Signs\"\n        }\n      ],\n      \"text\": \"Vital Signs\"\n    }\n  ],\n  \"code\": {\n
    \"coding\": [\n      {\n        \"system\":
    \"https://fhir.cerner.com/d242a518-4074-4bd4-a3a6-adfe0c5c1c51/codeSet/72\",\n        \"code\": \"703558\",\n
            \"display\": \"Temperature Oral\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\":
    \"Temperature Oral\"\n  },\n  \"subject\": {\n    \"reference\": \"Patient/12457979\"\n  },\n  \"encounter\": {\n
        \"reference\": \"Encounter/97787491\"\n  },\n  \"effectiveDateTime\": \"2020-07-04T07:15:00.000Z\",\n
    \"issued\": \"2021-05-06T20:07:14.000Z\",\n  \"performer\": [\n    {\n      \"extension\": [\n        {\n
    \"valueCodeableConcept\": {\n            \"coding\": [\n              {\n                \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n                \"code\": \"LA\",\n
    \"display\": \"legal authenticator\"\n              }\n            ],\n            \"text\": \"legal
    authenticator\"\n          },\n          \"url\": \"http://hl7.org/fhir/StructureDefinition/event-
    performerFunction\"\n        }\n      ],\n      \"reference\": \"Practitioner/1\"\n    }\n  ],\n  \"valueQuantity\":
     {\n    \"value\": 123,\n    \"unit\": \"mmHg\",\n    \"system\": \"http://unitsofmeasure.org\",\n    \"code\":
    \"mm[Hg]\"\n  }\n}\n"
}
```


Match All

Show Source

* object ObservationCreateBody

* object ObservationUpdateBody-allOf[1]


```
{
    "required":[
        "ID"
    ],
    "properties":{
        "ID":{
            "type":"string",
            "description":"The logical ID of the resource to update. The ID value must match the Observation/<ID>
    value.",
            "example":"p123412"
        },
        "identifier":{
            "type":"object",
            "description":"A unique identifier assigned to the observation.<br>\n\n<b>Note:</b> Identifier is required
    for <code>Vital Signs</code> and it cannot be modified. Identifier is optional for <code>Laboratory</code>.\n",
            "example":"{\n  \"system\":\"https://fhir.cerner.com/ceuuid\",\n
    \"value\":\"CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900\"\n}\n"
        }
    },
    "allOf":[
        {
            "$ref":"#/definitions/ObservationCreateBody"
        },
        {
            "type":"object"
        }
    ],
    "example":"{\n  \"resourceType\": \"Observation\",\n  \"id\": \"VS-197356031\",\n  \"identifier\": [\n    {\n
    \"system\": \"https://fhir.cerner.com/ceuuid\",\n      \"value\":
```

```
    \"CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900\"\n    }\n  ],\n  \"status\": \"corrected\",\n
    \"category\": [\n    {\n      \"coding\": [\n        {\n          \"system\":
    \"http://terminology.hl7.org/CodeSystem/observation-category\",\n          \"code\": \"vital-signs\",\n
    \"display\": \"Vital Signs\"\n        }\n      ],\n      \"text\": \"Vital Signs\"\n    }\n  ],\n  \"code\": {\n
    \"coding\": [\n      {\n        \"system\":
    \"https://fhir.cerner.com/d242a518-4074-4bd4-a3a6-adfe0c5c1c51/codeSet/72\",\n        \"code\": \"703558\",\n
           \"display\": \"Temperature Oral\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\":
    \"Temperature Oral\"\n  },\n  \"subject\": {\n    \"reference\": \"Patient/12457979\"\n  },\n  \"encounter\": {\n
        \"reference\": \"Encounter/97787491\"\n  },\n  \"effectiveDateTime\": \"2020-07-04T07:15:00.000Z\",\n
    \"issued\": \"2021-05-06T20:07:14.000Z\",\n  \"performer\": [\n    {\n      \"extension\": [\n        {\n
    \"valueCodeableConcept\": {\n            \"coding\": [\n              {\n                \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n                \"code\": \"LA\",\n
    \"display\": \"legal authenticator\"\n              }\n            ],\n            \"text\": \"legal
    authenticator\"\n          },\n          \"url\": \"http://hl7.org/fhir/StructureDefinition/event-
    performerFunction\"\n        }\n      ],\n      \"reference\": \"Practitioner/1\"\n    }\n  ],\n  \"valueQuantity\":
     {\n    \"value\": 123,\n    \"unit\": \"mmHg\",\n    \"system\": \"http://unitsofmeasure.org\",\n    \"code\":
    \"mm[Hg]\"\n  }\n}\n"
}
```


Example:


```
{
  "resourceType": "Observation",
  "id": "VS-197356031",
  "identifier": [
    {
      "system": "https://fhir.cerner.com/ceuuid",
      "value": "CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900"
    }
  ],
  "status": "corrected",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/d242a518-4074-4bd4-a3a6-adfe0c5c1c51/codeSet/72",
        "code": "703558",
        "display": "Temperature Oral",
        "userSelected": true
      }
    ],
    "text": "Temperature Oral"
  },
  "subject": {
    "reference": "Patient/12457979"
  },
  "encounter": {
    "reference": "Encounter/97787491"
  },
  "effectiveDateTime": "2020-07-04T07:15:00.000Z",
  "issued": "2021-05-06T20:07:14.000Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/1"
    }
  ],
  "valueQuantity": {
    "value": 123,
    "unit": "mmHg",
    "system": "http://unitsofmeasure.org",
    "code": "mm[Hg]"
  }
}
```

Nested Schema : identifier

Type:

object

A unique identifier assigned to the observation. Identifier is required for and it cannot be modified. Identifier is optional for .

Note:

Vital Signs

Laboratory

Example:

```
{
  "system":"https://fhir.cerner.com/ceuuid",
  "value":"CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900"
}
```

Nested Schema : ObservationCreateBody

Type:

object

Show Source

category(required): array

category

A code that classifies the general type of observation being made.

Note:

* VitalSigns, Laboratory, and Imaging categories are supported.

* Only one category may be supplied on create.

Example:

```
{
  "category": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/observation-category',
          "code": 'vital-signs',
          "display": 'Vital Signs'
        }
      ],
      "text": 'Vital Signs'
    }
  ]
}
```

code(required): object

code

Describes what was observed. Also called the observation name.

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8331-1'
      }
    ],
    "text": 'Temperature Oral'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8302-2'
      },
      {
        "system": 'http://loinc.org',
        "code": '3137-7'
      }
    ],
    "text": 'Height/Length Measured'
  }
}
```

```
{
  "code": {
    "coding": [
      {
```

```
        "system": 'http://loinc.org',
        "code": '96914-7'
      },
    "text": 'CT dose and image quality category'
  }
}
```

component: array

component

Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations. Component is required to create blood pressure.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8480-6'
      }
    ],
    "text": 'Systolic Blood Pressure Invasive'
  },
  "valueQuantity": {
    "value": 120,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
          "code": 'N'
        }
      ]
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 45.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "high": {
        "value": 75.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

component.code(required): object

component.code

Describes what was observed. Also called the observation name.

Example:

```
{
  "coding": [
    {
      "system": 'http://loinc.org',
      "code": '8480-6'
    }
  ],
  "text": 'Systolic Blood Pressure Invasive'
}
```

component.interpretation: array

component.interpretation

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
   "coding": [
     {
       "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
       "code": 'N'
     }
   ],
   "text": 'NORMAL'
 }
```

component.referenceRange: object

component.referenceRange

Guidance on how to interpret the value by comparison to a normal or recommended range.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "low": {
    "value": 45.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "high": {
    "value": 75.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal'
      }
    ],
    "text": 'Normal Range'
  }
}
```

component.value[x]:

The information determined as a result of making the observation, if the information has a simple value. Supports writing observations with only valueQuantity.

Note:

Example:

```
{
  "value": 120,
  "unit": 'mmHg',
  "system": 'http://unitsofmeasure.org',
  "code": 'mm[Hg]'
}
```

effective[x](required):

The date and time when the observation was made. Only a type of is supported.

Note:

dateTime

Example:

```
{
  "effectiveDateTime": "2020-07-07T20:37:00.000Z"
}
```

encounter: object

encounter

The encounter during which the observation was first made.

Example:

```
{
  "encounter": {
    "reference": "Encounter/631923"
```

```
  }
}
```

interpretation: array

interpretation

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
  "interpretation": [
    {
      "coding": [
        {
          "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',
          "code": '211',
          "display": 'LOW',
          "userSelected": true
        }
      ],
      "text": 'LOW'
    }
  ]
}
```

```
{
  "interpretation": [
    {
      "coding": [
        {
          {
            "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
            "code": 'L',
            "display": 'Low'
          }
        ],
        "text": 'LOW'
      }
    ]
  }
```

issued: string

The instant the observation was made.

Example:

```
{
  "issued": "2020-07-06T21:02:47.000Z"
}
```

note: array

note

Comments about the observation

Example:

```
{
  "note": [
    {
      "authorReference": {
        "reference": 'Practitioner/62955753'
      },
      "time": '2018-08-17T09:55:42.000Z',
      "text": 'Charted at Incorrect Time'
    }
  ]
}
```

performer: array

performer

Limited to "real" people rather than equipment.

Note:

* Must be a reference to a practitioner.

* Vital signs require a performer.

Example:

```
{
"performer": [
    {
        "reference": 'Practitioner/12724064'
      }
  ]
}
```

performer.extension: object

performer.extension

Distinguishes the performer's type of involvement in the event.

Note:

* Vital signs support and codes.

PPRF

* When only one performer is provided, the code must be LA.

* When multiple performers are provided, the code must be LA for one of the performers.

* Laboratory supports LA, PPRF, PRF, OP, and AUTHEN codes.

Example:

```
{
   "extension": [
     {
       "valueCodeableConcept": {
         "coding": [
           {
             "system": 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
             "code": 'LA',
             "display": 'legal authenticator'
           }
         ],
         "text": 'legal authenticator'
       },
       "url": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'
     }
   ],
 }
```

```
{
   "extension": [
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "https://terminology.hl7.org/CodeSystem-v2-0443.html",
                  "code": "OP",
                  "display": "ordering provider"
               }
            ],
            "text": "ordering provider"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
               }
            ],
            "text": "primary performer"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"PRF",
                  "display":"performer"
               }
            ],
            "text":"performer"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"AUTHEN",
                  "display":"authenticator"
               }
            ],
            "text":"authenticator"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
   ]
}
```

referenceRange: array

referenceRange

Provides guidance for interpretating the observation.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:

```
{
  "referenceRange": [
    {
      "low": {
        "value": 35.2,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "high": {
        "value": 36.7,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal',
            "display": 'Normal Range'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

referenceRange.high: object

referenceRange.high

The upper bound of the reference range.

Example:

```
{
  "low": {
    "value": 36.7,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```

referenceRange.low: object

referenceRange.low

The lower bound of the reference range.

Example:

```
{
  "low": {
    "value": 35.2,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```

referenceRange.text: string

The text-based reference range in an observation, which may be used when a quantitative range is not appropriate for an observation.

Example:

```
{
  "text": 'Negative'
}
```

referenceRange.type: object

referenceRange.type

Codes to indicate what part of the targeted reference population it applies to. Only a of normal is supported.

Note:

ReferenceRange.type

Example:

```
{
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal',
        "display": 'Normal Range'
      }
    ],
    "text": 'Normal Range'
  }
}
```

* resourceType(required): string Allowed Values: ["Observation"] The type of the FHIR resource.

status(required): string

Codes identifying the lifecycle stage of an event. The , , and status codes are supported.

Note:

final

amended

corrected

Example:

```
{
  "status": "final"
}
```

subject(required): object

subject

Indicates the patient who the observation record is associated with. Must only include references to a patient.

Note:

Example:

```
{
  "subject": {
    "reference": "Patient/631923"
  }
}
```

value[x](required):

The value of the observation.

Note:

* Vital signs support writing observations with valueQuantity.

* Laboratory supports valueQuantity, valueCodeableConcept, valueString, valueAttachment, and valueDateTime.

* Imaging supports valueQuantity and valueCodeableConcept.

Example:

```
{
  "valueQuantity":
  {
    "value": 162,
    "unit": 'cm',
    "system": 'http://unitsofmeasure.org',
    "code": 'cm'
  }
}
```

```
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": 'http://snomed.info/sct',
        "code": '373066001',
        "display": 'Yes (qualifier value)'
      }
    ],
    "text": 'Yes'
  }
}
```

```
{
  "valueString": '2'
}
```

```
{
  "valueDateTime": '2020-07-04T07:15:00Z'
}
```

```
{
```

```
"type":"object",
"required":[
    "resourceType",
    "status",
    "category",
    "code",
    "subject",
    "effective[x]",
    "value[x]",
    "component.code"
],
"properties":{
    "resourceType":{
        "description":"The type of the FHIR resource.\n",
        "type":"string",
        "enum":[
            "Observation"
        ]
    },
    "status":{
        "description":"Codes identifying the lifecycle stage of an event.<br>\n\n<b>Note:</b> The
<code>final</code>, <code>amended</code>, and <code>corrected</code> status codes are
supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"status\": \"final\"\n}</code></pre>\n",
        "type":"string"
    },
    "category":{
        "description":"A code that classifies the general type of observation being
made.<br>\n\n<b>Note:</b>\n<ul>\n  <li> <code>VitalSigns</code>, <code>Laboratory</code>, and <code>Imaging</code>
categories are supported. </li>\n  <li> Only one category may be supplied on create.
</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"category\": [\n    {\n      \"coding\": [\n        {\n
          \"system\": 'http://terminology.hl7.org/CodeSystem/observation-category',\n          \"code\": 'vital-
signs',\n          \"display\": 'Vital Signs'\n        }\n      ],\n      \"text\": 'Vital Signs'\n    }\n
]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    },
    "code":{
        "description":"Describes what was observed. Also called the observation
name.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8331-1'\n      }\n    ],\n    \"text\": 'Temperature Oral'\n
}\n}</code></pre>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8302-2'\n      },\n      {\n        \"system\": 'http://loinc.org',\n
        \"code\": '3137-7'\n      }\n    ],\n    \"text\": 'Height/Length Measured'\n  }\n}</code></pre>\n\n
<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": 'http://loinc.org',\n        \"code\":
'96914-7'\n      },\n    \"text\": 'CT dose and image quality category'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "subject":{
        "description":"Indicates the patient who the observation record is associated with.<br>\n\n<b>Note:</b> Must
```

```
 only include references to a patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
\"Patient/631923\"\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "encounter":{
        "description":"The encounter during which the observation was first
made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"encounter\": {\n    \"reference\": \"Encounter/631923\"\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "effective[x]":{
        "description":"The date and time when the observation was made.<br>\n\n<b>Note:</b> Only a type of
<code>dateTime</code> is supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"effectiveDateTime\":
\"2020-07-07T20:37:00.000Z\"\n}</code></pre>\n"
    },
    "issued":{
        "description":"The instant the observation was made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"issued\":
\"2020-07-06T21:02:47.000Z\"\n}</code></pre>\n",
        "type":"string"
    },
    "performer":{
        "description":"Limited to \"real\" people rather than equipment.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Must be a
reference to a practitioner.</li>\n  <li>Vital signs require a
performer.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n\"performer\": [\n    {\n        \"reference\":
'Practitioner/12724064'\n      }\n  ]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        }
    },
    "performer.extension":{
        "description":"Distinguishes the performer's type of involvement in the event.<br>\n\n<b>Note:</b>\n<ul>\n
   <li>Vital signs support <code>LA</code> and <code>PPRF</code> codes.\n      <ul>\n         <li>When only one
performer is provided, the code must be <code>LA</code>.</li>\n         <li>When multiple performers are provided,
the code must be <code>LA</code> for one of the performers.</li>\n      </ul>\n   </li>\n   <li>Laboratory supports
<code>LA</code>, <code>PPRF</code>, <code>PRF</code>, <code>OP</code>, and <code>AUTHEN</code>
codes.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n   \"extension\": [\n     {\n       \"valueCodeableConcept\":
 {\n         \"coding\": [\n           {\n             \"system\":
'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',\n             \"code\": 'LA',\n
\"display\": 'legal authenticator'\n           }\n         ],\n         \"text\": 'legal authenticator'\n       },\n
       \"url\": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'\n     }\n   ],\n
}</code></pre>\n\n<pre><code>{\n   \"extension\": [\n      {\n         \"valueCodeableConcept\": {\n
\"coding\": [\n               {\n                  \"system\":
\"https://terminology.hl7.org/CodeSystem-v2-0443.html\",\n                  \"code\": \"OP\",\n
\"display\": \"ordering provider\"\n               }\n            ],\n            \"text\": \"ordering provider\"\n
         },\n         \"url\": \"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n      },\n
{\n         \"valueCodeableConcept\": {\n            \"coding\": [\n               {\n                  \"system\":
\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n                  \"code\": \"PPRF\",\n
                  \"display\": \"primary performer\"\n               }\n            ],\n            \"text\":
\"primary performer\"\n         },\n         \"url\": \"http://hl7.org/fhir/StructureDefinition/event-
performerFunction\"\n      },\n      {\n         \"valueCodeableConcept\":{\n            \"coding\":[\n
               {\n                  \"system\":\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                  \"code\":\"PRF\",\n                  \"display\":\"performer\"\n               }\n            ],\n
            \"text\":\"performer\"\n         },\n         \"url\":\"http://hl7.org/fhir/StructureDefinition/event-
performerFunction\"\n      },\n      {\n         \"valueCodeableConcept\":{\n            \"coding\":[\n
               {\n                  \"system\":\"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                  \"code\":\"AUTHEN\",\n                  \"display\":\"authenticator\"\n               }\n
            ],\n            \"text\":\"authenticator\"\n         },\n
\"url\":\"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n      }\n   ]\n}</code></pre>\n",
        "type":"object"
    },
    "value[x]":{
        "description":"The value of the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Vital signs support writing
observations with <code>valueQuantity</code>.</li>\n  <li>Laboratory supports <code>valueQuantity</code>,
<code>valueCodeableConcept</code>, <code>valueString</code>, <code>valueAttachment</code>, and
<code>valueDateTime</code>.</li>\n  <li>Imaging supports <code>valueQuantity</code> and
<code>valueCodeableConcept</code>.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"valueQuantity\":\n  {\n
\"value\": 162,\n    \"unit\": 'cm',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'cm'\n
}\n}</code></pre>\n\n<pre><code>{\n  \"valueCodeableConcept\": {\n    \"coding\": [\n      {\n        \"system\":
'http://snomed.info/sct',\n        \"code\": '373066001',\n        \"display\": 'Yes (qualifier value)'\n      }\n
    ],\n    \"text\": 'Yes'\n  }\n}\n</code></pre>\n<pre><code>{\n  \"valueString\":
'2'\n}\n</code></pre>\n<pre><code>{\n  \"valueDateTime\": '2020-07-04T07:15:00Z'\n}</code></pre>\n"
    },
    "interpretation":{
```

```
        "description":"A categorical assessment of an observation value. For example, high, low,
normal.<br>\n\n<b>Note:</b> Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
\"interpretation\": [\n    {\n      \"coding\": [\n        {\n          \"system\":
'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',\n          \"code\": '211',\n
\"display\": 'LOW',\n          \"userSelected\": true\n        }\n      ],\n      \"text\": 'LOW'\n    }\n
]\n}</code></pre>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n      \"coding\": [\n        {\n          {\n
            \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n            \"code\":
 'L',\n            \"display\": 'Low'\n          }\n        ],\n        \"text\": 'LOW'\n      }\n    ]\n
}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    },
    "note":{
        "description":"Comments about the observation<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"note\": [\n    {\n
     \"authorReference\": {\n        \"reference\": 'Practitioner/62955753'\n      },\n      \"time\":
'2018-08-17T09:55:42.000Z',\n      \"text\": 'Charted at Incorrect Time'\n    }\n  ]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "authorReference":{
                    "type":"object",
                    "properties":{
                        "reference":{
                            "type":"string"
                        }
                    }
                },
                "time":{
                    "type":"string"
                },
                "text":{
                    "type":"string"
                }
            }
        }
    },
    "referenceRange":{
        "description":"Provides guidance for interpretating the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only
one referenceRange is supported. </li>\n  <li><code>ReferenceRange.text</code> is not supported when

<code>referenceRange.high</code>, <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type
of <code>normal</code> is supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"referenceRange\": [\n
{\n      \"low\": {\n        \"value\": 35.2,\n        \"unit\": 'degC',\n        \"system\":
'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n      \"high\": {\n        \"value\": 36.7,\n
        \"unit\": 'degC',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n
      \"type\": {\n        \"coding\": [\n          {\n            \"system\":
'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n            \"code\": 'normal',\n
\"display\": 'Normal Range'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object"
        }
    },
    "referenceRange.low":{
        "description":"The lower bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 35.2,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "value":{
                "type":"number"
            },
            "unit":{
                "type":"string"
            },
            "system":{
                "type":"string"
```

```
            },
            "code":{
                "type":"string"
            }
        }
    },
    "referenceRange.high":{
        "description":"The upper bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 36.7,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
}\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "value":{
                "type":"number"
            },
            "unit":{
                "type":"string"
            },
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    },
    "referenceRange.text":{
        "description":"The text-based reference range in an observation, which may be used when a quantitative range
 is not appropriate for an observation.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"text\":
'Negative'\n}</code></pre>\n",
        "type":"string"
    },
    "referenceRange.type":{
        "description":"Codes to indicate what part of the targeted reference population it applies
to.<br>\n\n<b>Note:</b> Only a <code>ReferenceRange.type</code> of normal is
supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": {\n    \"coding\": [\n      {\n        \"system\":
'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n        \"code\": 'normal',\n        \"display\":
'Normal Range'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "component":{
        "description":"Some observations have multiple component observations. These component observations are
expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic
component observations for blood pressure measurement and multiple component observations for genetics
observations.<br>\n\n<b>Note:</b> Component is required to create blood
pressure.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
'http://loinc.org',\n        \"code\": '8480-6'\n      }\n    ],\n    \"text\": 'Systolic Blood Pressure Invasive'\n
  },\n  \"valueQuantity\": {\n    \"value\": 120,\n    \"unit\": 'mmHg',\n    \"system\":
'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"interpretation\": [\n    {\n      \"coding\": [\n
        {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
\"code\": 'N'\n        }\n      ]\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n        \"value\":
 45.0,\n        \"unit\": 'mmHg',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n
      },\n      \"high\": {\n        \"value\": 75.0,\n        \"unit\": 'mmHg',\n        \"system\":
'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n      },\n      \"type\": {\n        \"coding\": [\n
          {\n            \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
\"code\": 'normal'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object"
        }
    },
    "component.code":{
        "description":"Describes what was observed. Also called the observation
name.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"coding\": [\n    {\n      \"system\": 'http://loinc.org',\n
\"code\": '8480-6'\n    }\n  ],\n  \"text\": 'Systolic Blood Pressure Invasive'\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
```

```
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "component.value[x]":{
        "description":"The information determined as a result of making the observation, if the information has a
simple value.<br>\n\n<b>Note:</b> Supports writing observations with only
valueQuantity.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"value\": 120,\n  \"unit\": 'mmHg',\n  \"system\":
'http://unitsofmeasure.org',\n  \"code\": 'mm[Hg]'\n}</code></pre>\n"
    },
    "component.interpretation":{
        "description":"A categorical assessment of an observation value. For example, high, low,
normal.<br>\n\n<b>Note:</b> Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
\"coding\": [\n     {\n       \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
       \"code\": 'N'\n     }\n   ],\n   \"text\": 'NORMAL'\n }</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    },
    "component.referenceRange":{
        "description":"Guidance on how to interpret the value by comparison to a normal or recommended
range.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one referenceRange is supported.</li>\n
<li><code>ReferenceRange.text</code> is not supported when <code>referenceRange.high</code>,
<code>referenceRange.low</code>, or both are provided. Only a referenceRange.type of <code>normal</code> is
supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n    \"value\": 45.0,\n    \"unit\":
'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"high\": {\n    \"value\":
75.0,\n    \"unit\": 'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n
\"type\": {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-
meaning',\n        \"code\": 'normal'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
},
"example":"{\n  \"resourceType\": \"Observation\",\n  \"status\": \"final\",\n  \"category\": [\n    {\n
\"coding\": [\n        {\n          \"system\": \"http://terminology.hl7.org/CodeSystem/observation-category\",\n
          \"code\": \"vital-signs\",\n          \"display\": \"Vital Signs\"\n        }\n      ],\n      \"text\":
\"Vital Signs\"\n    }\n  ],\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": \"http://loinc.org\",\n
        \"code\": \"8331-1\"\n      }\n    ],\n    \"text\": \"Temperature Oral\"\n  },\n  \"subject\": {\n
```

```
    \"reference\": \"Patient/12457981\"\n  },\n  \"encounter\": {\n    \"reference\": \"Encounter/97845408\"\n  },\n
    \"effectiveDateTime\": \"2020-04-03T19:21:00.000Z\",\n  \"issued\": \"2020-04-03T19:21:40.000Z\",\n  \"performer\":
    [\n    {\n      \"extension\": [\n        {\n          \"valueCodeableConcept\": {\n            \"coding\": [\n
                  {\n               \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ParticipationType\",\n
                    \"code\": \"LA\",\n                \"display\": \"legal authenticator\"\n             }\n
                ],\n            \"text\": \"legal authenticator\"\n          },\n          \"url\":
    \"http://hl7.org/fhir/StructureDefinition/event-performerFunction\"\n        }\n      ],\n      \"reference\":
    \"Practitioner/11638321\"\n    }\n  ],\n  \"valueQuantity\": {\n    \"value\": 37.2,\n    \"unit\": \"degC\",\n
    \"system\": \"http://unitsofmeasure.org\",\n    \"code\": \"Cel\"\n  },\n  \"interpretation\": [\n    {\n
    \"coding\": [\n        {\n          \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation\",\n          \"code\": \"N\",\n
    \"display\": \"Normal\"\n        }\n      ]\n    }\n  ],\n  \"note\": [\n    {\n      \"authorReference\": {\n
           \"reference\": \"Practitioner/11638321\"\n      },\n      \"time\": \"2020-09-03T15:03:10.000Z\",\n
    \"text\": \"No additional test require\"\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n
    \"value\": 35.8,\n        \"unit\": \"degC\",\n        \"system\": \"http://unitsofmeasure.org\",\n        \"code\":
     \"Cel\"\n      },\n      \"high\": {\n        \"value\": 37.3,\n        \"unit\": \"degC\",\n        \"system\":
    \"http://unitsofmeasure.org\",\n        \"code\": \"Cel\"\n      },\n      \"type\": {\n        \"coding\": [\n
             {\n            \"system\": \"http://terminology.hl7.org/CodeSystem/referencerange-meaning\",\n
    \"code\": \"normal\",\n            \"display\": \"Normal Range\"\n          }\n        ],\n        \"text\":
    \"Normal Range\"\n      }\n    }\n  ]\n}\n"
}
```


Example:


```
{
  "resourceType": "Observation",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "8331-1"
      }
    ],
    "text": "Temperature Oral"
  },
  "subject": {
    "reference": "Patient/12457981"
  },
  "encounter": {
    "reference": "Encounter/97845408"
  },
  "effectiveDateTime": "2020-04-03T19:21:00.000Z",
  "issued": "2020-04-03T19:21:40.000Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/11638321"
    }
  ],
  "valueQuantity": {
    "value": 37.2,
    "unit": "degC",
    "system": "http://unitsofmeasure.org",
    "code": "Cel"
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
          "code": "N",
          "display": "Normal"
        }
```

```
      ]
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/11638321"
      },
      "time": "2020-09-03T15:03:10.000Z",
      "text": "No additional test require"
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 35.8,
        "unit": "degC",
        "system": "http://unitsofmeasure.org",
        "code": "Cel"
      },
      "high": {
        "value": 37.3,
        "unit": "degC",
        "system": "http://unitsofmeasure.org",
        "code": "Cel"
      },
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
            "code": "normal",
            "display": "Normal Range"
          }
        ],
        "text": "Normal Range"
      }
    }
  ]
}
```

Nested Schema : ObservationUpdateBody-allOf[1]

Type:

object

Nested Schema : category

Type:

array

A code that classifies the general type of observation being made.

Note:

* VitalSigns, Laboratory, and Imaging categories are supported.

* Only one category may be supplied on create.

Example:

```
{
  "category": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/observation-category',
          "code": 'vital-signs',
          "display": 'Vital Signs'
        }
      ],
      "text": 'Vital Signs'
    }
  ]
}
```

Show Source

* Array of: object items


```
{
    "description":"A code that classifies the general type of observation being made.<br>\n\n<b>Note:</b>\n<ul>\n  <li>
    <code>VitalSigns</code>, <code>Laboratory</code>, and <code>Imaging</code> categories are supported. </li>\n  <li>
    Only one category may be supplied on create. </li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"category\": [\n
        {\n      \"coding\": [\n        {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/observation-
    category',\n          \"code\": 'vital-signs',\n          \"display\": 'Vital Signs'\n        }\n      ],\n
    \"text\": 'Vital Signs'\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
```

```
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : code

Type:

object

Describes what was observed. Also called the observation name.

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8331-1'
      }
    ],
    "text": 'Temperature Oral'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8302-2'
      },
      {
        "system": 'http://loinc.org',
        "code": '3137-7'
      }
    ],
    "text": 'Height/Length Measured'
  }
}
```

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '96914-7'
      },
    "text": 'CT dose and image quality category'
  }
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"Describes what was observed. Also called the observation
    name.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8331-1'\n      }\n    ],\n    \"text\": 'Temperature Oral'\n
    }\n}</code></pre>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8302-2'\n      },\n      {\n        \"system\": 'http://loinc.org',\n
            \"code\": '3137-7'\n      }\n    ],\n    \"text\": 'Height/Length Measured'\n  }\n}</code></pre>\n\n
    <pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\": 'http://loinc.org',\n        \"code\":
    '96914-7'\n      },\n    \"text\": 'CT dose and image quality category'\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
```

```
                    "type":"string"
                },
                "code":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : component

Type:

array

Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations. Component is required to create blood pressure.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": 'http://loinc.org',
        "code": '8480-6'
      }
    ],
    "text": 'Systolic Blood Pressure Invasive'
  },
  "valueQuantity": {
    "value": 120,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
          "code": 'N'
        }
      ]
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 45.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "high": {
        "value": 75.0,
        "unit": 'mmHg',
        "system": 'http://unitsofmeasure.org',
        "code": 'mm[Hg]'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Some observations have multiple component observations. These component observations are expressed as
     separate code value pairs that share the same attributes. Examples include systolic and diastolic component
    observations for blood pressure measurement and multiple component observations for genetics
    observations.<br>\n\n<b>Note:</b> Component is required to create blood
    pressure.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    'http://loinc.org',\n        \"code\": '8480-6'\n      }\n    ],\n    \"text\": 'Systolic Blood Pressure Invasive'\n
      },\n  \"valueQuantity\": {\n    \"value\": 120,\n    \"unit\": 'mmHg',\n    \"system\":
```

```
    'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"interpretation\": [\n    {\n      \"coding\": [\n
          {\n          \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n
    \"code\": 'N'\n        }\n      ]\n    }\n  ],\n  \"referenceRange\": [\n    {\n      \"low\": {\n        \"value\":
     45.0,\n        \"unit\": 'mmHg',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n
         },\n      \"high\": {\n        \"value\": 75.0,\n        \"unit\": 'mmHg',\n        \"system\":
    'http://unitsofmeasure.org',\n        \"code\": 'mm[Hg]'\n      },\n      \"type\": {\n        \"coding\": [\n
            {\n            \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
    \"code\": 'normal'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
    ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object"
    }
}
```


Nested Schema : component.code

Type:

object

Describes what was observed. Also called the observation name.

Example:

```
{
  "coding": [
    {
      "system": 'http://loinc.org',
      "code": '8480-6'
    }
  ],
  "text": 'Systolic Blood Pressure Invasive'
}
```

Show Source

* coding: array coding

* text: string


```
{
    "description":"Describes what was observed. Also called the observation name.<br>\n<b>Example:</b>\n\n<pre><code>{\n
      \"coding\": [\n    {\n      \"system\": 'http://loinc.org',\n      \"code\": '8480-6'\n    }\n  ],\n  \"text\":
    'Systolic Blood Pressure Invasive'\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : component.interpretation

Type:

array

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
   "coding": [
     {
       "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
       "code": 'N'
     }
   ],
   "text": 'NORMAL'
 }
```

Show Source

* Array of: object items


```
{
    "description":"A categorical assessment of an observation value. For example, high, low, normal.<br>\n\n<b>Note:</b>
     Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n   \"coding\": [\n     {\n
```

```
    \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n       \"code\": 'N'\n     }\n
    ],\n   \"text\": 'NORMAL'\n }</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : component.referenceRange

Type:

object

Guidance on how to interpret the value by comparison to a normal or recommended range.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:


```
{
  "low": {
    "value": 45.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "high": {
    "value": 75.0,
    "unit": 'mmHg',
    "system": 'http://unitsofmeasure.org',
    "code": 'mm[Hg]'
  },
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal'
      }
    ],
    "text": 'Normal Range'
  }
}
```


Show Source

* coding: array coding

* text: string


```
{
    "description":"Guidance on how to interpret the value by comparison to a normal or recommended
    range.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one referenceRange is supported.</li>\n
    <li><code>ReferenceRange.text</code> is not supported when <code>referenceRange.high</code>,
    <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type of <code>normal</code> is
    supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n    \"value\": 45.0,\n    \"unit\":
    'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n  \"high\": {\n    \"value\":
    75.0,\n    \"unit\": 'mmHg',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'mm[Hg]'\n  },\n
    \"type\": {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-
    meaning',\n        \"code\": 'normal'\n      }\n    ],\n    \"text\": 'Normal Range'\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
```

```
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : encounter

Type:

object

The encounter during which the observation was first made.

Example:

```
{
  "encounter": {
    "reference": "Encounter/631923"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The encounter during which the observation was first made.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n
    \"encounter\": {\n    \"reference\": \"Encounter/631923\"\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : interpretation

Type:

array

A categorical assessment of an observation value. For example, high, low, normal. Only one interpretation may be supplied.

Note:

Example:

```
{
  "interpretation": [
    {
      "coding": [
        {
          "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',
          "code": '211',
          "display": 'LOW',
          "userSelected": true
        }
      ],
      "text": 'LOW'
    }
  ]
}
```

```
{
  "interpretation": [
    {
      "coding": [
        {
          {
            "system": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',
            "code": 'L',
            "display": 'Low'
          }
        ],
        "text": 'LOW'
      }
    ]
  }
```

Show Source

* Array of: object items

```
{
    "description":"A categorical assessment of an observation value. For example, high, low, normal.<br>\n\n<b>Note:</b>
     Only one interpretation may be supplied.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n
          \"coding\": [\n        {\n          \"system\":
```

```
    'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/52',\n          \"code\": '211',\n
    \"display\": 'LOW',\n          \"userSelected\": true\n        }\n      ],\n      \"text\": 'LOW'\n    }\n
    ]\n}</code></pre>\n\n<pre><code>{\n  \"interpretation\": [\n    {\n      \"coding\": [\n        {\n          {\n
                \"system\": 'http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation',\n            \"code\":
     'L',\n            \"display\": 'Low'\n          }\n        ],\n        \"text\": 'LOW'\n      }\n    ]\n
    }</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : note

Type:

array

Comments about the observation

Example:


```
{
  "note": [
    {
      "authorReference": {
        "reference": 'Practitioner/62955753'
      },
      "time": '2018-08-17T09:55:42.000Z',
      "text": 'Charted at Incorrect Time'
    }
  ]
}
```


Show Source

* Array of: object items


```
{
    "description":"Comments about the observation<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"note\": [\n    {\n
    \"authorReference\": {\n        \"reference\": 'Practitioner/62955753'\n      },\n      \"time\":
    '2018-08-17T09:55:42.000Z',\n      \"text\": 'Charted at Incorrect Time'\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "authorReference":{
                "type":"object",
                "properties":{
                    "reference":{
                        "type":"string"
                    }
                }
            },
            "time":{
                "type":"string"
            },
            "text":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : performer

Type:

array

Limited to "real" people rather than equipment.

Note:

* Must be a reference to a practitioner.

* Vital signs require a performer.

Example:

```
{
"performer": [
    {
        "reference": 'Practitioner/12724064'
      }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Limited to \"real\" people rather than equipment.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Must be a
    reference to a practitioner.</li>\n  <li>Vital signs require a
    performer.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n\"performer\": [\n    {\n        \"reference\":
    'Practitioner/12724064'\n      }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : performer.extension

Type:

object

Distinguishes the performer's type of involvement in the event.

Note:

* Vital signs support and codes.

PPRF

* When only one performer is provided, the code must be LA.

* When multiple performers are provided, the code must be LA for one of the performers.

* Laboratory supports LA, PPRF, PRF, OP, and AUTHEN codes.

Example:

```
{
   "extension": [
     {
       "valueCodeableConcept": {
         "coding": [
           {
             "system": 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
             "code": 'LA',
             "display": 'legal authenticator'
           }
         ],
         "text": 'legal authenticator'
       },
       "url": 'http://hl7.org/fhir/StructureDefinition/event-performerFunction'
     }
   ],
 }
{
   "extension": [
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "https://terminology.hl7.org/CodeSystem-v2-0443.html",
                  "code": "OP",
                  "display": "ordering provider"
               }
            ],
            "text": "ordering provider"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept": {
            "coding": [
               {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code": "PPRF",
                  "display": "primary performer"
               }
```

```
            ],
            "text": "primary performer"
         },
         "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"PRF",
                  "display":"performer"
               }
            ],
            "text":"performer"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      },
      {
         "valueCodeableConcept":{
            "coding":[
               {
                  "system":"http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                  "code":"AUTHEN",
                  "display":"authenticator"
               }
            ],
            "text":"authenticator"
         },
         "url":"http://hl7.org/fhir/StructureDefinition/event-performerFunction"
      }
   ]
}
```


Nested Schema : referenceRange

Type:

array

Provides guidance for interpretating the observation.

Note:

* Only one referenceRange is supported.

* ReferenceRange.text is not supported when referenceRange.high, referenceRange.low, or both are provided. Only a referenceRange.type of normal is supported.

Example:


```
{
  "referenceRange": [
    {
      "low": {
        "value": 35.2,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "high": {
        "value": 36.7,
        "unit": 'degC',
        "system": 'http://unitsofmeasure.org',
        "code": 'Cel'
      },
      "type": {
        "coding": [
          {
            "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
            "code": 'normal',
            "display": 'Normal Range'
          }
        ],
        "text": 'Normal Range'
      }
    }
  ]
}
```


Show Source

* Array of: object items


```
{
    "description":"Provides guidance for interpretating the observation.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one
    referenceRange is supported. </li>\n  <li><code>ReferenceRange.text</code> is not supported when
    <code>referenceRange.high</code>, <code>referenceRange.low</code>, or both are provided. Only a referenceRange.type
    of <code>normal</code> is supported.</li>\n</ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"referenceRange\": [\n
    {\n      \"low\": {\n        \"value\": 35.2,\n        \"unit\": 'degC',\n        \"system\":
    'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n      \"high\": {\n        \"value\": 36.7,\n
           \"unit\": 'degC',\n        \"system\": 'http://unitsofmeasure.org',\n        \"code\": 'Cel'\n      },\n
         \"type\": {\n        \"coding\": [\n          {\n            \"system\":
```

```
    'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n           \"code\": 'normal',\n
    \"display\": 'Normal Range'\n          }\n        ],\n        \"text\": 'Normal Range'\n      }\n    }\n
    ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object"
    }
}
```


Nested Schema : referenceRange.high

Type:

object

The upper bound of the reference range.

Example:


```
{
  "low": {
    "value": 36.7,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```


Show Source

* code: string

* system: string

* unit: string

* value: number


```
{
    "description":"The upper bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 36.7,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "value":{
            "type":"number"
        },
        "unit":{
            "type":"string"
        },
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : referenceRange.low

Type:

object

The lower bound of the reference range.

Example:


```
{
  "low": {
    "value": 35.2,
    "unit": 'degC',
    "system": 'http://unitsofmeasure.org',
    "code": 'Cel'
  }
}
```


Show Source

* code: string

* system: string

* unit: string

* value: number


```
{
    "description":"The lower bound of the reference range.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"low\": {\n
    \"value\": 35.2,\n    \"unit\": 'degC',\n    \"system\": 'http://unitsofmeasure.org',\n    \"code\": 'Cel'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "value":{
            "type":"number"
        },
```

```
        "unit":{
            "type":"string"
        },
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : referenceRange.type

Type:

object

Codes to indicate what part of the targeted reference population it applies to. Only a of normal is supported.

Note:

ReferenceRange.type

Example:

```
{
  "type": {
    "coding": [
      {
        "system": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',
        "code": 'normal',
        "display": 'Normal Range'
      }
    ],
    "text": 'Normal Range'
  }
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"Codes to indicate what part of the targeted reference population it applies to.<br>\n\n<b>Note:</b>
    Only a <code>ReferenceRange.type</code> of normal is supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"type\":
    {\n    \"coding\": [\n      {\n        \"system\": 'http://terminology.hl7.org/CodeSystem/referencerange-meaning',\n
            \"code\": 'normal',\n        \"display\": 'Normal Range'\n      }\n    ],\n    \"text\": 'Normal Range'\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : subject

Type:

object

Indicates the patient who the observation record is associated with. Must only include references to a patient.

Note:

Example:

```
{
  "subject": {
    "reference": "Patient/631923"
  }
}
```

Show Source

* reference: string

```
{
    "description":"Indicates the patient who the observation record is associated with.<br>\n\n<b>Note:</b> Must only
```

```
    include references to a patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
    \"Patient/631923\"\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
```

```
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
```

```
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* authorReference: object authorReference

* text: string

* time: string

```
{
    "type":"object",
    "properties":{
        "authorReference":{
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        },
        "time":{
            "type":"string"
        },
        "text":{
            "type":"string"
        }
    }
}
```

Nested Schema : authorReference

Type:

object

Show Source

* reference: string

```
{
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* reference: string

```
{
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider

* System

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-196186655
-d '{
  "resourceType": "Observation",
  "id": "L-196186655",
  "status": "corrected",
  "meta": {
    "source": "43ba6d6a-1006-4546-bf28-48d84a98dbb1"
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ],
      "text": "Laboratory"
    }
  ],
  "code": {
    "coding": [
```

```
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/72",
        "code": "21704910",
        "display": "Potassium Level",
        "userSelected": true
      }
    ],
    "text": "Potassium Level"
  },
  "subject": {
    "reference": "Patient/12457981"
  },
  "encounter": {
    "reference": "Encounter/2673896"
  },
  "issued": "2020-07-30T20:42:00.000Z",
  "effectiveDateTime": "2020-07-30T20:42:00.000Z",
  "performer": [
    {
      "reference": "Practitioner/1477926"
    }
  ],
  "valueQuantity": {
    "value": 13.2,
    "unit": "mg/dL",
    "system": "http://unitsofmeasure.org",
    "code": "mg/dL"
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
          "code": "N",
          "display": "Normal"
        }
      ],
      "text": "Normal"
    }
  ],
  "referenceRange": [
    {
      "low": {
        "value": 12.0,
        "unit": "g/dL",
        "system": "http://unitsofmeasure.org",
        "code": "g/dL"
      },
      "high": {
        "value": 15.5,
        "unit": "g/dL",
        "system": "http://unitsofmeasure.org",
        "code": "g/dL"
      }
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/1477926"
      },
      "time": "2020-07-30T20:42:15.000Z",
      "text": "Specimen slightly hemolyzed."
    }
  ]
}
```


Example Response:


```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```


Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-imaging Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider

* System

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/L-206972569
-d '{
  "resourceType": "Observation",
  "id": "L-206972569",
  "identifier": [
      {
          "system": "https://fhir.cerner.com/ceuuid",
          "value": "CE87caf4b7-9397-4667-9897-702218017c9e-206972569-2024111310480300"
      }
  ],
  "category": [
      {
          "coding": [
              {
                  "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                  "code": "imaging",
                  "display": "Imaging"
              }
          ]
      }
  ],
  "code": {
      "coding": [
          {
              "system": "http://loinc.org",
              "code": "96914-7",
              "display": "CT dose and image quality category"
          }
      ],
      "text": "CT dose and image quality category"
  },
  "subject": {
      "reference": "Patient/12461981"
  },
  "encounter": {
      "reference": "Encounter/97701407"
  },
  "effectiveDateTime": "2023-11-16T08:00:00.000+00:00",
  "status": "corrected",
  "valueCodeableConcept": {
      "coding": [
          {
              "system": "http://loinc.org",
              "code": "LA31771-1",
              "display": "Full Body"
          }
      ]
  }
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Wed, 13 Nov 2024 10:54:41 GMT
Etag: W/"3"
Last-Modified: Wed, 13 Nov 2024 10:54:41 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-vitals-blood-pressure Response

This resource action is still under development. This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

Note:

* Provider

* System

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/BP-59857018-59857020
```

```
-d '{
  "resourceType": "Observation",
  "id": "BP-59857018-59857020",
  "identifier": [
    {
      "system": "https://fhir.cerner.com/ceuuid",
      "value": "CEfda49233-ccfa-4ed4-afbc-9f5082c2bf0c-59857018-2022111805415000"
    },
    {
      "system": "https://fhir.cerner.com/ceuuid",
      "value": "CEfda49233-ccfa-4ed4-afbc-9f5082c2bf0c-59857020-2022111805415000"
    }
  ],
  "status": "corrected",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "85354-9",
        "display": "Blood pressure panel with all children optional"
      }
    ],
    "text": "Blood pressure"
  },
  "subject": {
    "reference": "Patient/15116458"
  },
  "encounter": {
    "reference": "Encounter/15687784"
  },
  "effectiveDateTime": "2022-06-09T08:00:58.000Z",
  "issued": "2022-06-09T08:00:58.000Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/15441459"
    }
  ],
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "8480-6"
          }
        ],
        "text": "Systolic Blood Pressure Sitting"
      },
      "valueQuantity": {
        "value": 99,
        "unit": "mmHg",
        "system": "http://unitsofmeasure.org",
        "code": "mm[Hg]"
      },
      "interpretation": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
              "code": "N",
              "display": "Normal"
            }
```

```
          ]
        }
      ],
      "referenceRange": [
        {
          "low": {
            "value": 35,
            "unit": "mmHg",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          },
          "high": {
            "value": 37,
            "unit": "mmHg",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          },
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
                "code": "normal",
                "display": "Normal Range"
              }
            ],
            "text": "Normal Range"
          }
        }
      ]
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "8462-4"
          }
        ],
        "text": "Diastolic Blood Pressure Sitting"
      },
      "valueQuantity": {
        "value": 66,
        "unit": "mmHg",
        "system": "http://unitsofmeasure.org",
        "code": "mm[Hg]"
      },
      "interpretation": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
              "code": "N",
              "display": "Normal"
            }
          ]
        }
      ],
      "referenceRange": [
        {
          "low": {
            "value": 35,
            "unit": "mmHg",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          },
          "high": {
            "value": 37,
            "unit": "mmHg",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          },
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/referencerange-meaning",
                "code": "normal",
                "display": "Normal Range"
              }
            ],
            "text": "Normal Range"
          }
        }
      ]
    }
  ]
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-vitals-body Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider

* System

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-197356031
-d '{
  "resourceType": "Observation",
  "id": "VS-197356031",
  "identifier": [
    {
      "system": "https://fhir.cerner.com/ceuuid",
      "value": "CE87caf4b7-9397-4667-9897-702218017c9e-197356031-2021061619245900"
    }
  ],
  "status": "corrected",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/d242a518-4074-4bd4-a3a6-adfe0c5c1c51/codeSet/72",
        "code": "703558",
        "display": "Temperature Oral",
        "userSelected": true
      }
    ],
    "text": "Temperature Oral"
  },
  "subject": {
    "reference": "Patient/12457979"
  },
  "encounter": {
    "reference": "Encounter/97787491"
  },
  "effectiveDateTime": "2020-07-04T07:15:00.000Z",
  "issued": "2021-05-06T20:07:14.000Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/1"
    }
  ],
```

```
  "valueQuantity": {
    "value": 123,
    "unit": "mmHg",
    "system": "http://unitsofmeasure.org",
    "code": "mm[Hg]"
  }
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-vitals-pulse-oxymetry Response

This resource action is still under development. This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

Note:

* Provider

* System

Example Request:

```
PUT https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Observation/VS-CV-78693473-SECT-8098
-d '{
  "resourceType": "Observation",
  "id": "VS-CV-83059963-SECT-8098",
  "identifier": [
    {
      "system": "https://fhir.cerner.com/cvuuid",
      "value": "8344a9b7-b1b7-406e-b303-1009edd66459"
    }
  ],
  "status": "amended",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "2708-6",
        "display": "Oxygen saturation in Arterial blood"
      },
      {
        "system": "http://loinc.org",
        "code": "59408-5",
        "display": "Oxygen saturation in Arterial blood by Pulse oximetry"
      }
    ],
    "text": "SpO2"
  },
  "subject": {
    "reference": "Patient/12457979"
  },
  "encounter": {
    "reference": "Encounter/97787491"
  },
  "effectiveDateTime": "2022-06-05T08:00:38.022Z",
  "issued": "2022-06-05T08:00:56Z",
  "performer": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
```

```
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                "code": "LA",
                "display": "legal authenticator"
              }
            ],
            "text": "legal authenticator"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/event-performerFunction"
        }
      ],
      "reference": "Practitioner/15441459"
    }
  ],
  "valueQuantity": {
    "value": 44,
    "unit": "%",
    "system": "http://unitsofmeasure.org",
    "code": "%"
  },
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "3151-8",
            "display": "Inhaled oxygen flow rate"
          }
        ],
        "text": "Oxygen Flow Rate"
      },
      "valueQuantity": {
        "value": 55,
        "unit": "L/min",
        "system": "http://unitsofmeasure.org",
        "code": "L/min"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "3150-0",
            "display": "Inhaled oxygen concentration"
          }
        ],
        "text": "FIO2"
      },
      "valueQuantity": {
        "value": 66,
        "unit": "%",
        "system": "http://unitsofmeasure.org",
        "code": "%"
      }
    }
  ]
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 20 Aug 2019 21:17:04 GMT
Etag: W/"12809861"
Last-Modified: TTue, 20 Aug 2019 21:17:04 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
```

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

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OperationDefinition

File: r4-rest-api-v32/api-operationdefinition.html

Description

OperationDefinition

This topic describes information for the OperationDefinition resource.

Overview

The OperationDefinition resource defines an operation. This information includes the inputs, outputs, description, and other items necessary to actually run the operation.

This resource only exposes the custom operations that the server defines. Operations that are defined as part of another implementation guide are referenced directly instead of being duplicated in this server. While the HL7 FHIR specification also supports describing named queries with this resource, Oracle Cerner does not currently support any custom named queries.

Fields

* ID
* Name
* Title
* Status
* URL
* Kind
* Date
* Publisher
* Description
* Jurisdiction
* Code
* Resource
* System
* Type
* Instance
* Parameter

List of OperationDefinition IDs

Table - List of Operation Definitions

ID	Description
binary-autogen-ccd-if	Generates a copy of a patient's Continuity of Care Document (CCD) as a Binary.
organization-get-cg-for-mrcu	Retrieves details for a caregiver organization given a care unit organization ID.
group-export	Obtains a detailed set of FHIR resources of diverse resource types pertaining to all patients in specified group.
charge_item_modify	Modifies an existing ChargeItem resulting in a newly created ChargeItem.
charge_item_create	Creates a charge event, which can result in the creation of one or more ChargeItems.
charge_item_credit	Creates an offsetting ChargeItem for an existing debit ChargeItem.

OperationDefinition

The operations from the OperationDefinition category.

Get an operation definition by ID

Method: get Path: /OperationDefinition/{ID}

Get an operation definition by ID

File: r4-rest-api-v32/op-operationdefinition-id-get.html

Retrieves an individual operation definition by the operation definition ID.

Note: You can retrieve this resource by its defining URL or from the OperationDefinition resource located at the service root URL.

For example, both of the following URLs work:

```
https://fhir-ehr.cerner.com/r4/OperationDefinition/binary-autogen-ccd-if
```

```
https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/OperationDefinition/binary-autogen-ccd-if
```

Authorization Types

Authorization is not required.

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/OperationDefinition/binary-autogen-ccd-if
```

Example Response:


```
{
  "resourceType": "OperationDefinition",
  "id": "binary-autogen-ccd-if",
  "url": "https://fhir-ehr.cerner.com/r4/OperationDefinition/binary-autogen-ccd-if",
  "name": "GenerateContinuityOfCareDocument",
  "title": "Generate Continuity of Care Document (CCD)",
  "status": "active",
  "kind": "operation",
  "date": "2019-01-18",
  "publisher": "Cerner",
  "description": "Generates a copy of a patient's Continuity of Care Document (CCD) as a Binary",
  "code": "autogen-ccd-if",
  "resource": [
    "Binary"
  ],
  "system": false,
  "type": true,
  "instance": false,
  "parameter": [
    {
      "name": "patient",
      "use": "in",
      "min": 1,
      "max": "1",
      "documentation": "The id of the patient that is the subject of the generated Continuity of Care Document (CCD).",
      "type": "id"
    },
    {
      "name": "start",
      "use": "in",
      "min": 0,
      "max": "1",
      "documentation": "The start of the date range for which the Continuity of Care Document (CCD) is to be generated.
    If no start date is provided, then the document will be generated with all records up to the end or current date.",
      "type": "date"
    },
    {
      "name": "end",
      "use": "in",
      "min": 0,
      "max": "1",
      "documentation": "The end of the date range for which the Continuity of Care Document (CCD) is to be generated. If
     no end date is provided, then the document will be generated with all records up to the current date.",
      "type": "date"
    },
    {
      "name": "result",
      "use": "out",
      "min": 1,
      "max": "1",
      "documentation": "Binary for the generated Continuity of Care Document (CCD)",
      "type": "Binary"
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Organization

File: r4-rest-api-v32/api-organization.html

Description

Organization

This topic describes information for the Organization resource.

Overview

The Organization resource describes a grouping of people or business entities relevant to the healthcare process. Organizations include hospitals, employers, insurance companies, physicians' offices, rehabilitation facilities, laboratories, and so on.

This resource supports the US Core Organization Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The following fields are returned if valued:

* Organization Identifier

* Active

* Type

* Name

* Contact Information

* Address

Terminology Bindings

Organization.identifier.type

Description: A coded type for the identifier that can be used to determine which identifier to use for a specific purpose.

Details: Identifier Type Codes

System: http://terminology.hl7.org/CodeSystem/v2-0203

Details: For more information, see Code Set 334 Organization Alias Type under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR source id>/codeSet/334

Organization.type

Description: Indicates the type of Organization.

Details: OrganizationType

System: http://terminology.hl7.org/CodeSystem/organization-type

Details: For more information, see Code Set 278 Organization Type under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR source id>/codeSet/278

Organization

The operations from the Organization category.

Create an organization

Method: post Path: /Organization

Get a list of caregiver organizations

Method: get Path: /Organization/$get-cg-for-mrcu

Get a list of organizations

Method: get Path: /Organization

Get an organization by ID

Method: get Path: /Organization/{ID}

Create an organization

File: r4-rest-api-v32/op-organization-post.html

Creates a new organization.

Note: Only the body fields mentioned below are supported. Unsupported fields are ignored or can result in errors.

Authorization Types

This operation supports the following authorization types:

* Provider

* System

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

The body of the create operation

Root Schema : OrganizationCreate

Type:

object

A summary representation of the create (POST) operation for an organization.

Show Source

active: boolean

Whether the organization's record is still in active use. If set, value must be .

Note:

true

Example:

```
{
  "active": true
}
```

identifier: string

The identifier for the organization. If set, the provided identifier must meet the following conditions:

Note:

* Must contain the system, value, and type fields.

* Must contain only an NPI, OID, or TAXID system.

Example:

```
{
  "identifier": [
    {
      "system": "http://hl7.org/fhir/sid/us-npi",
      "value": "FHIR-NPI-10191",
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "NPI"
          }
        ]
      }
    }
  ]
}
```

name(required): string

The name used for the organization. At least one name must be provided and should be unique.

Note:

Example:

```
{
  "name": "example-organization"
}
```

* resourceType(required): string Allowed Values: ["Organization"] The type of the FHIR resource.

type: array

type

The organization type.

Example:

```
{
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/organization-type",
          "code": "govt"
        }
      ]
    }
  ]
}
```

```
{
    "description":"A summary representation of the create (POST) operation for an organization.",
    "type":"object",
    "required":[
        "resourceType",
        "name"
    ],
```

```
    "properties":{
        "resourceType":{
            "description":"The type of the FHIR resource.\n",
            "type":"string",
            "enum":[
                "Organization"
            ]
        },
        "identifier":{
            "description":"The identifier for the organization.<br>\n\n<b>Note:</b> If set, the provided identifier must
     meet the following conditions:\n    <ul>\n      <li>Must contain the <code>system</code>, <code>value</code>, and
    <code>type</code> fields.</li>\n      <li> Must contain only an NPI, OID, or TAXID system. </li>\n
    </ul>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"identifier\": [\n    {\n      \"system\":
    \"http://hl7.org/fhir/sid/us-npi\",\n      \"value\": \"FHIR-NPI-10191\",\n      \"type\": {\n        \"coding\":
    [\n          {\n            \"system\": \"http://terminology.hl7.org/CodeSystem/v2-0203\",\n            \"code\":
    \"NPI\"\n          }\n        ]\n      }\n    }\n  ]\n}</code></pre>\n",
            "type":"string"
        },
        "active":{
            "description":"Whether the organization's record is still in active use.<br>\n\n<b>Note:</b> If set, value
    must be <code>true</code>.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"active\": true\n}</code></pre>\n",
            "type":"boolean"
        },
        "name":{
            "description":"The name used for the organization.<br>\n\n<b>Note:</b> At least one name must be provided
    and should be unique.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"name\": \"example-
    organization\"\n}</code></pre>\n",
            "type":"string"
        },
        "type":{
            "description":"The organization type.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": [\n    {\n
    \"coding\": [\n        {\n          \"system\": \"http://terminology.hl7.org/CodeSystem/organization-type\",\n
              \"code\": \"govt\"\n        }\n      ]\n    }\n  ]\n}</code></pre>\n",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : type

Type:

array

The organization type.

Example:


```
{
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/organization-type",
          "code": "govt"
        }
      ]
    }
  ]
}
```


Show Source

* Array of: object items


```
{
    "description":"The organization type.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": [\n    {\n      \"coding\":
    [\n        {\n          \"system\": \"http://terminology.hl7.org/CodeSystem/organization-type\",\n
    \"code\": \"govt\"\n        }\n      ]\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
```

```
    "items":{
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

Default Response

The response header indicates the current version to use on a subsequent update.

Example Request:

```
POST https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization
```

Example Request Body:

```
{
    'resourceType': 'Organization',
    'identifier': [
      {
        'system': 'http://hl7.org/fhir/sid/us-npi',
        'value': 'FHIR-NPI-10191',
        'type': {
          'coding': [
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v2-0203',
              'code': 'NPI'
            }
          ]
        }
      }
    ],
    'active': true,
    'name': 'sandbox-fhir-org1',
    'type': [
      {
        'coding': [
          {
            'system': 'http://terminology.hl7.org/CodeSystem/organization-type',
            'code': 'govt'
          }
        ]
      }
    ]
  }
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Wed, 14 Aug 2019 17:23:14 GMT
  Etag: W/"6767735"
  Location: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/6767735
  Last-Modified: Wed, 14 Aug 2019 17:23:14 GMT
  Vary: Origin
  opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
  X-Request-Id: 11111111-1111-1111-1111-111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a list of caregiver organizations

File: r4-rest-api-v32/op-organization-get-cg-mrcu-get.html

Retrieves a list of caregiver organizations that meet supplied query parameters. Only a single _id parameter can be used.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Query Parameters

* _id(required): string The ID of the care unit organization. Example: _id=12345

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example $get-cg-for-mrcu operation.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/$get-cg-for-mrcu?_id=3304067
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': '02116a06-1bef-4397-9347-8d9ae1e98173',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/$get-cg-for-
    mrcu?_id=3304067'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/3304066',
        'resource': {
          'resourceType': 'Organization',
          'id': '3304066',
          'meta': {
            'versionId': '1',
            'lastUpdated': '2020-12-11T21:05:24Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Organization</b></p><p><b>Name</b>: s5gcq4ZbcYMSVC
     CORE test organization1ZybNk7kCr0</p><p><b>Status</b>: Active</p><p><b>TAXID</b>: zYuYtCHkmh</p><p><b>OID</b>:
    urn:oid:2.4.8.6.1.6</p></div>'
          },
          'identifier': [
            {
              'use': 'official',
              'system': 'urn:oid:2.16.840.1.113883.4.4',
              'value': 'zYuYtCHkmh'
            },
            {
              'use': 'official',
              'type': {
                'coding': [
                  {
                    'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/334',
                    'code': 'OID',
                    'display': 'OID',
                    'userSelected': true
                  },
                  {
                    'system': 'urn:ietf:rfc:3986',
                    'code': 'urn:oid',
                    'display': 'OID'
                  }
                ],
```

```
      'text': 'OID'
    },
    'system': 'urn:ietf:rfc:3986',
    'value': 'urn:oid:2.4.8.6.1.6'
  }
],
'active': true,
'type': [
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'HOSPITAL',
        'display': 'Hospital',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/organization-type',
        'code': 'prov',
        'display': 'Healthcare Provider'
      }
    ],
    'text': 'Hospital'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'CHARTACCESS',
        'display': 'Chart Access',
        'userSelected': true
      }
    ],
    'text': 'Chart Access'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'PHARMACY',
        'display': 'Pharmacy',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/organization-type',
        'code': 'dept',
        'display': 'Hospital Department'
      }
    ],
    'text': 'Pharmacy'
  }
],
'name': 's5gcq4ZbcYMSVC CORE test organization1ZybNk7kCr0',
'telecom': [
  {
    'system': 'sms',
    'value': '88793851',
    'use': 'mobile'
  },
  {
    'system': 'phone',
    'value': '13294723851',
    'use': 'work'
  },
  {
    'system': 'email',
    'value': 'external@cerner.com'
  }
],
'address': [
  {
    'use': 'work',
    'text': '6276 Rockriver Drive\\nKansas City, MO 64117\\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
    'use': 'billing',
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
```

```
            'postalCode': '64117',
            'country': 'US'
          }
        ]
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a list of organizations

File: r4-rest-api-v32/op-organization-get.html

Retrieves a list of organizations that meet supplied query parameters.

Note:

* Performing an organization search request without any parameters returns all available organizations, paginated.

* Searching all organizations is not recommended for sites that have a large number of organizations.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Query Parameters

* _count: number The maximum number of results to return. Defaults to 100.

* _id: string The logical resource ID. Example: _id=doc-1234

* address: string An address for the organization. All organizations to be returned whose street, name, state, or postal code starts with the given string. Example: address=1234

identifier: string

The NPI, OID, FederalTAXID, or other identifier for the organization. Example:

* Code details are required.

* NPI, OID, FederalTAXID: A system is optional. If a system is not provided, the search is performed across all systems supported by the Organization resource.

* Other Identifiers: When searching by a cernerSystem, system and value are both required for search.

* When a valid system is provided, the search is performed against the specific system.

* When an invalid system is provided, a 400 (Bad Request) status code is returned.

identifier=123456789

* name: string The name used for the organization. Example: name=aaaaaaa

type: string

The organization's type. Searching with proprietary codes or systems is not supported. Example:

`http://terminology.hl7.org/CodeSystem/organization-type|govt`

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example search by .

_id

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization?_id=675844
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': '572dbd7f-2505-41cd-a9ac-240db77545a1',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization?_id=675844'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/675844',
        'resource': {
          'resourceType': 'Organization',
          'id': '675844',
          'meta': {
            'versionId': '29',
            'lastUpdated': '2018-05-22T16:46:28Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Organization</b></p><p><b>Name</b>: Model
    Hospital</p><p><b>Status</b>: Active</p><p><b>TAXID</b>: 123456785</p></div>'
          },
          'identifier': [
            {
              'use': 'official',
              'type': {
                'coding': [
                  {
                    'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/334',
                    'code': 'TAXID',
                    'display': 'zzTax Identifier',
                    'userSelected': true
                  },
                  {
                    'system': 'http://terminology.hl7.org/CodeSystem/v2-0203',
                    'code': 'TAX',
                    'display': 'Tax ID number'
                  }
                ],
                'text': 'zzTax Identifier'
              },
              'system': 'urn:oid:2.16.840.1.113883.4.4',
              'value': '123456785'
            }
          ],
          'active': true,
          'type': [
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
                  'code': 'EMPLOYER',
                  'display': 'Employer',
                  'userSelected': true
                }
              ],
              'text': 'Employer'
            },
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
                  'code': 'PLACEOFSVC',
                  'display': 'Place of Service',
                  'userSelected': true
                },
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/organization-type',
                  'code': 'prov',
                  'display': 'Healthcare Provider'
                }
              ],
              'text': 'Place of Service'
            },
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
                  'code': 'CLIENT',
                  'display': 'Client',
```

```
        'userSelected': true
      }
    ],
    'text': 'Client'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'LABORATORY',
        'display': 'Exchange Lab',
        'userSelected': true
      }
    ],
    'text': 'Exchange Lab'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'FACILITY',
        'display': 'Facility',
        'userSelected': true
      }
    ],
    'text': 'Facility'
  }
],
'name': 'Model Hospital',
'telecom': [
  {
    'system': 'phone',
    'value': '5555550013',
    'use': 'work'
  }
],
'address': [
  {
    'use': 'work',
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
    'use': 'billing',
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
    'use': 'billing',
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
```

```
          }
        ]
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get an organization by ID

File: r4-rest-api-v32/op-organization-id-get.html

Retrieves an individual organization by the associated organization ID.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example retrieval of a organization record by resource ID.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Organization/675844
```

Example Response:

```
{
    'resourceType': 'Organization',
    'id': '675844',
    'meta': {
      'versionId': '29',
      'lastUpdated': '2018-05-22T16:46:28Z'
    },
    'text': {
      'status': 'generated',
      'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Organization</b></p><p><b>Name</b>: Model
    Hospital</p><p><b>Status</b>: Active</p><p><b>TAXID</b>: 123456785</p></div>'
    },
    'identifier': [
      {
        'use': 'official',
        'type': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/334',
              'code': 'TAXID',
              'display': 'zzTax Identifier',
              'userSelected': true
            },
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v2-0203',
              'code': 'TAX',
              'display': 'Tax ID number'
            }
          ],
          'text': 'zzTax Identifier'
```

```
    },
    'system': 'urn:oid:2.16.840.1.113883.4.4',
    'value': '123456785'
  }
],
'active': true,
'type': [
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'EMPLOYER',
        'display': 'Employer',
        'userSelected': true
      }
    ],
    'text': 'Employer'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'PLACEOFSVC',
        'display': 'Place of Service',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/organization-type',
        'code': 'prov',
        'display': 'Healthcare Provider'
      }
    ],
    'text': 'Place of Service'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'CLIENT',
        'display': 'Client',
        'userSelected': true
      }
    ],
    'text': 'Client'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'LABORATORY',
        'display': 'Exchange Lab',
        'userSelected': true
      }
    ],
    'text': 'Exchange Lab'
  },
  {
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/278',
        'code': 'FACILITY',
        'display': 'Facility',
        'userSelected': true
      }
    ],
    'text': 'Facility'
  }
],
'name': 'Model Hospital',
'telecom': [
  {
    'system': 'phone',
    'value': '5555550013',
    'use': 'work'
  }
],
'address': [
  {
    'use': 'work',
    'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
    'line': [
      '6276 Rockriver Drive'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64117',
    'country': 'US'
  },
  {
```

```
        'use': 'billing',
        'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
        'line': [
          '6276 Rockriver Drive'
        ],
        'city': 'Kansas City',
        'state': 'MO',
        'postalCode': '64117',
        'country': 'US'
      },
      {
        'use': 'billing',
        'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
        'line': [
          '6276 Rockriver Drive'
        ],
        'city': 'Kansas City',
        'state': 'MO',
        'postalCode': '64117',
        'country': 'US'
      },
      {
        'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
        'line': [
          '6276 Rockriver Drive'
        ],
        'city': 'Kansas City',
        'state': 'MO',
        'postalCode': '64117',
        'country': 'US'
      },
      {
        'text': '6276 Rockriver Drive\nKansas City, MO 64117\nUS',
        'line': [
          '6276 Rockriver Drive'
        ],
        'city': 'Kansas City',
        'state': 'MO',
        'postalCode': '64117',
        'country': 'US'
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Patient

File: r4-rest-api-v32/api-patient.html

Description

Patient

This topic describes information for the Patient resource.

Overview

The Patient resource provides general demographic information about a person receiving healthcare services from a specific organization. Common demographic fields include patient ID, patient name, gender, date of birth, address, phone number, primary language, and marital status. Additional concepts returned as extensions and not part of the base resource include time of birth, preferred contact, race, ethnicity, and birth sex. Oracle Health Millennium Platform is a patient-centric application; thus, many other resources include the patient ID in their queries. A person receiving care from multiple organizations may have data available in multiple patient resources in multiple FHIR servers.

This resource supports the US Core Patient Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The examples provided here are non-normative and replaying them in the public sandbox is not guaranteed to yield the results shown on the site. The field is present only when the gender identity is confidential.

Note:

meta.security

The following fields are returned if valued:

* Patient id

* Patient Identifier

* Active

* Patient name

* Telecom Information (may include phone and email)

* Gender (administrative)

* Extensions including birth time, preferred contact, birth sex, ethnicity, communication preference, race, and inline security label (for more information, see the Extensions section of this topic)

* Date of Birth

Deceased

* Boolean

* dateTime

* Address

* Marital status

Multiple Birth Information

* Boolean

* Integer

* Contact person (guardian, parent or emergency)

* Communication (preferred language)

Primary Care Providers

* Reference (Practitioner)

* Reference (Organization)

Terminology Bindings

Patient.identifier.type

Description: The type used to determine which identifier to use for a specific purpose.

Details: Identifier Type Codes

System: http://hl7.org/fhir/identifier-type

Details: v2 Identifier Type

System: http://terminology.hl7.org/CodeSystem/v2-0203

Details: For more information, see Code Set 4 Person Identifier (Alias) Types under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/4

Patient.maritalStatus

Description: The patient's most recent marital (civil) status.

Details: Marital Status Codes

System: http://hl7.org/fhir/marital-status

Details: v3.MaritalStatus

System: http://terminology.hl7.org/CodeSystem/v3-MaritalStatus

Details: v3.NullFlavor

System: http://terminology.hl7.org/CodeSystem/v3-NullFlavor

Details: For more information, see Code Set 38 Marital Status under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/38

Patient.contact.relationship

Description: The nature of the relationship between a patient and a contact person for that patient.

Details: PatientContactRelationship

System: http://hl7.org/fhir/patient-contactrelationship

Details: v2.0131

System: http://terminology.hl7.org/CodeSystem/v2-0131

Details: For more information, see Code Set 351 Person Relationship Types under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/351

Patient.communication.language

Description: The language that may be used to communicate with the patient about their health.

Details: CommonLanguages

System: http://hl7.org/fhir/languages

Details: IETF Identifying Languages

System: urn:ietf:bcp:47

Details:

System:

Details: For more information, see Code Set 36 Languages under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/36

Patient.meta.security

Description: These tags connect specific resources to the overall security policy and infrastructure.

Details: ValueSet

System: http://terminology.hl7.org/CodeSystem/v3-ActCode

Extensions

Patient Birth Time

Patient Preferred Contact

US Core Race

US Core Ethnicity

US Core Birth Sex

* US Core Gender Identity

* US Core Sex

* US Core Tribal Affiliation

Communication Preference

* Inline Security Label

Patient

The operations from the Patient category.

Create a patient

Method: post Path: /Patient

Get a list of patients

Method: get Path: /Patient

Get a patient by ID

Method: get Path: /Patient{ID}

Patch a patient

Method: patch Path: /Patient{ID}

Create a patient

File: r4-rest-api-v32/op-patient-post.html

Creates a new patient.

Authorization Types

This resource supports the following authorization types:

* Provider

* System

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : PatientCreateBody

Type:

object

Show Source

address: array

address

Addresses for the individual.

At least one of the following should be provided:

* line
* city
* district
* state
* postalCode
* country

Each address must not contain the text field.

birthDate: string

The date of birth for the individual.

Example:

1990-09-15

communication: array

communication

Maximum Number of Items:

Language which may be used to communicate with the patient about their health.

extension: array

extension

The Patient resource supports multiple extensions. See the StructureDefinition of each extension for further formatting details.

US Core Birth Sex

The sex of the patient at birth.

Example:


```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
  "valueCode": "F"
}
```


US Core Ethnicity

The ethnicity of the patient.

Example:


```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
  "extension": [
    {
      "url": "ombCategory",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2186-5",
        "display": "Non Hispanic or Latino"
      }
    }
  ]
}
```


US Core Race

The race of the patient.

Example:


```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
  "extension": [
    {
      "url": "ombCategory",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2028-9",
        "display": "Asian"
      }
    },
    {
      "url": "detailed",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2039-6",
        "display": "Japanese"
```

```
      }
    }
  ]
}
```

US Core Gender Identity

The US Core Gender Identity extension represents an individual's sense of being a man, woman, boy, girl, nonbinary, or something else, ascertained by asking them what that identity is.

Example:

```
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "446141000124107"
      }
    ]
  },
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
}
```

US Core Sex

The US Core Sex extension is used to reflect the documentation of a person's sex.

Example:

```
{
  "valueCode": "248153007",
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex"
}
```

US Core Tribal Affiliation

The US Core Tribal Affiliation extension represents a tribe or band with which a person associates and, optionally, whether they are enrolled.

Example:

```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation",
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
            "code": "472",
            "display": "Native Village of Napaskiak"
          }
        ],
        "text": "Native Village of Napaskiak"
      },
      "url": "tribalAffiliation"
    },
    {
      "valueBoolean": true,
      "url": "isEnrolled"
    }
  ]
}
```

Communication Preference

The communication preference for the patient.

Example:

```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
      "extension": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                    "code": "485602703",
                    "display": "Appointment Reminder",
                    "userSelected": false
                  }
                ],
                "text": "Appointment Reminder"
              },
              "url": "communication-type"
            },
            {
              "valueCodeableConcept": {
```

```
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                    "code": "4054467",
                    "display": "Phone",
                    "userSelected": false
                  }
                ]
                },
              "url": "contact-method"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                    "code": "163",
                    "display": "Business",
                    "userSelected": false
                  }
                ],
                "text": "Home"
              },
              "url": "contact-type"
            }
          ],
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
      "verified": "2019-12-26T15:41:52.000Z"
    }
  ]
}
```


gender: string

Allowed Values:

["male", "female", "other", "unknown"]

The gender that the patient is considered to have for administration and record keeping purposes.

generalPractitioner: array

generalPractitioner

Patient's nominated care provider.

identifier(required): array

identifier

Minimum Number of Items:

The identifiers for the patient.

At least one identifier must be provided. The provided identifier must be a reference to the Organization in which the patient is being enrolled.

Subsequent identifiers may be provided, but have the following constraints:

* Must not contain the assigner and use fields.

* Must contain the type, system, and value fields.

Example:


```
{
  "identifier": [
    {
      "assigner": {
        "reference": "Organization/619848"
      }
    },
    {
      "type": {
        "coding": [
          {
            "code": "MR",
            "system": "http://hl7.org/fhir/v2/0203"
          }
        ]
      },
      "system": "urn:oid:1.1.1.1.1.1",
      "value": "<unique identifier>",
      "period": {
        "start": "2016-01-02T00:00:00-05:00",
        "end": "2020-01-02T00:00:00-05:00"
      }
    }
  ]
}
```


maritalStatus: object

maritalStatus

The patient's most recent marital (civil) status.

name(required): array

name

Minimum Number of Items:

The name associated with the practitioner.

At least one name must be provided.

* It must have a use of official.

* If a period is provided, it must not have an end date.

* It must include both a given and a family name.

All provided names:

* Must not contain the text field.

* Must contain either a given or family name.

resourceType(required): string

Allowed Values:

["Patient"]

The type of the FHIR resource.

telecom: array

telecom

A contact detail (such as a telephone number or an email address) by which the individual may be contacted.

If an extension is provided:

* The extension.valueString field has a character limit of 100.

* The system field must be phone.

* The extension.url field must be http://hl7.org/fhir/StructureDefinition/contactpoint-extension.


```
{
    "type":"object",
    "required":[
        "resourceType",
        "identifier",
        "name"
    ],
    "properties":{
        "resourceType":{
            "description":"<p>The type of the FHIR resource.</p>\n",
            "type":"string",
            "enum":[
                "Patient"
            ]
        },
        "identifier":{
            "description":"<p>The identifiers for the patient.</p>\n<p>At least one identifier must be provided. The
    provided identifier must be a reference to the Organization in which the patient is being
    enrolled.</p>\n<p>Subsequent identifiers may be provided, but have the following constraints:\n  <ul>\n    <li>Must
    not contain the <code>assigner</code> and <code>use</code> fields. </li>\n    <li>Must contain the
    <code>type</code>, <code>system</code>, and <code>value</code> fields. </li>\n  </ul>\n</p>\n<p><b>Example:</b>\n
    <pre><code>{\n  \"identifier\": [\n    {\n      \"assigner\": {\n        \"reference\": \"Organization/619848\"\n
          }\n    },\n    {\n      \"type\": {\n        \"coding\": [\n          {\n            \"code\": \"MR\",\n
                \"system\": \"http://hl7.org/fhir/v2/0203\"\n          }\n        ]\n      },\n      \"system\":
    \"urn:oid:1.1.1.1.1.1\",\n      \"value\": \"&lt;unique identifier&gt;\",\n      \"period\": {\n        \"start\":
    \"2016-01-02T00:00:00-05:00\",\n        \"end\": \"2020-01-02T00:00:00-05:00\"\n      }\n    }\n  ]\n}\n
    </code></pre>\n</p>\n",
            "type":"array",
            "minItems":1,
            "items":{
                "type":"object",
                "properties":{
                    "assigner":{
                        "type":"string",
                        "description":"<p>May only be provided on the first identifier object. Contains a reference to
    the organization in which the patient is being enrolled.</p>\n",
                        "example":"Organization/619848"
                    },
                    "type":{
                        "type":"object",
                        "properties":{
                            "coding":{
                                "type":"array",
                                "minItems":1,
                                "maxItems":1,
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "code":{
                                            "type":"string",
```

```
                                    "example":"MR"
                                },
                                "system":{
                                    "type":"string",
                                    "example":"http://hl7.org/fhir/v2/0203"
                                }
                            }
                        }
                    }
                },
                "system":{
                    "type":"string",
                    "example":"urn:oid:1.1.1.1.1.1"
                },
                "value":{
                    "type":"string",
                    "description":"<p>Must be a unique value.</p>",
                    "example":"CW1234563"
                },
                "period":{
                    "type":"object",
                    "properties":{
                        "start":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>",
                            "example":"2019-12-01T00:00:00.000Z"
                        },
                        "end":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>",
                            "example":"2029-12-01T23:59:59.000Z"
                        }
                    }
                }
            }
        }
    },
    "name":{
        "description":"<p>The name associated with the practitioner.</p>\n<p>At least one name must be provided.\n
<ul>\n    <li>It must have a <code>use</code> of official. </li>\n    <li>If a <code>period</code> is provided, it
must not have an <code>end</code> date. </li>\n    <li>It must include both a <code>given</code> and a
<code>family</code> name. </li>\n  </ul>\n</p>\n<p>All provided names:\n  <ul>\n    <li>Must not contain the
<code>text</code> field. </li>\n    <li>Must contain either a <code>given</code> or <code>family</code> name.
</li>\n  </ul>\n</p>\n",
        "type":"array",
        "minItems":1,
        "items":{
            "type":"object",
            "required":[
                "use",
                "family"
            ],
            "properties":{
                "family":{
                    "type":"string",
                    "maxLength":100,
                    "example":"Williams"
                },
                "given":{
                    "type":"array",
                    "minItems":1,
                    "items":{
                        "type":"string",
                        "maxLength":100,
                        "example":"James McAfee"
                    }
                },
                "prefix":{
                    "type":"array",
                    "maxItems":1,
                    "items":{
                        "type":"string",
                        "maxLength":100,
                        "example":"Dr."
                    }
                },
                "suffix":{
                    "maxItems":1,
                    "type":"array",
                    "items":{
                        "type":"string",
                        "maxLength":100,
                        "example":"M.D."
                    }
                },
                "period":{
                    "type":"object",
                    "description":"<p>Must not include an <code>end</code> date.</p>",
```

```
                    "properties":{
                        "start":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>",
                            "example":"2020-12-01T00:00:00.000Z"
                        }
                    },
                    "use":{
                        "type":"string",
                        "enum":[
                            "usual",
                            "official",
                            "temp",
                            "nickname",
                            "anonymous",
                            "old",
                            "maiden"
                        ]
                    }
                }
            }
        },
        "telecom":{
            "description":"<p>A contact detail (such as a telephone number or an email address) by which the individual
may be contacted.</p>\n<p>If an extension is provided:\n  <ul>\n    <li>The <code>extension.valueString</code> field
 has a character limit of 100. </li>\n    <li>The <code>system</code> field must be <code>phone</code>. </li>\n
<li>The <code>extension.url</code> field must be <code>http://hl7.org/fhir/StructureDefinition/contactpoint-
extension</code>. </li>\n  </ul>\n</p>\n",
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "system",
                    "use",
                    "value"
                ],
                "properties":{
                    "system":{
                        "type":"string",
                        "enum":[
                            "phone",
                            "fax",
                            "email",
                            "pager",
                            "url",
                            "sms",
                            "other"
                        ]
                    },
                    "value":{
                        "type":"string",
                        "maxLength":100
                    },
                    "use":{
                        "type":"string",
                        "enum":[
                            "home",
                            "work",
                            "temp",
                            "old",
                            "mobile"
                        ]
                    },
                    "rank":{
                        "type":"integer"
                    },
                    "period":{
                        "type":"object",
                        "properties":{
                            "start":{
                                "type":"string",
                                "description":"<p>Must include a time component with a timezone.</p>\n",
                                "example":"2012-05-17T15:33:18.000Z"
                            },
                            "end":{
                                "type":"string",
                                "description":"<p>Must include a time component with a timezone.</p>\n",
                                "example":"2028-05-17T15:33:18.000Z"
                            }
                        }
                    }
                }
            }
        },
        "gender":{
            "description":"<p>The gender that the patient is considered to have for administration and record keeping
purposes.</p>\n",
            "type":"string",
```

```
        "enum":[
            "male",
            "female",
            "other",
            "unknown"
        ]
    },
    "birthDate":{
        "description":"<p>The date of birth for the individual.</p>\n",
        "type":"string",
        "example":"1990-09-15"
    },
    "address":{
        "description":"<p>Addresses for the individual.</p>\n<p>At least one of the following should be provided:\n
  <ul>\n    <li>\n      <code>line</code>\n    </li>\n    <li>\n      <code>city</code>\n    </li>\n    <li>\n
<code>district</code>\n    </li>\n    <li>\n      <code>state</code>\n    </li>\n    <li>\n
<code>postalCode</code>\n    </li>\n    <li>\n      <code>country</code>\n    </li>\n  </ul>\n</p>\n<p>Each address
must not contain the <code>text</code> field.</p>\n",
        "type":"array",
        "items":{
            "type":"object",
            "required":[
                "use"
            ],
            "properties":{
                "use":{
                    "type":"string",
                    "enum":[
                        "home",
                        "work",
                        "temp",
                        "old",
                        "billing"
                    ]
                },
                "type":{
                    "type":"string",
                    "enum":[
                        "postal",
                        "physical",
                        "both"
                    ]
                },
                "line":{
                    "type":"array",
                    "items":{
                        "type":"string",
                        "maxLength":100,
                        "example":"1212 Faircastle Drive"
                    },
                    "example":[
                        "1212 Faircastle Drive",
                        "Apartment 406"
                    ]
                },
                "city":{
                    "type":"string",
                    "maxLength":100,
                    "example":"KC"
                },
                "district":{
                    "type":"string",
                    "maxLength":100,
                    "example":"Jackson"
                },
                "state":{
                    "type":"string",
                    "maxLength":100,
                    "example":"KS"
                },
                "postalCode":{
                    "type":"string",
                    "maxLength":25,
                    "example":"64199"
                },
                "country":{
                    "type":"string",
                    "maxLength":100,
                    "example":"United States of America"
                },
                "period":{
                    "type":"object",
                    "properties":{
                        "start":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>\n",
                            "example":"2012-05-17T15:33:18.000Z"
                        },
                        "end":{
```

```
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>\n",
                        "example":"2032-05-17T15:33:18.000Z"
                    }
                }
            }
        }
    }
},
"maritalStatus":{
    "description":"<p>The patient's most recent marital (civil) status.</p>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string",
            "example":"Never Married"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string",
                        "example":"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"
                    },
                    "code":{
                        "type":"string",
                        "example":"S"
                    },
                    "display":{
                        "type":"string",
                        "example":"Never Married"
                    }
                }
            }
        }
    }
},
"communication":{
    "description":"<p>Language which may be used to communicate with the patient about their health.</p>\n",
    "type":"array",
    "maxItems":1,
    "items":{
        "type":"object",
        "properties":{
            "language":{
                "type":"object",
                "properties":{
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string",
                                    "example":"urn:ietf:bcp:47"
                                },
                                "code":{
                                    "type":"string",
                                    "example":"en"
                                },
                                "display":{
                                    "type":"string",
                                    "example":"English"
                                }
                            }
                        }
                    },
                    "text":{
                        "type":"string",
                        "example":"English"
                    }
                }
            },
            "preferred":{
                "type":"boolean"
            }
        }
    }
},
"generalPractitioner":{
    "description":"<p>Patient's nominated care provider.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "description":"<p>Must only include references to Practitioners.</p>",
        "properties":{
            "reference":{
```

```
                    "type":"string",
                    "example":"Practitioner/995932"
                }
            }
        }
    },
    "extension":{
        "description":"<p>The Patient resource supports multiple extensions. See the StructureDefinition of each
extension for further formatting details.</p>\n<b>US Core Birth Sex</b>\n<p>The sex of the patient at
birth.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-
core-birthsex\",\n  \"valueCode\": \"F\"\n}\n  </code></pre>\n</p>\n<b>US Core Ethnicity</b>\n<p>The ethnicity of
the patient.</p>\n<p><b>Example:</b>\n  <pre><code>\n{\n  \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity\",\n  \"extension\": [\n    {\n      \"url\":
\"ombCategory\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n
\"code\": \"2186-5\",\n        \"display\": \"Non Hispanic or Latino\"\n      }\n    }\n  ]\n}\n
</code></pre>\n</p>\n<b>US Core Race</b>\n<p>The race of the patient.</p>\n<p><b>Example:</b>\n  <pre><code>{\n
\"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race\",\n  \"extension\": [\n    {\n
\"url\": \"ombCategory\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n
        \"code\": \"2028-9\",\n        \"display\": \"Asian\"\n      }\n    },\n    {\n      \"url\":
\"detailed\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n        \"code\":
 \"2039-6\",\n        \"display\": \"Japanese\"\n      }\n    }\n  ]\n}\n  </code></pre>\n</p>\n<b>US Core Gender
Identity</b>\n<p>The US Core Gender Identity extension represents an individual's sense of being a man, woman, boy,
girl, nonbinary, or something else, ascertained by asking them what that identity is.</p>\n<p><b>Example:</b>\n
<pre><code>{\n  \"valueCodeableConcept\": {\n    \"coding\": [\n      {\n        \"system\":
\"http://snomed.info/sct\",\n        \"code\": \"446141000124107\"\n      }\n    ]\n  },\n  \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity\"\n}\n  </code></pre>\n</p>\n<b>US Core
Sex</b>\n<p>The US Core Sex extension is used to reflect the documentation of a person's
sex.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"valueCode\": \"248153007\",\n  \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex\"\n}\n  </code></pre>\n</p>\n<b>US Core Tribal
Affiliation</b>\n<p>The US Core Tribal Affiliation extension represents a tribe or band with which a person
associates and, optionally, whether they are enrolled.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation\",\n  \"extension\": [\n    {\n
\"valueCodeableConcept\": {\n        \"coding\": [\n          {\n            \"system\":
\"http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS\",\n            \"code\": \"472\",\n
\"display\": \"Native Village of Napaskiak\"\n          }\n        ],\n        \"text\": \"Native Village of
Napaskiak\"\n      },\n      \"url\": \"tribalAffiliation\"\n    },\n    {\n      \"valueBoolean\": true,\n
\"url\": \"isEnrolled\"\n    }\n  ]\n}\n  </code></pre>\n</p>\n<b>Communication Preference</b>\n<p>The communication
 preference for the patient.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"extension\": [\n    {\n      \"url\":
\"https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference\",\n      \"extension\": [\n
{\n          \"extension\": [\n            {\n              \"valueCodeableConcept\": {\n                \"coding\":
 [\n                  {\n                    \"system\":
\"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016\",\n                    \"code\":
\"485602703\",\n                    \"display\": \"Appointment Reminder\",\n                    \"userSelected\":
false\n                  }\n                ],\n                \"text\": \"Appointment Reminder\"\n
},\n              \"url\": \"communication-type\"\n            },\n            {\n
\"valueCodeableConcept\": {\n                \"coding\": [\n                  {\n                    \"system\":
\"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042\",\n                    \"code\":
\"4054467\",\n                    \"display\": \"Phone\",\n                    \"userSelected\": false\n
                }\n                ]\n              },\n              \"url\": \"contact-method\"\n
},\n            {\n              \"valueCodeableConcept\": {\n                \"coding\": [\n                  {\n
                    \"system\": \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43\",\n
                    \"code\": \"163\",\n                    \"display\": \"Business\",\n
\"userSelected\": false\n                  }\n                ],\n                \"text\": \"Home\"\n
},\n              \"url\": \"contact-type\"\n            }\n          ],\n      \"url\": \"https://fhir-
ehr.cerner.com/r4/StructureDefinition/communication-preference\",\n      \"verified\":
\"2019-12-26T15:41:52.000Z\"\n    }\n  ]\n}\n  </code></pre>\n</p>\n",
        "type":"array",
        "items":{
            "type":"object",
            "required":[
                "url"
            ],
            "properties":{
                "url":{
                    "type":"string"
                },
                "valueCode":{
                    "type":"string"
                }
            }
        }
    }
},
"example":"{\n  \"resourceType\": \"Patient\",\n  \"extension\": [\n    {\n      \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex\",\n      \"valueCode\": \"M\"\n    },\n    {\n
      \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race\",\n      \"extension\": [\n
{\n          \"url\": \"ombCategory\",\n          \"valueCoding\": {\n            \"system\":
\"urn:oid:2.16.840.1.113883.6.238\",\n            \"code\": \"2028-9\",\n            \"display\": \"Asian\"\n
          }\n        },\n        {\n          \"url\": \"detailed\",\n          \"valueCoding\": {\n
\"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n            \"code\": \"2039-6\",\n            \"display\":
\"Japanese\"\n          }\n        }\n      ]\n    },\n    {\n      \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity\",\n      \"extension\": [\n        {\n
          \"url\": \"ombCategory\",\n          \"valueCoding\": {\n            \"system\":
\"urn:oid:2.16.840.1.113883.6.238\",\n            \"code\": \"2186-5\",\n            \"display\": \"Non Hispanic or
Latino\"\n          }\n        }\n      ]\n    },\n    {\n      \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity\",\n      \"valueCodeableConcept\": {\n
        \"coding\": [\n          {\n            \"system\": \"http://snomed.info/sct\",\n            \"code\":
\"446141000124107\"\n          }\n        ]\n      }\n    },\n    {\n      \"url\":
\"http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex\",\n      \"valueCode\": \"248153007\"\n    },\n
```

```
    {\n      \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation\",\n
    \"extension\": [\n        {\n          \"valueCodeableConcept\": {\n            \"coding\": [\n              {\n
                  \"system\": \"http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS\",\n                \"code\":
    \"472\",\n                \"display\": \"Native Village of Napaskiak\",\n                \"userSelected\": true\n
                }\n            ],\n            \"text\": \"Native Village of Napaskiak\"\n          },\n
    \"url\": \"tribalAffiliation\"\n        },\n        {\n          \"valueBoolean\": true,\n          \"url\":
    \"isEnrolled\"\n        }\n      ]\n    },\n    {\n      \"url\": \"https://fhir-
    ehr.cerner.com/r4/StructureDefinition/communication-preference\",\n      \"extension\": [\n        {\n
    \"extension\": [\n            {\n              \"valueCodeableConcept\": {\n                \"coding\": [\n
                    {\n                  \"system\":
    \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016\",\n                    \"code\":
    \"485602703\",\n                    \"display\": \"Appointment Reminder\",\n                    \"userSelected\":
    false\n                  }\n                ],\n                \"text\": \"Appointment Reminder\"\n
    },\n              \"url\": \"communication-type\"\n            },\n            {\n
    \"valueCodeableConcept\": {\n                \"coding\": [\n                  {\n                    \"system\":
    \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042\",\n                    \"code\":
    \"4054467\",\n                    \"display\": \"Phone\",\n                    \"userSelected\": false\n
                  }\n                ]\n              },\n              \"url\": \"contact-method\"\n
    },\n            {\n              \"valueCodeableConcept\": {\n                \"coding\": [\n                  {\n
                      \"system\": \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43\",\n
                      \"code\": \"163\",\n                    \"display\": \"Business\",\n
    \"userSelected\": false\n                  }\n                ],\n                \"text\": \"Home\"\n
    },\n              \"url\": \"contact-type\"\n            }\n          ],\n          \"verified\":
    \"2019-12-26T15:41:52.000Z\"\n        }\n      ]\n    }\n  ],\n  \"identifier\": [\n    {\n      \"assigner\": {\n
          \"reference\": \"Organization/675844\"\n      }\n    }\n  ],\n  \"name\": [\n    {\n      \"use\":
    \"official\",\n      \"family\": \"Wolf\",\n      \"given\": [\n        \"Person\",\n        \"Name\"\n      ],\n
        \"period\": {\n        \"start\": \"2010-05-17T14:54:31.000Z\"\n      }\n    },\n    {\n      \"use\":
    \"usual\",\n      \"given\": [\n        \"Bigby\"\n      ],\n      \"period\": {\n        \"start\":
    \"2012-05-22T15:45:50.000Z\"\n      }\n    }\n  ],\n  \"telecom\": [\n    {\n      \"system\": \"phone\",\n
    \"value\": \"8168229121\",\n      \"use\": \"home\",\n      \"id\": \"CI-PH-29811920-0\",\n      \"extension\": [\n
          {\n          \"valueString\": \"12345\",\n          \"url\":
    \"http://hl7.org/fhir/StructureDefinition/contactpoint-extension\"\n        }\n      ],\n      \"period\": {\n
          \"start\": \"2012-05-17T15:33:18.000Z\"\n      }\n    }\n  ],\n  \"gender\": \"male\",\n  \"birthDate\":
    \"1990-09-15\",\n  \"address\": [\n    {\n      \"use\": \"home\",\n      \"line\": [\n        \"121212 Metcalf
    Drive\",\n        \"Apartment 403\"\n      ],\n      \"city\": \"Kansas City\",\n      \"district\": \"Jackson\",\n
          \"state\": \"KS\",\n      \"postalCode\": \"64199\",\n      \"country\": \"United States of America\",\n
    \"period\": {\n        \"start\": \"2012-05-17T15:33:18.000Z\"\n      }\n    }\n  ],\n  \"maritalStatus\": {\n
    \"coding\": [\n      {\n        \"system\": \"http://terminology.hl7.org/CodeSystem/v3-NullFlavor\",\n
    \"code\": \"UNK\",\n        \"display\": \"Unknown\"\n      }\n    ],\n    \"text\": \"Unknown\"\n  },\n
    \"communication\": [\n    {\n      \"language\": {\n        \"coding\": [\n          {\n            \"system\":
    \"urn:ietf:bcp:47\",\n            \"code\": \"en\",\n            \"display\": \"English\"\n          }\n        ],\n
          \"text\": \"English\"\n      },\n      \"preferred\": true\n    }\n  ],\n  \"generalPractitioner\": [\n
    {\n      \"reference\": \"Practitioner/4122622\"\n    }\n  ]\n}\n"
}
```


Example:


```
{
  "resourceType": "Patient",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
      "valueCode": "M"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2028-9",
            "display": "Asian"
          }
        },
        {
          "url": "detailed",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2039-6",
            "display": "Japanese"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2186-5",
            "display": "Non Hispanic or Latino"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity",
```

```
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "446141000124107"
        }
      ]
    }
  },
  {
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
    "valueCode": "248153007"
  },
  {
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation",
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
              "code": "472",
              "display": "Native Village of Napaskiak",
              "userSelected": true
            }
          ],
          "text": "Native Village of Napaskiak"
        },
        "url": "tribalAffiliation"
      },
      {
        "valueBoolean": true,
        "url": "isEnrolled"
      }
    ]
  },
  {
    "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
    "extension": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                  "code": "485602703",
                  "display": "Appointment Reminder",
                  "userSelected": false
                }
              ],
              "text": "Appointment Reminder"
            },
            "url": "communication-type"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                  "code": "4054467",
                  "display": "Phone",
                  "userSelected": false
                }
              ]
            },
            "url": "contact-method"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                  "code": "163",
                  "display": "Business",
                  "userSelected": false
                }

              ],
              "text": "Home"
            },
            "url": "contact-type"
          }
        ],
        "verified": "2019-12-26T15:41:52.000Z"
      }
    ]
  }
],
"identifier": [
```

```
    {
      "assigner": {
        "reference": "Organization/675844"
      }
    }
  ],
  "name": [
    {
      "use": "official",
      "family": "Wolf",
      "given": [
        "Person",
        "Name"
      ],
      "period": {
        "start": "2010-05-17T14:54:31.000Z"
      }
    },
    {
      "use": "usual",
      "given": [
        "Bigby"
      ],
      "period": {
        "start": "2012-05-22T15:45:50.000Z"
      }
    }
  ],
  "telecom": [
    {
      "system": "phone",
      "value": "8168229121",
      "use": "home",
      "id": "CI-PH-29811920-0",
      "extension": [
        {
          "valueString": "12345",
          "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
        }
      ],
      "period": {
        "start": "2012-05-17T15:33:18.000Z"
      }
    }
  ],
  "gender": "male",
  "birthDate": "1990-09-15",
  "address": [
    {
      "use": "home",
      "line": [
        "121212 Metcalf Drive",
        "Apartment 403"
      ],
      "city": "Kansas City",
      "district": "Jackson",
      "state": "KS",
      "postalCode": "64199",
      "country": "United States of America",
      "period": {
        "start": "2012-05-17T15:33:18.000Z"
      }
    }
  ],
  "maritalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-NullFlavor",
        "code": "UNK",
        "display": "Unknown"
      }
    ],
    "text": "Unknown"
  },
  "communication": [
    {
      "language": {
        "coding": [
          {
            "system": "urn:ietf:bcp:47",
            "code": "en",
            "display": "English"
          }
        ],
        "text": "English"
      },
      "preferred": true
    }
  ],
```

```
  "generalPractitioner": [
    {
      "reference": "Practitioner/4122622"
    }
  ]
}
```


Nested Schema : address

Type:

array

Addresses for the individual.

At least one of the following should be provided:

* line

* city

* district

* state

* postalCode

* country

Each address must not contain the text field.

Show Source

* Array of: object items


```
{
    "description":"<p>Addresses for the individual.</p>\n<p>At least one of the following should be provided:\n  <ul>\n
        <li>\n      <code>line</code>\n    </li>\n    <li>\n      <code>city</code>\n    </li>\n    <li>\n
    <code>district</code>\n    </li>\n    <li>\n      <code>state</code>\n    </li>\n    <li>\n
    <code>postalCode</code>\n    </li>\n    <li>\n      <code>country</code>\n    </li>\n  </ul>\n</p>\n<p>Each address
    must not contain the <code>text</code> field.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "use"
        ],
        "properties":{
            "use":{
                "type":"string",
                "enum":[
                    "home",
                    "work",
                    "temp",
                    "old",
                    "billing"
                ]
            },
            "type":{
                "type":"string",
                "enum":[
                    "postal",
                    "physical",
                    "both"
                ]
            },
            "line":{
                "type":"array",
                "items":{
                    "type":"string",
                    "maxLength":100,
                    "example":"1212 Faircastle Drive"
                },
                "example":[
                    "1212 Faircastle Drive",
                    "Apartment 406"
                ]
            },
            "city":{
                "type":"string",
                "maxLength":100,
                "example":"KC"
            },
            "district":{
                "type":"string",
                "maxLength":100,
                "example":"Jackson"
            },
            "state":{
                "type":"string",
                "maxLength":100,
                "example":"KS"
            },
            "postalCode":{
```

```
                "type":"string",
                "maxLength":25,
                "example":"64199"
            },
            "country":{
                "type":"string",
                "maxLength":100,
                "example":"United States of America"
            },
            "period":{
                "type":"object",
                "properties":{
                    "start":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>\n",
                        "example":"2012-05-17T15:33:18.000Z"
                    },
                    "end":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>\n",
                        "example":"2032-05-17T15:33:18.000Z"
                    }
                }
            }
        }
    }
}
```


Nested Schema : communication

Type:

array

Maximum Number of Items:

Language which may be used to communicate with the patient about their health.

Show Source

* Array of: object items


```
{
    "description":"<p>Language which may be used to communicate with the patient about their health.</p>\n",
    "type":"array",
    "maxItems":1,
    "items":{
        "type":"object",
        "properties":{
            "language":{
                "type":"object",
                "properties":{
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string",
                                    "example":"urn:ietf:bcp:47"
                                },
                                "code":{
                                    "type":"string",
                                    "example":"en"
                                },
                                "display":{
                                    "type":"string",
                                    "example":"English"
                                }
                            }
                        }
                    },
                    "text":{
                        "type":"string",
                        "example":"English"
                    }
                }
            },
            "preferred":{
                "type":"boolean"
            }
        }
    }
}
```


Nested Schema : extension

Type:

array

The Patient resource supports multiple extensions. See the StructureDefinition of each extension for further formatting details.

US Core Birth Sex

The sex of the patient at birth.

Example:

```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
  "valueCode": "F"
}
```

US Core Ethnicity

The ethnicity of the patient.

Example:

```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
  "extension": [
    {
      "url": "ombCategory",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2186-5",
        "display": "Non Hispanic or Latino"
      }
    }
  ]
}
```

US Core Race

The race of the patient.

Example:

```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
  "extension": [
    {
      "url": "ombCategory",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2028-9",
        "display": "Asian"
      }
    },
    {
      "url": "detailed",
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2039-6",
        "display": "Japanese"
      }
    }
  ]
}
```

US Core Gender Identity

The US Core Gender Identity extension represents an individual's sense of being a man, woman, boy, girl, nonbinary, or something else, ascertained by asking them what that identity is.

Example:

```
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "446141000124107"
      }
    ]
  },
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
}
```

US Core Sex

The US Core Sex extension is used to reflect the documentation of a person's sex.

Example:

```
{
  "valueCode": "248153007",
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex"
}
```

US Core Tribal Affiliation

The US Core Tribal Affiliation extension represents a tribe or band with which a person associates and, optionally, whether they are enrolled.

Example:


```
{
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation",
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
            "code": "472",
            "display": "Native Village of Napaskiak"
          }
        ],
        "text": "Native Village of Napaskiak"
      },
      "url": "tribalAffiliation"
    },
    {
      "valueBoolean": true,
      "url": "isEnrolled"
    }
  ]
}
```


Communication Preference

The communication preference for the patient.

Example:


```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
      "extension": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                    "code": "485602703",
                    "display": "Appointment Reminder",
                    "userSelected": false
                  }
                ],
                "text": "Appointment Reminder"
              },
              "url": "communication-type"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                    "code": "4054467",
                    "display": "Phone",
                    "userSelected": false
                  }
                ]
                },
              "url": "contact-method"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                    "code": "163",
                    "display": "Business",
                    "userSelected": false
                  }
                ],
                "text": "Home"
              },
              "url": "contact-type"
            }
          ],
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
      "verified": "2019-12-26T15:41:52.000Z"
    }
  ]
}
```


Show Source

* Array of: object items


```
{
    "description":"<p>The Patient resource supports multiple extensions. See the StructureDefinition of each extension
    for further formatting details.</p>\n<b>US Core Birth Sex</b>\n<p>The sex of the patient at
    birth.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-
    core-birthsex\",\n  \"valueCode\": \"F\"\n}\n  </code></pre>\n</p>\n<b>US Core Ethnicity</b>\n<p>The ethnicity of
    the patient.</p>\n<p><b>Example:</b>\n  <pre><code>\n{\n  \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity\",\n  \"extension\": [\n    {\n      \"url\":
    \"ombCategory\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n
    \"code\": \"2186-5\",\n        \"display\": \"Non Hispanic or Latino\"\n      }\n    }\n  ]\n}\n
    </code></pre>\n</p>\n<b>US Core Race</b>\n<p>The race of the patient.</p>\n<p><b>Example:</b>\n  <pre><code>{\n
    \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race\",\n  \"extension\": [\n    {\n
    \"url\": \"ombCategory\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n
           \"code\": \"2028-9\",\n        \"display\": \"Asian\"\n      }\n    },\n    {\n      \"url\":
    \"detailed\",\n      \"valueCoding\": {\n        \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n        \"code\":
     \"2039-6\",\n        \"display\": \"Japanese\"\n      }\n    }\n  ]\n}\n  </code></pre>\n</p>\n<b>US Core Gender
    Identity</b>\n<p>The US Core Gender Identity extension represents an individual's sense of being a man, woman, boy,
    girl, nonbinary, or something else, ascertained by asking them what that identity is.</p>\n<p><b>Example:</b>\n
    <pre><code>{\n  \"valueCodeableConcept\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://snomed.info/sct\",\n        \"code\": \"446141000124107\"\n      }\n    ]\n  },\n  \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity\"\n}\n  </code></pre>\n</p>\n<b>US Core
    Sex</b>\n<p>The US Core Sex extension is used to reflect the documentation of a person's
    sex.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"valueCode\": \"248153007\",\n  \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex\"\n}\n  </code></pre>\n</p>\n<b>US Core Tribal
    Affiliation</b>\n<p>The US Core Tribal Affiliation extension represents a tribe or band with which a person
    associates and, optionally, whether they are enrolled.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation\",\n  \"extension\": [\n    {\n
    \"valueCodeableConcept\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS\",\n            \"code\": \"472\",\n
    \"display\": \"Native Village of Napaskiak\"\n          }\n        ],\n        \"text\": \"Native Village of
    Napaskiak\"\n      },\n      \"url\": \"tribalAffiliation\"\n    },\n    {\n      \"valueBoolean\": true,\n
    \"url\": \"isEnrolled\"\n    }\n  ]\n}\n  </code></pre>\n</p>\n<b>Communication Preference</b>\n<p>The communication
     preference for the patient.</p>\n<p><b>Example:</b>\n  <pre><code>{\n  \"extension\": [\n    {\n      \"url\":
    \"https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference\",\n      \"extension\": [\n
    {\n          \"extension\": [\n            {\n              \"valueCodeableConcept\": {\n                \"coding\":
     [\n                {\n                  \"system\":
    \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016\",\n                  \"code\":
    \"485602703\",\n                  \"display\": \"Appointment Reminder\",\n                  \"userSelected\":
    false\n                }\n              ],\n              \"text\": \"Appointment Reminder\"\n
    },\n            \"url\": \"communication-type\"\n          },\n          {\n
    \"valueCodeableConcept\": {\n              \"coding\": [\n                {\n                  \"system\":
    \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042\",\n                  \"code\":
    \"4054467\",\n                  \"display\": \"Phone\",\n                  \"userSelected\": false\n
                }\n              ]\n            },\n            \"url\": \"contact-method\"\n
    },\n          {\n            \"valueCodeableConcept\": {\n              \"coding\": [\n                {\n
                  \"system\": \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43\",\n
                  \"code\": \"163\",\n                  \"display\": \"Business\",\n
    \"userSelected\": false\n                }\n              ],\n              \"text\": \"Home\"\n
    },\n            \"url\": \"contact-type\"\n          }\n        ],\n      \"url\": \"https://fhir-
    ehr.cerner.com/r4/StructureDefinition/communication-preference\",\n      \"verified\":
    \"2019-12-26T15:41:52.000Z\"\n    }\n  ]\n}\n  </code></pre>\n</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "url"
        ],
        "properties":{
            "url":{
                "type":"string"
            },
            "valueCode":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : generalPractitioner

Type:

array

Patient's nominated care provider.

Show Source

Array of: object

items

Must only include references to Practitioners.


```
{
    "description":"<p>Patient's nominated care provider.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "description":"<p>Must only include references to Practitioners.</p>",
        "properties":{
```

```
            "reference":{
                "type":"string",
                "example":"Practitioner/995932"
            }
        }
    }
}
```

Nested Schema : identifier

Type:

array

Minimum Number of Items:

The identifiers for the patient.

At least one identifier must be provided. The provided identifier must be a reference to the Organization in which the patient is being enrolled.

Subsequent identifiers may be provided, but have the following constraints:

* Must not contain the assigner and use fields.

* Must contain the type, system, and value fields.

Example:

```
{
  "identifier": [
    {
      "assigner": {
        "reference": "Organization/619848"
      }
    },
    {
      "type": {
        "coding": [
          {
            "code": "MR",
            "system": "http://hl7.org/fhir/v2/0203"
          }
        ]
      },
      "system": "urn:oid:1.1.1.1.1.1",
      "value": "<unique identifier>",
      "period": {
        "start": "2016-01-02T00:00:00-05:00",
        "end": "2020-01-02T00:00:00-05:00"
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"<p>The identifiers for the patient.</p>\n<p>At least one identifier must be provided. The provided
    identifier must be a reference to the Organization in which the patient is being enrolled.</p>\n<p>Subsequent
    identifiers may be provided, but have the following constraints:\n  <ul>\n    <li>Must not contain the
    <code>assigner</code> and <code>use</code> fields. </li>\n    <li>Must contain the <code>type</code>,
    <code>system</code>, and <code>value</code> fields. </li>\n  </ul>\n</p>\n<p><b>Example:</b>\n  <pre><code>{\n
    \"identifier\": [\n    {\n      \"assigner\": {\n        \"reference\": \"Organization/619848\"\n      }\n    },\n
        {\n      \"type\": {\n        \"coding\": [\n          {\n            \"code\": \"MR\",\n            \"system\":
     \"http://hl7.org/fhir/v2/0203\"\n          }\n        ]\n      },\n      \"system\": \"urn:oid:1.1.1.1.1.1\",\n
          \"value\": \"&lt;unique identifier&gt;\",\n      \"period\": {\n        \"start\":
    \"2016-01-02T00:00:00-05:00\",\n        \"end\": \"2020-01-02T00:00:00-05:00\"\n      }\n    }\n  ]\n}\n
    </code></pre>\n</p>\n",
    "type":"array",
    "minItems":1,
    "items":{
        "type":"object",
        "properties":{
            "assigner":{
                "type":"string",
                "description":"<p>May only be provided on the first identifier object. Contains a reference to the
    organization in which the patient is being enrolled.</p>\n",
                "example":"Organization/619848"
            },
            "type":{
                "type":"object",
                "properties":{
                    "coding":{
                        "type":"array",
                        "minItems":1,
                        "maxItems":1,
                        "items":{
                            "type":"object",
                            "properties":{
                                "code":{
```

```
                                    "type":"string",
                                    "example":"MR"
                                },
                                "system":{
                                    "type":"string",
                                    "example":"http://hl7.org/fhir/v2/0203"
                                }
                            }
                        }
                    }
                },
                "system":{
                    "type":"string",
                    "example":"urn:oid:1.1.1.1.1.1"
                },
                "value":{
                    "type":"string",
                    "description":"<p>Must be a unique value.</p>",
                    "example":"CW1234563"
                },
                "period":{
                    "type":"object",
                    "properties":{
                        "start":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>",
                            "example":"2019-12-01T00:00:00.000Z"
                        },
                        "end":{
                            "type":"string",
                            "description":"<p>Must include a time component with a timezone.</p>",
                            "example":"2029-12-01T23:59:59.000Z"
                        }
                    }
                }
            }
        }
}
```

Nested Schema : maritalStatus

Type:

object

The patient's most recent marital (civil) status.

Show Source

* coding: array coding

* text: string Example: Never Married

```
{
    "description":"<p>The patient's most recent marital (civil) status.</p>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string",
            "example":"Never Married"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string",
                        "example":"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"
                    },
                    "code":{
                        "type":"string",
                        "example":"S"
                    },
                    "display":{
                        "type":"string",
                        "example":"Never Married"
                    }
                }
            }
        }
    }
}
```

Nested Schema : name

Type:

array

Minimum Number of Items:

The name associated with the practitioner.

At least one name must be provided.

* It must have a use of official.

* If a period is provided, it must not have an end date.

* It must include both a given and a family name.

All provided names:

* Must not contain the text field.

* Must contain either a given or family name.

Show Source

* Array of: object items


```
{
    "description":"<p>The name associated with the practitioner.</p>\n<p>At least one name must be provided.\n  <ul>\n
        <li>It must have a <code>use</code> of official. </li>\n    <li>If a <code>period</code> is provided, it must
    not have an <code>end</code> date. </li>\n    <li>It must include both a <code>given</code> and a
    <code>family</code> name. </li>\n  </ul>\n</p>\n<p>All provided names:\n  <ul>\n    <li>Must not contain the
    <code>text</code> field. </li>\n    <li>Must contain either a <code>given</code> or <code>family</code> name.
    </li>\n  </ul>\n</p>\n",
    "type":"array",
    "minItems":1,
    "items":{
        "type":"object",
        "required":[
            "use",
            "family"
        ],
        "properties":{
            "family":{
                "type":"string",
                "maxLength":100,
                "example":"Williams"
            },
            "given":{
                "type":"array",
                "minItems":1,
                "items":{
                    "type":"string",
                    "maxLength":100,
                    "example":"James McAfee"
                }
            },
            "prefix":{
                "type":"array",
                "maxItems":1,
                "items":{
                    "type":"string",
                    "maxLength":100,
                    "example":"Dr."
                }
            },
            "suffix":{
                "maxItems":1,
                "type":"array",
                "items":{
                    "type":"string",
                    "maxLength":100,
                    "example":"M.D."
                }
            },
            "period":{
                "type":"object",
                "description":"<p>Must not include an <code>end</code> date.</p>",
                "properties":{
                    "start":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>",
                        "example":"2020-12-01T00:00:00.000Z"
                    }
                }
            },
            "use":{
                "type":"string",
                "enum":[
                    "usual",
                    "official",
                    "temp",
                    "nickname",
                    "anonymous",
                    "old",
                    "maiden"
                ]
            }
        }
```

```
    }
}
```

Nested Schema : telecom

Type:

array

A contact detail (such as a telephone number or an email address) by which the individual may be contacted.

If an extension is provided:

* The extension.valueString field has a character limit of 100.

* The system field must be phone.

* The extension.url field must be http://hl7.org/fhir/StructureDefinition/contactpoint-extension.

Show Source

* Array of: object items


```
{
    "description":"<p>A contact detail (such as a telephone number or an email address) by which the individual may be
    contacted.</p>\n<p>If an extension is provided:\n  <ul>\n    <li>The <code>extension.valueString</code> field has a
    character limit of 100. </li>\n    <li>The <code>system</code> field must be <code>phone</code>. </li>\n    <li>The
    <code>extension.url</code> field must be <code>http://hl7.org/fhir/StructureDefinition/contactpoint-
    extension</code>. </li>\n  </ul>\n</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "system",
            "use",
            "value"
        ],
        "properties":{
            "system":{
                "type":"string",
                "enum":[
                    "phone",
                    "fax",
                    "email",
                    "pager",
                    "url",
                    "sms",
                    "other"
                ]
            },
            "value":{
                "type":"string",
                "maxLength":100
            },
            "use":{
                "type":"string",
                "enum":[
                    "home",
                    "work",
                    "temp",
                    "old",
                    "mobile"
                ]
            },
            "rank":{
                "type":"integer"
            },
            "period":{
                "type":"object",
                "properties":{
                    "start":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>\n",
                        "example":"2012-05-17T15:33:18.000Z"
                    },
                    "end":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>\n",
                        "example":"2028-05-17T15:33:18.000Z"
                    }
                }
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* city: string Maximum Length: 100 Example: KC
* country: string Maximum Length: 100 Example: United States of America
* district: string Maximum Length: 100 Example: Jackson
* line: array line
* period: object period
* postalCode: string Maximum Length: 25 Example: 64199
* state: string Maximum Length: 100 Example: KS
* type: string Allowed Values: ["postal", "physical", "both"]
* use(required): string Allowed Values: ["home", "work", "temp", "old", "billing"]


```
{
    "type":"object",
    "required":[
        "use"
    ],
    "properties":{
        "use":{
            "type":"string",
            "enum":[
                "home",
                "work",
                "temp",
                "old",
                "billing"
            ]
        },
        "type":{
            "type":"string",
            "enum":[
                "postal",
                "physical",
                "both"
            ]
        },
        "line":{
            "type":"array",
            "items":{
                "type":"string",
                "maxLength":100,
                "example":"1212 Faircastle Drive"
            },
            "example":[
                "1212 Faircastle Drive",
                "Apartment 406"
            ]
        },
        "city":{
            "type":"string",
            "maxLength":100,
            "example":"KC"
        },
        "district":{
            "type":"string",
            "maxLength":100,
            "example":"Jackson"
        },
        "state":{
            "type":"string",
            "maxLength":100,
            "example":"KS"
        },
        "postalCode":{
            "type":"string",
            "maxLength":25,
            "example":"64199"
        },
        "country":{
            "type":"string",
            "maxLength":100,
            "example":"United States of America"
        },
        "period":{
            "type":"object",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>\n",
                    "example":"2012-05-17T15:33:18.000Z"
                },
                "end":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>\n",
                    "example":"2032-05-17T15:33:18.000Z"
                }
```

```
            }
        }
    }
}
```

Nested Schema : line

Type:

array

Show Source

* Array of: string Maximum Length: 100 Example: 1212 Faircastle Drive

```
{
    "type":"array",
    "items":{
        "type":"string",
        "maxLength":100,
        "example":"1212 Faircastle Drive"
    },
    "example":[
        "1212 Faircastle Drive",
        "Apartment 406"
    ]
}
```

Example:

```
[
    "1212 Faircastle Drive",
    "Apartment 406"
]
```

Nested Schema : period

Type:

object

Show Source

end: string

Must include a time component with a timezone.

Example:

2032-05-17T15:33:18.000Z

start: string

Must include a time component with a timezone.

Example:

2012-05-17T15:33:18.000Z

```
{
    "type":"object",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>\n",
            "example":"2012-05-17T15:33:18.000Z"
        },
        "end":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>\n",
            "example":"2032-05-17T15:33:18.000Z"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* language: object language

* preferred: boolean

```
{
    "type":"object",
    "properties":{
        "language":{
            "type":"object",
            "properties":{
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
```

```
                            "system":{
                                "type":"string",
                                "example":"urn:ietf:bcp:47"
                            },
                            "code":{
                                "type":"string",
                                "example":"en"
                            },
                            "display":{
                                "type":"string",
                                "example":"English"
                            }
                        }
                    }
                },
                "text":{
                    "type":"string",
                    "example":"English"
                }
            }
        },
        "preferred":{
            "type":"boolean"
        }
    }
}
```

Nested Schema : language

Type:

object

Show Source

* coding: array coding

* text: string Example: English

```
{
    "type":"object",
    "properties":{
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string",
                        "example":"urn:ietf:bcp:47"
                    },
                    "code":{
                        "type":"string",
                        "example":"en"
                    },
                    "display":{
                        "type":"string",
                        "example":"English"
                    }
                }
            }
        },
        "text":{
            "type":"string",
            "example":"English"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string",
                "example":"urn:ietf:bcp:47"
            },
            "code":{
                "type":"string",
                "example":"en"
            },
            "display":{
```

```
                "type":"string",
                "example":"English"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string Example: en

* display: string Example: English

* system: string Example: urn:ietf:bcp:47

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string",
            "example":"urn:ietf:bcp:47"
        },
        "code":{
            "type":"string",
            "example":"en"
        },
        "display":{
            "type":"string",
            "example":"English"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* url(required): string

* valueCode: string

```
{
    "type":"object",
    "required":[
        "url"
    ],
    "properties":{
        "url":{
            "type":"string"
        },
        "valueCode":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Must only include references to Practitioners.

Show Source

* reference: string Example: Practitioner/995932

```
{
    "type":"object",
    "description":"<p>Must only include references to Practitioners.</p>",
    "properties":{
        "reference":{
            "type":"string",
            "example":"Practitioner/995932"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

assigner: string

May only be provided on the first identifier object. Contains a reference to the organization in which the patient is being enrolled.

Example:

Organization/619848

* period: object period

* system: string Example: urn:oid:1.1.1.1.1.1

* type: object type

value: string

Must be a unique value.

Example:

CW1234563


```
{
    "type":"object",
    "properties":{
        "assigner":{
            "type":"string",
            "description":"<p>May only be provided on the first identifier object. Contains a reference to the
    organization in which the patient is being enrolled.</p>\n",
            "example":"Organization/619848"
        },
        "type":{
            "type":"object",
            "properties":{
                "coding":{
                    "type":"array",
                    "minItems":1,
                    "maxItems":1,
                    "items":{
                        "type":"object",
                        "properties":{
                            "code":{
                                "type":"string",
                                "example":"MR"
                            },
                            "system":{
                                "type":"string",
                                "example":"http://hl7.org/fhir/v2/0203"
                            }
                        }
                    }
                }
            }
        },
        "system":{
            "type":"string",
            "example":"urn:oid:1.1.1.1.1.1"
        },
        "value":{
            "type":"string",
            "description":"<p>Must be a unique value.</p>",
            "example":"CW1234563"
        },
        "period":{
            "type":"object",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2019-12-01T00:00:00.000Z"
                },
                "end":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2029-12-01T23:59:59.000Z"
                }
            }
        }
    }
}
```


Nested Schema : period

Type:

object

Show Source

end: string

Must include a time component with a timezone.

Example:

2029-12-01T23:59:59.000Z

start: string

Must include a time component with a timezone.

Example:

2019-12-01T00:00:00.000Z


```
{
    "type":"object",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2019-12-01T00:00:00.000Z"
        },
        "end":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2029-12-01T23:59:59.000Z"
        }
    }
}
```


Nested Schema : type

Type:

object

Show Source

* coding: array coding Minimum Number of Items: 1 Maximum Number of Items: 1


```
{
    "type":"object",
    "properties":{
        "coding":{
            "type":"array",
            "minItems":1,
            "maxItems":1,
            "items":{
                "type":"object",
                "properties":{
                    "code":{
                        "type":"string",
                        "example":"MR"
                    },
                    "system":{
                        "type":"string",
                        "example":"http://hl7.org/fhir/v2/0203"
                    }
                }
            }
        }
    }
}
```


Nested Schema : coding

Type:

array

Minimum Number of Items:

Maximum Number of Items:

Show Source

* Array of: object items


```
{
    "type":"array",
    "minItems":1,
    "maxItems":1,
    "items":{
        "type":"object",
        "properties":{
            "code":{
                "type":"string",
                "example":"MR"
            },
            "system":{
                "type":"string",
                "example":"http://hl7.org/fhir/v2/0203"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string Example: MR

* system: string Example: http://hl7.org/fhir/v2/0203

```
{
    "type":"object",
    "properties":{
        "code":{
            "type":"string",
            "example":"MR"
        },
        "system":{
            "type":"string",
            "example":"http://hl7.org/fhir/v2/0203"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string",
                "example":"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"
            },
            "code":{
                "type":"string",
                "example":"S"
            },
            "display":{
                "type":"string",
                "example":"Never Married"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string Example: S

* display: string Example: Never Married

* system: string Example: http://terminology.hl7.org/CodeSystem/v3-MaritalStatus

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string",
            "example":"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus"
        },
        "code":{
            "type":"string",
            "example":"S"
        },
        "display":{
            "type":"string",
            "example":"Never Married"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* family(required): string Maximum Length: 100 Example: Williams

* given: array given Minimum Number of Items: 1

period: object

period

Must not include an end date.

* prefix: array prefix Maximum Number of Items: 1

* suffix: array suffix Maximum Number of Items: 1

* use(required): string Allowed Values: ["usual", "official", "temp", "nickname", "anonymous", "old", "maiden"]


```
{
    "type":"object",
    "required":[
        "use",
        "family"
    ],
    "properties":{
        "family":{
            "type":"string",
            "maxLength":100,
            "example":"Williams"
        },
        "given":{
            "type":"array",
            "minItems":1,
            "items":{
                "type":"string",
                "maxLength":100,
                "example":"James McAfee"
            }
        },
        "prefix":{
            "type":"array",
            "maxItems":1,
            "items":{
                "type":"string",
                "maxLength":100,
                "example":"Dr."
            }
        },
        "suffix":{
            "maxItems":1,
            "type":"array",
            "items":{
                "type":"string",
                "maxLength":100,
                "example":"M.D."
            }
        },
        "period":{
            "type":"object",
            "description":"<p>Must not include an <code>end</code> date.</p>",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2020-12-01T00:00:00.000Z"
                }
            }
        },
        "use":{
            "type":"string",
            "enum":[
                "usual",
                "official",
                "temp",
                "nickname",
                "anonymous",
                "old",
                "maiden"
            ]
        }
    }
}
```


Nested Schema : given

Type:

array

Minimum Number of Items:

Show Source

* Array of: string Maximum Length: 100 Example: James McAfee


```
{
    "type":"array",
    "minItems":1,
    "items":{
```

```
        "type":"string",
        "maxLength":100,
        "example":"James McAfee"
    }
}
```

Nested Schema : period

Type:

object

Must not include an end date.

Show Source

start: string

Must include a time component with a timezone.

Example:

2020-12-01T00:00:00.000Z

```
{
    "type":"object",
    "description":"<p>Must not include an <code>end</code> date.</p>",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2020-12-01T00:00:00.000Z"
        }
    }
}
```

Nested Schema : prefix

Type:

array

Maximum Number of Items:

Show Source

* Array of: string Maximum Length: 100 Example: Dr.

```
{
    "type":"array",
    "maxItems":1,
    "items":{
        "type":"string",
        "maxLength":100,
        "example":"Dr."
    }
}
```

Nested Schema : suffix

Type:

array

Maximum Number of Items:

Show Source

* Array of: string Maximum Length: 100 Example: M.D.

```
{
    "maxItems":1,
    "type":"array",
    "items":{
        "type":"string",
        "maxLength":100,
        "example":"M.D."
    }
}
```

Nested Schema : items

Type:

object

Show Source

* period: object period

* rank: integer

* system(required): string Allowed Values: ["phone", "fax", "email", "pager", "url", "sms", "other"]

* use(required): string Allowed Values: ["home", "work", "temp", "old", "mobile"]

* value(required): string Maximum Length: 100

```
{
    "type":"object",
```

```
    "required":[
        "system",
        "use",
        "value"
    ],
    "properties":{
        "system":{
            "type":"string",
            "enum":[
                "phone",
                "fax",
                "email",
                "pager",
                "url",
                "sms",
                "other"
            ]
        },
        "value":{
            "type":"string",
            "maxLength":100
        },
        "use":{
            "type":"string",
            "enum":[
                "home",
                "work",
                "temp",
                "old",
                "mobile"
            ]
        },
        "rank":{
            "type":"integer"
        },
        "period":{
            "type":"object",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>\n",
                    "example":"2012-05-17T15:33:18.000Z"
                },
                "end":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>\n",
                    "example":"2028-05-17T15:33:18.000Z"
                }
            }
        }
    }
}
```


Nested Schema : period

Type:

object

Show Source

end: string

Must include a time component with a timezone.

Example:

2028-05-17T15:33:18.000Z

start: string

Must include a time component with a timezone.

Example:

2012-05-17T15:33:18.000Z


```
{
    "type":"object",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>\n",
            "example":"2012-05-17T15:33:18.000Z"
        },
        "end":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>\n",
            "example":"2028-05-17T15:33:18.000Z"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

Location: string

The URL location of the created entity.

Example: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/5786010

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Default Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider

* System

Example Request:

```
POST https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Wed, 27 Mar 2019 17:23:14 GMT
Etag: W/"0"
Location: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/5786010
Last-Modified: Wed, 27 Mar 2019 17:23:14 GMT
Vary: Origin
X-Request-Id: 11111111111111111111111111111111
opc-request-id: /4B7BCF043117301F1B647EF553C7F250/A326F573558AD7570F63BFC19D88739A
```

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Get a list of patients

File: r4-rest-api-v32/op-patient-get.html

Retrieves a list of patients that meet supplied query parameters.

Note:

* A 422 (Unprocessable Content) status code is returned when more than 1000 patients qualify for the search criteria.

* The name, family, and given parameters support the :exact modifier and search for current names only, based on the name's period.

* Oracle Cerner does not recommend combining the family or given parameters with the name parameter when searching for a patient. Whenever possible, use the :exact modifier.

Request

Query Parameters

_count: integer

Minimum Value:

The maximum number of results to return.

Example: _count=15

_id: array[string]

Minimum Number of Items:

The logical resource ID. This parameter is required if no other parameter is used.

Example: _id=12345

_revinclude: string

The Provenance resource entries to be returned as part of the bundle.

* May be provided once with the value of Provenance:target. Example: _revinclude=Provenance:target.

* Requires the OAuth2 token to include the Provenance scope corresponding with the current Patient scope, such as user/Provenance.read, patient/Provenance.read, or system/Provenance.read.

Allowed Values:

["Provenance:target"]

address-postalcode: string

The postal code in the address details of the patient. May be provided once with a single value.

Example: address-postalcode=12345

birthdate: array[string]

Collection Format:

multi

Minimum Number of Items:

Maximum Number of Items:

The patient's date of birth.

* May be provided once using one of the following prefixes to imply a date range: ge, le, gt, lt, or eq.

* May be provided twice to indicate a date range and must contain one prefix each of le and ge. Example: birthdate=ge2001-03-13&birthdate=le2001-05-01.

* Must not be provided with a time component.

email: string

The patient's email. May be provided once with a single value.

Example: email=example@example.com

family: array[string]

Minimum Number of Items:

Maximum Number of Items:

The start of the family name of the patient. May be provided once with a single value.

The :exact modifier is supported and searches for current names based on the name's period.

Example: family=Smith

gender: string

The patient's gender. May be provided once with a single value.

Allowed Values:

["male", "female", "other", "unknown"]

given: array[string]

Minimum Number of Items:

Maximum Number of Items:

The start of the given name of the patient. May be provided once with a single value.

The :exact modifier is supported and searches for current names based on the name's period.

Searching by the given name requires the family parameter.

Example: given=Sally

identifier: array[string]

Minimum Number of Items:

Maximum Number of Items:

A patient identifier.

* May be provided once with a single value.

* Requires a code, but the system is optional. If the system is not provided, the search is performed across all systems supported by the Patient resource.

* Searches against the specific system, if it is provided.

* Accepts a Social Security Number (SSN), but the value is not returned in a matching response.

Example: identifier=urn:oid:1.1.1.1.1.1|1022228

name: array[string]

Minimum Number of Items:

Maximum Number of Items:

The start of the given name or the family name of the patient. May be provided once with a single value.

The :exact modifier is supported and searches for current names based on the name's period.

Example: name=Riley

phone: string

The patient's phone number. May be provided once with a single value.

Example: phone=1234567891

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?family=smart&given=joe&birthdate=1990-09-15
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "b8e08a98-849f-4544-9fa8-985aa445e31b",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?_id=12724067"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/12724067",
      "resource": {
        "resourceType": "Patient",
        "id": "12724067",
        "meta": {
          "versionId": "10",
          "lastUpdated": "2020-07-06T21:21:22.000Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Patient</b></p><p><b>Name</b>: SMART,
    JOE</p><p><b>Status</b>: Active</p><p><b>DOB</b>: Apr 29, 1976</p><p><b>Administrative Gender</b>:
    Male</p><p><b>Marital Status</b>: Married</p></div>"
        },
        "extension": [
          {
            "id": "59434424",
            "extension": [
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                      "code": "485602703",
                      "display": "Appointment Reminders",
                      "userSelected": true
                    },
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/communication-topic",
                      "code": "appointment-reminder",
                      "display": "Appointment Reminder",
                      "userSelected": false
                    }
                  ],
                  "text": "Appointment Reminders"
                },
                "url": "communication-type"
              },
              {
```

```
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
            "code": "495085513",
            "display": "Fax",
            "userSelected": true
          },
          {
            "system": "http://hl7.org/fhir/contact-point-system",
            "code": "fax",
            "display": "Fax",
            "userSelected": false
          }
        ],
        "text": "Fax"
      },
      "url": "contact-method"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
            "code": "163",
            "display": "Business",
            "userSelected": true
          },
          {
            "system": "http://hl7.org/fhir/contact-point-use",
            "code": "work",
            "display": "Work",
            "userSelected": false
          }
        ],
        "text": "Business"
      },
      "url": "contact-type"
    },
    {
      "valueDateTime": "2019-04-13T20:00:00.000Z",
      "url": "verified-datetime"
    }
  ],
  "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2106-3",
        "display": "White",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "White",
      "url": "text"
    }
  ],
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2186-5",
        "display": "Non Hispanic or Latino",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "Not Hispanic, Latino, or Spanish Origin",
      "url": "text"
    }
  ],
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"
},
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "446141000124107",
        "display": "Identifies as female gender (finding)",
```

```
          "userSelected": false
        }
      ],
      "text": "Identifies as female"
    },
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
  },
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
              "code": "557",
              "display": "Umkumiut Native Village",
              "userSelected": true
            }
          ],
          "text": "Umkumiut Native Village"
        },
        "url": "tribalAffiliation"
      }
    ],
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation"
  },
  {
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
    "valueCode": "248153007"
  }
],
"identifier": [
  {
    "id": "CI-490016886-0",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
        }
      ],
      "text": "MRN"
    },
    "system": "urn:oid:2.16.840.1.113883.6.1000",
    "value": "6931",
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-490058771-1",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
        }
      ],
      "text": "MRN"
    },
    "system": "urn:oid:2.16.840.1.113883.6.1000",
    "value": "6978",
    "period": {
      "end": "2020-07-06T21:21:25.000Z"
    }
  },
  {
    "id": "CI-490059574-3",
    "use": "usual",
    "type": {
      "coding": [
```

```
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "670843",
              "display": "Messaging",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "U",
              "display": "Unspecified identifier",
              "userSelected": false
            }
          ],
          "text": "Messaging"
        },
        "_system": {
          "extension": [
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "value": "3C36293A3B964994AD8E6C0305F3330A",
        "period": {
          "start": "2020-06-30T20:08:26.000Z"
        }
      },
      {
        "id": "CI-490058805-4",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "670843",
              "display": "Messaging",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "U",
              "display": "Unspecified identifier",
              "userSelected": false
            }
          ],
          "text": "Messaging"
        },
        "_system": {
          "extension": [
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "value": "43DA797A657B47548F258A9B50EB41F5",
        "period": {
          "start": "2020-06-12T16:03:32.000Z"
        }
      },
      {
        "id": "CI-490059570-5",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "2553236771",
              "display": "Federated Person Principal",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "AN",
              "display": "Account number",
              "userSelected": false
            }
          ],
          "text": "Federated Person Principal"
        },
        "system": "urn:oid:2.16.840.1.113883.3.13.6",
        "value": "URN:CERNER:IDENTITY-
FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:AN7TD9A62CV8Z53Z",
        "period": {
          "start": "2020-06-30T20:08:25.000Z"
        }
      },
      {
```

```
        "id": "CI-490058801-6",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "2553236771",
              "display": "Federated Person Principal",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "AN",
              "display": "Account number",
              "userSelected": false
            }
          ],
          "text": "Federated Person Principal"
        },
        "system": "urn:oid:2.16.840.1.113883.3.13.6",
        "value": "URN:CERNER:IDENTITY-
FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:KR8KC9MI9EQ8KC23",
        "period": {
          "start": "2020-06-12T16:03:29.000Z"
        }
      }
    ],
    "active": true,
    "name": [
      {
        "id": "CI-12724067-0",
        "use": "official",
        "text": "SMART, JOE",
        "family": "SMART",
        "given": [
          "JOE"
        ],
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      },
      {
        "id": "CI-490059796-0",
        "use": "old",
        "text": "SMART, STEPHEN ALLEN",
        "family": "SMART",
        "given": [
          "STEPHEN",
          "ALLEN"
        ],
        "period": {
          "end": "2020-07-06T21:21:26.000Z"
        }
      }
    ],
    "telecom": [
      {
        "id": "CI-PH-29811920-0",
        "extension": [
          {
            "valueUrl": "(816)888-8886",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          },
          {
            "valueString": "12345",
            "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
          }
        ],
        "system": "phone",
        "value": "8168888886",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      },
      {
        "id": "CI-EM-29822662-0",
        "system": "email",
        "value": "joesmart@yopmail.com",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2020-03-30T19:31:11.000Z"
        }
      }
    ],
    "gender": "male",
    "birthDate": "1976-04-29",
```

```
        "deceasedBoolean": false,
        "address": [
          {
            "id": "CI-24313553-0",
            "use": "home",
            "text": "12345 Main St\\nKansas city, MO 64116\\nUS",
            "line": [
              "12345 Main St"
            ],
            "city": "Kansas city",
            "district": "Jackson",
            "state": "MO",
            "postalCode": "64116",
            "country": "US",
            "period": {
              "start": "2019-12-26T15:13:36.000Z"
            }
          }
        ],
        "maritalStatus": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/38",
              "code": "309237",
              "display": "Married",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
              "code": "M",
              "display": "Married",
              "userSelected": false
            }
          ],
          "text": "Married"
        },
        "communication": [
          {
            "language": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36",
                  "code": "151",
                  "display": "English",
                  "userSelected": true
                },
                {
                  "system": "urn:ietf:bcp:47",
                  "code": "en",
                  "display": "English",
                  "userSelected": false
                }
              ],
              "text": "English"
            },
            "preferred": true
          }
        ],
        "generalPractitioner": [
          {
            "id": "CI-490017023-0",
            "reference": "Practitioner/4122622",
            "display": "Cerner Test, Physician - Hospitalist Cerner"
          },
          {
            "reference": "Organization/675844",
            "display": "Example Active Practice"
          }
        ]
      }
    }
  ]
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

x-confidential-response Response

Example of patients with confidential gender identity.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?_id=12724067
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "b8e08a98-849f-4544-9fa8-985aa445e31b",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?_id=12724067"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/12724067",
      "resource": {
        "resourceType": "Patient",
        "id": "12724067",
        "meta": {
          "versionId": "10",
          "lastUpdated": "2020-07-06T21:21:22.000Z",
          "security": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode"
              "code": "PROCESSINLINELABEL"
              "display": "process inline security label"
            }
          ]
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Patient</b></p><p><b>Meta Security</b>: process inline
    security label</p><p><b>Name</b>: SMART, JOE</p><p><b>Status</b>: Active</p><p><b>DOB</b>: Apr 29,
   1976</p><p><b>Administrative Gender</b>: Male</p><p><b>Marital Status</b>: Married</p></div>"
        },
        "extension": [
          {
            "id": "59434424",
            "extension": [
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                      "code": "485602703",
                      "display": "Appointment Reminders",
                      "userSelected": true
                    },
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/communication-topic",
                      "code": "appointment-reminder",
                      "display": "Appointment Reminder",
                      "userSelected": false
                    }
                  ],
                  "text": "Appointment Reminders"
                },
                "url": "communication-type"
              },
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                      "code": "495085513",
                      "display": "Fax",
                      "userSelected": true
                    },
                    {
                      "system": "http://hl7.org/fhir/contact-point-system",
                      "code": "fax",
                      "display": "Fax",
                      "userSelected": false
                    }
                  ],
                  "text": "Fax"
                },
                "url": "contact-method"
              },
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                      "code": "163",
                      "display": "Business",
                      "userSelected": true
                    },
```

```
          {
            "system": "http://hl7.org/fhir/contact-point-use",
            "code": "work",
            "display": "Work",
            "userSelected": false
          }
        ],
        "text": "Business"
      },
      "url": "contact-type"
    },
    {
      "valueDateTime": "2019-04-13T20:00:00.000Z",
      "url": "verified-datetime"
    }
  ],
  "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2106-3",
        "display": "White",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "White",
      "url": "text"
    }
  ],
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2186-5",
        "display": "Non Hispanic or Latino",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "Not Hispanic, Latino, or Spanish Origin",
      "url": "text"
    }
  ],
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"
},
{
  "valueCodeableConcept": {
    "extension": [
      {
        "valueCoding": {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
          "code": "PRS",
          "display": "patient requested information sensitivity",
          "userSelected": false
        },
        "url": "http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-inline-sec-label"
      }
    ],
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "446141000124107",
        "display": "Identifies as female gender (finding)",
        "userSelected": false
      }
    ],
    "text": "Identifies as female"
  },
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
},
{
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
            "code": "557",
            "display": "Umkumiut Native Village",
            "userSelected": true
```

```
            }
          ],
          "text": "Umkumiut Native Village"
        },
        "url": "tribalAffiliation"
      }
    ],
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation"
  },
  {
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
    "valueCode": "248153007"
  }
],
"identifier": [
  {
    "id": "CI-490016886-0",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
        }
      ],
      "text": "MRN"
    },
    "system": "urn:oid:2.16.840.1.113883.6.1000",
    "value": "6931",
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-490058771-1",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
        }
      ],
      "text": "MRN"
    },
    "system": "urn:oid:2.16.840.1.113883.6.1000",
    "value": "6978",
    "period": {
      "end": "2020-07-06T21:21:25.000Z"
    }
  },
  {
    "id": "CI-490059574-3",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "670843",
          "display": "Messaging",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "U",
          "display": "Unspecified identifier",
          "userSelected": false
        }
      ],
      "text": "Messaging"
    },
    "_system": {
      "extension": [
```

```
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "value": "3C36293A3B964994AD8E6C0305F3330A",
        "period": {
          "start": "2020-06-30T20:08:26.000Z"
        }
      },
      {
        "id": "CI-490058805-4",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "670843",
              "display": "Messaging",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "U",
              "display": "Unspecified identifier",
              "userSelected": false
            }
          ],
          "text": "Messaging"
        },
        "_system": {
          "extension": [
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "value": "43DA797A657B47548F258A9B50EB41F5",
        "period": {
          "start": "2020-06-12T16:03:32.000Z"
        }
      },
      {
        "id": "CI-490059570-5",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "2553236771",
              "display": "Federated Person Principal",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "AN",
              "display": "Account number",
              "userSelected": false
            }
          ],
          "text": "Federated Person Principal"
        },
        "system": "urn:oid:2.16.840.1.113883.3.13.6",
        "value": "URN:CERNER:IDENTITY-
FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:AN7TD9A62CV8Z53Z",
        "period": {
          "start": "2020-06-30T20:08:25.000Z"
        }
      },
      {
        "id": "CI-490058801-6",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "2553236771",
              "display": "Federated Person Principal",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "AN",
              "display": "Account number",
              "userSelected": false
            }
          ],
```

```
          "text": "Federated Person Principal"
        },
        "system": "urn:oid:2.16.840.1.113883.3.13.6",
        "value": "URN:CERNER:IDENTITY-
FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:KR8KC9MI9EQ8KC23",
        "period": {
          "start": "2020-06-12T16:03:29.000Z"
        }
      }
    ],
    "active": true,
    "name": [
      {
        "id": "CI-12724067-0",
        "use": "official",
        "text": "SMART, JOE",
        "family": "SMART",
        "given": [
          "JOE"
        ],
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      },
      {
        "id": "CI-490059796-0",
        "use": "old",
        "text": "SMART, STEPHEN ALLEN",
        "family": "SMART",
        "given": [
          "STEPHEN",
          "ALLEN"
        ],
        "period": {
          "end": "2020-07-06T21:21:26.000Z"
        }
      }
    ],
    "telecom": [
      {
        "id": "CI-PH-29811920-0",
        "extension": [
          {
            "valueUrl": "(816)888-8886",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          },
          {
            "valueString": "12345",
            "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
          }
        ],
        "system": "phone",
        "value": "8168888886",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      },
      {
        "id": "CI-EM-29822662-0",
        "system": "email",
        "value": "joesmart@yopmail.com",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2020-03-30T19:31:11.000Z"
        }
      }
    ],
    "gender": "male",
    "birthDate": "1976-04-29",
    "deceasedBoolean": false,
    "address": [
      {
        "id": "CI-24313553-0",
        "use": "home",
        "text": "12345 Main St\\nKansas city, MO 64116\\nUS",
        "line": [
          "12345 Main St"
        ],
        "city": "Kansas city",
        "district": "Jackson",
        "state": "MO",
        "postalCode": "64116",
        "country": "US",
        "period": {
          "start": "2019-12-26T15:13:36.000Z"
        }
```

```
        }
      ],
      "maritalStatus": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/38",
            "code": "309237",
            "display": "Married",
            "userSelected": true
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
            "code": "M",
            "display": "Married",
            "userSelected": false
          }
        ],
        "text": "Married"
      },
      "communication": [
        {
          "language": {
            "coding": [
              {
                "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36",
                "code": "151",
                "display": "English",
                "userSelected": true
              },
              {
                "system": "urn:ietf:bcp:47",
                "code": "en",
                "display": "English",
                "userSelected": false
              }
            ],
            "text": "English"
          },
          "preferred": true
        }
      ],
      "generalPractitioner": [
        {
          "id": "CI-490017023-0",
          "reference": "Practitioner/4122622",
          "display": "Cerner Test, Physician - Hospitalist Cerner"
        }
      ]
    }
  }
 ]
}
```


Headers

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

x-patient-revinclude Response

This operation supports the following authorization types:

* Provider

* System

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?_id=629928&_revinclude=Provenance:target
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "65ee4c79-3514-4eab-b606-10ad56fb0d5d",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient?_id=629928&_revinclude=Provenance%3Atarget"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/629928",
      "resource": {
        "resourceType": "Patient",
        "id": "629928",
        "meta": {
          "versionId": "36",
```

```
      "lastUpdated": "2020-09-17T16:35:12.000Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Patient</b></p><p><b>Name</b>: England,\n
              Brad N</p><p><b>Status</b>: Active</p><p><b>DOB</b>: Jan 20, 1999</p>\n
<p><b>Administrative Gender</b>: Unknown</p></div>"
    },
    "identifier": [
      {
        "extension": [
          {
            "valueCode": "unknown",
            "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
          }
        ]
      }
    ],
    "active": true,
    "name": [
      {
        "id": "CI-629928-0",
        "use": "official",
        "text": "England, Brad N",
        "family": "England",
        "given": [
          "Brad",
          "N"
        ],
        "period": {
          "start": "2004-01-27T00:00:00.000Z"
        }
      }
    ],
    "telecom": [
      {
        "id": "CI-PH-46477780-0",
        "extension": [
          {
            "valueUrl": "2232323",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          }
        ],
        "system": "phone",
        "value": "2232323",
        "rank": 1,
        "period": {
          "start": "2021-01-11T06:00:00.000Z"
        }
      },
      {
        "id": "CI-PH-9543543-1",
        "extension": [
          {
            "valueUrl": "1234567890",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          },
          {
            "valueString": "345",
            "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
          }
        ],
        "system": "phone",
        "value": "1234567890",
        "use": "home",
        "rank": 1,
        "period": {
          "start": "2014-01-20T15:48:39.000Z",
          "end": "2014-03-20T15:48:39.000Z"
        }
      },
      {
        "id": "CI-PH-9543573-2",
        "extension": [
          {
            "valueUrl": "1234567890",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          },
          {
            "valueString": "345",
            "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
          }
        ],
        "system": "phone",
        "value": "1234567890",
        "use": "home",
        "rank": 1,
        "period": {
          "start": "2014-01-20T15:48:39.000Z",
```

```
      "end": "2014-03-20T15:48:39.000Z"
    }
  }
],
"gender": "unknown",
"birthDate": "1999-01-20",
"contact": [
  {
    "relationship": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "83266815",
            "display": "Authorized Representative",
            "userSelected": true
          }
        ],
        "text": "Authorized Representative"
      }
    ],
    "name": {
      "id": "CI-170778458-0",
      "use": "official",
      "text": "MARSTON, Dutch",
      "family": "MARSTON",
      "given": [
        "Dutch"
      ],
      "period": {
        "start": "2020-08-26T14:44:08.000Z"
      }
    },
    "telecom": [
      {
        "system": "email",
        "value": "kaishwaryaramesh@gmail.com",
        "use": "home",
        "rank": 1,
        "period": {
          "start": "2020-10-09T07:29:32.000Z"
        }
      },
      {
        "system": "email",
        "value": "abigail.testpatient@gmail.com",
        "use": "home",
        "rank": 2,
        "period": {
          "start": "2020-08-26T14:45:59.000Z"
        }
      },
      {
        "system": "email",
        "value": "abigail.testpatient@gmail.com",
        "use": "home",
        "rank": 3,
        "period": {
          "start": "2020-08-26T16:47:31.000Z"
        }
      },
      {
        "system": "email",
        "value": "abigail.testpatient@gmail.com",
        "use": "home",
        "rank": 4,
        "period": {
          "start": "2020-08-27T20:27:31.000Z"
        }
      },
      {
        "system": "email",
        "value": "abigail.testpatient@gmail.com",
        "use": "home",
        "rank": 5,
        "period": {
          "start": "2020-08-28T15:24:19.000Z"
        }
      },
      {
        "system": "email",
        "value": "abigail.testpatient@gmail.com",
        "use": "home",
        "rank": 6,
        "period": {
          "start": "2020-08-31T15:16:18.000Z"
        }
      },
      {
```

```
            "system": "email",
            "value": "abigail.testpatient@gmail.com",
            "use": "home",
            "rank": 7,
            "period": {
              "start": "2020-09-10T20:41:05.000Z"
            }
          },
          {
            "system": "email",
            "value": "abigail.testpatient@gmail.com",
            "use": "home",
            "rank": 8,
            "period": {
              "start": "2020-09-11T15:52:04.000Z"
            }
          }
        ],
        "gender": "male",
        "period": {
          "start": "2020-08-27T20:27:31.000Z"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/ptntdemo-17390433",
  "resource": {
    "resourceType": "Provenance",
    "id": "ptntdemo-17390433",
    "meta": {
      "versionId": "17390433",
      "lastUpdated": "2021-06-03T11:18:59Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p>\n
<p><b>Target</b>:</p><ul><li>Patient/629928</li></ul>\n              <p><b>Recorded</b>: Jun  3, 2021  6:18 A.M.
CDT</p>\n              <p><b>Agents</b>:</p><dl><dt>organizationName PrH</dt>\n              <dd><b>Agent Type</b>:
Author</dd>\n              <dd><b>Agent Role</b>: Source</dd></dl><p><b>Entity Source</b>:
DocumentReference/S-68607</p></div>"
    },
    "target": [
      {
        "reference": "Patient/629928"
      }
    ],
    "recorded": "2021-06-03T11:18:59Z",
    "agent": [
      {
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
              "code": "author"
            }
          ],
          "text": "Author"
        },
        "role": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                "code": "SOURCE"
              }
            ],
            "text": "Source"
          }
        ],
        "who": {
          "reference": "Organization/59917604",
          "display": "organizationName PrH"
        }
      }
    ],
    "entity": [
      {
        "role": "source",
        "what": {
          "reference": "DocumentReference/S-68607"
        }
      }
    ]
  },
```

```
      "search": {
        "mode": "include"
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

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Get a patient by ID

File: r4-rest-api-v32/op-patientid-get.html

Retrieves a patient by the associated ID.

Oracle Health Millennium Platform supports the ability to logically merge a patient record into another patient record when both records are describing the same patient. This action is known as a patient combine. If necessary, you can perform a patient uncombine to undo this merging. When the requested patient record is combined into another record, an inactive patient entry is returned with a link to the current patient entry. Entries for combined patients are only returned when retrieving the entries directly by ID. They are not returned when searching with other parameters.

Request

Path Parameters

ID(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/12724067
```

Example Response:

```
{
  "resourceType": "Patient",
  "id": "12724067",
  "meta": {
    "versionId": "10",
    "lastUpdated": "2020-07-06T21:21:22.000Z"
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Patient</b></p><p><b>Name</b>: SMART,
    JOE</p><p><b>Status</b>: Active</p><p><b>DOB</b>: Apr 29, 1976</p><p><b>Administrative Gender</b>:
    Male</p><p><b>Marital Status</b>: Married</p></div>"
  },
  "extension": [
    {
      "id": "59434424",
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                "code": "485602703",
                "display": "Appointment Reminders",
                "userSelected": true
              },
              {
```

```
            "system": "http://terminology.hl7.org/CodeSystem/communication-topic",
            "code": "appointment-reminder",
            "display": "Appointment Reminder",
            "userSelected": false
          }
        ],
        "text": "Appointment Reminders"
      },
      "url": "communication-type"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
            "code": "495085513",
            "display": "Fax",
            "userSelected": true
          },
          {
            "system": "http://hl7.org/fhir/contact-point-system",
            "code": "fax",
            "display": "Fax",
            "userSelected": false
          }
        ],
        "text": "Fax"
      },
      "url": "contact-method"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
            "code": "163",
            "display": "Business",
            "userSelected": true
          },
          {
            "system": "http://hl7.org/fhir/contact-point-use",
            "code": "work",
            "display": "Work",
            "userSelected": false
          }
        ],
        "text": "Business"
      },
      "url": "contact-type"
    },
    {
      "valueDateTime": "2019-04-13T20:00:00.000Z",
      "url": "verified-datetime"
    }
  ],
  "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2106-3",
        "display": "White",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "White",
      "url": "text"
    }
  ],
  "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
},
{
  "extension": [
    {
      "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2186-5",
        "display": "Non Hispanic or Latino",
        "userSelected": false
      },
      "url": "ombCategory"
    },
    {
      "valueString": "Not Hispanic, Latino, or Spanish Origin",
      "url": "text"
```

```
      }
    ],
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"
  },
  {
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "446141000124107",
          "display": "Identifies as female gender (finding)",
          "userSelected": false
        }
      ],
      "text": "Identifies as female"
    },
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
  },
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
              "code": "468",
              "display": "Naknek Native Village",
              "userSelected": true
            }
          ],
          "text": "Naknek Native Village"
        },
        "url": "tribalAffiliation"
      },
      {
        "valueBoolean": false,
        "url": "isEnrolled"
      }
    ],
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation"
  },
  {
    "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
    "valueCode": "248153007"
  }
],
"identifier": [
  {
    "id": "CI-490016886-0",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
        }
      ],
      "text": "MRN"
    },
    "system": "urn:oid:2.16.840.1.113883.6.1000",
    "value": "6931",
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-490058771-1",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "10",
          "display": "MRN",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "MR",
          "display": "Medical record number",
          "userSelected": false
```

```
      }
    ],
    "text": "MRN"
  },
  "system": "urn:oid:2.16.840.1.113883.6.1000",
  "value": "6978",
  "period": {
    "end": "2020-07-06T21:21:25.000Z"
  }
},
{
  "id": "CI-490059574-3",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "670843",
        "display": "Messaging",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "U",
        "display": "Unspecified identifier",
        "userSelected": false
      }
    ],
    "text": "Messaging"
  },
  "_system": {
    "extension": [
      {
        "valueCode": "unknown",
        "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
      }
    ]
  },
  "value": "3C36293A3B964994AD8E6C0305F3330A",
  "period": {
    "start": "2020-06-30T20:08:26.000Z"
  }
},
{
  "id": "CI-490058805-4",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "670843",
        "display": "Messaging",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "U",
        "display": "Unspecified identifier",
        "userSelected": false
      }
    ],
    "text": "Messaging"
  },
  "_system": {
    "extension": [
      {
        "valueCode": "unknown",
        "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
      }
    ]
  },
  "value": "43DA797A657B47548F258A9B50EB41F5",
  "period": {
    "start": "2020-06-12T16:03:32.000Z"
  }
},
{
  "id": "CI-490059570-5",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "2553236771",
        "display": "Federated Person Principal",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
```

```
          "code": "AN",
          "display": "Account number",
          "userSelected": false
        }
      ],
      "text": "Federated Person Principal"
    },
    "system": "urn:oid:2.16.840.1.113883.3.13.6",
    "value": "URN:CERNER:IDENTITY-
  FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:AN7TD9A62CV8Z53Z",
    "period": {
      "start": "2020-06-30T20:08:25.000Z"
    }
  },
  {
    "id": "CI-490058801-6",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "2553236771",
          "display": "Federated Person Principal",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "AN",
          "display": "Account number",
          "userSelected": false
        }
      ],
      "text": "Federated Person Principal"
    },
    "system": "urn:oid:2.16.840.1.113883.3.13.6",
    "value": "URN:CERNER:IDENTITY-
  FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:KR8KC9MI9EQ8KC23",
    "period": {
      "start": "2020-06-12T16:03:29.000Z"
    }
  }
],
"active": true,
"name": [
  {
    "id": "CI-12724067-0",
    "use": "official",
    "text": "SMART, JOE",
    "family": "SMART",
    "given": [
      "JOE"
    ],
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-490059796-0",
    "use": "old",
    "text": "SMART, STEPHEN ALLEN",
    "family": "SMART",
    "given": [
      "STEPHEN",
      "ALLEN"
    ],
    "period": {
      "end": "2020-07-06T21:21:26.000Z"
    }
  }
],
"telecom": [
  {
    "id": "CI-PH-29811920-0",
    "extension": [
      {
        "valueUrl": "(816)888-8886",
        "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
      },
      {
        "valueString": "12345",
        "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
      }
    ],
    "system": "phone",
    "value": "8168888886",
    "use": "home",
    "rank": "1",
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
```

```
    }
  },
  {
    "id": "CI-EM-29822662-0",
    "system": "email",
    "value": "joesmart@yopmail.com",
    "use": "home",
    "rank": "1",
    "period": {
      "start": "2020-03-30T19:31:11.000Z"
    }
  }
],
"gender": "male",
"birthDate": "1976-04-29",
"deceasedBoolean": false,
"address": [
  {
    "id": "CI-24313553-0",
    "use": "home",
    "text": "12345 Main St\\nKansas city, MO 64116\\nUS",
    "line": [
      "12345 Main St"
    ],
    "city": "Kansas city",
    "district": "Jackson",
    "state": "MO",
    "postalCode": "64116",
    "country": "US",
    "period": {
      "start": "2019-12-26T15:13:36.000Z"
    }
  }
],
"maritalStatus": {
  "coding": [
    {
      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/38",
      "code": "309237",
      "display": "Married",
      "userSelected": true
    },
    {
      "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
      "code": "M",
      "display": "Married",
      "userSelected": false
    }
  ],
  "text": "Married"
},
[
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12791125-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "2553243305",
          "display": "Authorized Representative",
          "userSelected": true
        }
      ],
      "text": "Authorized Representative"
    }
  ],
  "name": {
    "id": "CI-12791125-0",
    "use": "official",
    "text": "RPTEST, CHILD"
  },
  "period": {
    "start": "2019-12-26T16:06:26.000Z"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12856285-12724067"
      },
```

```
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1152",
          "display": "Emergency Contact",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0131",
          "code": "C",
          "display": "Emergency Contact",
          "userSelected": false
        }
      ],
      "text": "Emergency Contact"
    }
  ],
  "name": {
    "id": "CI-12856285-1",
    "use": "official",
    "text": "Pari, Mary L"
  },
  "period": {
    "start": "2023-09-05T07:11:57.000Z"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12724071-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1152",
          "display": "Emergency Contact",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0131",
          "code": "C",
          "display": "Emergency Contact",
          "userSelected": false
        }
      ],
      "text": "Emergency Contact"
    }
  ],
  "name": {
    "id": "CI-12724071-2",
    "use": "official",
    "text": "SMART, VALERIE"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774384-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774384-3",
```

```
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774387-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774387-4",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774419-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774419-5",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774501-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774501-6",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
```

```
    "extension": [
      {
        "valueReference": {
          "reference": "RelatedPerson/12774519-12724067"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
      }
    ],
    "relationship": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1153",
            "display": "Family Member",
            "userSelected": true
          }
        ],
        "text": "Family Member"
      }
    ],
    "name": {
      "id": "CI-12774519-7",
      "use": "official",
      "text": "NOT, SET"
    }
  },
  {
    "extension": [
      {
        "valueReference": {
          "reference": "RelatedPerson/12774534-12724067"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
      }
    ],
    "relationship": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1153",
            "display": "Family Member",
            "userSelected": true
          }
        ],
        "text": "Family Member"
      }
    ],
    "name": {
      "id": "CI-12774534-8",
      "use": "official",
      "text": "NOT, SET"
    }
  },
  {
    "extension": [
      {
        "valueReference": {
          "reference": "RelatedPerson/12774747-12724067"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
      }
    ],
    "relationship": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1153",
            "display": "Family Member",
            "userSelected": true
          }
        ],
        "text": "Family Member"
      }
    ],
    "name": {
      "id": "CI-12774747-9",
      "use": "official",
      "text": "NOT, SET"
    }
  },
  {
    "extension": [
      {
        "valueReference": {
          "reference": "RelatedPerson/12774743-12724067"
        },
```

```
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774743-10",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774418-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774418-11",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774533-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774533-12",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12856287-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
```

```
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1156",
            "display": "Guardian",
            "userSelected": true
          }
        ],
        "text": "Guardian"
      }
    ],
    "name": {
      "id": "CI-12856287-13",
      "use": "official",
      "text": "PAUL, Park L"
    },
    "period": {
      "start": "2023-09-05T07:12:00.000Z"
    }
  }
]
  "communication": [
    {
      "language": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36",
            "code": "151",
            "display": "English",
            "userSelected": true
          },
          {
            "system": "urn:ietf:bcp:47",
            "code": "en",
            "display": "English",
            "userSelected": false
          }
        ],
        "text": "English"
      },
      "preferred": true
    }
  ],
  "generalPractitioner": [
    {
      "id": "CI-490017023-0",
      "reference": "Practitioner/4122622",
      "display": "Cerner Test, Physician - Hospitalist Cerner"
    },
    {
      "reference": "Organization/675844",
      "display": "Example Active Practice"
    }
  ]
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

x-confidential-response Response

Example of patient with confidential gender identity.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/12724067
```

Example Response:


```
{
  "resourceType": "Patient",
  "id": "12724067",
  "meta": {
    "versionId": "10",
    "lastUpdated": "2020-07-06T21:21:22.000Z",
    "security": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode"
        "code": "PROCESSINLINELABEL"
        "display": "process inline security label"
      }
    ]
  },
  "text": {
    "status": "generated",
```

```
  "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Patient</b></p><p><b>Meta Security</b>: process inline
  security label</p><p><b>Name</b>: SMART, JOE</p><p><b>Status</b>: Active</p><p><b>DOB</b>: Apr 29,
  1976</p><p><b>Administrative Gender</b>: Male</p><p><b>Marital Status</b>: Married</p></div>"
},
"extension": [
  {
    "id": "59434424",
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
              "code": "485602703",
              "display": "Appointment Reminders",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/communication-topic",
              "code": "appointment-reminder",
              "display": "Appointment Reminder",
              "userSelected": false
            }
          ],
          "text": "Appointment Reminders"
        },
        "url": "communication-type"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
              "code": "495085513",
              "display": "Fax",
              "userSelected": true
            },
            {
              "system": "http://hl7.org/fhir/contact-point-system",
              "code": "fax",
              "display": "Fax",
              "userSelected": false
            }
          ],
          "text": "Fax"
        },
        "url": "contact-method"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
              "code": "163",
              "display": "Business",
              "userSelected": true
            },
            {
              "system": "http://hl7.org/fhir/contact-point-use",
              "code": "work",
              "display": "Work",
              "userSelected": false
            }
          ],
          "text": "Business"
        },
        "url": "contact-type"
      },
      {
        "valueDateTime": "2019-04-13T20:00:00.000Z",
        "url": "verified-datetime"
      }
    ],
    "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
  },
  {
    "extension": [
      {
        "valueCoding": {
          "system": "urn:oid:2.16.840.1.113883.6.238",
          "code": "2106-3",
          "display": "White",
          "userSelected": false
        },
        "url": "ombCategory"
      },
      {
        "valueString": "White",
        "url": "text"
```

```
        }
      ],
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
    },
    {
      "extension": [
        {
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2186-5",
            "display": "Non Hispanic or Latino",
            "userSelected": false
          },
          "url": "ombCategory"
        },
        {
          "valueString": "Not Hispanic, Latino, or Spanish Origin",
          "url": "text"
        }
      ],
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"
    },
    {
      "valueCodeableConcept": {
        "extension": [
          {
            "valueCoding": {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
              "code": "PRS",
              "display": "patient requested information sensitivity",
              "userSelected": false
            },
            "url": "http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-inline-sec-label"
          }
        ],
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "446141000124107",
            "display": "Identifies as female gender (finding)",
            "userSelected": false
          }
        ],
        "text": "Identifies as female"
      },
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity"
    },
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
                "code": "468",
                "display": "Naknek Native Village",
                "userSelected": true
              }
            ],
            "text": "Naknek Native Village"
          },
          "url": "tribalAffiliation"
        },
        {
          "valueBoolean": false,
          "url": "isEnrolled"
        }
      ],
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
      "valueCode": "248153007"
    }
  ],
  "identifier": [
    {
      "id": "CI-490016886-0",
      "use": "usual",
      "type": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
            "code": "10",
            "display": "MRN",
            "userSelected": true
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
```

```
        "code": "MR",
        "display": "Medical record number",
        "userSelected": false
      }
    ],
    "text": "MRN"
  },
  "system": "urn:oid:2.16.840.1.113883.6.1000",
  "value": "6931",
  "period": {
    "start": "2019-12-26T15:14:12.000Z"
  }
},
{
  "id": "CI-490058771-1",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "10",
        "display": "MRN",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "MR",
        "display": "Medical record number",
        "userSelected": false
      }
    ],
    "text": "MRN"
  },
  "system": "urn:oid:2.16.840.1.113883.6.1000",
  "value": "6978",
  "period": {
    "end": "2020-07-06T21:21:25.000Z"
  }
},
{
  "id": "CI-490059574-3",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "670843",
        "display": "Messaging",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "U",
        "display": "Unspecified identifier",
        "userSelected": false
      }
    ],
    "text": "Messaging"
  },
  "_system": {
    "extension": [
      {
        "valueCode": "unknown",
        "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
      }
    ]
  },
  "value": "3C36293A3B964994AD8E6C0305F3330A",
  "period": {
    "start": "2020-06-30T20:08:26.000Z"
  }
},
{
  "id": "CI-490058805-4",
  "use": "usual",
  "type": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
        "code": "670843",
        "display": "Messaging",
        "userSelected": true
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "U",
        "display": "Unspecified identifier",
        "userSelected": false
      }
```

```
      ],
      "text": "Messaging"
    },
    "_system": {
      "extension": [
        {
          "valueCode": "unknown",
          "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
        }
      ]
    },
    "value": "43DA797A657B47548F258A9B50EB41F5",
    "period": {
      "start": "2020-06-12T16:03:32.000Z"
    }
  },
  {
    "id": "CI-490059570-5",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "2553236771",
          "display": "Federated Person Principal",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "AN",
          "display": "Account number",
          "userSelected": false
        }
      ],
      "text": "Federated Person Principal"
    },
    "system": "urn:oid:2.16.840.1.113883.3.13.6",
    "value": "URN:CERNER:IDENTITY-
  FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:AN7TD9A62CV8Z53Z",
    "period": {
      "start": "2020-06-30T20:08:25.000Z"
    }
  },
  {
    "id": "CI-490058801-6",
    "use": "usual",
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
          "code": "2553236771",
          "display": "Federated Person Principal",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
          "code": "AN",
          "display": "Account number",
          "userSelected": false
        }
      ],
      "text": "Federated Person Principal"
    },
    "system": "urn:oid:2.16.840.1.113883.3.13.6",
    "value": "URN:CERNER:IDENTITY-
  FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:KR8KC9MI9EQ8KC23",
    "period": {
      "start": "2020-06-12T16:03:29.000Z"
    }
  }
],
"active": true,
"name": [
  {
    "id": "CI-12724067-0",
    "use": "official",
    "text": "SMART, JOE",
    "family": "SMART",
    "given": [
      "JOE"
    ],
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-490059796-0",
    "use": "old",
    "text": "SMART, STEPHEN ALLEN",
```

```
    "family": "SMART",
    "given": [
      "STEPHEN",
      "ALLEN"
    ],
    "period": {
      "end": "2020-07-06T21:21:26.000Z"
    }
  }
],
"telecom": [
  {
    "id": "CI-PH-29811920-0",
    "extension": [
      {
        "valueUrl": "(816)888-8886",
        "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
      },
      {
        "valueString": "12345",
        "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
      }
    ],
    "system": "phone",
    "value": "8168888886",
    "use": "home",
    "rank": "1",
    "period": {
      "start": "2019-12-26T15:14:12.000Z"
    }
  },
  {
    "id": "CI-EM-29822662-0",
    "system": "email",
    "value": "joesmart@yopmail.com",
    "use": "home",
    "rank": "1",
    "period": {
      "start": "2020-03-30T19:31:11.000Z"
    }
  }
],
"gender": "male",
"birthDate": "1976-04-29",
"deceasedBoolean": false,
"address": [
  {
    "id": "CI-24313553-0",
    "use": "home",
    "text": "12345 Main St\\nKansas city, MO 64116\\nUS",
    "line": [
      "12345 Main St"
    ],
    "city": "Kansas city",
    "district": "Jackson",
    "state": "MO",
    "postalCode": "64116",
    "country": "US",
    "period": {
      "start": "2019-12-26T15:13:36.000Z"
    }
  }
],
"maritalStatus": {
  "coding": [
    {
      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/38",
      "code": "309237",
      "display": "Married",
      "userSelected": true
    },
    {
      "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
      "code": "M",
      "display": "Married",
      "userSelected": false
    }
  ],
  "text": "Married"
},
[
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12791125-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
```

```
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "2553243305",
          "display": "Authorized Representative",
          "userSelected": true
        }
      ],
      "text": "Authorized Representative"
    }
  ],
  "name": {
    "id": "CI-12791125-0",
    "use": "official",
    "text": "RPTEST, CHILD"
  },
  "period": {
    "start": "2019-12-26T16:06:26.000Z"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12856285-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1152",
          "display": "Emergency Contact",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0131",
          "code": "C",
          "display": "Emergency Contact",
          "userSelected": false
        }
      ],
      "text": "Emergency Contact"
    }
  ],
  "name": {
    "id": "CI-12856285-1",
    "use": "official",
    "text": "Pari, Mary L"
  },
  "period": {
    "start": "2023-09-05T07:11:57.000Z"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12724071-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1152",
          "display": "Emergency Contact",
          "userSelected": true
        },
        {
          "system": "http://terminology.hl7.org/CodeSystem/v2-0131",
          "code": "C",
          "display": "Emergency Contact",
          "userSelected": false
        }
      ],
      "text": "Emergency Contact"
    }
  ],
  "name": {
```

```
    "id": "CI-12724071-2",
    "use": "official",
    "text": "SMART, VALERIE"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774384-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774384-3",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774387-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true

        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774387-4",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774419-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774419-5",
    "use": "official",
    "text": "NOT, SET"
  }
},
```

```
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774501-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774501-6",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774519-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774519-7",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774534-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774534-8",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774747-12724067"
```

```
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774747-9",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774743-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774743-10",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774418-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
    {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
          "code": "1153",
          "display": "Family Member",
          "userSelected": true
        }
      ],
      "text": "Family Member"
    }
  ],
  "name": {
    "id": "CI-12774418-11",
    "use": "official",
    "text": "NOT, SET"
  }
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "RelatedPerson/12774533-12724067"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
    }
  ],
  "relationship": [
```

```
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1153",
            "display": "Family Member",
            "userSelected": true
          }
        ],
        "text": "Family Member"
      }
    ],
    "name": {
      "id": "CI-12774533-12",
      "use": "official",
      "text": "NOT, SET"
    }
  },
  {
    "extension": [
      {
        "valueReference": {
          "reference": "RelatedPerson/12856287-12724067"

        },
        "url": "http://hl7.org/fhir/StructureDefinition/patient-relatedPerson"
      }
    ],
    "relationship": [
      {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351",
            "code": "1156",
            "display": "Guardian",
            "userSelected": true
          }
        ],
        "text": "Guardian"
      }
    ],
    "name": {
      "id": "CI-12856287-13",
      "use": "official",
      "text": "PAUL, Park L"
    },
    "period": {
      "start": "2023-09-05T07:12:00.000Z"
    }
  }
]
  "communication": [
    {
      "language": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36",
            "code": "151",
            "display": "English",
            "userSelected": true
          },
          {
            "system": "urn:ietf:bcp:47",
            "code": "en",
            "display": "English",
            "userSelected": false
          }
        ],
        "text": "English"
      },
      "preferred": true
    }
  ],
  "generalPractitioner": [
    {
      "id": "CI-490017023-0",
      "reference": "Practitioner/4122622",
      "display": "Cerner Test, Physician - Hospitalist Cerner"
    }
  ]
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

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Patch a patient

File: r4-rest-api-v32/op-patientid-patch.html

Add a new identifier for the patient. Test the value of the of the identifier at the given in the list of patient identifiers. Remove the identifier at the given in the list of patient identifiers. Requires a test operation to be provided for the of the identifier intended to be removed. Replace the of the identifier at the given in the list of patient identifiers. Requires a test operation to be provided for the of the identifier whose is intended to be replaced. Replace the of the identifier at the given in the list of patient identifiers. Requires a test operation to be provided for the of the identifier whose is intended to be replaced. Replace the of the identifier at the given in the list of patient identifiers. Test the value of the of the name at the given in the list of patient names. This operation is only supported for the first name in the list of patient names (== 0). Replace the of the name at the given in the list of patient names. Replace the of the name at the given in the list of patient names. Replace the of the name at the given in the list of patient names. Replace the of the name at the given in the list of patient names. Replace the of the name at the given in the list of patient names. Add a new contact detail for the patient. Must contain the , , and fields, and must include a time component with a timezone when specifying a . Test the value of the of the telecom at the given in the list of patient telecoms. Remove the telecom at the given in the list of patient telecoms. Requires a test operation to be provided for the of the telecom intended to be removed. Replace the of the telecom at the given in the list of patient telecoms. Requires a test operation to be provided for the of the telecom whose is intended to be replaced. Replace the of the telecom at the given in the list of patient telecoms. Requires a test operation to be provided for the of the telecom whose is intended to be replaced. Replaces the phone extension of the telecom at the given in the list of patient telecoms. Replace the of the telecom at the given in the list of patient telecoms. Replace the gender that the patient is considered to have for administration and record keeping purposes. Replace the date of birth for the individual. Add a new address for the patient. Test the value of the of the address at the given in the list of patient addresses. Remove the address at the given in the list of patient addresses. Requires a test operation to be provided for the of the address intended to be removed. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the of the address at the given in the list of patient addresses. Replace the patient's most recent marital (civil) status. Replace the language which may be used to communicate with the patient about his health. value should a list with only a single element. Add a new nominated care provider. Must be a reference to a . Test the value of the of the general practitioner at the given in the list of the patient's nominated care providers. Remove the general practitoner at the given in the list of the patient's nominated care providers. Requires a test operation to be provided for the of the general practitioner intended to be removed. Replace the value of an extension that adds additional information about the patient. Test the value of the of the Communication Preference at the given in the list of the patient's extensions. Remove the CommunicationPreference at the given in the list of the patient's extensions. Requires a test operation to be provided for the of the Communication Preference intended to be removed. Add a new Communication Preference extension for the patient. Must be a valid extension. Replace the CommunicationPreference at the given in the list of the patient's extensions. At a time only one communication Preference can be changed.

Add Identifier

Description:

Path | Operation | Value Type
/identifier/- | add | Identifier

Example:

```
{
  "path": "/identifier/-",
  "op": "add",
  "value": {
    "type": {
      "coding": [
        {
          "code": "MR",
          "system": "http://hl7.org/fhir/v2/0203"
        }
      ]
    },
    "system": "urn:oid:1.1.1.1.1.1",
    "value": "<unique identifier>",
    "period": {
      "start": "2016-01-02T00:00:00-05:00",
      "end": "2020-01-02T00:00:00-05:00"
    }
  }
}
```

Note:

* Must not contain the assigner and use fields.

* Must contain the type, system, and value fields, and must include a time component with a timezone when specifying a period.

Test Identifier ID

Description:

Identifier.id

{index}

Path	Operation	Value Type
/identifier/{index}/id	test	string

Example:

```
{
  "path": "/identifier/0/id",
  "op": "test",
  "value": "CI-9926250-1"
}
```

Remove Identifier

Description:

{index}

Path	Operation	Value Type
/identifier/{index}	remove	N/A

Example:

```
{
  "path": "/identifier/1",
  "op": "remove"
}
```

Note:

Identifier.id

Replace Identifier System

Description:

Identifier.system

{index}

Path	Operation	Value Type
/identifier/{index}/system	replace	uri

Example:

```
{
  "path": "/identifier/1/system",
  "op": "replace",
  "value": "urn:oid:2.16.840.1.113883.3.13.6"
}
```

Note:

Identifier.id

system

Replace Identifier Value

Description:

Identifier.value

{index}

Path	Operation	Value Type
/identifier/{index}/value	replace	string

Example:

```
{
  "path": "/identifier/0/value",
  "op": "replace",
  "value": "<unique identifier>"
}
```

Note:

Identifier.id

value

Replace Identifier Period

Description:

Identifier.period

{index}

Path | Operation | Value Type
/identifier/{index}/period | replace | Period

Example:

```
{
  "path": "/identifier/1/period",
  "op": "replace",
  "value": {
    "start": "2016-01-02T00:00:00-05:00",
    "end": "2022-01-02T00:00:00-05:00"
  }
}
```

Note:

* Requires a test operation to be provided for the Identifier.id of the identifier whose period is intended to be replaced.

* The start and end when provided must include a time component with a timezone.

* When either start or end are not provided its value will be unset.

Test NameID

Description:

HumanName.id

{index}

Path | Operation | Value Type
/name/{index}/id | test | string

Example:

```
{
  "path": "/name/0/id",
  "op": "test",
  "value": "CI-5786010-0"
}
```

Note:

{index}

Replace Name Family

Description:

HumanName.family

{index}

Path | Operation | Value Type
/name/{index}/family | replace | String

Example:

```
{
  "path": "/name/0/family",
  "op": "replace",
  "value": "Potter"
}
```

Note:

* Requires a test operation to be provided for the HumanName.id of the name whose family is intended to be replaced.

* This operation is only supported for the first name in the list of patient names ({index} == 0).

Replace Name Given

Description:

HumanName.given

{index}

Path | Operation | Value Type
/name/{index}/given | replace | String

Example:

```
{
  "path": "/name/0/given",
  "op": "replace",
  "value": [
    "Harry",
    "James"
  ]
}
```

Note:

* Requires a test operation to be provided for the HumanName.id of the name whose given is intended to be replaced.

* This operation is only supported for the first name in the list of patient names ({index} == 0).

* If fewer values for given are provided in the list than previously set, the additional older values will be unset.

Replace Name Prefix

Description:

HumanName.prefix

{index}

Path	Operation	Value Type
/name/{index}/prefix	replace	String

Example:

```
{
  "path": "/name/0/prefix",
  "op": "replace",
  "value": "Mr."
}
```

Note:

* Requires a test operation to be provided for the HumanName.id of the name whose prefix is intended to be replaced.

* This operation is only supported for the first name in the list of patient names ({index} == 0).

* If no prefix value is provided the prefix on the name will be unset.

Replace Name Suffix

Description:

HumanName.suffix

{index}

Path	Operation	Value Type
/name/{index}/suffix	replace	String

Example:

```
{
  "path": "/name/2/suffix",
  "op": "replace",
  "value": "Jr."
}
```

Note:

* Requires a test operation to be provided for the HumanName.id of the name whose suffix is intended to be replaced.

* This operation is only supported for the first name in the list of patient names ({index} == 0).

* If no suffix value is provided the suffix on the name will be unset.

Replace Name Period

Description:

HumanName.period

{index}

Path	Operation	Value Type
/name/{index}/period	replace	Period

Example:

```
{
  "path": "/name/0/period",
  "op": "replace",
  "value": {
    "start": "2016-01-02T00:00:00-05:00"
  }
}
```

Note:

* Requires a test operation to be provided for the HumanName.id of the name whose period is intended to be replaced.

* This operation is only supported for the first name in the list of patient names ({index} == 0).

* Must not contain the end field.

* The start must include a time component with a timezone.

Add Telecom

Description:

Path	Operation	Value Type
/telecom/-	add	ContactPoint

Example:

```
{
```

```
  "path": "/telecom/-",
  "op": "add",
  "value": {
    "system": "phone",
    "value": "8168229121",
    "use": "home",
    "rank": 1,
    "extension": [
      {
        "valueString": "12345",
        "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
      }
    ],
    "period": {
      "start": "2012-05-17T15:33:18.000Z"
    }
  }
}
```

Note:

system

use

value

period

Test Telecom ID

Description:

ContactPoint.id

{index}

Path	Operation	Value Type
/telecom/{index}/id	test	string

Example:

```
{
  "path": "/telecom/0/id",
  "op": "test",
  "value": "CI-PH-2807950-0"
}
```

Remove Telecom

Description:

{index}

Path	Operation	Value Type
/telecom/{index}	remove	N/A

Example:

```
{
  "path": "/telecom/1",
  "op": "remove"
}
```

Note:

ContactPoint.id

Replace Telecom Value

Description:

ContactPoint.value

{index}

Path	Operation	Value Type
/telecom/{index}/value	replace	string

Example:

```
{
  "path": "/telecom/1/value",
  "op": "replace",
  "value": "987-654-3210"
}
```

Note:

ContactPoint.id

value

Replace Telecom Rank

Description:

ContactPoint.rank

{index}

Path	Operation	Value Type
/telecom/{index}/rank	replace	positiveInt

Example:

```
{
  "path": "/telecom/1/rank",
  "op": "replace",
  "value": 2
}
```

Note:

ContactPoint.id

rank

Replace Telecom Extension

Description:

{index}

Path	Operation	Value Type
/telecom/{index}/extension	replace	Extension

Example:

```
{
  "path": "/telecom/1/extension",
  "op": "replace",
  "value": [
    {
      "valueString": "12345",
      "url": "http://hl7.org/fhir/StructureDefinition/contactpoint-extension"
    }
  ]
}
```

Note:

* Requires a test operation to be provided for the ContactPoint.id of the telecom whose extension value is intended to be replaced.

* The system field of the telecom whose extension is intended to be replaced must be 'phone'.

* If no valueString is provided, the phone extension will be unset.

* The url value must be http://hl7.org/fhir/StructureDefinition/contactpoint-extension.

Replace Telecom Period

Description:

ContactPoint.period

{index}

Path	Operation	Value Type
/telecom/{index}/period	replace	Period

Example:

```
{
  "path": "/telecom/1/period",
  "op": "replace",
  "value": {
    "start": "2016-01-02T00:00:00-05:00",
    "end": "2022-01-02T00:00:00-05:00"
  }
}
```

Note:

* Requires a test operation to be provided for the ContactPoint.id of the telecom whose period is intended to be replaced.

* The start and end when provided must include a time component with a timezone.

* When either start or end are not provided its value will be unset.

Replace Gender

Description:

Path	Operation	Value Type
/gender	replace	code

Example:

```
{
```

```
  "path": "/gender",
  "op": "replace",
  "value": "female"
}
```

Replace Birth Date

Description:

Path | Operation | Value Type
/birthDate | replace | date

Example:

```
{
  "path": "/birthDate",
  "op": "replace",
  "value": "1990-09-15"
}
```

Add Address

Description:

Path | Operation | Value Type
/address/- | add | Address

Example:

```
{
  "path": "/address/-",
  "op": "add",
  "value": {
    "use": "home",
    "line": [
      "1212 Faircastle Drive",
      "Apartment 406"
    ],
    "city": "KC",
    "district": "Jackson",
    "state": "KS",
    "postalCode": "64199",
    "country": "United States of America",
    "period": {
      "start": "2012-05-17T15:33:18.000Z"
    }
  }
}
```

Note:

* Must not contain the text field.

* Must contain the use field, and must include a time component with a timezone when specifying a period.

Test Address ID

Description:

Address.id

{index}

Path | Operation | Value Type
/address/{index}/id | test | string

Example:

```
{
  "path": "/address/0/id",
  "op": "test",
  "value": "CI-2671902-0"
}
```

Remove Address

Description:

{index}

Path | Operation | Value Type
/address/{index} | remove | N/A

Example:

```
{
  "path": "/address/1",
  "op": "remove"
}
```

Note:

Address.id

Replace Address Line

Description:

Address.line

{index}

Path	Operation	Value Type
/address/{index}/line	replace	string

Example:

```
{
  "path": "/address/1/line",
  "op": "replace",
  "value": [
    "1212 Faircastle Drive",
    "Apartment 410"
  ]
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose line is intended to be replaced.

* If no line is provided, all the lines on the address will be unset.

* If fewer values for line are provided in the list than previously set, the additional older values will be unset.

Replace Address City

Description:

Address.city

{index}

Path	Operation	Value Type
/address/{index}/city	replace	string

Example:

```
{
  "path": "/address/1/city",
  "op": "replace",
  "value": "St. Louis"
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose city is intended to be replaced.

* If no city is provided, the city on the address will be unset.

Replace Address District

Description:

Address.district

{index}

Path	Operation	Value Type
/address/{index}/district	replace	string

Example:

```
{
  "path": "/address/1/district",
  "op": "replace",
  "value": "Johnson"
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose district is intended to be replaced.

* If no district is provided, the district on the address will be unset.

Replace Address State

Description:

Address.state

{index}

Path	Operation	Value Type
/address/{index}/state	replace	string

Example:

```
{
  "path": "/address/1/state",
  "op": "replace",
  "value": "MO"
```

```
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose state is intended to be replaced.

* If no state is provided, the state on the address will be unset.

Replace Address Postal Code

Description:

Address.postalCode

{index}

Path	Operation	Value Type
/address/{index}/postalCode	replace	string

Example:

```
{
  "path": "/address/1/postalCode",
  "op": "replace",
  "value": "66209"
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose postalCode is intended to be replaced.

* If no postalCode is provided, the postalCode on the address will be unset.

Replace Address Country

Description:

Address.country

{index}

Path	Operation	Value Type
/address/{index}/country	replace	string

Example:

```
{
  "path": "/address/1/country",
  "op": "replace",
  "value": "Canada"
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose country is intended to be replaced.

* If no country is provided, the country on the address will be unset.

Replace Address Period

Description:

Address.period

{index}

Path	Operation	Value Type
/address/{index}/period	replace	Period

Example:

```
{
  "path": "/address/1/period",
  "op": "replace",
  "value": {
    "start": "2016-01-02T00:00:00-05:00",
    "end": "2022-01-02T00:00:00-05:00"
  }
}
```

Note:

* Requires a test operation to be provided for the Address.id of the address whose period is intended to be replaced.

* The start and end when provided must include a time component with a timezone.

* When either start or end are not provided its value will be unset.

Replace Marital Status

Description:

Path	Operation	Value Type
/maritalStatus	replace	CodeableConcept

Example:

```
{
  "path": "/maritalStatus",
  "op": "replace",
  "value": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
        "code": "S"
      }
    ]
  }
}
```

Replace Communication

Description:

Path	Operation	Value Type
/communication	replace	BackboneElement

Example:

```
{
  "path": "/communication",
  "op": "replace",
  "value": [
    {
      "language": {
        "coding": [
          {
            "system": "urn:ietf:bcp:47",
            "code": "en",
          }
        ]
      },
      "preferred": true
    }
  ]
}
```

Note:

Patient.communication

Add General Practitioner

Description:

Path	Operation	Value Type
/generalPractitioner/-	add	Reference

Example:

```
{
  "path": "/generalPractitioner/-",
  "op": "add",
  "value": {
    "reference": "Practitioner/995932"
  }
}
```

Note:

Practitioner

Test General Practitioner ID

Description:

Reference.id

{index}

Path	Operation	Value Type
/generalPractitioner/{index}/id	test	string

Example:

```
{
  "path": "/generalPractitioner/1/id",
  "op": "test",
  "value": "CI-312732-2"
}
```

Remove General Practitioner

Description:

{index}

Path	Operation	Value Type
/generalPractitioner/{index}	remove	N/A

Example:

```
{
  "path": "/generalPractitioner/1",
  "op": "remove"
}
```

Note:

Reference.id

Replace Extension

Description:

Path	Operation	Value Type
/extension	replace	Extension

Example:

```
{
  "path": "/extension",
  "op": "replace",
  "value": [
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
      "valueCode": "M"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2106-3",
            "display": "White"
          }
        },
        {
          "url": "detailed",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2110-5",
            "display": "English"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2135-2",
            "display": "Hispanic or Latino"
          }
        },
        {
          "url": "detailed",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2156-8",
            "display": "Costa Rican"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "446141000124107"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
      "valueCode": "248153007"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation",
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
```

```
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
                "code": "468",
                "display": "Naknek Native Village"
              }
            ],
            "text": "Naknek Native Village"
          },
          "url": "tribalAffiliation"
        },
        {
          "valueBoolean": false,
          "url": "isEnrolled"
        }
      ]
    },
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
      "extension": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                    "code": "2572136733",
                    "display": "Appointment Reminder",
                    "userSelected": false
                  }
                ],
                "text": "Appointment Reminder"
              },
              "url": "communication-type"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                    "code": "4054467",
                    "display": "Phone",
                    "userSelected": false
                  }
                ]
              },
              "url": "contact-method"
            },
            {
              "valueCodeableConcept": {
                "coding": [
                    {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                    "code": "163",
                    "display": "Business",
                    "userSelected": false
                  }
                ],
                "text": "Home"
              },
              "url": "contact-type"
            },
            {
              "valueDateTime": "2019-12-26T15:41:52.000Z",
              "url": "verified-datetime"
            }
          ],
          "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
        }
      ]
    }
  ]
}
```


Note:

Supported extensions that can be replaced:

* US Core Birth Sex
* US Core Race
* US Core Ethnicity
* US Core Gender Identity
* US Core Sex
* US Core Tribal Affiliation

Test Communication Preference ID

Description:

CommunicationPreference.id

{index}

Path	Operation	Value Type
/extension/{index}/id	test	string

Example:

```
{
  "path": "/extension/1/id",
  "op": "test",
  "value": "151534123"
}
```

Remove Communication PreferenceID

Description:

{index}

Path	Operation	Value Type
/extension/{index}	remove	N/A

Example:

```
{
  "path": "/extension/1",
  "op": "remove"
}
```

Note:

CommunicationPreference.id

Add Communication Preference

Description:

Path	Operation	Value Type
/extension/-	add	Communication Preference

Example:

```
{
  "path": "/extension/-",
  "op": "add",
  "value": {
    "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
    "extension": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4640016",
                  "code": "485602703",
                  "display": "Appointment Reminder",
                  "userSelected": false
                }
              ],
              "text": "Appointment Reminder"
            },
            "url": "communication-type"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/23042",
                  "code": "4054467",
                  "display": "Phone",
                  "userSelected": false
                }
              ]
            },
            "url": "contact-method"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                  {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/43",
                  "code": "163",
                  "display": "Business",
                  "userSelected": false
                }
              ],
```

```
            "text": "Home"
          },
          "url": "contact-type"
        },
        {
          "valueDateTime": "2019-12-26T15:41:52.000Z",
          "url": "verified-datetime"
        }
      ],
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference"
    }
  ]
 }
}
```

Note:

CommunicationPreference

Replace Communication PreferenceID

Description:

{index}

Path	Operation	Value Type
/extension/{index}	replace	Communication Preference

Example:

```
{
  "path": "/extension/0",
  "op": "replace",
  "value": {
    "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/communication-preference",
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/dacc6494-e336-45ad-8729-b789ff8663c6/codeSet/4640016",
              "code": "2572145011",
              "display": "Clinical Communications",
              "userSelected": true
            }
          ]
        },
        "url": "communication-type"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/34fc7274-1120-4899-937d-a9ea1becc848/codeSet/23042",
              "code": "4054468"
            }
          ]
        },
        "url": "contact-method"
      }
    ]
  }
}
```

Note:

* Requires a test operation to be provided for the CommunicationPreference.id of the Communication Preference intended to be removed.

* Test operation path and replace extension path index should be same.

Request

Supported Media Types

* application/fhir+json

Path Parameters

ID(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* If-Match(required): string The current version of the Patient resource.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

content-type(required): string

Specifies the content type of the request entity body.

Example: application/json-patch+json

Body ()

Root Schema : PatientPatchBody

Type:

array

Minimum Number of Items:

Show Source

* Array of: object items


```
{
    "type":"array",
    "minItems":1,
    "items":{
        "type":"object",
        "required":[
            "op",
            "path"
        ],
        "properties":{
            "op":{
                "type":"string",
                "enum":[
                    "add",
                    "remove",
                    "replace",
                    "test"
                ]
            },
            "path":{
                "type":"string"
            },
            "value":{
                "type":"string"
            }
        }
    },
    "example":"[\n  {\n    \"op\": \"add\",\n    \"path\": \"/identifier/-\",\n    \"value\": {\n      \"type\": {\n
            \"coding\": [\n          {\n            \"code\": \"MR\",\n            \"system\":
    \"http://hl7.org/fhir/v2/0203\"\n          }\n        ]\n      },\n      \"system\": \"2.16.840.1.113883.6.1000\",\n
          \"value\": \"THIS:IS:A:UNIQUE:IDENTIFIER\",\n      \"period\": {\n        \"start\":
    \"2018-01-02T00:00:00-05:00\",\n        \"end\": \"2022-01-02T00:00:00-05:00\"\n      }\n    }\n  },\n  {\n
    \"path\": \"/name/0/id\",\n    \"op\": \"test\",\n    \"value\": \"CI-12724067-0\"\n  },\n  {\n    \"path\":
    \"/name/0/given\",\n    \"op\": \"replace\",\n    \"value\": [\n      \"Joe\",\n      \"Smarty\"\n    ]\n  },\n  {\n
        \"path\": \"/telecom/0/id\",\n    \"op\": \"test\",\n    \"value\": \"CI-PH-29811920-0\"\n  },\n  {\n
    \"path\": \"/telecom/0\",\n    \"op\": \"remove\"\n  },\n  {\n    \"path\": \"/maritalStatus\",\n    \"op\":
    \"replace\",\n    \"value\": {\n      \"coding\": [\n        {\n          \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus\",\n          \"code\": \"D\"\n        }\n      ]\n    }\n
      },\n  {\n    \"path\": \"/communication\",\n    \"op\": \"replace\",\n    \"value\": [\n      {\n
    \"language\": {\n          \"coding\": [\n            {\n              \"system\":
    \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36\",\n              \"code\": \"151\"\n
              }\n          ]\n        },\n        \"preferred\": true\n      }\n    ]\n  },\n  {\n    \"path\":
    \"/extension\",\n    \"op\": \"replace\",\n    \"value\": [\n      {\n        \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex\",\n        \"valueCode\": \"M\"\n      },\n
          {\n        \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race\",\n        \"extension\":
    [\n          {\n            \"url\": \"ombCategory\",\n            \"valueCoding\": {\n              \"system\":
    \"urn:oid:2.16.840.1.113883.6.238\",\n              \"code\": \"2106-3\",\n              \"display\": \"White\"\n
              }\n          },\n          {\n            \"url\": \"detailed\",\n            \"valueCoding\": {\n
                \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n              \"code\": \"2110-5\",\n
    \"display\": \"English\"\n            }\n          }\n        ]\n      },\n      {\n        \"url\":
    \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity\",\n        \"extension\": [\n          {\n
              \"url\": \"ombCategory\",\n            \"valueCoding\": {\n              \"system\":
    \"urn:oid:2.16.840.1.113883.6.238\",\n              \"code\": \"2135-2\",\n              \"display\": \"Hispanic or
    Latino\"\n            }\n          },\n          {\n            \"url\": \"detailed\",\n            \"valueCoding\":
     {\n              \"system\": \"urn:oid:2.16.840.1.113883.6.238\",\n              \"code\": \"2156-8\",\n
                \"display\": \"Costa Rican\"\n            }\n          }\n        ]\n      },\n      {\n
    \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity\",\n
    \"valueCodeableConcept\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://snomed.info/sct\",\n            \"code\": \"446141000124107\"\n          }\n        ]\n      },\n      {\n
          \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex\",\n        \"valueCode\":
    \"248153007\"\n      },\n      {\n        \"url\": \"http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-
    affiliation\",\n        \"extension\": [\n          {\n            \"valueCodeableConcept\": {\n
    \"coding\": [\n                {\n                  \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS\",\n                  \"code\": \"472\",\n
                    \"display\": \"Native Village of Napaskiak\"\n                }\n              ],\n
    \"text\": \"Native Village of Napaskiak\"\n            },\n            \"url\": \"tribalAffiliation\"\n
    },\n          {\n            \"valueBoolean\": true,\n            \"url\": \"isEnrolled\"\n          }\n        ]\n
          }\n    ]\n  }\n]\n"
}
```


Example:


```
[
  {
    "op": "add",
```

```
  "path": "/identifier/-",
  "value": {
    "type": {
      "coding": [
        {
          "code": "MR",
          "system": "http://hl7.org/fhir/v2/0203"
        }
      ]
    },
    "system": "2.16.840.1.113883.6.1000",
    "value": "THIS:IS:A:UNIQUE:IDENTIFIER",
    "period": {
      "start": "2018-01-02T00:00:00-05:00",
      "end": "2022-01-02T00:00:00-05:00"
    }
  }
},
{
  "path": "/name/0/id",
  "op": "test",
  "value": "CI-12724067-0"
},
{
  "path": "/name/0/given",
  "op": "replace",
  "value": [
    "Joe",
    "Smarty"
  ]
},
{
  "path": "/telecom/0/id",
  "op": "test",
  "value": "CI-PH-29811920-0"
},
{
  "path": "/telecom/0",
  "op": "remove"
},
{
  "path": "/maritalStatus",
  "op": "replace",
  "value": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
        "code": "D"
      }
    ]
  }
},
{
  "path": "/communication",
  "op": "replace",
  "value": [
    {
      "language": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36",
            "code": "151"
          }
        ]
      },
      "preferred": true
    }
  ]
},
{
  "path": "/extension",
  "op": "replace",
  "value": [
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
      "valueCode": "M"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2106-3",
            "display": "White"
          }
        },
        {
```

```
            "url": "detailed",
            "valueCoding": {
              "system": "urn:oid:2.16.840.1.113883.6.238",
              "code": "2110-5",
              "display": "English"
            }
          }
        ]
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
        "extension": [
          {
            "url": "ombCategory",
            "valueCoding": {
              "system": "urn:oid:2.16.840.1.113883.6.238",
              "code": "2135-2",
              "display": "Hispanic or Latino"
            }
          },
          {
            "url": "detailed",
            "valueCoding": {
              "system": "urn:oid:2.16.840.1.113883.6.238",
              "code": "2156-8",
              "display": "Costa Rican"
            }
          }
        ]
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity",
        "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "446141000124107"
          }
        ]
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex",
        "valueCode": "248153007"
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation",
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS",
                  "code": "472",
                  "display": "Native Village of Napaskiak"
                }
              ],
              "text": "Native Village of Napaskiak"
            },
            "url": "tribalAffiliation"
          },
          {
            "valueBoolean": true,
            "url": "isEnrolled"
          }
        ]
      }
    ]
  }
]
```

Nested Schema : items

Type:

object

Show Source

* op(required): string Allowed Values: ["add", "remove", "replace", "test"]

* path(required): string

* value: string

```
{
    "type":"object",
    "required":[
        "op",
        "path"
    ],
    "properties":{
```

```
        "op":{
            "type":"string",
            "enum":[
                "add",
                "remove",
                "replace",
                "test"
            ]
        },
        "path":{
            "type":"string"
        },
        "value":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/json-patch+json

Default Response

This operation supports the following authorization types: The ETag response header indicates the current If-Match version to use on a subsequent update.

* Provider

* System

Example Request:

```
PATCH https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Patient/12724067
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Mon, 06 Apr 2020 19:00:43 GMT
  Etag: W/"1"
  Last-Modified: Mon, 06 Apr 2020 19:00:43 GMT
  Vary: Origin
  X-Request-Id: 11111111111111111111111111111111
```

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Person

File: r4-rest-api-v32/api-person.html

Description

Person

This topic describes information for the Person resource.

Overview

The Person resource identifies an individual outside of a specific healthcare context and provides a mechanism to associate person resources across different facilities or organizations.

The following fields are returned if valued:

ID

Identifier

Name

Telecom

Gender

Date of birth

Address

Active

Terminology Bindings

Person.identifier.type

Description: The type used to determine which identifier to use for a specific purpose.

Details: v2 Identifier Type

System: http://terminology.hl7.org/CodeSystem/v2-0203

Details: For more information, see Code Set 4 Person Identifier (Alias) Types under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/4

Person

The operations from the Person category.

Get a list of persons

Method: get Path: /Person

Get a person by ID

Method: get Path: /Person/{ID}

Get a list of persons

File: r4-rest-api-v32/op-person-get.html

Retrieves a list of persons that meet supplied query parameters.

Request

Query Parameters

_id: array[string]

Minimum Number of Items:

The logical resource ID. This parameter is required if the identifier parameter is not used.

Example: _id=12345

identifier: string

The person identifier. This parameter is required if the _id parameter is not used.

The value of the identifier parameter must include both a system and a code.

Example: identifier=urn:oid:2.16.840.1.113883.3.13.6|01022228

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Person/2572382193
```

Example Response:

```
{
  "resourceType": "Bundle",
  "id": "f2800374-292d-4508-bd3f-b4da814d1b5f",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-
    0e701af7583d/Person?identifier=urn%3Aoid%3A2.16.840.1.113883.6.1000%7C6931"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Person/12724067",
      "resource": {
        "resourceType": "Person",
        "id": "12724067",
        "meta": {
          "versionId": "10",
          "lastUpdated": "2020-07-06T21:21:22.000Z"
        },
```

```
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Person</b></p><p><b>Status</b>:
Active</p><p><b>Name</b>: SMART, JOE</p><p><b>DOB</b>: Apr 29, 1976</p><p><b>Administrative Gender</b>:
Male</p></div>"
    },
    "identifier": [
      {
        "id": "CI-490525912-3",
        "use": "usual",
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
              "code": "10",
              "display": "MRN",
              "userSelected": true
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "MR",
              "display": "Medical record number",
              "userSelected": false
            }
          ],
          "text": "MRN"
        },
        "system": "urn:oid:2.16.840.1.113883.6.1000",
        "value": "31577"
      }
    ],
    "name": [
      {
        "id": "CI-12724067-0",
        "use": "official",
        "text": "SMART, JOE",
        "family": "SMART",
        "given": [
          "JOE"
        ],
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      }
    ],
    "telecom": [
      {
        "id": "CI-PH-29811920-0",
        "extension": [
          {
            "valueUrl": "(816)888-8886",
            "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
          }
        ],
        "system": "phone",
        "value": "8168888886",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2019-12-26T15:14:12.000Z"
        }
      },
      {
        "id": "CI-EM-29822662-0",
        "system": "email",
        "value": "joesmart@example.com",
        "use": "home",
        "rank": "1",
        "period": {
          "start": "2020-03-30T19:31:11.000Z"
        }
      }
    ],
    "gender": "male",
    "birthDate": "1976-04-29",
    "address": [
      {
        "id": "CI-24313553-0",
        "use": "home",
        "text": "12345 Main St\\nKansas City, MO 64116\\nUS",
        "line": [
          "12345 Main St"
        ],
        "city": "Kansas City",
        "district": "Jackson",
        "state": "MO",
        "postalCode": "64116",
        "country": "US",
        "period": {
```

```
            "start": "2019-12-26T15:13:36.000Z"
          }
        }
      ],
      "active": true
    }
  }
 ]
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

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Get a person by ID

File: r4-rest-api-v32/op-person-id-get.html

Retrieves a person by the associated ID.

Oracle Health Millennium Platform supports the ability to logically merge a person record into another person record when both records are describing the same person. This merge is known as a person combine. If necessary, you can perform a person uncombine to undo the merge. When the requested person record is combined into another record, an inactive person entry is returned with a link to the current person entry. Entries for combined patients are only returned when retrieving the entries directly by ID. They are not returned when searching with other parameters.

Request

Path Parameters

ID(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Person/2572382193
```

Example Response:


```
{
  "resourceType": "Person",
  "id": "12724067",
  "meta": {
    "versionId": "10",
    "lastUpdated": "2020-07-06T21:21:22.000Z"
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Person</b></p><p><b>Status</b>: Active</p><p><b>Name</b>:
    SMART, JOE</p><p><b>DOB</b>: Apr 29, 1976</p><p><b>Administrative Gender</b>: Male</p></div>"
  },
  "identifier": [
    {
      "id": "CI-490525912-3",
      "use": "usual",
      "type": {
        "coding": [
          {
            "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4",
            "code": "10",
```

```
            "display": "MRN",
            "userSelected": true
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "MR",
            "display": "Medical record number",
            "userSelected": false
          }
        ],
        "text": "MRN"
      },
      "system": "urn:oid:2.16.840.1.113883.6.1000",
      "value": "31577"
    }
  ],
  "name": [
    {
      "id": "CI-12724067-0",
      "use": "official",
      "text": "SMART, JOE",
      "family": "SMART",
      "given": [
        "JOE"
      ],
      "period": {
        "start": "2019-12-26T15:14:12.000Z"
      }
    }
  ],
  "telecom": [
    {
      "id": "CI-PH-29811920-0",
      "extension": [
        {
          "valueUrl": "(816)888-8886",
          "url": "http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address"
        }
      ],
      "system": "phone",
      "value": "8168888886",
      "use": "home",
      "rank": "1",
      "period": {
        "start": "2019-12-26T15:14:12.000Z"
      }
    },
    {
      "id": "CI-EM-29822662-0",
      "system": "email",
      "value": "joesmart@example.com",
      "use": "home",
      "rank": "1",
      "period": {
        "start": "2020-03-30T19:31:11.000Z"
      }
    }
  ],
  "gender": "male",
  "birthDate": "1976-04-29",
  "address": [
    {
      "id": "CI-24313553-0",
      "use": "home",
      "text": "12345 Main St\\nKansas City, MO 64116\\nUS",
      "line": [
        "12345 Main St"
      ],
      "city": "Kansas City",
      "district": "Jackson",
      "state": "MO",
      "postalCode": "64116",
      "country": "US",
      "period": {
        "start": "2019-12-26T15:13:36.000Z"
      }
    }
  ],
  "active": true
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

x-person-combines Response

Person Combines Example

Oracle Health Millennium Platform supports the ability to logically merge a person record into another person record when both records are describing the same person. This merge is known as a person combine. If necessary, you can perform a person uncombine to undo the merge. When the requested person record is combined into another record, an inactive person entry is returned with a link to the current person entry. Entries for combined patients are only returned when retrieving the entries directly by ID. They are not returned when searching with other parameters.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Person/12742633
```

Example Response:

```
{
    "resourceType": "Person",
    "id": "12742633",
    "meta": {
      "versionId": "0"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Person</b></p><p><b>Status</b>:
    Inactive</p><p><b>Link</b>: Person/12724067</p></div>"
    },
    "active": false,
    "link": [
      {
        "target": {
          "reference": "Person/12724067"
        }
      }
    ]
  }
```

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Practitioner

File: r4-rest-api-v32/api-practitioner.html

Description

Practitioner

This topic describes information for the Practitioner resource.

Overview

The Practitioner resource provides information about a person formally involved in the care of a patient on behalf of a healthcare facility. Practitioners include but are not limited to physicians, nurses, pharmacists, therapists, technologists, and social workers.

Not all practitioners have access to the electronic health record (EHR) but can be referenced by other resources to indicate that they are in some way involved in a patient's care.

This resource supports the following HL7 FHIR US Core Implementation Guide STU 6.1.0 profile:

* US Core Practitioner Profile

Note:

The provider's location and contact information can be found with the practitioner.Address and practitioner.Telecom fields.

The following fields are returned if valued:

Practitioner ID

Identifier

Active

Name

Telecom

Address

Gender

Qualification

Is Physician Extension (for more information, see the Extensions section of this topic)

Terminology Bindings

Practitioner.identifier.type

Description: The type used to determine which identifier to use for a specific purpose.

Details: Identifier Type Codes

System: http://hl7.org/fhir/identifier-type

Details: v2 Identifier Type

System: http://terminology.hl7.org/CodeSystem/v2-0203

Practitioner.qualification.code

Description: A coded representation of the qualification.

Details: Qualification codes

System: http://terminology.hl7.org/CodeSystem/v2-0360|2.7

Extensions

All URLs for custom extensions are defined as https://fhir-ehr.cerner.com/r4/StructureDefinition/{id}

ID | Value[x] Type | Description
is-physician | boolean | Indicates whether the practitioner is a physician.

Practitioner

The operations from the Practitioner category.

Create a practitioner

Method: post Path: /Practitioner/

Get a list of practitioners

Method: get Path: /Practitioner/

Get a practitioner by ID

Method: get Path: /Practitioner/{id}

Create a practitioner

File: r4-rest-api-v32/op-practitioner-post.html

Creates a practitioner to be referenced by other resources. This API is not used for user provisioning.

Authorization Types

This resource supports the following authorization types:

* Provider

* System

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : PractitionerCreateBody

Type:

object

Show Source

active: boolean

Allowed Values:

[true]

Whether this practitioner record is in active use.

identifier: array

identifier

The identifier for the practitioner.

Must be a unique identifier.

Must not contain the id or assigner fields.

name(required): array

name

Minimum Number of Items:

Maximum Number of Items:

The name associated with the practitioner.

Must not contain the text field.

resourceType(required): string

Allowed Values:

["Practitioner"]

The type of the FHIR resource.


```
{
    "type":"object",
    "required":[
        "resourceType",
        "name"
    ],
    "properties":{
        "resourceType":{
            "description":"<p>The type of the FHIR resource.</p>\n",
            "type":"string",
            "enum":[
                "Practitioner"
            ]
        },
        "name":{
            "description":"<p>The name associated with the practitioner.</p>\n<p>Must not contain the <code>text</code>
    field.</p>\n",
            "type":"array",
            "minItems":1,
            "maxItems":1,
            "items":{
                "type":"object",
                "required":[
                    "given",
                    "family"
                ],
                "properties":{
                    "family":{
                        "type":"string",
                        "example":"Williams"
                    },
                    "given":{
                        "type":"array",
                        "minItems":1,
                        "items":{
                            "type":"string",
                            "example":"James McAfee"
                        }
                    },
                    "prefix":{
                        "type":"array",
                        "maxItems":1,
                        "items":{
                            "type":"string",
                            "example":"Dr."
                        }
                    },
                    "suffix":{
                        "maxItems":1,
                        "type":"array",
                        "items":{
                            "type":"string",
                            "example":"M.D."
                        }
                    },
                    "period":{
                        "type":"object",
                        "description":"<p>Must not include an <code>end</code> date.</p>",
                        "properties":{
                            "start":{
                                "type":"string",
                                "description":"<p>Must include a time component with a timezone.</p>",
                                "example":"2020-12-01T00:00:00.000Z"
                            }
                        }
                    }
                }
            }
        },
        "identifier":{
            "description":"<p>The identifier for the practitioner.</p>\n<p>Must be a unique identifier.<p>\n<p>Must not
    contain the <code>id</code> or <code>assigner</code> fields.</p>\n",
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "system",
                    "value"
                ],
                "properties":{
                    "type":{
                        "type":"object",
```

```
                    "properties":{
                        "coding":{
                            "type":"array",
                            "minItems":1,
                            "maxItems":1,
                            "items":{
                                "type":"object",
                                "properties":{
                                    "code":{
                                        "type":"string",
                                        "example":"DEA"
                                    },
                                    "system":{
                                        "type":"string",
                                        "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
                                    }
                                }
                            }
                        }
                    },
                    "system":{
                        "type":"string",
                        "example":"urn:oid:2.16.840.1.113883.4.814"
                    },
                    "value":{
                        "type":"string",
                        "example":"CW1234563"
                    },
                    "period":{
                        "type":"object",
                        "properties":{
                            "start":{
                                "type":"string",
                                "description":"<p>Must include a time component with a timezone.</p>",
                                "example":"2019-12-01T00:00:00.000Z"
                            },
                            "end":{
                                "type":"string",
                                "description":"<p>Must include a time component with a timezone.</p>",
                                "example":"2029-12-01T23:59:59.000Z"
                            }
                        }
                    },
                    "use":{
                        "type":"string",
                        "enum":[
                            "official"
                        ]
                    }
                }
            }
        },
        "active":{
            "description":"<p>Whether this practitioner record is in active use.</p>\n",
            "type":"boolean",
            "enum":[
                true
            ]
        }
    },
    "example":{
        "resourceType":"Practitioner",
        "active":true,
        "name":[
            {
                "family":"Williams",
                "given":[
                    "Rory",
                    "James"
                ],
                "prefix":[
                    "Dr."
                ],
                "suffix":[
                    "M.D."
                ],
                "period":{
                    "start":"2019-12-01T00:00:00.000Z"
                }
            }
        ],
        "identifier":[
            {
                "type":{
                    "coding":[
                        {
                            "code":"DEA",
                            "system":"http://terminology.hl7.org/CodeSystem/v2-0203"
```

```
                        }
                    ]
                },
                "system":"urn:oid:2.16.840.1.113883.4.814",
                "value":"CW1234563",
                "period":{
                    "start":"2019-12-01T00:00:00.000Z",
                    "end":"2029-12-01T23:59:59.000Z"
                }
            }
        ]
    }
}
```

Example:

```
{
    "resourceType":"Practitioner",
    "active":true,
    "name":[
        {
            "family":"Williams",
            "given":[
                "Rory",
                "James"
            ],
            "prefix":[
                "Dr."
            ],
            "suffix":[
                "M.D."
            ],
            "period":{
                "start":"2019-12-01T00:00:00.000Z"
            }
        }
    ],
    "identifier":[
        {
            "type":{
                "coding":[
                    {
                        "code":"DEA",
                        "system":"http://terminology.hl7.org/CodeSystem/v2-0203"
                    }
                ]
            },
            "system":"urn:oid:2.16.840.1.113883.4.814",
            "value":"CW1234563",
            "period":{
                "start":"2019-12-01T00:00:00.000Z",
                "end":"2029-12-01T23:59:59.000Z"
            }
        }
    ]
}
```

Nested Schema : identifier

Type:

array

The identifier for the practitioner.

Must be a unique identifier.

Must not contain the id or assigner fields.

Show Source

* Array of: object items

```
{
    "description":"<p>The identifier for the practitioner.</p>\n<p>Must be a unique identifier.<p>\n<p>Must not contain
    the <code>id</code> or <code>assigner</code> fields.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "system",
            "value"
        ],
        "properties":{
            "type":{
                "type":"object",
                "properties":{
                    "coding":{
                        "type":"array",
                        "minItems":1,
                        "maxItems":1,
                        "items":{
                            "type":"object",
```

```
                            "properties":{
                                "code":{
                                    "type":"string",
                                    "example":"DEA"
                                },
                                "system":{
                                    "type":"string",
                                    "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
                                }
                            }
                        }
                    }
                }
            },
            "system":{
                "type":"string",
                "example":"urn:oid:2.16.840.1.113883.4.814"
            },
            "value":{
                "type":"string",
                "example":"CW1234563"
            },
            "period":{
                "type":"object",
                "properties":{
                    "start":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>",
                        "example":"2019-12-01T00:00:00.000Z"
                    },
                    "end":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>",
                        "example":"2029-12-01T23:59:59.000Z"
                    }
                }
            },
            "use":{
                "type":"string",
                "enum":[
                    "official"
                ]
            }
        }
    }
}
```


Nested Schema : name

Type:

array

Minimum Number of Items:

Maximum Number of Items:

The name associated with the practitioner.

Must not contain the text field.

Show Source

* Array of: object items


```
{
    "description":"<p>The name associated with the practitioner.</p>\n<p>Must not contain the <code>text</code>
    field.</p>\n",
    "type":"array",
    "minItems":1,
    "maxItems":1,
    "items":{
        "type":"object",
        "required":[
            "given",
            "family"
        ],
        "properties":{
            "family":{
                "type":"string",
                "example":"Williams"
            },
            "given":{
                "type":"array",
                "minItems":1,
                "items":{
                    "type":"string",
                    "example":"James McAfee"
                }
            },
            "prefix":{
                "type":"array",
```

```
                "maxItems":1,
                "items":{
                    "type":"string",
                    "example":"Dr."
                }
            },
            "suffix":{
                "maxItems":1,
                "type":"array",
                "items":{
                    "type":"string",
                    "example":"M.D."
                }
            },
            "period":{
                "type":"object",
                "description":"<p>Must not include an <code>end</code> date.</p>",
                "properties":{
                    "start":{
                        "type":"string",
                        "description":"<p>Must include a time component with a timezone.</p>",
                        "example":"2020-12-01T00:00:00.000Z"
                    }
                }
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* period: object period

* system(required): string Example: urn:oid:2.16.840.1.113883.4.814

* type: object type

* use: string Allowed Values: ["official"]

* value(required): string Example: CW1234563


```
{
    "type":"object",
    "required":[
        "system",
        "value"
    ],
    "properties":{
        "type":{
            "type":"object",
            "properties":{
                "coding":{
                    "type":"array",
                    "minItems":1,
                    "maxItems":1,
                    "items":{
                        "type":"object",
                        "properties":{
                            "code":{
                                "type":"string",
                                "example":"DEA"
                            },
                            "system":{
                                "type":"string",
                                "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
                            }
                        }
                    }
                }
            }
        },
        "system":{
            "type":"string",
            "example":"urn:oid:2.16.840.1.113883.4.814"
        },
        "value":{
            "type":"string",
            "example":"CW1234563"
        },
        "period":{
            "type":"object",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2019-12-01T00:00:00.000Z"
```

```
                },
                "end":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2029-12-01T23:59:59.000Z"
                }
            }
        },
        "use":{
            "type":"string",
            "enum":[
                "official"
            ]
        }
    }
}
```

Nested Schema : period

Type:

object

Show Source

end: string

Must include a time component with a timezone.

Example:

2029-12-01T23:59:59.000Z

start: string

Must include a time component with a timezone.

Example:

2019-12-01T00:00:00.000Z

```
{
    "type":"object",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2019-12-01T00:00:00.000Z"
        },
        "end":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2029-12-01T23:59:59.000Z"
        }
    }
}
```

Nested Schema : type

Type:

object

Show Source

* coding: array coding Minimum Number of Items: 1 Maximum Number of Items: 1

```
{
    "type":"object",
    "properties":{
        "coding":{
            "type":"array",
            "minItems":1,
            "maxItems":1,
            "items":{
                "type":"object",
                "properties":{
                    "code":{
                        "type":"string",
                        "example":"DEA"
                    },
                    "system":{
                        "type":"string",
                        "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
                    }
                }
            }
        }
    }
}
```

Nested Schema : coding

Type:

array

Minimum Number of Items:

Maximum Number of Items:

Show Source

* Array of: object items


```
{
    "type":"array",
    "minItems":1,
    "maxItems":1,
    "items":{
        "type":"object",
        "properties":{
            "code":{
                "type":"string",
                "example":"DEA"
            },
            "system":{
                "type":"string",
                "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string Example: DEA

* system: string Example: http://terminology.hl7.org/CodeSystem/v2-0203


```
{
    "type":"object",
    "properties":{
        "code":{
            "type":"string",
            "example":"DEA"
        },
        "system":{
            "type":"string",
            "example":"http://terminology.hl7.org/CodeSystem/v2-0203"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* family(required): string Example: Williams

* given(required): array given Minimum Number of Items: 1

period: object

period

Must not include an end date.

* prefix: array prefix Maximum Number of Items: 1

* suffix: array suffix Maximum Number of Items: 1


```
{
    "type":"object",
    "required":[
        "given",
        "family"
    ],
    "properties":{
        "family":{
            "type":"string",
            "example":"Williams"
        },
        "given":{
            "type":"array",
            "minItems":1,
            "items":{
                "type":"string",
                "example":"James McAfee"
            }
        },
        "prefix":{
            "type":"array",
```

```
            "maxItems":1,
            "items":{
                "type":"string",
                "example":"Dr."
            }
        },
        "suffix":{
            "maxItems":1,
            "type":"array",
            "items":{
                "type":"string",
                "example":"M.D."
            }
        },
        "period":{
            "type":"object",
            "description":"<p>Must not include an <code>end</code> date.</p>",
            "properties":{
                "start":{
                    "type":"string",
                    "description":"<p>Must include a time component with a timezone.</p>",
                    "example":"2020-12-01T00:00:00.000Z"
                }
            }
        }
    }
}
```


Nested Schema : given

Type:

array

Minimum Number of Items:

Show Source

* Array of: string Example: James McAfee


```
{
    "type":"array",
    "minItems":1,
    "items":{
        "type":"string",
        "example":"James McAfee"
    }
}
```


Nested Schema : period

Type:

object

Must not include an end date.

Show Source

start: string

Must include a time component with a timezone.

Example:

2020-12-01T00:00:00.000Z


```
{
    "type":"object",
    "description":"<p>Must not include an <code>end</code> date.</p>",
    "properties":{
        "start":{
            "type":"string",
            "description":"<p>Must include a time component with a timezone.</p>",
            "example":"2020-12-01T00:00:00.000Z"
        }
    }
}
```


Nested Schema : prefix

Type:

array

Maximum Number of Items:

Show Source

* Array of: string Example: Dr.


```
{
    "type":"array",
    "maxItems":1,
    "items":{
        "type":"string",
        "example":"Dr."
```

```
    }
}
```

Nested Schema : suffix

Type:

array

Maximum Number of Items:

Show Source

* Array of: string Example: M.D.

```
{
    "maxItems":1,
    "type":"array",
    "items":{
        "type":"string",
        "example":"M.D."
    }
}
```

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Response

Supported Media Types

* application/fhir+json

201 Response

Created

Headers

ETag: string

The identifier for a specific version of a resource. This identifier is formatted as W/"" and used to specify the If-Match header value on subsequent updates.

Location: string

The URL of the created entity.

Example: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Practitioner/7118008

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]

```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
```

```
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
```

```
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
```

```
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Get a list of practitioners

File: r4-rest-api-v32/op-practitioner-get.html

Retrieves a list of practitioners that meet supplied query parameters.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Query Parameters

_count: integer

Minimum Value:

The maximum number of results to return.

Example: _count=15

_id: array[string]

Minimum Number of Items:

The logical resource ID. This parameter is required if the identifier, family, name, or active parameter is not used.

Example: _id=12345

active: boolean

The indication of whether this practitioner's record is in active use. This parameter is required if _id, identifier, family, or name is not used.

family: string

The start of the family name of the practitioner. This parameter is required if _id, identifier, name, or active is not used.

Example: family=Smith

given: string

The start of the given name of the practitioner.

Searching by the given name requires the family parameter.

Example: given=Sally

identifier: string

The practitioner identifier. This parameter is required if the _id, family, name, or active parameter is not used.

The value of the identifier parameter must include both a system and a code.

Example: identifier=http://hl7.org/fhir/sid/us-npi|4326587548

name: string

Minimum Length:

The start of the given name or the family name of the practitioner. This parameter is required if _id, identifier, family,, or active is not used.

Example: name=Riley

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Bundle",
    "id":"7ec264d2-c7c7-49b2-8478-22351c82db73",
    "type":"searchset",
    "link":[
```

```
    {
        "relation":"self",
        "url":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Practitioner?_id=109413936"
    }
],
"entry":[
    {
        "fullUrl":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Practitioner/109413936",
        "resource":{
            "resourceType":"Practitioner",
            "id":"109413936",
            "meta":{
                "versionId":"0",
                "lastUpdated":"2019-03-07T20:40:34.000Z"
            },
            "text":{
                "status":"generated",
                "div":"<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Practitioner</b></p><p><b>Name</b>: Lombardi,
 Falco Shine</p><p><b>Identifiers</b>: DOCUPIN: F88788, DOCDEA: 887887887, SPI: 16611661, NPI: 6656656, PRSNLPRIMID:
 12332122</p><p><b>Gender</b>: Male</p><p><b>Status</b>: Active</p></div>"
            },
            "extension":[
                {
                    "valueBoolean":true,
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/is-physician"
                }
            ],
            "identifier":[
                {
                    "use":"usual",
                    "type":{
                        "coding":[
                            {
                                "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                                "code":"PRN",
                                "display":"Provider number"
                            }
                        ],
                        "text":"DOCUPIN"
                    },
                    "_system":{
                        "extension":[
                            {
                                "url":"http://hl7.org/fhir/StructureDefinition/data-absent-reason",
                                "valueCode":"unknown"
                            }
                        ]
                    },
                    "value":"F88788",
                    "period":{
                        "start":"2019-03-07T20:40:35.000Z"
                    }
                },
                {
                    "use":"usual",
                    "type":{
                        "coding":[
                            {
                                "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                                "code":"DEA",
                                "display":"Drug Enforcement Administration registration number"
                            }
                        ],
                        "text":"DOCDEA"
                    },
                    "system":"urn:oid:2.16.840.1.113883.4.814",
                    "value":"887887887",
                    "period":{
                        "start":"2019-03-07T20:40:35.000Z"
                    }
                },
                {
                    "use":"usual",
                    "type":{
                        "coding":[
                            {
                                "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                                "code":"PRN",
                                "display":"Provider number"
                            }
                        ],
                        "text":"SureScripts Prescriber Index"
                    },
                    "system":"urn:oid:2.16.840.1.113883.3.2054",
                    "value":"16611661",
                    "period":{
                        "start":"2019-04-12T18:24:23.000Z"
                    }
                },
```

```
                {
                    "use":"usual",
                    "type":{
                        "coding":[
                            {
                                "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                                "code":"NPI",
                                "display":"National provider identifier"
                            }
                        ],
                        "text":"National Provider Identifier"
                    },
                    "system":"http://hl7.org/fhir/sid/us-npi",
                    "value":"6656656",
                    "period":{
                        "start":"2019-03-07T20:40:35.000Z"
                    }
                },
                {
                    "use":"usual",
                    "type":{
                        "text":"Personnel Primary Identifier"
                    },
                    "system":"urn:oid:1.2.243.58",
                    "value":"12332122",
                    "period":{
                        "start":"2019-04-12T18:24:23.000Z"
                    }
                }
            ],
            "active":true,
            "name":[
                {
                    "use":"usual",
                    "text":"Lombardi, Falco Shine",
                    "family":"Lombardi",
                    "given":[
                        "Falco",
                        "Shine"
                    ],
                    "prefix":[
                        "Dr."
                    ],
                    "suffix":[
                        "M.D."
                    ],
                    "period":{
                        "start":"2019-03-07T20:40:35.000Z"
                    }
                }
            ],
            "telecom":[
                {
                    "system":"phone",
                    "value":"7861231234",
                    "use":"work"
                },
                {
                    "system":"email",
                    "value":"falco.lombardi@sfox.com",
                    "use":"work"
                }
            ],
            "address":[
                {
                    "use":"work",
                    "text":"111 Corneria Dr.\\nTallahassee, FL 32304\\nUSA",
                    "line":[
                        "111 Corneria Dr."
                    ],
                    "city":"Tallahassee",
                    "district":"Leon",
                    "state":"FL",
                    "postalCode":"32304",
                    "country":"USA"
                }
            ],
            "gender":"male",
            "qualification":[
                {
                    "code":{
                        "coding":[
                            {

"system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600",
                                "code":"677091",
                                "display":"MD",
                                "userSelected":true
                            }
```

```
                            ],
                            "text":"MD"
                        },
                        "period":{
                            "start":"2014-07-09T06:00:00.000Z"
                        }
                    },
                    {
                        "code":{
                            "coding":[
                                {
    "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600",
                                    "code":"677079",
                                    "display":"PhD",
                                    "userSelected":true
                                }
                            ],
                            "text":"PhD"
                        },
                        "period":{
                            "start":"2014-07-09T06:00:00.000Z"
                        }
                    }
                ]
            }
        }
    ]
}
```

Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]

```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
```

```
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Get a practitioner by ID

File: r4-rest-api-v32/op-practitioner-id-get.html

Retrieves a practitioner by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Practitioner",
    "id":"109413936",
    "meta":{
        "versionId":"0",
        "lastUpdated":"2019-03-07T20:40:34.000Z"
    },
    "text":{
        "status":"generated",
        "div":"<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Practitioner</b></p><p><b>Name</b>: Lombardi, Falco
    Shine</p><p><b>Identifiers</b>: DOCUPIN: F88788, DOCDEA: 887887887, SPI: 16611661, NPI: 6656656, PRSNLPRIMID:
    12332122</p><p><b>Gender</b>: Male</p><p><b>Status</b>: Active</p></div>"
    },
    "extension":[
        {
            "valueBoolean":true,
            "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/is-physician"
        }
    ],
    "identifier":[
        {
            "use":"usual",
            "type":{
                "coding":[
                    {
                        "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                        "code":"PRN",
                        "display":"Provider number"
                    }
                ],
                "text":"DOCUPIN"
            },
            "_system":{
                "extension":[
                    {
                        "url":"http://hl7.org/fhir/StructureDefinition/data-absent-reason",
                        "valueCode":"unknown"
                    }
                ]
            },
            "value":"F88788",
            "period":{
                "start":"2019-03-07T20:40:35.000Z"
            }
        },
        {
            "use":"usual",
            "type":{
                "coding":[
                    {
                        "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                        "code":"DEA",
                        "display":"Drug Enforcement Administration registration number"
                    }
                ],
                "text":"DOCDEA"
            },
            "system":"urn:oid:2.16.840.1.113883.4.814",
            "value":"887887887",
            "period":{
                "start":"2019-03-07T20:40:35.000Z"
            }
        },
        {
            "use":"usual",
            "type":{
                "coding":[
                    {
                        "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                        "code":"PRN",
```

```
                    "display":"Provider number"
                }
            ],
            "text":"SureScripts Prescriber Index"
        },
        "system":"urn:oid:2.16.840.1.113883.3.2054",
        "value":"16611661",
        "period":{
            "start":"2019-04-12T18:24:23.000Z"
        }
    },
    {
        "use":"usual",
        "type":{
            "coding":[
                {
                    "system":"http://terminology.hl7.org/CodeSystem/v2-0203",
                    "code":"NPI",
                    "display":"National provider identifier"
                }
            ],
            "text":"National Provider Identifier"
        },
        "system":"http://hl7.org/fhir/sid/us-npi",
        "value":"6656656",
        "period":{
            "start":"2019-03-07T20:40:35.000Z"
        }
    },
    {
        "use":"usual",
        "type":{
            "text":"Personnel Primary Identifier"
        },
        "system":"urn:oid:1.2.243.58",
        "value":"12332122",
        "period":{
            "start":"2019-04-12T18:24:23.000Z"
        }
    }
],
"active":true,
"name":[
    {
        "use":"usual",
        "text":"Lombardi, Falco Shine",
        "family":"Lombardi",
        "given":[
            "Falco",
            "Shine"
        ],
        "prefix":[
            "Dr."
        ],
        "suffix":[
            "M.D."
        ],
        "period":{
            "start":"2019-03-07T20:40:35.000Z"
        }
    }
],
"telecom":[
    {
        "system":"phone",
        "value":"7861231234",
        "use":"work"
    },
    {
        "system":"email",
        "value":"falco.lombardi@sfox.com",
        "use":"work"
    }
],
"address":[
    {
        "use":"work",
        "text":"111 Corneria Dr.\\nTallahassee, FL 32304\\nUSA",
        "line":[
            "111 Corneria Dr."
        ],
        "city":"Tallahassee",
        "district":"Leon",
        "state":"FL",
        "postalCode":"32304",
        "country":"USA"
    }
],
"gender":"male",
```

```
    "qualification":[
        {
            "code":{
                "coding":[
                    {
                        "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600",
                        "code":"677091",
                        "display":"MD",
                        "userSelected":true
                    }
                ],
                "text":"MD"
            },
            "period":{
                "start":"2014-07-09T06:00:00.000Z"
            }
        },
        {
            "code":{
                "coding":[
                    {
                        "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600",
                        "code":"677079",
                        "display":"PhD",
                        "userSelected":true
                    }
                ],
                "text":"PhD"
            },
            "period":{
                "start":"2014-07-09T06:00:00.000Z"
            }
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
```

```
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
```

```
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
```

```
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Procedure

File: r4-rest-api-v32/api-procedure.html

Description

Procedure

This topic describes information for the Procedure resource.

Overview

The Procedure resource returns medical and surgical procedures performed on or for a patient during their lifetime. Historical procedures and procedures recorded during a specific visit are returned. Surgical procedures from finalized surgical cases are returned as free-text procedures if the product is configured to write procedures from finalized cases to procedure history.

This resource supports the US Core Procedure Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The examples provided here are non-normative and replaying them in the public sandbox is not guaranteed to yield the results shown on the site. The field is present only for confidential resources.

Note:

meta.security

Fields

ID

Status

Category

Code

Subject

Reference (Patient)

Encounter

Performed Period

DateTime

Period

Who recorded

Reference (Practitioner)

Who performed

Reference (Practitioner)

Location

Reason procedure performed

Reference (Condition)

Comment/Note

Annotation Author

Reference (Practitioner)

Terminology Bindings

Procedure.status

Description: The codes identifying the lifecycle stage of an event.

Details: EventStatus

System: http://hl7.org/fhir/event-status

Procedure.category

Description: A code that classifies the procedure for searching, sorting, and display purposes.

Details: Procedure Category Codes (SNOMED CT)

System: https://snomed.info/sct

Details: US Core Category

System: http://hl7.org/fhir/us/core/CodeSystem/us-core-category

Procedure.code

Description: A code to identify a specific procedure.

Details: Procedure Codes (SNOMED CT)

System: https://snomed.info/sct

Details: CPT

System: https://www.ama-assn.org/go/cpt

Details: ICD-10-PCS

System: https://www.icd10data.com/icd10pcs

Procedure.performer.function

Description: A code that distinguishes the performer's type of involvement in the procedure.

Details: Procedure Performer Role Codes (SNOMED CT)

System: https://snomed.info/sct

Details: For more information, see Code Set 338 Procedure Personnel Relationship Type under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/388

Procedure.meta.security

Description: These tags connect specific resources to the overall security policy and infrastructure.

Details: ValueSet

System: http://terminology.hl7.org/CodeSystem/v3-ActCode

Procedure

The operations from the Procedure category.

Create a procedure

Method: post Path: /Procedure

Get a list of procedures

Method: get Path: /Procedure

Get a procedure by ID

Method: get Path: /Procedure/{ID}

Create a procedure

File: r4-rest-api-v32/op-procedure-post.html

Creates a new procedure.

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : ProcedureCreateBody

Type:

object

Show Source

basedOn: object

basedOn

A reference to a resource that contains details of the request for this procedure. Must only include references to ServiceRequest.

Note:

Example:

```
{
  "basedOn": {
    "reference":"ServiceRequest/361330415"
  }
}
```

category: object

category

A code that classifies the procedure for searching, sorting, and display purposes.

Example:

```
{
  "category": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
        "code": "sdoh"
      }
    ]
  }
}
```

code(required): object

code

A code to identify a specific procedure. Must be provided.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "119954001",
```

```
        "display": "Adenoid excision (procedure)",
        "userSelected": true
      }
    ],
    "text": "Adenoid excision (text)"
  }
}
```

encounter: object

encounter

Indicates the encounter when the procedure was performed. Must only include references to Encounter.

Note:

Example:

```
{
  "encounter": {
    "reference":"Encounter/26070829"
  }
}
```

location: object

location

The location where the procedure actually happened. Must only include references to Location.

Note:

Example:

```
{
  "location": {
    "reference":"Location/1693915",
    "display": "Baseline East"
  }
}
```

* meta: object meta The Procedure metadata.

note: array

note

Any other notes and comments about the procedure.

Example:

```
{
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/4594011",
        "display": "Leroux, Erik"
      },
      "text": "Test Comment"
    }
  ]
}
```

performed[x]:

Estimated or actual date or date and time when the procedure was performed.

Note:

* Only a type of dateTime is supported.

* The dateTime is converted to a GMT date.

Example:

```
{
  "performedDateTime": "2015-10-14T13:13:20-06:00"
}
```

performedPeriod: object

performedPeriod

Estimated or actual period when the procedure was performed.

Note:

* Only a period with valid start and end dateTimes is supported.

* Either performedPeriod or performedDateTime can be supplied. If both are present, performedPeriod will take precedence.

* The dateTimes are converted to a GMT date.

Example:

```
{
  "performedPeriod": {
    "start": "2015-10-14T13:13:20-06:00",
    "end": "2015-10-14T14:12:20-06:00"
  }
```

```
}
```

performer: array

performer

Limited to "real" people rather than equipment.

Example:

```
{
  "performer": [
    {
      "function": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "223366009",
            "display": "Healthcare professional (occupation)",
            "userSelected": true
          }
        ],
        "text": "Healthcare professional (occupation)"
      },
      "actor": {
        "reference": "Practitioner/2150097",
        "display": "Strecker, Bradley"
      }
    }
  ]
}
```

performer.actor(required): object

performer.actor

The practitioner who was involved in the procedure. Must only include references to Practitioner.

Note:

Example:

```
{
  "actor": {
    "reference": "Practitioner/2150097",
    "display": "Strecker, Bradley"
  }
}
```

performer.function: object

performer.function

Distinguishes the type of involvement of the performer in the procedure.

Example:

```
{
  "function": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "223366009",
        "display": "Healthcare professional (occupation)",
        "userSelected": true
      }
    ],
    "text": "Healthcare professional (occupation)"
  }
}
```

```
{
  "severity": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/388",
        "code": "1207",
        "display": "Anesthesiologist",
        "userSelected": true
      }
    ],
    "text": "Anesthesiologist"
  }
}
```

* resourceType(required): string The type of the FHIR resource. Must be Procedure.

status(required): string

Codes that identify the lifecycle stage of an event. Only a status code is supported.

Note:

completed

Example:

```
{
  "status": "completed"
}
```

subject(required): object

subject

Indicates the patient who the procedure record is associated with. Must only include references to Patient.

Note:

Example:

```
{
  "subject": {
    "reference": "Patient/631923",
    "display": "Nilsson, Christine"
  }
}
```

```
{
    "type":"object",
    "required":[
        "resourceType",
        "status",
        "code",
        "subject",
        "performer.actor"
    ],
    "properties":{
        "resourceType":{
            "description":"The type of the FHIR resource. Must be Procedure.\n",
            "type":"string"
        },
        "meta":{
            "description":"The Procedure metadata.",
            "type":"object",
            "properties":{
                "security":{
                    "description":"The Procedure confidentiality.",
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "description":"The confidentiality system.",
                                "type":"string",
                                "example":"http://terminology.hl7.org/CodeSystem/v3-ActCode"
                            },
                            "code":{
                                "description":"The confidentiality label.",
                                "type":"string",
                                "example":"PHY"
                            }
                        }
                    }
                }
            },
            "example":"{\n  \"security\": [\n    {\n      \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n      \"code\": \"PHY\"\n    },\n    {\n      \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n      \"code\": \"PRS\"\n    }\n  ]\n}\n"
        },
        "status":{
            "description":"Codes that identify the lifecycle stage of an event.<br>\n\n<b>Note:</b> Only a status code
    <code>completed</code> is supported.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"status\":
    \"completed\"\n}\n</code></pre>\n",
            "type":"string"
        },
        "basedOn":{
            "description":"A reference to a resource that \ncontains details of the request for this procedure.<br>\n\n
    <b>Note:</b> Must only include references to ServiceRequest.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"basedOn\":
    {\n    \"reference\":\"ServiceRequest/361330415\"\n  }\n}\n</code></pre>\n",
            "type":"object"
        },
        "category":{
            "description":"A code that classifies the procedure for searching, \nsorting, and display
    purposes.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"category\": {\n    \"coding\": [\n      {\n        \"system\":
     \"http://hl7.org/fhir/us/core/CodeSystem/us-core-category\",\n        \"code\": \"sdoh\"\n      }\n    ]\n
    }\n}\n</code></pre>\n",
            "type":"object"
        },
        "code":{
            "description":"A code to identify a specific procedure.<br>\n\n<b>Note:</b> Must be
    provided.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://snomed.info/sct\",\n        \"code\": \"119954001\",\n        \"display\": \"Adenoid excision
    (procedure)\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\": \"Adenoid excision (text)\"\n
    }\n}\n</code></pre>\n",
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
```

```
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "subject":{
        "description":"Indicates the patient who the procedure record is associated with.<br>\n\n<b>Note:</b> Must
only include references to Patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
\"Patient/631923\",\n    \"display\": \"Nilsson, Christine\"\n  }\n}\n</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "encounter":{
        "description":"Indicates the encounter when the procedure was performed.<br>\n\n<b>Note:</b> Must only
include references to Encounter.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"encounter\": {\n
\"reference\":\"Encounter/26070829\"\n  }\n}\n</code></pre>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "performed[x]":{
        "description":"Estimated or actual date or date and time when the procedure was
performed.<br>\n\n<b>Note:</b>\n<ul>\n  <li> Only a type of dateTime is supported. </li>\n  <li> The dateTime is
converted to a GMT date. </li>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"performedDateTime\":
\"2015-10-14T13:13:20-06:00\"\n}\n</code></pre>\n"
    },
    "performedPeriod":{
        "description":"Estimated or actual period when the procedure was performed.<br>\n\n<b>Note:</b>\n<ul>\n
<li> Only a period with valid start and end dateTimes is supported. </li>\n  <li> Either performedPeriod or
performedDateTime can be supplied. If both are present, performedPeriod will take precedence. </li>\n  <li> The
dateTimes are converted to a GMT date. </li>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"performedPeriod\": {\n
    \"start\": \"2015-10-14T13:13:20-06:00\",\n    \"end\": \"2015-10-14T14:12:20-06:00\"\n  }\n}\n</code></pre>\n",
        "type":"object",
        "properties":{
            "start":{
                "type":"string"
            },
            "end":{
                "type":"string"
            }
        }
    },
    "performer":{
        "description":"Limited to \"real\" people rather than equipment.<br>\n<b>Example:</b>\n\n<pre><code>{\n
\"performer\": [\n    {\n      \"function\": {\n        \"coding\": [\n          {\n            \"system\":
\"http://snomed.info/sct\",\n            \"code\": \"223366009\",\n            \"display\": \"Healthcare
professional (occupation)\",\n            \"userSelected\": true\n          }\n        ],\n        \"text\":
\"Healthcare professional (occupation)\"\n      },\n      \"actor\": {\n        \"reference\":
\"Practitioner/2150097\",\n        \"display\": \"Strecker, Bradley\"\n      }\n    }\n  ]\n}\n</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object"
        }
    },
    "performer.function":{
        "description":"Distinguishes the type of involvement of the performer in the
procedure.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"function\": {\n    \"coding\": [\n      {\n        \"system\":
\"http://snomed.info/sct\",\n        \"code\": \"223366009\",\n        \"display\": \"Healthcare professional
(occupation)\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\": \"Healthcare professional
(occupation)\"\n  }\n}\n</code></pre>\n\n<pre><code>{\n  \"severity\": {\n    \"coding\": [\n      {\n
\"system\": \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/388\",\n        \"code\":
\"1207\",\n        \"display\": \"Anesthesiologist\",\n        \"userSelected\": true\n      }\n    ],\n
\"text\": \"Anesthesiologist\"\n  }\n}\n</code></pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
```

```
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "performer.actor":{
            "description":"The practitioner who was involved in the procedure.<br>\n\n<b>Note:</b> Must only include
    references to Practitioner.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"actor\": {\n    \"reference\":
    \"Practitioner/2150097\",\n    \"display\": \"Strecker, Bradley\"\n  }\n}\n</code></pre>\n",
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        },
        "location":{
            "description":"The location where the procedure actually happened.<br>\n\n<b>Note:</b> Must only include
    references to Location.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"location\": {\n
    \"reference\":\"Location/1693915\",\n    \"display\": \"Baseline East\"\n  }\n}\n</code></pre>\n",
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        },
        "note":{
            "description":"Any other notes and comments about the procedure.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"note\": [\n    {\n      \"authorReference\": {\n        \"reference\": \"Practitioner/4594011\",\n
    \"display\": \"Leroux, Erik\"\n      },\n      \"text\": \"Test Comment\"\n    }\n  ]\n}\n</code></pre>\n",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "authorReference":{
                        "type":"object",
                        "properties":{
                            "reference":{
                                "type":"string"
                            }
                        }
                    },
                    "time":{
                        "type":"string"
                    },
                    "text":{
                        "type":"string"
                    }
                }
            }
        }
    },
    "example":"{\n  \"resourceType\": \"Procedure\",\n  \"status\": \"completed\",\n  \"meta\": {\n    \"security\": [\n
          {\n        \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n        \"code\": \"PHY\"\n
          },\n      {\n        \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n        \"code\":
    \"PRS\"\n      }\n    ]\n  },\n  \"subject\": {\n    \"reference\": \"Patient/12724066\"\n  },\n  \"code\": {\n
    \"coding\": [\n      {\n        \"system\": \"http://snomed.info/sct\",\n        \"code\": \"119954001\",\n
    \"display\": \"Adenoid excision (procedure)\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\":
    \"Adenoid excision (text)\"\n  },\n  \"encounter\": {\n    \"reference\": \"Encounter/97953477\"\n  },\n
    \"location\": {\n    \"display\": \"Baseline East\"\n  },\n  \"performedDateTime\": \"2019-04-05T20:21:46Z\",\n
    \"performedPeriod\": {\n    \"start\": \"2019-04-05T20:21:46Z\",\n    \"end\": \"2019-04-05T21:21:46Z\"\n  },\n
    \"performer\": [\n    {\n      \"function\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://snomed.info/sct\",\n            \"code\": \"223366009\",\n            \"display\": \"Healthcare
    professional (occupation)\",\n            \"userSelected\": true\n          }\n        ],\n        \"text\":
    \"Healthcare professional (occupation)\"\n      },\n      \"actor\": {\n        \"reference\":
    \"Practitioner/4122630\"\n      }\n    }\n  ],\n  \"note\": [\n    {\n      \"text\": \"Procedure comment\",\n
    \"time\": \"2019-04-05T20:21:46Z\",\n      \"authorReference\": {\n        \"reference\": \"Practitioner/4122630\"\n
          }\n    }\n  ]\n}\n"
}
```


Example:


```
{
  "resourceType": "Procedure",
  "status": "completed",
  "meta": {
    "security": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PHY"
```

```
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PRS"
      }
    ]
  },
  "subject": {
    "reference": "Patient/12724066"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "119954001",
        "display": "Adenoid excision (procedure)",
        "userSelected": true
      }
    ],
    "text": "Adenoid excision (text)"
  },
  "encounter": {
    "reference": "Encounter/97953477"
  },
  "location": {
    "display": "Baseline East"
  },
  "performedDateTime": "2019-04-05T20:21:46Z",
  "performedPeriod": {
    "start": "2019-04-05T20:21:46Z",
    "end": "2019-04-05T21:21:46Z"
  },
  "performer": [
    {
      "function": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "223366009",
            "display": "Healthcare professional (occupation)",
            "userSelected": true
          }
        ],
        "text": "Healthcare professional (occupation)"
      },
      "actor": {
        "reference": "Practitioner/4122630"
      }
    }
  ],
  "note": [
    {
      "text": "Procedure comment",
      "time": "2019-04-05T20:21:46Z",
      "authorReference": {
        "reference": "Practitioner/4122630"
      }
    }
  ]
}
```

Nested Schema : basedOn

Type:

object

A reference to a resource that contains details of the request for this procedure. Must only include references to ServiceRequest.

Note:

Example:

```
{
  "basedOn": {
    "reference":"ServiceRequest/361330415"
  }
}
```

Nested Schema : category

Type:

object

A code that classifies the procedure for searching, sorting, and display purposes.

Example:

```
{
  "category": {
    "coding": [
```

```
      {
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
        "code": "sdoh"
      }
    ]
  }
}
```

Nested Schema : code

Type:

object

A code to identify a specific procedure. Must be provided.

Note:

Example:

```
{
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "119954001",
        "display": "Adenoid excision (procedure)",
        "userSelected": true
      }
    ],
    "text": "Adenoid excision (text)"
  }
}
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"A code to identify a specific procedure.<br>\n\n<b>Note:</b> Must be
    provided.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"code\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://snomed.info/sct\",\n        \"code\": \"119954001\",\n        \"display\": \"Adenoid excision
    (procedure)\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\": \"Adenoid excision (text)\"\n
    }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : encounter

Type:

object

Indicates the encounter when the procedure was performed. Must only include references to Encounter.

Note:

Example:

```
{
  "encounter": {
    "reference":"Encounter/26070829"
  }
}
```

Show Source

* reference: string

```
{
    "description":"Indicates the encounter when the procedure was performed.<br>\n\n<b>Note:</b> Must only include
    references to Encounter.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"encounter\": {\n
    \"reference\":\"Encounter/26070829\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
```

```
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : location

Type:

object

The location where the procedure actually happened. Must only include references to Location.

Note:

Example:

```
{
  "location": {
    "reference":"Location/1693915",
    "display": "Baseline East"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The location where the procedure actually happened.<br>\n\n<b>Note:</b> Must only include references
    to Location.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"location\": {\n    \"reference\":\"Location/1693915\",\n
        \"display\": \"Baseline East\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : meta

Type:

object

The Procedure metadata.

Show Source

* security: array security The Procedure confidentiality.

```
{
    "description":"The Procedure metadata.",
    "type":"object",
    "properties":{
        "security":{
            "description":"The Procedure confidentiality.",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "description":"The confidentiality system.",
                        "type":"string",
                        "example":"http://terminology.hl7.org/CodeSystem/v3-ActCode"
                    },
                    "code":{
                        "description":"The confidentiality label.",
                        "type":"string",
                        "example":"PHY"
                    }
                }
            }
        }
    },
    "example":"{\n  \"security\": [\n    {\n      \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n
          \"code\": \"PHY\"\n    },\n    {\n      \"system\": \"http://terminology.hl7.org/CodeSystem/v3-ActCode\",\n
          \"code\": \"PRS\"\n    }\n  ]\n}\n"
}
```

Example:

```
{
  "security": [
    {
      "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
      "code": "PHY"
    },
    {
      "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
      "code": "PRS"
    }
  ]
}
```

Nested Schema : note

Type:

array

Any other notes and comments about the procedure.

Example:

```
{
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/4594011",
        "display": "Leroux, Erik"
      },
      "text": "Test Comment"
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Any other notes and comments about the procedure.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"note\":
    [\n    {\n      \"authorReference\": {\n        \"reference\": \"Practitioner/4594011\",\n        \"display\":
    \"Leroux, Erik\"\n      },\n      \"text\": \"Test Comment\"\n    }\n  ]\n}\n</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "authorReference":{
                "type":"object",
                "properties":{
                    "reference":{
                        "type":"string"
                    }
                }
            },
            "time":{
                "type":"string"
            },
            "text":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : performedPeriod

Type:

object

Estimated or actual period when the procedure was performed.

Note:

* Only a period with valid start and end dateTimes is supported.

* Either performedPeriod or performedDateTime can be supplied. If both are present, performedPeriod will take precedence.

* The dateTimes are converted to a GMT date.

Example:

```
{
  "performedPeriod": {
    "start": "2015-10-14T13:13:20-06:00",
    "end": "2015-10-14T14:12:20-06:00"
  }
}
```

Show Source

* end: string

* start: string

```
{
    "description":"Estimated or actual period when the procedure was performed.<br>\n\n<b>Note:</b>\n<ul>\n  <li> Only a
     period with valid start and end dateTimes is supported. </li>\n  <li> Either performedPeriod or performedDateTime
    can be supplied. If both are present, performedPeriod will take precedence. </li>\n  <li> The dateTimes are
    converted to a GMT date. </li>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"performedPeriod\": {\n    \"start\":
     \"2015-10-14T13:13:20-06:00\",\n    \"end\": \"2015-10-14T14:12:20-06:00\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
        "start":{
            "type":"string"
        },
        "end":{
            "type":"string"
```

```
        }
    }
}
```

Nested Schema : performer

Type:

array

Limited to "real" people rather than equipment.

Example:

```
{
  "performer": [
    {
      "function": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "223366009",
            "display": "Healthcare professional (occupation)",
            "userSelected": true
          }
        ],
        "text": "Healthcare professional (occupation)"
      },
      "actor": {
        "reference": "Practitioner/2150097",
        "display": "Strecker, Bradley"
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"Limited to \"real\" people rather than equipment.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"performer\": [\n    {\n      \"function\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://snomed.info/sct\",\n            \"code\": \"223366009\",\n            \"display\": \"Healthcare
    professional (occupation)\",\n            \"userSelected\": true\n          }\n        ],\n        \"text\":
    \"Healthcare professional (occupation)\"\n      },\n      \"actor\": {\n        \"reference\":
    \"Practitioner/2150097\",\n        \"display\": \"Strecker, Bradley\"\n      }\n    }\n  ]\n}\n</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object"
    }
}
```

Nested Schema : performer.actor

Type:

object

The practitioner who was involved in the procedure. Must only include references to Practitioner.

Note:

Example:

```
{
  "actor": {
    "reference": "Practitioner/2150097",
    "display": "Strecker, Bradley"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The practitioner who was involved in the procedure.<br>\n\n<b>Note:</b> Must only include references
    to Practitioner.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"actor\": {\n    \"reference\":
    \"Practitioner/2150097\",\n    \"display\": \"Strecker, Bradley\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : performer.function

Type:

object

Distinguishes the type of involvement of the performer in the procedure.

Example:

```
{
  "function": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "223366009",
        "display": "Healthcare professional (occupation)",
        "userSelected": true
      }
    ],
    "text": "Healthcare professional (occupation)"
  }
}
```

```
{
  "severity": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/388",
        "code": "1207",
        "display": "Anesthesiologist",
        "userSelected": true
      }
    ],
    "text": "Anesthesiologist"
  }
}
```


Show Source

* coding: array coding

* text: string


```
{
    "description":"Distinguishes the type of involvement of the performer in the
    procedure.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"function\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://snomed.info/sct\",\n        \"code\": \"223366009\",\n        \"display\": \"Healthcare professional
    (occupation)\",\n        \"userSelected\": true\n      }\n    ],\n    \"text\": \"Healthcare professional
    (occupation)\"\n  }\n}\n</code></pre>\n\n<pre><code>{\n  \"severity\": {\n    \"coding\": [\n      {\n
    \"system\": \"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/388\",\n        \"code\":
    \"1207\",\n        \"display\": \"Anesthesiologist\",\n        \"userSelected\": true\n      }\n    ],\n
    \"text\": \"Anesthesiologist\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : subject

Type:

object

Indicates the patient who the procedure record is associated with. Must only include references to Patient.

Note:

Example:


```
{
  "subject": {
    "reference": "Patient/631923",
    "display": "Nilsson, Christine"
  }
}
```


Show Source

* reference: string


```
{
    "description":"Indicates the patient who the procedure record is associated with.<br>\n\n<b>Note:</b> Must only
    include references to Patient.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"subject\": {\n    \"reference\":
    \"Patient/631923\",\n    \"display\": \"Nilsson, Christine\"\n  }\n}\n</code></pre>\n",
    "type":"object",
    "properties":{
```

```
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : security

Type:

array

The Procedure confidentiality.

Show Source

* Array of: object items

```
{
    "description":"The Procedure confidentiality.",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "description":"The confidentiality system.",
                "type":"string",
                "example":"http://terminology.hl7.org/CodeSystem/v3-ActCode"
            },
            "code":{
                "description":"The confidentiality label.",
                "type":"string",
                "example":"PHY"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string The confidentiality label. Example: PHY

* system: string The confidentiality system. Example: http://terminology.hl7.org/CodeSystem/v3-ActCode

```
{
    "type":"object",
    "properties":{
        "system":{
            "description":"The confidentiality system.",
            "type":"string",
            "example":"http://terminology.hl7.org/CodeSystem/v3-ActCode"
        },
        "code":{
            "description":"The confidentiality label.",
            "type":"string",
            "example":"PHY"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* authorReference: object authorReference

* text: string

* time: string

```
{
    "type":"object",
    "properties":{
        "authorReference":{
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        },
        "time":{
            "type":"string"
        },
        "text":{
            "type":"string"
        }
    }
}
```

Nested Schema : authorReference

Type:

object

Show Source

* reference: string

```
{
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
```

```
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

201 Response

OK

Headers

* ETag: string The identifier for a specific version of a resource. This identifier is formatted as W/"" and is used to specify the If-Match header value on subsequent updates.

* Location: string The URL location of the created entity. Example: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/642774

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

Default Response

The ETag response header indicates the current If-Match version to use on a subsequent update.

Example Request:

```
POST https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Mon, 06 Apr 2020 19:00:43 GMT
  Etag: W/"1"
  Location: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/17228728
  Last-Modified: Mon, 06 Apr 2020 19:00:43 GMT
  Vary: Origin
  X-Request-Id: 11111111111111111111111111111111
```

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Get a list of procedures

File: r4-rest-api-v32/op-procedure-get.html

Retrieves a list of procedures that meet supplied query parameters.

Note: Currently, the patient/Provenance.read scope is not supported and _revinclude cannot be used for patient persona.

Request

Query Parameters

* _id: string The logical resource ID. This parameter is required if patient or subject is not used. Example: _id=7891

_lastUpdated: string

The date range in which the procedures was last updated. Example: or

* This parameter may be provided up to two times, and must use the eq, ge, gt, le, or lt prefixes.

* When a value is provided without a prefix, an implied eq prefix is used.

* When provided twice, the lower value must have a ge or gt prefix and the higher value must have an le or lt prefix.

* The _lastUpdated parameter may not be provided at the same time as the date parameter.

_lastUpdated=gt2014-09-24

_lastUpdated=gt2014-09-24T12:00:00.000Z

_revinclude: string

The Provenance resource entries to be returned as part of the bundle. Example:

* May be provided once with the Provenance:target value. For example, _revinclude=Provenance:target.

* May be provided in combination with the _id or patient parameter. For example, _id=570007845&_revinclude=Provenance:target or patient=12345&_revinclude=Provenance:target.

* When provided in a request to a closed endpoint, the OAuth2 token must include the user/Provenance.read scope.

_revinclude=Provenance:target

date: string

The date range that the procedure's or falls within. Example:

performedPeriod

performedDateTime

* May be provided once with a prefix and time component to imply a date range. Example: date=gt2015-09-24T00:00:00.000Z

* May otherwise be provided twice with le, lt, ge, or gt prefixes and a time component to search for procedures within a specific range. The date and prefix pairs must create a closed range.

* The date parameter may not be provided at the same time as the _lastUpdated parameter.

date=gt2015-09-24T12:00:00.000Z&date=le2020-07-15T16:00:00.000Z

* patient: string Who the procedure is for. This parameter is required if _id or subject is not used. Example: 12345

* subject: string Who the procedure is for. This parameter is required if _id or patient is not used. Example: Patient/12345

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

Back to Top

Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?patient=12724066
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "53ac99d1-c76b-4a07-90f8-73d13a01459b",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?patient=12724066"
    }
  ],
  "entry": [
    {
      "fullUrl": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572581295",
      "resource": {
        "resourceType": "Procedure",
        "id": "2572581295",
        "meta": {
          "versionId": "0",
          "lastUpdated": "2020-09-25T20:13:56Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Patient</b>: SMART,
    NANCY</p><p><b>Procedure</b>: Adenoid excision (text)</p><p><b>Category</b>: SDOH</p><p><b>Date</b>: Apr  5,
    2019</p><p><b>Status</b>: Completed</p><p><b>Location</b>: Baseline East</p><p><b>Recorder</b>: SYSTEM, SYSTEM
    Cerner</p><p><b>Performer</b>:</p><ul><li>Cerner Test, Physician - Women's Health Cerner</li></ul></div>"
```

```
    },
    "status": "completed",
    "category": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "sdoh",
          "display": "SDOH",
          "userSelected": false
        }
      ]
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "119954001",
          "display": "Adenoid excision (procedure)",
          "userSelected": true
        }
      ],
      "text": "Adenoid excision (text)"
    },
    "subject": {
      "reference": "Patient/12724066",
      "display": "SMART, NANCY"
    },
    "encounter": {
      "reference": "Encounter/97953477"
    },
    "performedDateTime": "2019-04-05",
    "recorder": {
      "reference": "Practitioner/1",
      "display": "SYSTEM, SYSTEM Cerner"
    },
    "performer": [
      {
        "actor": {
          "reference": "Practitioner/4122630",
          "display": "Cerner Test, Physician - Women's Health Cerner"
        }
      }
    ],
    "location": {
      "display": "Baseline East"
    },
    "note": [
      {
        "authorReference": {
          "reference": "Practitioner/4122630",
          "display": "Cerner Test, Physician - Women's Health Cerner"
        },
        "time": "2019-04-05T20:21:46Z",
        "text": "Procedure comment"
      }
    ]
  }
},
{
  "fullUrl": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572476121",
  "resource": {
    "resourceType": "Procedure",
    "id": "2572476121",
    "meta": {
      "versionId": "0",
      "lastUpdated": "2020-08-24T08:42:34Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Patient</b>: SMART,
NANCY</p><p><b>Procedure</b>: Adenoid excision (text)</p><p><b>Category</b>: Education</p><p><b>Status</b>:
Completed</p><p><b>Recorder</b>: SYSTEM, SYSTEM Cerner</p></div>"
    },
    "status": "completed",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "409073007",
          "display": "Education (procedure)",
          "userSelected": false
        }
      ]
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "119954001",
```

```
              "display": "Adenoid excision (procedure)",
              "userSelected": true
            }
          ],
          "text": "Adenoid excision (text)"
        },
        "subject": {
          "reference": "Patient/12724066",
          "display": "SMART, NANCY"
        },
        "encounter": {
          "reference": "Encounter/97953477"
        },
        "_performedDateTime": {
          "extension": [
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "recorder": {
          "reference": "Practitioner/1",
          "display": "SYSTEM, SYSTEM Cerner"
        }
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-response Response

This operation supports the following authorization types:

Searching confidential procedures.

* Provider

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?patient=12724066
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "53ac99d1-c76b-4a07-90f8-73d13a01459b",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?patient=12724066"
    }
  ],
  "entry": [
    {
      "fullUrl": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572581295",
      "resource": {
        "resourceType": "Procedure",
        "id": "2572581295",
        "meta": {
          "versionId": "0",
          "lastUpdated": "2020-09-25T20:13:56Z",
          "security": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
              "code": "PHY",
              "display": "physician requested information sensitivity",
              "userSelected": false
            },
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
              "code": "PRS",
              "display": "patient requested information sensitivity",
              "userSelected": false
            }
          ]
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Meta Security</b>: physician
```

```
requested information sensitivity, patient requested information sensitivity</p><p><b>Patient</b>: SMART,
NANCY</p><p><b>Procedure</b>: Adenoid excision (text)</p><p><b>Category</b>: SDOH</p><p><b>Date</b>: Apr  5,
2019</p><p><b>Status</b>: Completed</p><p><b>Location</b>: Baseline East</p><p><b>Recorder</b>: SYSTEM, SYSTEM
Cerner</p><p><b>Performer</b>:</p><ul><li>Cerner Test, Physician - Women's Health Cerner</li></ul></div>"
    },
    "status": "completed",
    "category": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "sdoh",
          "display": "SDOH",
          "userSelected": false
        }
      ]
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "119954001",
          "display": "Adenoid excision (procedure)",
          "userSelected": true
        }
      ],
      "text": "Adenoid excision (text)"
    },
    "subject": {
      "reference": "Patient/12724066",
      "display": "SMART, NANCY"
    },
    "encounter": {
      "reference": "Encounter/97953477"
    },
    "performedDateTime": "2019-04-05",
    "recorder": {
      "reference": "Practitioner/1",
      "display": "SYSTEM, SYSTEM Cerner"
    },
    "performer": [
      {
        "actor": {
          "reference": "Practitioner/4122630",
          "display": "Cerner Test, Physician - Women's Health Cerner"
        }
      }
    ],
    "location": {
      "display": "Baseline East"
    },
    "note": [
      {
        "authorReference": {
          "reference": "Practitioner/4122630",
          "display": "Cerner Test, Physician - Women's Health Cerner"
        },
        "time": "2019-04-05T20:21:46Z",
        "text": "Procedure comment"
      }
    ]
  }
},
{
  "fullUrl": "http://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572476121",
  "resource": {
    "resourceType": "Procedure",
    "id": "2572476121",
    "meta": {
      "versionId": "0",
      "lastUpdated": "2020-08-24T08:42:34Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Patient</b>: SMART,
NANCY</p><p><b>Procedure</b>: Adenoid excision (text)</p><p><b>Category</b>: Education</p><p><b>Status</b>:
Completed</p><p><b>Recorder</b>: SYSTEM, SYSTEM Cerner</p></div>"
    },
    "status": "completed",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "409073007",
          "display": "Education (procedure)",
          "userSelected": false
        }
      ]
    },
    "code": {
```

```
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "119954001",
              "display": "Adenoid excision (procedure)",
              "userSelected": true
            }
          ],
          "text": "Adenoid excision (text)"
        },
        "subject": {
          "reference": "Patient/12724066",
          "display": "SMART, NANCY"
        },
        "encounter": {
          "reference": "Encounter/97953477"
        },
        "_performedDateTime": {
          "extension": [
            {
              "valueCode": "unknown",
              "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
            }
          ]
        },
        "recorder": {
          "reference": "Practitioner/1",
          "display": "SYSTEM, SYSTEM Cerner"
        }
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-revinclude Response

This operation supports the following authorization types:

* Provider

* System

Example Request:

```
GET https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?_id=570007845&_revinclude=Provenance:target
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "e448772d-7265-48b2-a942-99002b44337d",
  "type": "searchset",
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure?patient=744104&_revinclude=Provenance%3Atarget"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/570007845",
      "resource": {
        "resourceType": "Procedure",
        "id": "570007845",
        "meta": {
          "versionId": "0",
          "lastUpdated": "2021-05-28T15:02:50Z"
        },
        "text": {
          "status": "generated",
          "div": "ProcedurePatient: HEATHERJSMITHProcedure: Adenoid excision (text)<p><b>Category</b>: SDOH</p>Date: Apr
      5, 2019Status: CompletedLocation: Baseline EastRecorder: Powerchart, UserPerformer:Smith MD, John"
        },
        "status": "completed",
        "category": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
              "code": "sdoh",
              "display": "SDOH",
              "userSelected": false
            }
```

```
      ]
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "119954001",
          "display": "Adenoid excision (procedure)",
          "userSelected": true
        }
      ],
      "text": "Adenoid excision (text)"
    },
    "subject": {
      "reference": "Patient/744104",
      "display": "HEATHERJSMITH"
    },
    "encounter": {
      "reference": "Encounter/18245662"
    },
    "performedDateTime": "2019-04-05",
    "recorder": {
      "reference": "Practitioner/1255969",
      "display": "Powerchart, User"
    },
    "performer": [
      {
        "actor": {
          "reference": "Practitioner/2523932",
          "display": "Smith MD, John"
        }
      }
    ],
    "location": {
      "display": "Baseline East"
    },
    "note": [
      {
        "authorReference": {
          "reference": "Practitioner/2523932",
          "display": "Smith MD, John"
        },
        "time": "2019-04-05T20:21:46Z",
        "text": "Procedure comment"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/proc-13511261",
  "resource": {
    "resourceType": "Provenance",
    "id": "proc-13511261",
    "meta": {
      "versionId": "13511261",
      "lastUpdated": "2021-05-28T10:09:59-05:00"
    },
    "text": {
      "status": "generated",
      "div": "ProvenanceTarget:Procedure/570007845Recorded: Oct 25, 2019  9:57 A.M. CDTAgents:UnknownAgent Type:
TransmitterAgent Role: SourceEntity Source: DocumentReference/S-201"
    },
    "target": [
      {
        "reference": "Procedure/570007845"
      }
    ],
    "recorded": "2019-10-25T09:57:43-05:00",
    "agent": [
      {
        "type": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
              "code": "transmitter"
            }
          ],
          "text": "Transmitter"
        },
        "role": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                "code": "SOURCE"
              }
```

```
                ],
                "text": "Source"
              }
            ],
            "who": {
              "reference": "Organization/619848",
              "display": "Unknown"
            }
          }
        ],
        "entity": [
          {
            "role": "source",
            "what": {
              "reference": "DocumentReference/S-201"
            }
          }
        ]
      },
      "search": {
        "mode": "include"
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a procedure by ID

File: r4-rest-api-v32/op-procedure-id-get.html

Retrieves an individual procedure by the associated procedure ID.

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572382193
```

Example Response:


```
{
  "resourceType": "Procedure",
  "id": "2572382193",
  "meta": {
    "versionId": "0",
    "lastUpdated": "2020-06-11T04:02:54Z"
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Patient</b>: SMART,
    NANCY</p><p><b>Procedure</b>: Appendectomy;</p><p><b>Category</b>: SDOH</p><p><b>Date</b>: 2002</p><p><b>Status</b>:
     Completed</p><p><b>Recorder</b>: Cerner Test, Physician - Women's Health Cerner</p></div>"
  },
  "status": "completed",
  "category": {
    "coding": [
      {
```

```
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
        "code": "sdoh",
        "display": "SDOH",
        "userSelected": false
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://www.ama-assn.org/go/cpt",
        "code": "44950",
        "display": "Appendectomy;",
        "userSelected": true
      }
    ],
    "text": "Appendectomy;"
  },
  "subject": {
    "reference": "Patient/12724066",
    "display": "SMART, NANCY"
  },
  "encounter": {
    "reference": "Encounter/97953477"
  },
  "performedDateTime": "2002",
  "recorder": {
    "reference": "Practitioner/4122630",
    "display": "Cerner Test, Physician - Women's Health Cerner"
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-confidential-response Response

This operation supports the following authorization types:

Reading confidential procedures.

* Provider

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/2572382193
```

Example Response:


```
{
  "resourceType": "Procedure",
  "id": "2572382193",
  "meta": {
    "versionId": "0",
    "lastUpdated": "2020-06-11T04:02:54Z",
    "security": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PHY",
        "display": "physician requested information sensitivity",
        "userSelected": false
      },
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
        "code": "PRS",
        "display": "patient requested information sensitivity",
        "userSelected": false
      }
    ]
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Meta Security</b>: physician
    requested information sensitivity, patient requested information sensitivity</p><p><b>Patient</b>: SMART,
    NANCY</p><p><b>Procedure</b>: Appendectomy;</p><p><b>Category</b>: SDOH</p><p><b>Date</b>: 2002</p><p><b>Status</b>:
     Completed</p><p><b>Recorder</b>: Cerner Test, Physician - Women's Health Cerner</p></div>"
  },
  "status": "completed",
  "category": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
        "code": "sdoh",
        "display": "SDOH",
        "userSelected": false
```

```
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://www.ama-assn.org/go/cpt",
        "code": "44950",
        "display": "Appendectomy;",
        "userSelected": true
      }
    ],
    "text": "Appendectomy;"
  },
  "subject": {
    "reference": "Patient/12724066",
    "display": "SMART, NANCY"
  },
  "encounter": {
    "reference": "Encounter/97953477"
  },
  "performedDateTime": "2002",
  "recorder": {
    "reference": "Practitioner/4122630",
    "display": "Cerner Test, Physician - Women's Health Cerner"
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-error-in-status Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET /r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Procedure/1788134987
```

Example Response:


```
{
  "resourceType": "Procedure",
  "id": "1788134987",
  "meta": {
    "versionId": "0",
    "lastUpdated": "2019-09-05T22:38:38Z"
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Procedure</b></p><p><b>Patient</b>: MARSTON,
    JACK</p><p><b>Procedure</b>: Error</p><p><b>Status</b>: Entered in Error</p></div>"
  },
  "status": "entered-in-error",
  "code": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
        "code": "error",
        "display": "Error"
      }
    ],
    "text": "Error"
  },
  "subject": {
    "reference": "Patient/166045489",
    "display": "MARSTON, JACK"
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Back to Top

Provenance

File: r4-rest-api-v32/api-provenance.html

Description

Provenance

This topic describes information for the Provenance resource.

Overview

The Provenance resource tracks the source of external information and is used to assess the trustworthiness and reliability of a resource. Sources can be an organization author or organization transmitter. The target field references a resource that this Provenance entry is related to. Entities are Continuity of Care Documents (CCDs) generated through the DocumentReference resource.

This resource supports the US Core Provenance Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The following fields are returned if valued:

Provenance ID

Target

Recorded

Agent

Agent type

Agent role

Agent who

Agent on behalf of (only when Agent.who is of type Practitioner)

Entity role

Entity what

Terminology Bindings

Provenance.entity.role

Description: How the entity was used during the activity.

Details: Provenance Entity Role

System: http://hl7.org/fhir/provenance-entity-role

Provenance

The operations from the Provenance category.

Create a provenance

Method: post Path: /Provenance

Get a list of provenances

Method: get Path: /Provenance

Get a provenance by ID

Method: get Path: /Provenance/{ID}

Create a provenance

File: r4-rest-api-v32/op-provenance-post.html

Creates a new provenance.

Note:

* Only the body fields mentioned below are supported. Unsupported fields are ignored or can result in errors.

* The Provenance.write scope must be provided to create a provenance with an observation measurement target.

Authorization Types

This operation supports the following authorization types:

* Provider

* System

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

The body of the create operation

Root Schema : ProvenanceCreate

Type:

object

A summary representation of the create (POST) operation for a provenance.

Show Source

agent(required): array

agent

An actor with a role in an activity who can be assigned some degree of responsibility for the activity. Provide at least one agent.

Note:

Example:

```
{
  "agent": [
    {
      "type": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
            "code": "transmitter"
          }
        ]
      },
      "role": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
              "code": "SOURCE"
            }
          ]
        }
      ],
      "who": {
        "reference": "Organization/619848"
      }
    }
  ]
}
```

entity: array

entity

An entity used in this activity.

Example:

```
{
  "entity": [
    {
      "role": "source",
      "what": {
        "reference": "DocumentReference/S-201"
      }
    }
  ]
}
```

recorded(required): string

The time at which the activity was recorded.

Example:

```
{
  "recorded": "2019-10-25T14:57:43Z"
}
```

* resourceType(required): string Allowed Values: ["Provenance"] The type of the FHIR resource.

target(required): object

target

The references that were generated or updated by the activity described in this resource.

Note:

* Only one target may be provided.

* The following resources are supported:
* AllergyIntolerance
* CarePlan
* CareTeam
* Condition
* DiagnosticReport
* DocumentReference
* Encounter
* Goal
* Immunization
* MedicationRequest
* Observation
* Patient
* Procedure

Example:

```
{
  "target": [
    {
      "reference": "DocumentReference/881049"
    }
  ]
}
```

```
{
    "description":"A summary representation of the create (POST) operation for a provenance.",
    "type":"object",
    "required":[
        "resourceType",
        "target",
        "recorded",
        "agent",
        "agent.who",
        "entity.role",
        "entity.what"
    ],
    "properties":{
        "resourceType":{
            "description":"The type of the FHIR resource.\n",
            "type":"string",
            "enum":[
                "Provenance"
            ]
        },
        "target":{
            "description":"The references that were generated or updated by the activity described in this
    resource.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one target may be provided.</li>\n  <li>The following resources are
    supported:</li>\n    <ul>\n        <li>AllergyIntolerance</li>\n        <li>CarePlan</li>\n
    <li>CareTeam</li>\n        <li>Condition</li>\n        <li>DiagnosticReport</li>\n
    <li>DocumentReference</li>\n        <li>Encounter</li>\n        <li>Goal</li>\n        <li>Immunization</li>\n
            <li>MedicationRequest</li>\n        <li>Observation</li>\n        <li><a
    href=\"https://docs.oracle.com/en/industries/health/millennium-platform-apis/millennium-authorization-
    framework/#requesting-authorization-on-behalf-of-a-user\">Patient</a></li>\n        <li>Procedure</li>\n
    </ul>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"target\": [\n    {\n      \"reference\":
    \"DocumentReference/881049\"\n    }\n  ]\n}</code></pre>\n",
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        },
        "recorded":{
            "description":"The time at which the activity was recorded.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"recorded\": \"2019-10-25T14:57:43Z\"\n}</code></pre>\n",
            "type":"string"
        },
        "agent":{
            "description":"An actor with a role in an activity who can be assigned some degree of responsibility for the
     activity.<br>\n\n<b>Note:</b> Provide at least one agent.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"agent\": [\n
        {\n      \"type\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type\",\n            \"code\":
    \"transmitter\"\n          }\n        ]\n      },\n      \"role\": [\n        {\n          \"coding\": [\n
                {\n              \"system\": \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n
                  \"code\": \"SOURCE\"\n            }\n          ]\n        }\n      ],\n      \"who\": {\n
    \"reference\": \"Organization/619848\"\n      }\n    }\n  ]\n}</code></pre>\n",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
```

```
                "agent.type":{
                    "description":"The participation the agent had with respect to the
activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": {\n    \"coding\": [\n      {\n        \"system\":
\"http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type\",\n        \"code\": \"transmitter\"\n
      }\n    ]\n  }\n}</code></pre>\n",
                    "type":"object",
                    "properties":{
                        "text":{
                            "type":"string"
                        },
                        "coding":{
                            "type":"array",
                            "items":{
                                "type":"object",
                                "properties":{
                                    "system":{
                                        "type":"string"
                                    },
                                    "code":{
                                        "type":"string"
                                    }
                                }
                            }
                        }
                    }
                },
                "agent.role":{
                    "description":"The function of the agent with respect to the
activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"role\": [\n    {\n      \"coding\": [\n        {\n
\"system\": \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n          \"code\": \"SOURCE\"\n
        }\n      ]\n    }\n  ]\n}</code></pre>\n",
                    "type":"object",
                    "properties":{
                        "text":{
                            "type":"string"
                        },
                        "coding":{
                            "type":"array",
                            "items":{
                                "type":"object",
                                "properties":{
                                    "system":{
                                        "type":"string"
                                    },
                                    "code":{
                                        "type":"string"
                                    }
                                }
                            }
                        }
                    }
                },
                "agent.who":{
                    "description":"The individual, device, or organization that participated in the
event.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"who\": {\n    \"reference\": \"Organization/619848\"\n
}\n}</code></pre>\n",
                    "type":"object",
                    "properties":{
                        "reference":{
                            "type":"string"
                        }
                    }
                }
            }

        }
    },
    "entity":{
        "description":"An entity used in this activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"entity\": [\n
{\n      \"role\": \"source\",\n      \"what\": {\n        \"reference\": \"DocumentReference/S-201\"\n      }\n
}\n  ]\n}</code></pre>\n",
        "type":"array",
        "items":{
            "type":"object",
            "properties":{
                "entity.role":{
                    "description":"How the entity was used during the
activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"role\": \"source\"\n}</code></pre>\n",
                    "type":"string"
                },
                "entity.what":{
                    "description":"The identity of the entity used.<br>\n<b>Example:</b>\n\n<pre><code>{\n
\"what\": {\n    \"reference\": \"DocumentReference/S-201\"\n  }\n}</code></pre>\n",
                    "type":"object",
                    "properties":{
                        "reference":{
                            "type":"string"
                        }
                    }
                }
```

```
            }
        }
    }
}
```

Nested Schema : agent

Type:

array

An actor with a role in an activity who can be assigned some degree of responsibility for the activity. Provide at least one agent.

Note:

Example:

```
{
  "agent": [
    {
      "type": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
            "code": "transmitter"
          }
        ]
      },
      "role": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
              "code": "SOURCE"
            }
          ]
        }
      ],
      "who": {
        "reference": "Organization/619848"
      }
    }
  ]
}
```

Show Source

* Array of: object items

```
{
    "description":"An actor with a role in an activity who can be assigned some degree of responsibility for the
    activity.<br>\n\n<b>Note:</b> Provide at least one agent.<br>\n\n<b>Example:</b>\n\n<pre><code>{\n  \"agent\": [\n
        {\n      \"type\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type\",\n            \"code\":
    \"transmitter\"\n          }\n        ]\n      },\n      \"role\": [\n        {\n          \"coding\": [\n
                {\n              \"system\": \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n
                  \"code\": \"SOURCE\"\n            }\n          ]\n        }\n      ],\n      \"who\": {\n
    \"reference\": \"Organization/619848\"\n      }\n    }\n  ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "agent.type":{
                "description":"The participation the agent had with respect to the
    activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type\",\n        \"code\": \"transmitter\"\n
          }\n    ]\n  }\n}</code></pre>\n",
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "agent.role":{
                "description":"The function of the agent with respect to the
```

```
    activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"role\": [\n    {\n      \"coding\": [\n        {\n
    \"system\": \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n          \"code\": \"SOURCE\"\n
            }\n      ]\n    }\n  ]\n}</code></pre>\n",
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "agent.who":{
                "description":"The individual, device, or organization that participated in the
    event.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"who\": {\n    \"reference\": \"Organization/619848\"\n
    }\n}</code></pre>\n",
                "type":"object",
                "properties":{
                    "reference":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : entity

Type:

array

An entity used in this activity.

Example:


```
{
  "entity": [
    {
      "role": "source",
      "what": {
        "reference": "DocumentReference/S-201"
      }
    }
  ]
}
```


Show Source

* Array of: object items


```
{
    "description":"An entity used in this activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"entity\": [\n    {\n
    \"role\": \"source\",\n      \"what\": {\n        \"reference\": \"DocumentReference/S-201\"\n      }\n    }\n
    ]\n}</code></pre>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "entity.role":{
                "description":"How the entity was used during the activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"role\": \"source\"\n}</code></pre>\n",
                "type":"string"
            },
            "entity.what":{
                "description":"The identity of the entity used.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"what\": {\n
        \"reference\": \"DocumentReference/S-201\"\n  }\n}</code></pre>\n",
                "type":"object",
                "properties":{
                    "reference":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : target

Type:

object

The references that were generated or updated by the activity described in this resource.

Note:

* Only one target may be provided.
* The following resources are supported:
* AllergyIntolerance
* CarePlan
* CareTeam
* Condition
* DiagnosticReport
* DocumentReference
* Encounter
* Goal
* Immunization
* MedicationRequest
* Observation
* Patient
* Procedure

Example:

```
{
  "target": [
    {
      "reference": "DocumentReference/881049"
    }
  ]
}
```

Show Source

* reference: string


```
{
    "description":"The references that were generated or updated by the activity described in this
    resource.<br>\n\n<b>Note:</b>\n<ul>\n  <li>Only one target may be provided.</li>\n  <li>The following resources are
    supported:</li>\n    <ul>\n        <li>AllergyIntolerance</li>\n        <li>CarePlan</li>\n
    <li>CareTeam</li>\n        <li>Condition</li>\n        <li>DiagnosticReport</li>\n
    <li>DocumentReference</li>\n        <li>Encounter</li>\n        <li>Goal</li>\n        <li>Immunization</li>\n
            <li>MedicationRequest</li>\n        <li>Observation</li>\n        <li><a
    href=\"https://docs.oracle.com/en/industries/health/millennium-platform-apis/millennium-authorization-
    framework/#requesting-authorization-on-behalf-of-a-user\">Patient</a></li>\n        <li>Procedure</li>\n
    </ul>\n</ul>\n\n\n<b>Example:</b>\n\n<pre><code>{\n  \"target\": [\n    {\n      \"reference\":
    \"DocumentReference/881049\"\n    }\n  ]\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

agent.role: object

agent.role

The function of the agent with respect to the activity.

Example:

```
{
  "role": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
          "code": "SOURCE"
        }
      ]
    }
  ]
}
```

agent.type: object

agent.type

The participation the agent had with respect to the activity.

Example:

```
{
  "type": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
        "code": "transmitter"
      }
    ]
  }
}
```

agent.who: object

agent.who

The individual, device, or organization that participated in the event.

Example:

```
{
  "who": {
    "reference": "Organization/619848"
  }
}
```

```
{
    "type":"object",
    "properties":{
        "agent.type":{
            "description":"The participation the agent had with respect to the
    activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"type\": {\n    \"coding\": [\n      {\n        \"system\":
    \"http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type\",\n        \"code\": \"transmitter\"\n
          }\n    ]\n  }\n}</code></pre>\n",
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "agent.role":{
            "description":"The function of the agent with respect to the
    activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"role\": [\n    {\n      \"coding\": [\n        {\n
    \"system\": \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n          \"code\": \"SOURCE\"\n
            }\n      ]\n    }\n  ]\n}</code></pre>\n",
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "agent.who":{
            "description":"The individual, device, or organization that participated in the
    event.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"who\": {\n    \"reference\": \"Organization/619848\"\n
    }\n}</code></pre>\n",
            "type":"object",
```

```
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : agent.role

Type:

object

The function of the agent with respect to the activity.

Example:

```
{
  "role": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
          "code": "SOURCE"
        }
      ]
    }
  ]
}
```

Show Source

* coding: array coding

* text: string


```
{
    "description":"The function of the agent with respect to the activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"role\": [\n    {\n      \"coding\": [\n        {\n          \"system\":
    \"http://terminology.hl7.org/CodeSystem/contractsignertypecodes\",\n          \"code\": \"SOURCE\"\n        }\n
          ]\n    }\n  ]\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : agent.type

Type:

object

The participation the agent had with respect to the activity.

Example:

```
{
  "type": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
        "code": "transmitter"
      }
    ]
  }
}
```

Show Source

* coding: array coding

* text: string


```
{
    "description":"The participation the agent had with respect to the activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n
      \"type\": {\n    \"coding\": [\n      {\n        \"system\": \"http://hl7.org/fhir/us/core/CodeSystem/us-core-
    provenance-participant-type\",\n        \"code\": \"transmitter\"\n      }\n    ]\n  }\n}</code></pre>\n",
```

```
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : agent.who

Type:

object

The individual, device, or organization that participated in the event.

Example:

```
{
  "who": {
    "reference": "Organization/619848"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The individual, device, or organization that participated in the
    event.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"who\": {\n    \"reference\": \"Organization/619848\"\n
    }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
```

```
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

entity.role: string

How the entity was used during the activity.

Example:

```
{
  "role": "source"
}
```

entity.what: object

entity.what

The identity of the entity used.

Example:

```
{
  "what": {
    "reference": "DocumentReference/S-201"
  }
}
```

```
{
    "type":"object",
    "properties":{
        "entity.role":{
            "description":"How the entity was used during the activity.<br>\n<b>Example:</b>\n\n<pre><code>{\n
    \"role\": \"source\"\n}</code></pre>\n",
            "type":"string"
        },
        "entity.what":{
            "description":"The identity of the entity used.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"what\": {\n
    \"reference\": \"DocumentReference/S-201\"\n  }\n}</code></pre>\n",
```

```
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : entity.what

Type:

object

The identity of the entity used.

Example:

```
{
  "what": {
    "reference": "DocumentReference/S-201"
  }
}
```

Show Source

* reference: string

```
{
    "description":"The identity of the entity used.<br>\n<b>Example:</b>\n\n<pre><code>{\n  \"what\": {\n
    \"reference\": \"DocumentReference/S-201\"\n  }\n}</code></pre>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

Default Response

The response header indicates the current version to use on a subsequent update.

Example Request:

```
POST https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance
```

Example Request Body:

```
{
    'resourceType': 'Provenance',
    'target': [
      {
        'reference': 'DocumentReference/881049'
      }
    ],
    'recorded': '2019-10-25T14:57:43Z',
    'agent': [
      {
        'type': {
          'coding': [
            {
              'system': 'http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type',
              'code': 'transmitter'
            }
          ]
        },
        'role': [
          {
            'coding': [
              {
                'system': 'http://terminology.hl7.org/CodeSystem/contractsignertypecodes',
                'code': 'SOURCE'
              }
            ]
          }
        ],
        'who': {
          'reference': 'Organization/619848'
        }
      }
    ],
    'entity': [
      {
        'role': 'source',
```

```
      'what': {
        'reference': 'DocumentReference/S-201'
      }
    }
  ]
}
```

Example Response:

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 31 Mar 2020 15:37:25 GMT
Etag: W/"881057"
Location: https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/doc-881057
Last-Modified: Tue, 31 Mar 2020 15:37:25 GMT
Vary: Origin
opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
X-Request-Id: 11111111-1111-1111-1111-111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a list of provenances

File: r4-rest-api-v32/op-provenance-get.html

Retrieves a list of provenances that meet supplied query parameters.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Query Parameters

_id: string

The logical resource ID. Example:

* This parameter is required if the target parameter is not used.

* May be a list separated by commas.

* Must not include any other parameters.

_id=doc-1234

target: string

The target references, which can be version-specific. The following resources are supported: Example:

* AllergyIntolerance

* CarePlan

* CareTeam

* Condition

* Coverage

* Device

* DiagnosticReport

* DocumentReference

* Encounter

* Goal

* Immunization

* MedicationDispense

* MedicationRequest
* NutritionOrder
* Observation
* Patient
* Procedure
* RelatedPerson
* ServiceRequest

target=DocumentReference/66547

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.
* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example search by .

_id

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance?_id=doc-881057
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': '160ad230-6a5c-4813-96e2-2b36be102145',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance?_id=doc-881057'
      }
    ],
    'entry': [
      'fullUrl': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/doc-881057',
      'resource': {
        'resourceType': 'Provenance',
        'id': 'doc-881057',
        'meta': {
          'versionId': '881057',
          'lastUpdated': '2019-10-25T14:57:43Z'
        },
        'text': {
          'status': 'generated',
          'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>Documen
    tReference/881049</li></ul><p><b>Recorded</b>: Oct 25, 2019  9:57 A.M. CDT</p><p><b>Agents</b>:</p><dl><dt>Baseline
    East Medical Center</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>: Source</dd></dl><p><b>Entity
    Source</b>: DocumentReference/S-202</p></div>'
        },
        'target': [
          {
            'reference': 'DocumentReference/881049'
          }
        ],
        'recorded': '2019-10-25T14:57:43Z',
        'agent': [
          {
            'type': {
              'coding': [
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/provenance-participant-type',
                  'code': 'author'
                }
              ],
              'text': 'Author'
            },
            'role': [
              {
                'coding': [
                  {
                    'system': 'http://terminology.hl7.org/CodeSystem/contractsignertypecodes',
                    'code': 'SOURCE'
                  }
```

```
              ],
              'text': 'Source'
            }
          ],
          'who': {
            'reference': 'Organization/619848',
            'display': 'Baseline East Medical Center'
          }
        }
      ],
      'entity': [
        {
          'role': 'source',
          'what': {
            'reference': 'DocumentReference/S-202'
          }
        }
      ]
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth Response

Example search using patient authorization.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance?_id=all-1595070
```

Example Response:

```
{
    'resourceType': 'Provenance',
    'id': 'all-1595070',
    'meta': {
      'versionId': '1595070',
      'lastUpdated': '2019-12-09T16:49:43Z'
    },
    'text': {
      'status': 'generated',
      'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>AllergyIn
    tolerance/S-1595069</li></ul><p><b>Recorded</b>: Dec  9, 2019 10:49 A.M.
    CST</p><p><b>Agents</b>:</p><dl><dt>Unknown</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>:
    Source</dd></dl><p><b>Entity Source</b>: DocumentReference/S-200</p></div>'
    },
    'target': [
      {
        'reference': 'AllergyIntolerance/S-1595069'
      }
    ],
    'recorded': '2019-12-09T16: 49: 43Z',
    'agent': [
      {
        'type': {
          'coding': [
            {
              'system': 'http: //terminology.hl7.org/CodeSystem/provenance-participant-type',
              'code': 'author'
            }
          ],
          'text': 'Author'
        },
        'role': [
          {
            'coding': [
              {
                'system': 'http: //terminology.hl7.org/CodeSystem/contractsignertypecodes',
                'code': 'SOURCE'
              }
            ],
            'text': 'Source'
          }
        ],
        'who': {
          'reference': 'Organization/50',
          'display': 'Unknown'
        }
      }
    ],
    'entity': [
      {
        'role': 'source',
```

```
        'what': {
          'reference': 'DocumentReference/S-200'
        }
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a provenance by ID

File: r4-rest-api-v32/op-provenance-id-get.html

Retrieves an individual provenance by the associated provenance ID.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example retrieval of a provenance record by resource ID.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/doc-881057
```

Example Response:

```
{
    'resourceType': 'Provenance',
    'id': 'doc-881057',
    'meta': {
      'versionId': '881057',
      'lastUpdated': '2019-10-25T14:57:43Z'
    },
    'text': {
      'status': 'generated',
      'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>DocumentRef
    erence/881049</li></ul><p><b>Recorded</b>: Oct 25, 2019  9:57 A.M. CDT</p><p><b>Agents</b>:</p><dl><dt>Baseline East
     Medical Center</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>: Source</dd></dl><p><b>Entity
    Source</b>: DocumentReference/S-202</p></div>'
    },
    'target': [
      {
        'reference': 'DocumentReference/881049'
      }
    ],
    'recorded': '2019-10-25T14:57:43Z',
    'agent': [
      {
        'type': {
          'coding': [
            {
              'system': 'http://terminology.hl7.org/CodeSystem/provenance-participant-type',
              'code': 'author'
            }
          ],
          'text': 'Author'
```

```
        },
        'role': [
          {
            'coding': [
              {
                'system': 'http://terminology.hl7.org/CodeSystem/contractsignertypecodes',
                'code': 'SOURCE'
              }
            ],
            'text': 'Source'
          }
        ],
        'who': {
          'reference': 'Organization/619848',
          'display': 'Baseline East Medical Center'
        }
      }
    ],
    'entity': [
      {
        'role': 'source',
        'what': {
          'reference': 'DocumentReference/S-202'
        }
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth Response

Example retrieval of a provenance record using patient authorization.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/all-1595070
```

Example Response:

```
{
    'resourceType': 'Bundle',
    'id': '3ddac8f3-0c3a-41fd-bd0a-1daeb6a42db9',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance?_id=all-1595070'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/all-1595070',
        'resource': {
          'resourceType': 'Provenance',
          'id': 'all-1595070',
          'meta': {
            'versionId': '1595070',
            'lastUpdated': '2019-12-09T16:49:43Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>All
    ergyIntolerance/S-1595069</li></ul><p><b>Recorded</b>: Dec  9, 2019 10:49 A.M.
    CST</p><p><b>Agents</b>:</p><dl><dt>Unknown</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>:
    Source</dd></dl><p><b>Entity Source</b>: DocumentReference/S-200</p></div>'
          },
          'target': [
            {
              'reference': 'AllergyIntolerance/S-1595069'
            }
          ],
          'recorded': '2019-12-09T16: 49: 43Z',
          'agent': [
            {
              'type': {
                'coding': [
                  {
                    'system': 'http: //terminology.hl7.org/CodeSystem/provenance-participant-type',
                    'code': 'author'
                  }
                ],
                'text': 'Author'
              },
```

```
            'role': [
              {
                'coding': [
                  {
                    'system': 'http: //terminology.hl7.org/CodeSystem/contractsignertypecodes',
                    'code': 'SOURCE'
                  }
                ],
                'text': 'Source'
              }
            ],
            'who': {
              'reference': 'Organization/50',
              'display': 'Unknown'
            }
          }
        ],
        'entity': [
          {
            'role': 'source',
            'what': {
              'reference': 'DocumentReference/S-200'
            }
          }
        ]
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Questionnaire

File: r4-rest-api-v32/api-questionnaire.html

Description

Questionnaire

This topic describes information for the Questionnaire resource.

Overview

The Questionnaire resource defines and organizes questions and the allowed answers to the questions typically used to collect patient healthcare information. A questionnaire is a snapshot in time and should be retrieved before each use. Examples of questionnaires are forms used to collect a patient's social history or family member history. Currently, only social history is supported. In order to find a questionnaire for a specific patient, first search QuestionnaireResponse by patient. Then use the 'questionnaire' field from the QuestionnaireResponse to retrieve the original questionnaire.

The following fields are returned if valued:

Questionnaire id

Name

Title

Status

Patient Subject Type

Item

Link id

Extension (for more information, see the Extensions section of this topic)

Text

Type

Item

Link id

Text

Type

Required

Repeats

Max Length

Extension (for more information, see the Extensions section of this topic)

Answer Option

Terminology Bindings

Questionnaire.item.item.answerOption.valueCoding

Description: The value of one of the permitted answers for a "choice" or "open-choice" question.

Details: For more information, see Nomenclature under the Proprietary Systems section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR source id>/nomenclature

Extensions

* Max Decimal Places

Custom Extensions

URLs for custom extensions are defined as https://fhir-ehr.cerner.com/r4/StructureDefinition/{id}

Table - Custom Extensions

ID | Value[x] Type | Description
note-allowed | boolean | Whether a comment is allowed.

Questionnaire

The operations from the Questionnaire category.

Get a list of questionnaires

Method: get Path: /Questionnaire/

Get a questionnaire by ID

Method: get Path: /Questionnaire/{id}

Get a list of questionnaires

File: r4-rest-api-v32/op-questionnaire-get.html

Retrieves a list of questionnaires that meet supplied query parameters.

Authorization Types

This resource supports the following authorization types:

* Patient
* Provider
* System

Request

Query Parameters

_id(required): array[string]

Minimum Number of Items:

The logical resource ID.

Example: _id=12345

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.
* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)

```
{
    "resourceType":"Bundle",
    "id":"902e1b49-b9c4-4786-b90b-c9612d478f2a",
    "type":"searchset",
    "link":[
        {
            "relation":"self",
            "url":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire?_id=SH-12508041"
        }
    ],
    "entry":[
        {
            "fullUrl":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
            "resource":{
                "resourceType":"Questionnaire",
                "id":"SH-12508041",
                "name":"Social_History",
                "title":"Social History",
                "status":"active",
                "subjectType":[
                    "Patient"
                ],
                "item":[
                    {
                        "linkId":"93",
                        "extension":[
                            {
                                "valueBoolean":true,
                                "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note-allowed"
                            }
                        ],
                        "text":"Tobacco",
                        "type":"group",
                        "item":[
                            {
                                "linkId":"93-6930356",
                                "text":"Tobacco use per day:",
                                "type":"text",
                                "required":false,
                                "repeats":false,
                                "maxLength":255
                            },
                            {
                                "linkId":"93-6930316",
                                "text":"Number of years:",
                                "type":"quantity",
                                "required":false,
                                "repeats":false
                            },
                            {
                                "linkId":"93-6930360",
                                "text":"Started at age:",
                                "type":"quantity",
                                "required":false,
                                "repeats":false
                            },
                            {
                                "linkId":"93-6930364",
                                "text":"Stopped at age:",
                                "type":"quantity",
                                "required":false,
                                "repeats":false
                            },
                            {
                                "linkId":"93-6930372",
                                "text":"Ready to change:",
                                "type":"choice",
                                "required":false,
                                "repeats":false,
                                "answerOption":[
                                    {
                                        "valueCoding":{

    "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                            "code":"960439",
                                            "display":"Yes"
                                        }
                                    },
                                    {
                                        "valueCoding":{

    "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                            "code":"960441",
                                            "display":"No"
                                        }
                                    }
                                ]
                            },
                            {
```

```
                                "linkId":"93-6930376",
                                "text":"Concerns about tobacco use in household:",
                                "type":"open-choice",
                                "required":false,
                                "repeats":false,
                                "maxLength":255,
                                "answerOption":[
                                    {
                                        "valueCoding":{

    "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                            "code":"960439",
                                            "display":"Yes"
                                        }
                                    },
                                    {
                                        "valueCoding":{

    "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                            "code":"960441",
                                            "display":"No"
                                        }
                                    }
                                ]
                            },
                            {
                                "linkId":"93-6930380",
                                "text":"Total pack years:",
                                "type":"quantity",
                                "required":false,
                                "repeats":false

                            },
                            {
                                "extension":[
                                    {
                                        "valueInteger":2,
                                        "url":"http://hl7.org/fhir/StructureDefinition/maxDecimalPlaces"
                                    }
                                ],
                                "linkId":"93-4356884",
                                "text":"Weight",
                                "type":"quantity",
                                "required":false,
                                "repeats":false,
                                "maxLength":6
                            }
                        ]
                    }
                ]
            }
        }
    ]
}
```

Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]

```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
```

```
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
```

```
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Get a questionnaire by ID

File: r4-rest-api-v32/op-questionnaire-id-get.html

Retrieves a questionnaire by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Questionnaire",
    "id":"SH-12508041",
    "name":"Social_History",
    "title":"Social History",
    "status":"active",
    "subjectType":[
        "Patient"
    ],
    "item":[
        {
            "linkId":"93",
            "extension":[
                {
                    "valueBoolean":true,
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note-allowed"
                }
            ],
            "text":"Tobacco",
            "type":"group",
            "item":[
                {
                    "linkId":"93-6930356",
                    "text":"Tobacco use per day:",
                    "type":"text",
                    "required":false,
                    "repeats":false,
                    "maxLength":255
                },
                {
                    "linkId":"93-6930316",
                    "text":"Number of years:",
                    "type":"quantity",
                    "required":false,
                    "repeats":false
                },
                {
                    "linkId":"93-6930360",
                    "text":"Started at age:",
                    "type":"quantity",
                    "required":false,
                    "repeats":false
```

```
                },
                {
                    "linkId":"93-6930364",
                    "text":"Stopped at age:",
                    "type":"quantity",
                    "required":false,
                    "repeats":false
                },
                {
                    "linkId":"93-6930372",
                    "text":"Ready to change:",
                    "type":"choice",
                    "required":false,
                    "repeats":false,
                    "answerOption":[
                        {
                            "valueCoding":{
                                "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                "code":"960439",
                                "display":"Yes"
                            }
                        },
                        {
                            "valueCoding":{
                                "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                "code":"960441",
                                "display":"No"
                            }
                        }
                    ]
                },
                {
                    "linkId":"93-6930376",
                    "text":"Concerns about tobacco use in household:",
                    "type":"open-choice",
                    "required":false,
                    "repeats":false,
                    "maxLength":255,
                    "answerOption":[
                        {
                            "valueCoding":{
                                "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                "code":"960439",
                                "display":"Yes"
                            }
                        },
                        {
                            "valueCoding":{
                                "system":"https://fhir.cerner.com/a758f80e-aa74-4118-80aa-98cc75846c76/nomenclature",
                                "code":"960441",
                                "display":"No"
                            }
                        }
                    ]
                },
                {
                    "linkId":"93-6930380",
                    "text":"Total pack years:",
                    "type":"quantity",
                    "required":false,
                    "repeats":false
                },
                {
                    "extension":[
                        {
                            "valueInteger":2,
                            "url":"http://hl7.org/fhir/StructureDefinition/maxDecimalPlaces"
                        }
                    ],
                    "linkId":"93-4356884",
                    "text":"Weight",
                    "type":"quantity",
                    "required":false,
                    "repeats":false,
                    "maxLength":6
                }
            ]
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : issue

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string


```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
```

```
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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QuestionnaireResponse

File: r4-rest-api-v32/api-questionnaireresponse.html

Description

QuestionnaireResponse

This topic describes information for the QuestionnaireResponse resource.

Overview

The QuestionnaireResponse resource is a collection of answers to a given questionnaire typically used to request patient healthcare information. A given QuestionnaireResponse resource must belong to a corresponding Questionnaire resource. An example of a questionnaire is a form used to collect a patient's social history. Currently, only social history is supported. The resource should be leveraged as a snapshot in time and new data should be consistently retrieved through the API rather than stored within an application.

This resource supports the US Core QuestionnaireResponse Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The examples provided here are non-normative and replaying them in the public sandbox is not guaranteed to yield the results shown on the site. The field is present only when confidential answers are present within the resource. Confidential answers contain extensions with their applied confidential label. See the for more information.

Note:

meta.security

HL7 FHIR Inline Security Label extension

The following fields are returned if valued:

ID

Questionnaire

Status

Subject

Authored

Item

ID

Link ID

Is Modifiable extension (for more information, see the Extensions section of this topic)

Note extension (for more information, see the Extensions section of this topic)

Text

Item

Link id

Text

Choice Answer extension (for more information, see the Extensions section of this topic)

* Inline Security Label extension

valueString answers

valueCoding answers

valueQuantity answers

Terminology Bindings

QuestionnaireResponse.answer.valueCoding

Description: An answer from a list of available coding options.

Details: For more information, see Nomenclature under the Proprietary Systems section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR source id>/nomenclature

QuestionnaireResponse.meta.security

Description: These tags connect specific resources to the overall security policy and infrastructure.

Details: ValueSet

System: http://terminology.hl7.org/CodeSystem/v3-ActCode

Extensions

Inline Security Label

Custom Extensions

URLs for custom extensions are defined as https://fhir-ehr.cerner.com/r4/StructureDefinition/{id}

ID | Value[x] Type | Description
is-modifiable | Boolean | Indication of whether data is modifiable or not. Only present when false.
choice-answer | Boolean | Indicates answers come from a list of options.
note | Annotation | Additional details about a given QuestionnaireResponse group item. Includes author and date/time information.

QuestionnaireResponse

The operations from the QuestionnaireResponse category.

Get a list of questionnaire responses

Method: get Path: /QuestionnaireResponse/

Get a questionnaire response by ID

Method: get Path: /QuestionnaireResponse/{id}

Update a questionnaire response

Method: put Path: /QuestionnaireResponse/{id}

Get a list of questionnaire responses

File: r4-rest-api-v32/op-questionnaireresponse-get.html

Retrives a list of questionnaire responses that meet supplied query parameters.

Authorization Types

This resource supports the following authorization types:

* Patient

* Provider

* System

Request

Query Parameters

_id: array[string]

Minimum Number of Items:

The logical resource ID.

Example: _id=12345

patient: array[string]

Minimum Number of Items:

The logical resource ID of the patient the questionnaire response is for. This parameter is required if the _id parameter is not used.

Example: patient=12724066

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Bundle",
    "id":"902e1b49-b9c4-4786-b90b-c9612d478f2a",
    "type":"searchset",
    "link":[
        {
            "relation":"self",
            "url":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse?_id=SH-12508041"
        }
    ],
    "entry":[
        {
            "fullUrl":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse/SH-12508041",
            "resource":{
                "resourceType":"QuestionnaireResponse",
                "id":"SH-12508041",
                "questionnaire":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
                "status":"in-progress",
                "subject":{
                    "reference":"Patient/12508041"
                },
                "authored":"2019-10-25T14:57:43Z",
                "item":[
                    {
                        "id":"111",
                        "linkId":"93",
                        "text":"Tobacco",
                        "extension":[
                            {
                                "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                                "valueAnnotation":{
                                    "authorReference":{
                                        "reference":"Practitioner/109413936",
                                        "display":"Lombardi, Falco Shine"
                                    },
                                    "time":"2021-06-15T22:15:26.000Z",
                                    "text":"category/group comment"
                                }
                            }
                        ],
                        "item":[
                            {
                                "linkId":"93-123",
                                "text":"Number of years:",
                                "answer":[
                                    {
                                        "valueQuantity":{
                                            "value":"20",
                                            "unit":"year (iso1000)",
                                            "system":"http://unitsofmeasure.org",
                                            "code":"a"
                                        }
                                    }
                                ]
                            },
                            {
                                "linkId":"93-456",
                                "text":"Tobacco use per day:",
                                "answer":[
                                    {
                                        "valueString":"Answer text"
                                    }
                                ]
                            },
                            {
                                "linkId":"93-789",
                                "text":"Ready to change:",
                                "extension":[
                                    {
                                        "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                                        "valueBoolean":true
                                    }
```

```
                                ],
                                "answer":[
                                    {
                                        "valueCoding":{
    "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                                            "code":"960439",
                                            "display":"Yes"
                                        }
                                    },
                                    {
                                        "valueString":"Other answer text"
                                    }
                                ]
                            }
                        ]
                    }
                ]
            }
        }
    ]
}
```

Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]

```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
```

```
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        },
                        "location":{
                            "type":"array",
                            "items":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string


```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : location

Type:

array

Show Source

* Array of: string


```
{
```

```
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

x-confidential-resource Response

Example response containing confidential information.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse?patient=12508041
```

Headers

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)

```
{
    "resourceType":"Bundle",
    "id":"902e1b49-b9c4-4786-b90b-c9612d478f2a",
    "type":"searchset",
    "link":[
        {
            "relation":"self",
            "url":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse?patient=12508041"
        }
    ],
    "entry":[
        {
            "fullUrl":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse/SH-12508041",
            "resource":{
                "resourceType":"QuestionnaireResponse",
                "id":"SH-12508041,",
                "meta":{
                    "lastUpdated":"2019-10-25T14:57:43Z",
                    "security":[
                        {
                            "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
```

```
                        "code":"PROCESSINLINELABEL",
                        "display":"process inline security label"
                    }
                ]
            },
            "text":{
                "status":"generated",
                "div":"<div xmlns=\\"http://www.w3.org/1999/xhtml\\"><p><b>Questionnaire Response</b></p><p><b>Meta
Security</b>: process inline security label</p><p><b>Patient</b>: 12508041</p><p><b>Type</b>: Social
History</p><p><b>Status</b>: In Progress</p><p><b>Authored Date Time</b>: Apr 24, 2025  3:50 P.M.
UTC</p><p><b>Questionnaire</b>: https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041</p></div>"
            },
            "questionnaire":"https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
            "status":"in-progress",
            "subject":{
                "reference":"Patient/12508041"
            },
            "authored":"2019-10-25T14:57:43Z",
            "item":[
                {
                    "id":"111",
                    "linkId":"93",
                    "text":"Tobacco",
                    "extension":[
                        {
                            "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                            "valueAnnotation":{
                                "authorReference":{
                                    "reference":"Practitioner/109413936",
                                    "display":"Lombardi, Falco Shine"
                                },
                                "time":"2021-06-15T22:15:26.000Z",
                                "text":"category/group comment"
                            }
                        }
                    ],
                    "item":[
                        {
                            "linkId":"93-123",
                            "text":"Number of years:",
                            "answer":[
                                {
                                    "valueQuantity":{
                                        "value":"20",
                                        "unit":"year (iso1000)",
                                        "system":"http://unitsofmeasure.org",
                                        "code":"a"
                                    }
                                }
                            ],
                            "extension":[
                                {
                                    "valueCoding":{
                                        "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
                                        "code":"PHY",
                                        "display":"physician requested information sensitivity",
                                        "userSelected":false
                                    },
                                    "url":"http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-
inline-sec-label"
                                }
                            ]
                        },
                        {
                            "linkId":"93-456",
                            "text":"Tobacco use per day:",
                            "answer":[
                                {
                                    "valueString":"Answer text"
                                }
                            ]
                        },
                        {
                            "linkId":"93-789",
                            "text":"Ready to change:",
                            "extension":[
                                {
                                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                                    "valueBoolean":true
                                }
                            ],
                            "answer":[
                                {
                                    "valueCoding":{

"system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                                        "code":"960439",
```

```
                                        "display":"Yes"
                                    }
                                },
                                {
                                    "valueString":"Other answer text"
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```


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Get a questionnaire response by ID

File: r4-rest-api-v32/op-questionnaireresponse-id-get.html

Retrieves a questionnaire response by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"QuestionnaireResponse",
    "id":"SH-12508041",
    "questionnaire":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
    "status":"in-progress",
    "subject":{
        "reference":"Patient/12508041"
    },
    "authored":"2019-10-25T14:57:43Z",
    "item":[
        {
            "id":"111",
            "linkId":"93",
            "text":"Tobacco",
            "extension":[
                {
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                    "valueAnnotation":{
                        "authorReference":{
                            "reference":"Practitioner/109413936",
```

```
                        "display":"Lombardi, Falco Shine"
                    },
                    "time":"2021-06-15T22:15:26.000Z",
                    "text":"category/group comment"
                }
            }
        ],
        "item":[
            {
                "linkId":"93-123",
                "text":"Number of years:",
                "answer":[
                    {
                        "valueQuantity":{
                            "value":"20",
                            "unit":"year (iso1000)",
                            "system":"http://unitsofmeasure.org",
                            "code":"a"
                        }
                    }
                ]
            },
            {
                "linkId":"93-456",
                "text":"Tobacco use per day:",
                "answer":[
                    {
                        "valueString":"Answer text"
                    }
                ]
            },
            {
                "linkId":"93-789",
                "text":"Ready to change:",
                "extension":[
                    {
                        "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                        "valueBoolean":true
                    }
                ],
                "answer":[
                    {
                        "valueCoding":{
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                            "code":"960439",
                            "display":"Yes"
                        }
                    },
                    {
                        "valueString":"Other answer text"
                    }
                ]
            }
        ]
    }
]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
```

```
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
```

```
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : location

Type:

array

Show Source

* Array of: string


```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


x-confidential-resource Response

Example response containing confidential information.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/QuestionnaireResponse/SH-12508041
```

Headers

* X-Request-Id: string Oracle troubleshooting identifier.

* opc-request-id: string Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"QuestionnaireResponse",
    "id":"SH-12508041,",
    "meta":{
        "lastUpdated":"2019-10-25T14:57:43Z",
        "security":[
            {
                "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
                "code":"PROCESSINLINELABEL",
                "display":"process inline security label"
            }
        ]
    },
    "text":{
        "status":"generated",
        "div":"<div xmlns=\\"http://www.w3.org/1999/xhtml\\"><p><b>Questionnaire Response</b></p><p><b>Meta
    Security</b>: process inline security label</p><p><b>Patient</b>: 12508041</p><p><b>Type</b>: Social
    History</p><p><b>Status</b>: In Progress</p><p><b>Authored Date Time</b>: Apr 24, 2025  3:50 P.M.
    UTC</p><p><b>Questionnaire</b>: https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041</p></div>"
    },
    "questionnaire":"https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
    "status":"in-progress",
    "subject":{
        "reference":"Patient/12508041"
    },
    "authored":"2019-10-25T14:57:43Z",
    "item":[
        {
            "id":"111",
            "linkId":"93",
            "text":"Tobacco",
            "extension":[
                {
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                    "valueAnnotation":{
                        "authorReference":{
                            "reference":"Practitioner/109413936",
                            "display":"Lombardi, Falco Shine"
                        },
                        "time":"2021-06-15T22:15:26.000Z",
                        "text":"category/group comment"
                    }
                }
            ],
            "item":[
                {
                    "linkId":"93-123",
                    "text":"Number of years:",
                    "answer":[
                        {
                            "valueQuantity":{
                                "value":"20",
                                "unit":"year (iso1000)",
                                "system":"http://unitsofmeasure.org",
                                "code":"a"
                            }
                        }
                    ],
                    "extension":[
                        {
                            "valueCoding":{
                                "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
                                "code":"PHY",
                                "display":"physician requested information sensitivity",
                                "userSelected":false
                            },
                            "url":"http://hl7.org/fhir/uv/security-label-ds4p/StructureDefinition/extension-inline-sec-
    label"
                        }
                    ]
                },
                {
                    "linkId":"93-456",
                    "text":"Tobacco use per day:",
                    "answer":[
                        {
```

```
                        "valueString":"Answer text"
                    }
                ]
            },
            {
                "linkId":"93-789",
                "text":"Ready to change:",
                "extension":[
                    {
                        "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                        "valueBoolean":true
                    }
                ],
                "answer":[
                    {
                        "valueCoding":{
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                            "code":"960439",
                            "display":"Yes"
                        }
                    },
                    {
                        "valueString":"Other answer text"
                    }
                ]
            }
        ]
    }
]
}
```

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Update a questionnaire response

File: r4-rest-api-v32/op-questionnaireresponse-id-put.html

Updates a questionnaire response.

Note:

* When an item is returned with the is-modifiable extension as false, that item cannot be updated unless the update is passed in without an ID.

* An update should only be performed directly after a questionnaire response is retrieved. If an item.id is provided from the GET request, it must be provided on the subsequent update and must match the item.id returned from the GET.

Request

Supported Media Types

* application/fhir+json

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* If-Match(required): string The current version of the QuestionnaireResponse resource.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

Root Schema : QuestionnaireResponseUpdateBody

Type:

object

Show Source

id(required): string

The logical ID of the resource.

Example:

SH-12508041

item(required): array

item

A group or question item from the original questionnaire for which answers are provided.

questionnaire(required): string

The questionnaire that defines and organizes the questions for which answers are being provided.

Example:

https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041

resourceType(required): string

Allowed Values:

["QuestionnaireResponse"]

The type of the FHIR resource.

status(required): string

Allowed Values:

["in-progress"]

The position of the questionnaire response within its overall lifecycle.

subject(required): object

subject

The subject of the questionnaire response. This subject is the patient that the answers apply to, but is not necessarily the source of information.


```
{
    "type":"object",
    "required":[
        "resourceType",
        "id",
        "questionnaire",
        "status",
        "subject",
        "item"
    ],
    "properties":{
        "resourceType":{
            "description":"<p>The type of the FHIR resource.</p>\n",
            "type":"string",
            "enum":[
                "QuestionnaireResponse"
            ]
        },
        "id":{
            "description":"<p>The logical ID of the resource.</p>\n",
            "type":"string",
            "example":"SH-12508041"
        },
        "questionnaire":{
            "description":"<p>The questionnaire that defines and organizes the questions for which answers are being
    provided.</p>\n",
            "type":"string",
            "example":"https://fhir-ehr-
    code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041"
        },
        "status":{
            "description":"<p>The position of the questionnaire response within its overall lifecycle.</p>\n",
            "type":"string",
            "enum":[
                "in-progress"
            ]
        },
        "subject":{
            "description":"<p>The subject of the questionnaire response. This subject is the patient that the answers
    apply to, but is not necessarily the source of information.</p>\n",
            "type":"object",
            "properties":{
                "reference":{
                    "type":"string",
                    "example":"Patient/12508041"
                }
            }
        },
        "item":{
            "description":"<p>A group or question item from the original questionnaire for which answers are
    provided.</p>\n",
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "linkId"
                ],
                "properties":{
                    "id":{
                        "type":"string",
                        "description":"<p>Unique ID for inter-element referencing.</p>\n<p><b>Note:</b> If an
    <code>item.id</code> is returned on a read, it must be provided on an update.</p>\n",
                        "example":"111"
                    },
```

```
                "linkId":{
                    "description":"<p>The item from the questionnaire that corresponds with this item in the
QuestionnaireResponse resource.</p>\n",
                    "example":"93"
                },
                "extension":{
                    "description":"<p>Additional details about a given questionnaire response group
item.</p>\n<p><b>Note:</b> Notes can only be added, not replaced or deleted.</p>\n",
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "url":{
                                "type":"string",
                                "enum":[
                                    "https://fhir-ehr.cerner.com/r4/StructureDefinition/note"
                                ]
                            },
                            "valueAnnotation":{
                                "type":"object",
                                "required":[
                                    "text"
                                ],
                                "properties":{
                                    "text":{
                                        "type":"string",
                                        "example":"category/group comment"
                                    }
                                }
                            }
                        }
                    }
                },
                "item":{
                    "type":"array",
                    "description":"<p>Questions nested beneath a group.</p>\n",
                    "items":{
                        "type":"object",
                        "required":[
                            "linkId",
                            "answer"
                        ],
                        "properties":{
                            "linkId":{
                                "type":"string",
                                "description":"<p>The item from the questionnaire that corresponds with this item in
 the QuestionnaireResponse resource.</p>\n",
                                "example":"93-123"
                            },
                            "extension":{
                                "description":"<p>Whether the answers come from a list of
options.</p>\n<p><b>Note:</b> Extension must be provided when answering a choice or open choice type of
question.</p>\n",
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "required":[
                                        "url",
                                        "valueBoolean"
                                    ],
                                    "properties":{
                                        "url":{
                                            "type":"string",
                                            "enum":[
                                                "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
                                            ]
                                        },
                                        "valueBoolean":{
                                            "type":"boolean",
                                            "enum":[
                                                true
                                            ]
                                        }
                                    }
                                }
                            },
                            "answer":{
                                "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only
 answer value types of String, Quantity, and Coding are supported.</p>\n",
                                "type":"array",
                                "items":{
                                    "description":"<p>Single-valued answer to the question.</p>\n",
                                    "type":"object",
                                    "properties":{
                                        "valueString":{
                                            "type":"string",
                                            "minLength":1,
                                            "example":"Answer text"
                                        },
```

```
                                        "valueCoding":{
                                            "type":"object",
                                            "required":[
                                                "system",
                                                "code"
                                            ],
                                            "properties":{
                                                "system":{
                                                    "type":"string",

"example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                                                },
                                                "code":{
                                                    "type":"string",
                                                    "example":"960439"
                                                }
                                            }
                                        },
                                        "valueQuantity":{
                                            "type":"object",
                                            "required":[
                                                "value"
                                            ],
                                            "properties":{
                                                "value":{
                                                    "type":"number",
                                                    "example":20
                                                }
                                            }
                                        }
                                    }
                                }
                            }
                        }
                    }
                }
            }
        }
    }
},
"example":{
    "resourceType":"QuestionnaireResponse",
    "id":"SH-12508041",
    "questionnaire":"https://fhir-ehr-
code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
    "status":"in-progress",
    "subject":{
        "reference":"Patient/12508041"
    },
    "item":[
        {
            "id":"111",
            "linkId":"93",
            "extension":[
                {
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                    "valueAnnotation":{
                        "text":"category/group comment"
                    }
                }
            ],
            "item":[
                {
                    "linkId":"93-123",
                    "answer":[
                        {
                            "valueQuantity":{
                                "value":"20",
                                "unit":"year (iso1000)",
                                "system":"http://unitsofmeasure.org",
                                "code":"a"
                            }
                        }
                    ]
                },
                {
                    "linkId":"93-456",
                    "answer":[
                        {
                            "valueString":"Answer text"
                        }
                    ]
                },
                {
                    "linkId":"93-789",
                    "extension":[
                        {
                            "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                            "valueBoolean":true
```

```
                            }
                        ],
                        "answer":[
                            {
                                "valueCoding":{

    "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                                    "code":"960439"
                                }
                            },
                            {
                                "valueString":"Other answer text"
                            }
                        ]
                    }
                ]
            }
        ]
    }
}
```

Example:


```
{
    "resourceType":"QuestionnaireResponse",
    "id":"SH-12508041",
    "questionnaire":"https://fhir-ehr-
    code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Questionnaire/SH-12508041",
    "status":"in-progress",
    "subject":{
        "reference":"Patient/12508041"
    },
    "item":[
        {
            "id":"111",
            "linkId":"93",
            "extension":[
                {
                    "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/note",
                    "valueAnnotation":{
                        "text":"category/group comment"
                    }
                }
            ],
            "item":[
                {
                    "linkId":"93-123",
                    "answer":[
                        {
                            "valueQuantity":{
                                "value":"20",
                                "unit":"year (iso1000)",
                                "system":"http://unitsofmeasure.org",
                                "code":"a"
                            }
                        }
                    ]
                },
                {
                    "linkId":"93-456",
                    "answer":[
                        {
                            "valueString":"Answer text"
                        }
                    ]
                },
                {
                    "linkId":"93-789",
                    "extension":[
                        {
                            "url":"https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer",
                            "valueBoolean":true
                        }
                    ],
                    "answer":[
                        {
                            "valueCoding":{
                                "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature",
                                "code":"960439"
                            }
                        },
                        {
                            "valueString":"Other answer text"
                        }
                    ]
                }
            ]
        }
    ]
}
```

Nested Schema : item

Type:

array

A group or question item from the original questionnaire for which answers are provided.

Show Source

* Array of: object items


```
{
    "description":"<p>A group or question item from the original questionnaire for which answers are provided.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "linkId"
        ],
        "properties":{
            "id":{
                "type":"string",
                "description":"<p>Unique ID for inter-element referencing.</p>\n<p><b>Note:</b> If an
    <code>item.id</code> is returned on a read, it must be provided on an update.</p>\n",
                "example":"111"
            },
            "linkId":{
                "description":"<p>The item from the questionnaire that corresponds with this item in the
    QuestionnaireResponse resource.</p>\n",
                "example":"93"
            },
            "extension":{
                "description":"<p>Additional details about a given questionnaire response group
    item.</p>\n<p><b>Note:</b> Notes can only be added, not replaced or deleted.</p>\n",
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "url":{
                            "type":"string",
                            "enum":[
                                "https://fhir-ehr.cerner.com/r4/StructureDefinition/note"
                            ]
                        },
                        "valueAnnotation":{
                            "type":"object",
                            "required":[
                                "text"
                            ],
                            "properties":{
                                "text":{
                                    "type":"string",
                                    "example":"category/group comment"
                                }
                            }
                        }
                    }
                }
            },
            "item":{
                "type":"array",
                "description":"<p>Questions nested beneath a group.</p>\n",
                "items":{
                    "type":"object",
                    "required":[
                        "linkId",
                        "answer"
                    ],
                    "properties":{
                        "linkId":{
                            "type":"string",
                            "description":"<p>The item from the questionnaire that corresponds with this item in the
    QuestionnaireResponse resource.</p>\n",
                            "example":"93-123"
                        },
                        "extension":{
                            "description":"<p>Whether the answers come from a list of options.</p>\n<p><b>Note:</b>
    Extension must be provided when answering a choice or open choice type of question.</p>\n",
                            "type":"array",
                            "items":{
                                "type":"object",
                                "required":[
                                    "url",
                                    "valueBoolean"
                                ],
                                "properties":{
                                    "url":{
                                        "type":"string",
                                        "enum":[
                                            "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
```

```
                                    ]
                                },
                                "valueBoolean":{
                                    "type":"boolean",
                                    "enum":[
                                        true
                                    ]
                                }
                            }
                        }
                    },
                    "answer":{
                        "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only answer
    value types of String, Quantity, and Coding are supported.</p>\n",
                        "type":"array",
                        "items":{
                            "description":"<p>Single-valued answer to the question.</p>\n",
                            "type":"object",
                            "properties":{
                                "valueString":{
                                    "type":"string",
                                    "minLength":1,
                                    "example":"Answer text"
                                },
                                "valueCoding":{
                                    "type":"object",
                                    "required":[
                                        "system",
                                        "code"
                                    ],
                                    "properties":{
                                        "system":{
                                            "type":"string",

    "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                                        },
                                        "code":{
                                            "type":"string",
                                            "example":"960439"
                                        }
                                    }
                                },
                                "valueQuantity":{
                                    "type":"object",
                                    "required":[
                                        "value"
                                    ],
                                    "properties":{
                                        "value":{
                                            "type":"number",
                                            "example":20
                                        }
                                    }
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : subject

Type:

object

The subject of the questionnaire response. This subject is the patient that the answers apply to, but is not necessarily the source of information.

Show Source

* reference: string Example: Patient/12508041


```
{
    "description":"<p>The subject of the questionnaire response. This subject is the patient that the answers apply to,
    but is not necessarily the source of information.</p>\n",
    "type":"object",
    "properties":{
        "reference":{
            "type":"string",
            "example":"Patient/12508041"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

extension: array

extension

Additional details about a given questionnaire response group item.

Note: Notes can only be added, not replaced or deleted.

id: string

Unique ID for inter-element referencing.

Note: If an item.id is returned on a read, it must be provided on an update.

Example:

111

item: array

item

Questions nested beneath a group.

linkId(required):

The item from the questionnaire that corresponds with this item in the QuestionnaireResponse resource.

Example:


```
{
    "type":"object",
    "required":[
        "linkId"
    ],
    "properties":{
        "id":{
            "type":"string",
            "description":"<p>Unique ID for inter-element referencing.</p>\n<p><b>Note:</b> If an <code>item.id</code>
    is returned on a read, it must be provided on an update.</p>\n",
            "example":"111"
        },
        "linkId":{
            "description":"<p>The item from the questionnaire that corresponds with this item in the
    QuestionnaireResponse resource.</p>\n",
            "example":"93"
        },
        "extension":{
            "description":"<p>Additional details about a given questionnaire response group item.</p>\n<p><b>Note:</b>
    Notes can only be added, not replaced or deleted.</p>\n",
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "url":{
                        "type":"string",
                        "enum":[
                            "https://fhir-ehr.cerner.com/r4/StructureDefinition/note"
                        ]
                    },
                    "valueAnnotation":{
                        "type":"object",
                        "required":[
                            "text"
                        ],
                        "properties":{
                            "text":{
                                "type":"string",
                                "example":"category/group comment"
                            }
                        }
                    }
                }
            }
        },
        "item":{
            "type":"array",
            "description":"<p>Questions nested beneath a group.</p>\n",
            "items":{
                "type":"object",
                "required":[
                    "linkId",
                    "answer"
                ],
                "properties":{
                    "linkId":{
                        "type":"string",
                        "description":"<p>The item from the questionnaire that corresponds with this item in the
```

```
    QuestionnaireResponse resource.</p>\n",
                        "example":"93-123"
                    },
                    "extension":{
                        "description":"<p>Whether the answers come from a list of options.</p>\n<p><b>Note:</b>
    Extension must be provided when answering a choice or open choice type of question.</p>\n",
                        "type":"array",
                        "items":{
                            "type":"object",
                            "required":[
                                "url",
                                "valueBoolean"
                            ],
                            "properties":{
                                "url":{
                                    "type":"string",
                                    "enum":[
                                        "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
                                    ]
                                },
                                "valueBoolean":{
                                    "type":"boolean",
                                    "enum":[
                                        true
                                    ]
                                }
                            }
                        }
                    },
                    "answer":{
                        "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only answer
    value types of String, Quantity, and Coding are supported.</p>\n",
                        "type":"array",
                        "items":{
                            "description":"<p>Single-valued answer to the question.</p>\n",
                            "type":"object",
                            "properties":{
                                "valueString":{
                                    "type":"string",
                                    "minLength":1,
                                    "example":"Answer text"
                                },
                                "valueCoding":{
                                    "type":"object",
                                    "required":[
                                        "system",
                                        "code"
                                    ],
                                    "properties":{
                                        "system":{
                                            "type":"string",

    "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                                        },
                                        "code":{
                                            "type":"string",
                                            "example":"960439"
                                        }
                                    }
                                },
                                "valueQuantity":{
                                    "type":"object",
                                    "required":[
                                        "value"
                                    ],
                                    "properties":{
                                        "value":{
                                            "type":"number",
                                            "example":20
                                        }
                                    }
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : extension

Type:

array

Additional details about a given questionnaire response group item.

Note: Notes can only be added, not replaced or deleted.

Show Source

* Array of: object items


```
{
    "description":"<p>Additional details about a given questionnaire response group item.</p>\n<p><b>Note:</b> Notes can
     only be added, not replaced or deleted.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "url":{
                "type":"string",
                "enum":[
                    "https://fhir-ehr.cerner.com/r4/StructureDefinition/note"
                ]
            },
            "valueAnnotation":{
                "type":"object",
                "required":[
                    "text"
                ],
                "properties":{
                    "text":{
                        "type":"string",
                        "example":"category/group comment"
                    }
                }
            }
        }
    }
}
```


Nested Schema : item

Type:

array

Questions nested beneath a group.

Show Source

* Array of: object items


```
{
    "type":"array",
    "description":"<p>Questions nested beneath a group.</p>\n",
    "items":{
        "type":"object",
        "required":[
            "linkId",
            "answer"
        ],
        "properties":{
            "linkId":{
                "type":"string",
                "description":"<p>The item from the questionnaire that corresponds with this item in the
    QuestionnaireResponse resource.</p>\n",
                "example":"93-123"
            },
            "extension":{
                "description":"<p>Whether the answers come from a list of options.</p>\n<p><b>Note:</b> Extension must
    be provided when answering a choice or open choice type of question.</p>\n",
                "type":"array",
                "items":{
                    "type":"object",
                    "required":[
                        "url",
                        "valueBoolean"
                    ],
                    "properties":{
                        "url":{
                            "type":"string",
                            "enum":[
                                "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
                            ]
                        },
                        "valueBoolean":{
                            "type":"boolean",
                            "enum":[
                                true
                            ]
                        }
                    }
                }
            },
            "answer":{
                "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only answer value types
    of String, Quantity, and Coding are supported.</p>\n",
                "type":"array",
                "items":{
```

```
                    "description":"<p>Single-valued answer to the question.</p>\n",
                    "type":"object",
                    "properties":{
                        "valueString":{
                            "type":"string",
                            "minLength":1,
                            "example":"Answer text"
                        },
                        "valueCoding":{
                            "type":"object",
                            "required":[
                                "system",
                                "code"
                            ],
                            "properties":{
                                "system":{
                                    "type":"string",

    "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                                },
                                "code":{
                                    "type":"string",
                                    "example":"960439"
                                }
                            }
                        },
                        "valueQuantity":{
                            "type":"object",
                            "required":[
                                "value"
                            ],
                            "properties":{
                                "value":{
                                    "type":"number",
                                    "example":20
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* url: string Allowed Values: ["https://fhir-ehr.cerner.com/r4/StructureDefinition/note"]

* valueAnnotation: object valueAnnotation

```
{
    "type":"object",
    "properties":{
        "url":{
            "type":"string",
            "enum":[
                "https://fhir-ehr.cerner.com/r4/StructureDefinition/note"
            ]
        },
        "valueAnnotation":{
            "type":"object",
            "required":[
                "text"
            ],
            "properties":{
                "text":{
                    "type":"string",
                    "example":"category/group comment"
                }
            }
        }
    }
}
```

Nested Schema : valueAnnotation

Type:

object

Show Source

* text(required): string Example: category/group comment

```
{
    "type":"object",
```

```
    "required":[
        "text"
    ],
    "properties":{
        "text":{
            "type":"string",
            "example":"category/group comment"
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

answer(required): array

answer

The respondent's answer to the question.

Note: Only answer value types of String, Quantity, and Coding are supported.

extension: array

extension

Whether the answers come from a list of options.

Note: Extension must be provided when answering a choice or open choice type of question.

linkId(required): string

The item from the questionnaire that corresponds with this item in the QuestionnaireResponse resource.

Example:

93-123


```
{
    "type":"object",
    "required":[
        "linkId",
        "answer"
    ],
    "properties":{
        "linkId":{
            "type":"string",
            "description":"<p>The item from the questionnaire that corresponds with this item in the
    QuestionnaireResponse resource.</p>\n",
            "example":"93-123"
        },
        "extension":{
            "description":"<p>Whether the answers come from a list of options.</p>\n<p><b>Note:</b> Extension must be
    provided when answering a choice or open choice type of question.</p>\n",
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "url",
                    "valueBoolean"
                ],
                "properties":{
                    "url":{
                        "type":"string",
                        "enum":[
                            "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
                        ]
                    },
                    "valueBoolean":{
                        "type":"boolean",
                        "enum":[
                            true
                        ]
                    }
                }
            }
        },
        "answer":{
            "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only answer value types of
    String, Quantity, and Coding are supported.</p>\n",
            "type":"array",
            "items":{
                "description":"<p>Single-valued answer to the question.</p>\n",
                "type":"object",
                "properties":{
                    "valueString":{
                        "type":"string",
                        "minLength":1,
                        "example":"Answer text"
```

```
                    },
                    "valueCoding":{
                        "type":"object",
                        "required":[
                            "system",
                            "code"
                        ],
                        "properties":{
                            "system":{
                                "type":"string",
                                "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                            },
                            "code":{
                                "type":"string",
                                "example":"960439"
                            }
                        }
                    },
                    "valueQuantity":{
                        "type":"object",
                        "required":[
                            "value"
                        ],
                        "properties":{
                            "value":{
                                "type":"number",
                                "example":20
                            }
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : answer

Type:

array

The respondent's answer to the question.

Note: Only answer value types of String, Quantity, and Coding are supported.

Show Source

Array of: object

items

Single-valued answer to the question.

```
{
    "description":"<p>The respondent's answer to the question.</p>\n<p><b>Note:</b> Only answer value types of String,
    Quantity, and Coding are supported.</p>\n",
    "type":"array",
    "items":{
        "description":"<p>Single-valued answer to the question.</p>\n",
        "type":"object",
        "properties":{
            "valueString":{
                "type":"string",
                "minLength":1,
                "example":"Answer text"
            },
            "valueCoding":{
                "type":"object",
                "required":[
                    "system",
                    "code"
                ],
                "properties":{
                    "system":{
                        "type":"string",
                        "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                    },
                    "code":{
                        "type":"string",
                        "example":"960439"
                    }
                }
            },
            "valueQuantity":{
                "type":"object",
                "required":[
                    "value"
                ],
                "properties":{
                    "value":{
                        "type":"number",
```

```
                    "example":20
                }
            }
        }
    }
}
```

Nested Schema : extension

Type:

array

Whether the answers come from a list of options.

Note: Extension must be provided when answering a choice or open choice type of question.

Show Source

* Array of: object items

```
{
    "description":"<p>Whether the answers come from a list of options.</p>\n<p><b>Note:</b> Extension must be provided
    when answering a choice or open choice type of question.</p>\n",
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "url",
            "valueBoolean"
        ],
        "properties":{
            "url":{
                "type":"string",
                "enum":[
                    "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
                ]
            },
            "valueBoolean":{
                "type":"boolean",
                "enum":[
                    true
                ]
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Single-valued answer to the question.

Show Source

* valueCoding: object valueCoding

* valueQuantity: object valueQuantity

* valueString: string Minimum Length: 1 Example: Answer text

```
{
    "description":"<p>Single-valued answer to the question.</p>\n",
    "type":"object",
    "properties":{
        "valueString":{
            "type":"string",
            "minLength":1,
            "example":"Answer text"
        },
        "valueCoding":{
            "type":"object",
            "required":[
                "system",
                "code"
            ],
            "properties":{
                "system":{
                    "type":"string",
                    "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
                },
                "code":{
                    "type":"string",
                    "example":"960439"
                }
            }
        },
        "valueQuantity":{
            "type":"object",
            "required":[
                "value"
```

```
            ],
            "properties":{
                "value":{
                    "type":"number",
                    "example":20
                }
            }
        }
    }
}
```

Nested Schema : valueCoding

Type:

object

Show Source

* code(required): string Example: 960439

* system(required): string Example: https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature

```
{
    "type":"object",
    "required":[
        "system",
        "code"
    ],
    "properties":{
        "system":{
            "type":"string",
            "example":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/nomenclature"
        },
        "code":{
            "type":"string",
            "example":"960439"
        }
    }
}
```

Nested Schema : valueQuantity

Type:

object

Show Source

* value(required): number Example: 20

```
{
    "type":"object",
    "required":[
        "value"
    ],
    "properties":{
        "value":{
            "type":"number",
            "example":20
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* url(required): string Allowed Values: ["https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"]

* valueBoolean(required): boolean Allowed Values: [true]

```
{
    "type":"object",
    "required":[
        "url",
        "valueBoolean"
    ],
    "properties":{
        "url":{
            "type":"string",
            "enum":[
                "https://fhir-ehr.cerner.com/r4/StructureDefinition/choice-answer"
            ]
        },
        "valueBoolean":{
            "type":"boolean",
            "enum":[
                true
            ]
        }
    }
```

```
}
```

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

ETag: string

The identifier for a specific version of a resource. This will be formatted as W/"" and is used to specify the If-Match header value on subsequent updates.

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]

```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
```

```
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        },
                        "location":{
                            "type":"array",
                            "items":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string
* details: object details
* location: array location
* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding
* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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RelatedPerson

File: r4-rest-api-v32/api-relatedperson.html

Description

RelatedPerson

The following topic describes information for the RelatedPerson resource.

Overview

The RelatedPerson resource provides information about a person who is involved in the care of a patient but has no formal responsibility. Related persons typically have a personal relationship with the patient such as a spouse, relative, friend, guardian, or attorney. Related persons are commonly a source of information about a patient.

The distinction between practitioner and related person is that a practitioner cares for multiple patients on behalf of a healthcare facility where a related person performs care tasks for a specific patient and is not associated with any healthcare facility.

This resource supports the US Core RelatedPerson Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The following fields are returned if valued:

RelatedPerson ID

RelatedPerson Identifier

Active

Patient

Relationship

Related person's name

Telecom information (may include phone and email)

Gender (administrative)

Date of birth

Address

Communication (preferred language)

Extensions including related person encounter, relationship level, period, and relation (for more information, see the Extensions section of this topic)

Terminology Bindings

RelatedPerson.identifier.type

Description: The type that determines which identifier to use for a specific purpose.

Details: Identifier Type Codes

System: http://terminology.hl7.org/CodeSystem/v2-0203

Details: For more information, see Code Set 4 Person Identifier (Alias) Types under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/4

RelatedPerson.communication.language

Description: The language used to communicate with a related person about a patient's health.

Details: CommonLanguages

System: urn:ietf:bcp:47

Details: For more information, see Code Set 36 Languages under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/36

RelatedPerson.relationship

Description: The nature of the relationship between a patient and the related person.

Details: PatientRelationshipType

System: http://terminology.hl7.org/CodeSystem/v2-0131

System: http://terminology.hl7.org/CodeSystem/v3-RoleCode

Details: For more information, see Code Set 351 Person Relationship Types under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/351

RelatedPerson.relationship.extension[x].valueCodeableConcept

Description: The related person's familial relationship to the patient. This binding is used in relation extensions.

Details: PatientRelationshipType

System: http://terminology.hl7.org/CodeSystem/v2-0131

System: http://terminology.hl7.org/CodeSystem/v3-RoleCode

Details: For more information, see Code Set 40 Person Relationships under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/40

RelatedPerson.extension[x].valueCodeableConcept (for more information, see the Extensions section of this topic).

Description: The resource's relationship to the patient or encounter level. This binding is used in relationship-level extensions.

Details: Relationship Level

System: http://hl7.org/fhir/resource-types

Extensions

All URLs for custom extensions are defined as https://fhir-ehr.cerner.com/r4/StructureDefinition/{id}

Table - Extensions

ID	Value[x] Type	Description
related-person-encounter	reference	Reference to the encounter associated with the encounter-level related person.
relationship-level	CodeableConcept	The resource's relationship to the patient or encounter.
period	Period	A time period defined by a start and end date and time.
relation	CodeableConcept	The related person's familial relationship to the patient.

RelatedPerson

Create a related person record

Method: post Path: /RelatedPerson

Get a list of related person records

Method: get Path: /RelatedPerson

Get a related person record by ID

Method: get Path: /RelatedPerson/{ID}

Patch a related person record

Method: patch Path: /RelatedPerson/{ID}

Create a related person record

File: r4-rest-api-v32/op-relatedperson-post.html

This operation supports the following authorization types:

Creates a new encounter-level or patient-level related person record.

Note:

Only the body fields mentioned below are supported. Unsupported fields are ignored, or can result in errors.

* Provider

* System

Request

Supported Media Types

* application/fhir+json

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

The body of the create operation.

Root Schema : RelatedPersonCreate

Type:

object

A summary representation of the Create (POST) operation for RelatedPerson.

Show Source

active: boolean

Note: If set, the value must be true. Example:

Whether this related person record is in active use.

```
{
  "active": true
}
```

address: string

Note: Example:

Addresses for the individual.

* Each address must contain the use field.

* Each address must not contain the text field.

* When specifying a period, the fields must include a time component with a timezone.

* If line is provided, only the first four values are used. Additional values are ignored.

```
{
  "address": [
    {
      "use": "home",
      "line": [
         "1212 Faircastle Drive",
         "Apartment 406"
      ],
      "city": "KC",
      "district": "Jackson",
      "state": "KS",
      "postalCode": "64199",
      "country": "United States of America",
      "period": {
        "start": "2012-05-17T15:33:18.000Z"
      }
    }
  ]
```

```
}
```

birthDate: string

Example:

The date of birth for the individual.

```
{
  "birthDate": "1990-09-15"
}
```

communication:

Note: Example:

Language which may be used to communicate with the related person.

* Only one communication may be provided.

* communication.prefered if provided should be true.

```
{
  "communication": [
    {
      "language": {
        "coding": [
          {
            "system": "urn:ietf:bcp:47",
          "code": "en",
            "display": "English"
          }
        ]
      },
      "preferred": true
    }
  ]
}
```

gender: string

Example:

Administrative gender is the gender that the related person is considered to have for administration and record keeping purposes.

```
{
  "gender": "male"
}
```

identifier: string

Note: Example:

The identifiers for the related person.

* Each identifier must not contain the use fields.

* Each identifier must contain the type, system, and value fields.

* When specifying a period, the fields must include a time component with a timezone.

```
{
  "identifier": [
    {
      "type": {
        "coding": [
          {
            "code": "MR",
            "system": "http://hl7.org/fhir/v2/0203"
          }
        ]
      },
      "system": "urn:oid:5.5.5.5.5.5",
      "value": "<UNIQUE IDENTIFIER>",
      "period": {
        "start": "2016-01-02T00:00:00-05:00",
        "end": "2020-01-02T00:00:00-05:00"
      }
    }
  ]
}
```

name(required): string

Note: Example:

The name associated with the individual.

* Exactly one name must be provided. The provided name must meet the following conditions:

* Must have a use of 'official'.

* Cannot contain the text field.

* Must contain either the given or family fields.

* Up to two given names may be provided. If additional given names are required they must be provided in the second value as a single string separated by blank spaces.

* Only one prefix and one suffix may be provided.

* period.end is not supported.

* When specifying a period, the fields must include a time component with a timezone.


```
{
  "name": [
    {
      "use": "official",
      "family": "Doe",
      "given": [
        "John",
        "Smith"
      ],
      "period": {
        "start": "2010-05-17T14:54:31.000Z"
      }
    }
  ]
}
```


patient(required): object

patient

Example:

Patient that the individual is related to for the encounter-level RelatedPerson.


```
{
  "patient": {
    "reference": "Patient/12724066"
  }
}
```


Related Person Encounter: object

Related Person Encounter

Note: Example:

A reference to an encounter.

* The valueReference.reference must be an encounter reference.


```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter",
      "valueReference": {
        "reference": "Encounter/97939518"
      }
    }
  ]
}
```


relationship(required): object

relationship

Note: Example:

The nature of the relationship between a patient and the related person.

* When specifying a period, the fields must include a time component with a timezone.

* The coding field can contain codings from the standard or proprietary system, but not both.

* Only one coding may be provided.


```
  {
    "relationship":
    [
      {
        "coding":
          [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
              "code": "FAMMEMB"
            }
          ],
        "extension":
          [
            {
              "valuePeriod":
                {
                  "start": "2016-12-19T16:44:25.000Z",
                  "end": "2022-05-01T16:56:56.000Z"
                },
              "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/period"
            },
            {
```

```
            "valueCodeableConcept":
              {
                "coding":
                  [
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                      "code": "SIS"
                    }
                  ]
              },
            "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/relation"
          }
        ]
    }
  ]
}
```

Relationship Level: object

Relationship Level

Note: Example:

The resource's relationship to either the patient or encounter level.

* The valueCodeableConcept.coding[0].code can be Encounter or Patient.

```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/resource-types",
            "code": "Encounter",
          }
        ]
      }
    }
  ]
}
```

* resourceType(required): string Allowed Values: ["RelatedPerson"] The type of the FHIR resource.

telecom: string

Note: Example:

A contact detail, for example, a telephone number or an email address, by which the individual may be contacted.

* Each telecom must contain the system, use, and value fields.

* System must be a phone or an email.

* When specifying a period, the fields must include a time component with a timezone.

```
{
  "telecom": [
    {
      "system": "phone",
      "value": "8168229121",
      "use": "home",
      "period": {
        "start": "2012-05-17T15:33:18.000Z"
      }
    },
    {
      "system": "email",
      "value": "testemail@example.com",
      "use": "home",
      "period": {
        "start": "2008-01-10T15:33:18.000Z"
      }
    }
  ]
}
```

```
{
    "description":"A summary representation of the Create (POST) operation for RelatedPerson.",
    "type":"object",
    "required":[
        "resourceType",
        "relationship",
        "patient",
        "name"
    ],
    "properties":{
        "resourceType":{
            "description":"The type of the FHIR resource.\n",
            "type":"string",
            "enum":[
                "RelatedPerson"
            ]
```

```
    },
    "relationship":{
        "description":"<p>The nature of the relationship between a patient and the related
person.</p>\n\nNote:\n<ul>\n  <li> When specifying a <code>period</code>, the fields must include a time component
with a timezone. </li>\n  <li> The <code>coding</code> field can contain codings from the standard or proprietary
system, but not both. </li>\n  <li> Only one <code>coding</code> may be provided.
</li>\n</ul>\n\n\nExample:\n\n<pre>\n  <code>\n  {\n    \"relationship\":\n    [\n      {\n        \"coding\":\n
          [\n            {\n              \"system\": \"http://terminology.hl7.org/CodeSystem/v3-RoleCode\",\n
              \"code\": \"FAMMEMB\"\n            }\n          ],\n        \"extension\":\n          [\n
{\n              \"valuePeriod\":\n                {\n                  \"start\": \"2016-12-19T16:44:25.000Z\",\n
                \"end\": \"2022-05-01T16:56:56.000Z\"\n              },\n              \"url\": \"https://fhir-
ehr.cerner.com/r4/StructureDefinition/period\"\n            },\n            {\n
\"valueCodeableConcept\":\n                {\n                  \"coding\":\n                    [\n
                    {\n                      \"system\":
\"http://terminology.hl7.org/CodeSystem/v3-RoleCode\",\n                        \"code\": \"SIS\"\n
                    }\n                    ]\n                },\n              \"url\": \"https://fhir-
ehr.cerner.com/r4/StructureDefinition/relation\"\n            }\n          ]\n      }\n    ]\n  }\n
</code>\n</pre>\n",
        "type":"object",
        "properties":{
            "text":{
                "type":"string"
            },
            "coding":{
                "type":"array",
                "items":{
                    "type":"object",
                    "properties":{
                        "system":{
                            "type":"string"
                        },
                        "code":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    },
    "patient":{
        "description":"<p>Patient that the individual is related to for the encounter-level
RelatedPerson.</p>\nExample:\n\n<pre><code>{\n  \"patient\": {\n    \"reference\": \"Patient/12724066\"\n
}\n}\n\n</pre></code>\n",
        "type":"object",
        "properties":{
            "reference":{
                "type":"string"
            }
        }
    },
    "name":{
        "description":"<p>The name associated with the individual.</p>\n\nNote:\n<ul>\n  <li> Exactly one name must
be provided. The provided name must meet the following conditions:\n    <ul>\n      <li> Must have a
<code>use</code> of 'official'. </li>\n      <li> Cannot contain the <code>text</code> field. </li>\n      <li> Must
 contain either the <code>given</code> or <code>family</code> fields. </li>\n      <li> Up to two <code>given</code>
 names may be provided. If additional given names are required they must be provided in the second value as a single
 string separated by blank spaces. </li>\n      <li> Only one <code>prefix</code> and one <code>suffix</code> may be
 provided. </li>\n      <li> <code>period.end</code> is not supported. </li>\n      <li> When specifying a
<code>period</code>, the fields must include a time component with a timezone. </li>\n    </ul>\n
</li>\n</ul>\n\n\nExample:\n\n<pre><code>\n{\n  \"name\": [\n    {\n      \"use\": \"official\",\n      \"family\":
\"Doe\",\n      \"given\": [\n        \"John\",\n        \"Smith\"\n      ],\n      \"period\": {\n
\"start\": \"2010-05-17T14:54:31.000Z\"\n      }\n    }\n  ]\n}\n\n</pre></code>\n",
        "type":"string"
    },
    "Related Person Encounter":{
        "description":"<p>A reference to an encounter.</p>\n\nNote:\n<ul>\n  <li>The
<code>valueReference.reference</code> must be an encounter reference.</li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n
\"extension\": [\n    {\n      \"url\": \"https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-
encounter\",\n      \"valueReference\": {\n        \"reference\": \"Encounter/97939518\"\n      }\n    }\n
]\n}\n\n</pre></code>\n",
        "type":"object",
        "properties":{
            "url":{
                "type":"string"
            }
        }
    },
    "Relationship Level":{
        "description":"<p>The resource's relationship to either the patient or encounter level.</p>\n\nNote:\n<ul>\n
  <li> The <code>valueCodeableConcept.coding[0].code</code> can be <a target=\"_blank\"
href=\"https://hl7.org/fhir/R4/encounter.html\"><code>Encounter</code></a> or <a target=\"_blank\"
href=\"https://hl7.org/fhir/R4/patient.html\"><code>Patient</code></a>. </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n
  \"extension\": [\n    {\n      \"url\": \"https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-
level\",\n      \"valueCodeableConcept\": {\n        \"coding\": [\n          {\n            \"system\":
\"http://hl7.org/fhir/resource-types\",\n            \"code\": \"Encounter\",\n          }\n        ]\n      }\n
}\n  ]\n}\n\n</pre></code>\n",
        "type":"object",
        "properties":{
            "url":{
```

```
                    "type":"string"
                }
            }
        },
        "identifier":{
            "description":"<p>The identifiers for the related person.</p>\n\nNote:\n<ul>\n    <li> Each identifier must
    not contain the <code>use</code> fields. </li>\n    <li> Each identifier must contain the <code>type</code>,
    <code>system</code>, and <code>value</code> fields. </li>\n    <li> When specifying a <code>period</code>, the
    fields must include a time component with a timezone. </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n  \"identifier\":
    [\n    {\n      \"type\": {\n        \"coding\": [\n          {\n            \"code\": \"MR\",\n
    \"system\": \"http://hl7.org/fhir/v2/0203\"\n          }\n        ]\n      },\n      \"system\":
    \"urn:oid:5.5.5.5.5.5\",\n      \"value\": \"&lt;UNIQUE IDENTIFIER&gt;\",\n      \"period\": {\n        \"start\":
    \"2016-01-02T00:00:00-05:00\",\n        \"end\": \"2020-01-02T00:00:00-05:00\"\n      }\n    }\n
    ]\n}\n\n</pre></code>\n",
            "type":"string"

        },
        "address":{
            "description":"<p>Addresses for the individual.</p>\n\nNote:\n<ul>\n  <li> Each address must contain the
    <code>use</code> field. </li>\n  <li> Each address must not contain the <code>text</code> field. </li>\n  <li> When
    specifying a <code>period</code>, the fields must include a time component with a timezone. </li>\n  <li> If
    <code>line</code> is provided, only the first four values are used. Additional values are ignored.
    </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n  \"address\": [\n    {\n      \"use\": \"home\",\n      \"line\": [\n
             \"1212 Faircastle Drive\",\n          \"Apartment 406\"\n      ],\n      \"city\": \"KC\",\n
    \"district\": \"Jackson\",\n      \"state\": \"KS\",\n      \"postalCode\": \"64199\",\n      \"country\": \"United
    States of America\",\n      \"period\": {\n        \"start\": \"2012-05-17T15:33:18.000Z\"\n      }\n    }\n
    ]\n}\n\n</pre></code>\n",
            "type":"string"
        },
        "telecom":{
            "description":"<p>A contact detail, for example, a telephone number or an email address, by which the
    individual may be contacted.</p>\n\nNote:\n<ul>\n  <li> Each telecom must contain the <code>system</code>,
    <code>use</code>, and <code>value</code> fields. </li>\n  <li> System must be a <code>phone</code> or an
    <code>email</code>. </li>\n  <li> When specifying a <code>period</code>, the fields must include a time component
    with a timezone. </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n  \"telecom\": [\n    {\n      \"system\": \"phone\",\n
          \"value\": \"8168229121\",\n      \"use\": \"home\",\n      \"period\": {\n        \"start\":
    \"2012-05-17T15:33:18.000Z\"\n      }\n    },\n    {\n      \"system\": \"email\",\n      \"value\":
    \"testemail@example.com\",\n      \"use\": \"home\",\n      \"period\": {\n        \"start\":
    \"2008-01-10T15:33:18.000Z\"\n      }\n    }\n  ]\n}\n\n</pre></code>\n",
            "type":"string"
        },
        "active":{
            "description":"<p>Whether this related person record is in active use.</p>\n\nNote:\nIf set, the value must
    be true.\n\nExample:\n<pre><code>{\n  \"active\": true\n}\n\n</pre></code>\n",
            "type":"boolean"
        },
        "gender":{
            "description":"<p>Administrative gender is the gender that the related person is considered to have for
    administration and record keeping purposes.</p>\nExample:\n\n<pre><code>{\n  \"gender\":
    \"male\"\n}\n\n</pre></code>\n",
            "type":"string"
        },
        "birthDate":{
            "description":"<p>The date of birth for the individual.</p>\nExample:\n\n<pre><code>{\n  \"birthDate\":
    \"1990-09-15\"\n}\n\n</pre></code>\n",
            "type":"string"
        },
        "communication":{
            "description":"<p>Language which may be used to communicate with the related person.</p>\n\nNote:\n<ul>\n
    <li> Only one communication may be provided. </li>\n  <li> <code>communication.prefered</code> if provided should be
     <code>true</code>. </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n  \"communication\": [\n    {\n      \"language\":
    {\n        \"coding\": [\n          {\n            \"system\": \"urn:ietf:bcp:47\",\n          \"code\": \"en\",\n
              \"display\": \"English\"\n          }\n        ]\n      },\n      \"preferred\": true\n    }\n
    ]\n}\n\n</pre></code>\n"
        }
    }
}
```


Nested Schema : patient

Type:

object

Example:

Patient that the individual is related to for the encounter-level RelatedPerson.


```
{
  "patient": {
    "reference": "Patient/12724066"
  }
}
```


Show Source

* reference: string


```
{
    "description":"<p>Patient that the individual is related to for the encounter-level
    RelatedPerson.</p>\nExample:\n\n<pre><code>{\n  \"patient\": {\n    \"reference\": \"Patient/12724066\"\n
    }\n}\n\n</pre></code>\n",
    "type":"object",
```

```
    "properties":{
        "reference":{
            "type":"string"
        }
    }
}
```

Nested Schema : Related Person Encounter

Type:

object

Note: Example:

A reference to an encounter.

* The valueReference.reference must be an encounter reference.


```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter",
      "valueReference": {
        "reference": "Encounter/97939518"
      }
    }
  ]
}
```


Show Source

* url: string


```
{
    "description":"<p>A reference to an encounter.</p>\n\nNote:\n<ul>\n  <li>The <code>valueReference.reference</code>
    must be an encounter reference.</li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n  \"extension\": [\n    {\n      \"url\":
     \"https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter\",\n      \"valueReference\": {\n
            \"reference\": \"Encounter/97939518\"\n      }\n    }\n  ]\n}\n\n</pre></code>\n",
    "type":"object",
    "properties":{
        "url":{
            "type":"string"
        }
    }
}
```


Nested Schema : relationship

Type:

object

Note: Example:

The nature of the relationship between a patient and the related person.

* When specifying a period, the fields must include a time component with a timezone.

* The coding field can contain codings from the standard or proprietary system, but not both.

* Only one coding may be provided.


```
  {
    "relationship":
    [
      {
        "coding":
          [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
              "code": "FAMMEMB"
            }
          ],
        "extension":
          [
            {
              "valuePeriod":
                {
                  "start": "2016-12-19T16:44:25.000Z",
                  "end": "2022-05-01T16:56:56.000Z"
                },
              "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/period"
            },
            {
              "valueCodeableConcept":
                {
                  "coding":
                    [
                      {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                        "code": "SIS"
                      }
                    ]
                },
```

```
            "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/relation"
          }
        ]
      }
    ]
  }
```

Show Source

* coding: array coding

* text: string

```
{
    "description":"<p>The nature of the relationship between a patient and the related person.</p>\n\nNote:\n<ul>\n
    <li> When specifying a <code>period</code>, the fields must include a time component with a timezone. </li>\n  <li>
    The <code>coding</code> field can contain codings from the standard or proprietary system, but not both. </li>\n
    <li> Only one <code>coding</code> may be provided. </li>\n</ul>\n\n\nExample:\n\n<pre>\n  <code>\n  {\n
    \"relationship\":\n    [\n      {\n        \"coding\":\n          [\n            {\n              \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-RoleCode\",\n              \"code\": \"FAMMEMB\"\n            }\n
          ],\n        \"extension\":\n          [\n            {\n              \"valuePeriod\":\n
    {\n                \"start\": \"2016-12-19T16:44:25.000Z\",\n                \"end\":
    \"2022-05-01T16:56:56.000Z\"\n              },\n              \"url\": \"https://fhir-
    ehr.cerner.com/r4/StructureDefinition/period\"\n            },\n            {\n
    \"valueCodeableConcept\":\n              {\n                \"coding\":\n                  [\n
                    {\n                      \"system\":
    \"http://terminology.hl7.org/CodeSystem/v3-RoleCode\",\n                      \"code\": \"SIS\"\n
                    }\n                    ]\n              },\n              \"url\": \"https://fhir-
    ehr.cerner.com/r4/StructureDefinition/relation\"\n            }\n          ]\n      }\n    ]\n  }\n
    </code>\n</pre>\n",
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : Relationship Level

Type:

object

Note: Example:

The resource's relationship to either the patient or encounter level.

* The valueCodeableConcept.coding[0].code can be Encounter or Patient.

```
{
  "extension": [
    {
      "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/resource-types",
            "code": "Encounter",
          }
        ]
      }
    }
  ]
}
```

Show Source

* url: string

```
{
    "description":"<p>The resource's relationship to either the patient or encounter level.</p>\n\nNote:\n<ul>\n  <li>
    The <code>valueCodeableConcept.coding[0].code</code> can be <a target=\"_blank\"
    href=\"https://hl7.org/fhir/R4/encounter.html\"><code>Encounter</code></a> or <a target=\"_blank\"
    href=\"https://hl7.org/fhir/R4/patient.html\"><code>Patient</code></a>. </li>\n</ul>\n\n\nExample:\n\n<pre><code>{\n
      \"extension\": [\n    {\n      \"url\": \"https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-
    level\",\n      \"valueCodeableConcept\": {\n        \"coding\": [\n          {\n            \"system\":
    \"http://hl7.org/fhir/resource-types\",\n            \"code\": \"Encounter\",\n          }\n        ]\n      }\n
    }\n  ]\n}\n\n</pre></code>\n",
    "type":"object",
```

```
    "properties":{
        "url":{
            "type":"string"
        }
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/fhir+json

Default Response

: : : The response header indicates the current version to use on a subsequent update.

Example create request and response for a patient-level related person record.

Example Request

```
POST https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson
```

Example Request Body

```
{
    'resourceType': 'RelatedPerson',
    'extension': [
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://hl7.org/fhir/resource-types',
              'code': 'Patient'
            }
          ]
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
      }
    ],
    'identifier': [
      {
        'type': {
          'coding': [
            {
```

```
          'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
          'code': '22'
        }
      ]
    },
    'system': 'urn:oid:2.16.840.1.113883.3.42.10001.100001.12',
    'value': '1000000105'
  }
],
'active': true,
'patient': {
  'reference': 'Patient/12724068'
},
'relationship': [
  {
    'extension': [
      {
        'valuePeriod': {
          'start': '2019-12-26T16:06:26.000Z'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
              'code': 'MTH'
            }
          ]
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ],
    'coding': [
      {
        'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
        'code': 'GT'
      }
    ]
  }
],
'name': [
  {
    'use': 'official',
    'family': 'SMART',
    'given': [
      'NANCY'
    ],
    'period': {
      'start': '2019-12-26T15:12:34.000Z'
    }
  }
],
'telecom': [
  {
    'extension': [
      {
        'valueUrl': '13215555555',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '13215555555',
    'use': 'work',
    'period': {
      'start': '2021-03-22T14:12:37.000Z'
    }
  }
],
'gender': 'female',
'birthDate': '1980-08-11',
'address': [
  {
    'use': 'home',
    'line': [
      '3P'
    ],
    'city': 'KANSAS CITY',
    'district': 'Jackson',
    'state': 'MO',
    'postalCode': '641510001',
    'country': 'US',
    'period': {
      'start': '2020-12-08T15:05:44.000Z'
    }
  }
],
'communication': [
```

```
      {
        'language': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
              'code': '151'
            },
            {
              'system': 'urn:ietf:bcp:47',
              'code': 'en'
            }
          ]
        },
        'preferred': 'true'
      }
    ]
  }
```

Example Response

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Fri, 16 Jul 2021 17:23:14 GMT
  Etag: W/"0"
  Location: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/12724066-12724068
  Last-Modified: Fri, 16 Jul 2021 17:23:13 GMT
  Vary: Origin
  X-Request-Id: 11111111-1111-1111-1111-111111111111
  opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-create-encounter-level Response

: : : The response header indicates the current version to use on a subsequent update.

Example create request and response for an encounter-level related person record.

Example Request

```
POST https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson
```

Example Request Body

```
{
    'resourceType': 'RelatedPerson',
    'extension': [
      {
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter',
        'valueReference': {
          'reference': 'Encounter/97939518'
        }
      },
      {
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level',
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://hl7.org/fhir/resource-types',
              'code': 'Encounter'
            }
          ]
        }
      }
    ],
    'patient': {
      'reference': 'Patient/12724066'
    },
    'relationship': [
      {
        'coding': [
          {
            'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351',
            'code': '1152',
            'display': 'Emergency Contact'
          }
        ],
```

```
    'extension': [
      {
        'valuePeriod': {
          'start': '2016-12-19T16:44:25.000Z',
          'end': '2022-05-01T16:56:56.000Z'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
              'code': 'SIS'
            }
          ]
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ]
  }
],
'name': [
  {
    'use': 'official',
    'family': 'Doe',
    'given': [
      'John',
      'Smith'
    ],
    'period': {
      'start': '2010-05-17T14:54:31.000Z'
    }
  }
],
'communication': [
  {
    'language': {
      'coding': [
        {
          'system': 'urn:ietf:bcp:47',
          'code': 'en',
          'display': 'English'
        }
      ]
    },
    'preferred': true
  }
],
'identifier': [
  {
    'type': {
      'coding': [
        {
          'code': 'AN',
          'system': 'http://terminology.hl7.org/CodeSystem/v2-0203'
        }
      ]
    },
    'system': 'urn:oid:5.5.5.5.5.5',
    'value': '',
    'period': {
      'start': '2016-01-02T00:00:00-05:00',
      'end': '2020-01-02T00:00:00-05:00'
    }
  }
],
'address': [
  {
    'use': 'home',
    'line': [
      '1212 Faircastle Drive',
      'Apartment 406'
    ],
    'city': 'KC',
    'district': 'Jackson',
    'state': 'KS',
    'postalCode': '64199',
    'country': 'United States of America',
    'period': {
      'start': '2012-05-17T15:33:18.000Z'
    }
  }
],
'gender': 'male',
'birthDate': '1990-09-15',
'telecom': [
  {
    'system': 'phone',
```

```
        'value': '8168229121',
        'use': 'home',
        'period': {
          'start': '2012-05-17T15:33:18.000Z'
        }
      },
      {
        'system': 'email',
        'value': 'testemail@example.com',
        'use': 'home',
        'period': {
          'start': '2008-01-10T15:33:18.000Z'
        }
      }
    ]
  }
```

Example Response

```
  Cache-Control: no-cache
  Content-Length: 0
  Content-Type: application/fhir+json
  Date: Wed, 27 Mar 2019 17:23:14 GMT
  Etag: W/"0"
  Location: https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/E-12747484-97939518
  Last-Modified: Thu, 17 Dec 2020 16:37:42 GMT
  Vary: Origin
  X-Request-Id: 11111111-1111-1111-1111-111111111111
  opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a list of related person records

File: r4-rest-api-v32/op-relatedperson-get.html

This operation supports the following authorization types:

Retrieves related person records that meet supplied query parameters.

Note:

-relationship-level is required for patient and identifier searches.

-relationship-level can only be provided with patient or identifier.

-relationship-level can only be set to Patient.

* Provider

* Patient

* System

Request

Query Parameters

* -encounter: string A reference to an encounter associated with the related person. This parameter is required if the identifier, _id, and patient parameters are not used. Example: -encounter=97697434

-relationship-level: string

The level of relationship for the related person. This parameter is required if either of the or parameters are used. Example:

patient

-encounter

```
-relationship-level=http://hl7.org/fhir/resource-types|Patient
```

* _id: string The logical resource ID. This parameter is required if the patient, identifier, and -encounter parameters are not used. Example: _id=1234-5678

_revinclude: string

The Provenance resource entries to be returned as part of the bundle. Example:

* May be provided once with the Provenance:target value. Example: _revinclude=Provenance:target.

* May be provided with the _id or patient parameters. Example: _id=214938095&_revinclude=Provenance:target.

* When _revinclude is provided in a request to the closed endpoint, the OAuth2 token must include the scope corresponding with the authorization type, such as user/Provenance.read, patient/Provenance.read, or system/Provenance.read.

_revinclude=Provenance:target

* identifier: string A related person identifier. This parameter is required if the patient, _id, and -encounter parameters are not used. When provided, the identifier query parameter must include both a system and a code. Example: identifier=urn:oid:2.16.840.1.113883.3.13.6|RANDES011E8442-2E86-4A00-8394-EFC6A3434A8F

* patient: string A reference to a patient associated with the related person. This parameter is required if the identifier, _id, and -encounter parameters are not used. Example: patient=14067892

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

: :

Example search by patient and -relationship-level=Patient.

Example Request

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?patient=12724066&-relationship-
    level=Patient
```

Example Response


```
{
    'resourceType': 'Bundle',
    'id': '106abf01-6358-44d4-8f66-37526caa6ab4',
    'type': 'searchset',
    'total': 2,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?patient=12724066'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/12724066-12724068',
        'resource': {
          'resourceType': 'RelatedPerson',
          'id': '12724066-12724068',
          'meta': {
            'versionId': 145,
            'lastUpdated': '2021-03-22T14:12:37.000Z'
          },
          'text': {
            'status': 'extensions',
            'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Related Person</b></p><p><b>Name</b>: SMART,
    NANCY</p><p><b>DOB</b>: Aug 11, 1980</p><p><b>Gender</b>: female</p><p><b>Patient</b>: SMART,
    HAILEY</p><p><b>Status</b>: Active</p><p><b>Relationship</b>: Default Guarantor</p><p><b>Relationship</b>:
    Child</p><p><b>Relationship</b>: Family Member</p><p><b>Relationship Level</b>: Patient</p></div>'
          },
          'extension': [
            {
              'valueCodeableConcept': {
                'coding': [
                  {
                    'system': 'http://hl7.org/fhir/resource-types',
                    'code': 'Patient',
                    'display': 'Patient'
                  }
                ],
                'text': 'Patient'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
            }
          ],
          'identifier': [
            {
              'id': 'CI-490060520-1',
              'use': 'usual',
              'type': {
                'coding': [
```

```
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
              'code': '22',
              'display': 'Military Id',
              'userSelected': true
            }
          ],
          'text': 'Military Id'
        },
        'system': 'urn:oid:2.16.840.1.113883.3.42.10001.100001.12',
        'value': '1000000105'
      },
      {
        'id': 'CI-490068325-2',
        'use': 'usual',
        'type': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
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              'display': 'mother',
              'userSelected': false
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            'text': 'Mother'
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    'use': 'official',
    'text': 'SMART, NANCY',
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    'given': [
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      'id': '12724066-12724069',
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      },
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        'status': 'extensions',
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NANCY</p><p><b>DOB</b>: Aug 11, 1980</p><p><b>Gender</b>: female</p><p><b>Patient</b>: SMART,
TIMMY</p><p><b>Status</b>: Active</p><p><b>Relationship</b>: Authorized Representative</p><p><b>Relationship</b>:
Default Guarantor</p><p><b>Relationship</b>: Child</p><p><b>Relationship</b>: Family Member</p><p><b>Relationship
Level</b>: Patient</p></div>'
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                'display': 'Military Id',
                'userSelected': true
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            'text': 'Military Id'
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                  'userSelected': true
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              'display': 'mother',
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        'display': 'Default Guarantor',
        'userSelected': true
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        'userSelected': false
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        'userSelected': true
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        'display': 'family member',
        'userSelected': false
      }
    ],
    'text': 'Family Member'
  }
],
'name': [
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    'text': 'SMART, NANCY',
    'family': 'SMART',
    'given': [
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    'extension': [
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        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
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    'extension': [
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            'use': 'mobile',
            'period': {
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            }
          },
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            'id': 'CI-EM-29822894-0',
            'system': 'email',
            'value': 'workemail@gmail.com',
            'use': 'home',
            'period': {
              'start': '2020-06-30T19:47:10.000Z'
            }
          }
        ],
        'gender': 'female',
        'birthDate': '1980-08-11',
        'address': [
          {
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            'text': '3P\nKANSAS CITY, MO 641510001\nUS',
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            'district': 'Jackson',
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              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
                  'code': '151',
                  'display': 'English',
                  'userSelected': true
                },
                {
                  'system': 'urn:ietf:bcp:47',
                  'code': 'en',
                  'display': 'English',
                  'userSelected': false
                }
              ],
              'text': 'English'
            },
            'preferred': 'true'
          }
        ]
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-search-by-encounter Response

: :

Example search by -encounter.

Example Request

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?-encounter=97697434
```

Example Response


```
{
    'resourceType': 'Bundle',
    'id': '8de80541-8f9c-4168-a308-bce0c576e945',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?-encounter=97697434'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/E-12457994-97697434',
        'resource': {
          'resourceType': 'RelatedPerson',
          'id': 'E-12457994-97697434',
          'meta': {
            'versionId': '21',
            'lastUpdated': '2018-10-30T14:17:47.000Z'
          },
          'text': {
            'status': 'extensions',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Related Person</b></p><p><b>Name</b>:MATERNITY,
    TESTTHREE</p><p><b>DOB</b>: Jun 20, 1987</p><p><b>Gender</b>:female</p><p><b>Patient</b>: NEWBORN,
    TESTTHREE</p><p><b>Status</b>:Active</p><p><b>Relationship</b>: Default
    Guarantor</p><p><b>Relationship</b>:Child</p><p><b>Relationship Level</b>: Encounter</p><p><b>Encounter</b>:
    97697434</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Encounter/97697434'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter'
            },
            {
              'valueCodeableConcept': {
                'coding': [
                  {
                    'system': 'http://hl7.org/fhir/resource-types',
                    'code': 'Encounter',
                    'display': 'Encounter'
                  }
                ],
                'text': 'Encounter'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
            }
          ],
          'active': true,
          'patient': {
            'reference': 'Patient/12457995',
            'display': 'NEWBORN, TESTTHREE'
          },
          'relationship': [
            {
              'id': 'CI-97982588-0',
              'extension': [
                {
                  'valuePeriod': {
                    'start': '2016-12-08T21:53:53.000Z'
                  },
                  'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
                },
                {
                  'valueCodeableConcept': {
                    'coding': [
                      {
                        'system': 'https://fhir.cerner.com/2c400054-42d8-4e74-87b7-80b5bd5fde9f/codeSet/40',
                        'code': '156',
                        'display': 'Mother',
                        'userSelected': true
                      },
                      {
                        'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
                        'code': 'MTH',
                        'display': 'mother',
                        'userSelected': false
                      }
```

```
          ],
          'text': 'Mother'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ],
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351',
        'code': '1150',
        'display': 'Default Guarantor',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
        'code': 'GT',
        'display': 'Guarantor',
        'userSelected': false
      }
    ],
    'text': 'Default Guarantor'
  }
],
'name': [
  {
    'id': 'CI-12457994-0',
    'use': 'official',
    'text': 'MATERNITY, TESTTHREE',
    'family': 'MATERNITY',
    'given': [
      'TESTTHREE'
    ],
    'period': {
      'start': '2016-12-08T21:28:22.000Z'
    }
  }
],
'telecom': [
  {
    'id': 'CI-PH-29583893-0',
    'extension': [
      {
        'valueUrl': '816-888-8886',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '4264758753',
    'use': 'home',
    'period': {
      'start': '2016-12-08T21:28:22.000Z'
    }
  }
],
'gender': 'female',
'birthDate': '1987-06-20',
'address': [
  {
    'id': 'CI-24079465-0',
    'use': 'home',
    'text': '3343 Genessee St\nKansas City, MO 64111\nUS',
    'line': [
      '3343 Genessee St'
    ],
    'city': 'Kansas City',
    'state': 'MO',
    'postalCode': '64111',
    'country': 'US',
    'period': {
      'start': '2016-12-08T21:05:59.000Z'
    }
  }
],
'communication': [
  {
    'language': {
      'coding': [
        {
          'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
          'code': '151',
          'display': 'English',
          'userSelected': true
        },
        {
          'system': 'urn:ietf:bcp:47',
          'code': 'en',
          'display': 'English',
          'userSelected': false
        }
```

```
              ],
              'text': 'English'
            },
            'preferred': 'true'
          }
        ]
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-search-with-_revinclude Response

Example search by using . : :

_id

_revinclude

Example Request

```
GET https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?_id=12510003-
    12541997&_revinclude=Provenance:target
```

Example Response

```
{
    'resourceType': 'Bundle',
    'id': '1d559294-3824-45c4-b5d8-b516e538e9ea',
    'type': 'searchset',
    'total': 1,
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson?_id=12510003-
    12541997&_revinclude=Provenance%3Atarget'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/12510003-12541997',
        'resource': {
          'resourceType': 'RelatedPerson',
          'id': '12510003-12541997',
          'meta': {
            'versionId': '52',
            'lastUpdated': '2019-08-28T21:37:08.000Z'
          },
          'text': {
            'status': 'extensions',
            'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Related Person</b></p><p><b>Name</b>: ZZTEST,
    BRIAN</p><p><b>DOB</b>: Nov 17, 1987</p><p><b>Gender</b>: male</p><p><b>Patient</b>: ZZTEST,
    MSPQ</p><p><b>Status</b>: Active</p><p><b>Relationship Level</b>: Patient</p></div>'
          },
          'extension': [
            {
              'valueCodeableConcept': {
                'coding': [
                  {
                    'system': 'http://hl7.org/fhir/resource-types',
                    'code': 'Patient',
                    'display': 'Patient'
                  }
                ],
                'text': 'Patient'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
            }
          ],
          'active': true,
          'patient': {
            'reference': 'Patient/12541997',
            'display': 'ZZTEST, MSPQ'
          },
          'name': [
            {
              'id': 'CI-12510003-0',
              'use': 'official',
              'text': 'ZZTEST, BRIAN',
              'family': 'ZZTEST',
              'given': [
                'BRIAN'
```

```
          ],
          'period': {
            'start': '2017-02-28T17:26:34.000Z'
          }
        }
      ],
      'telecom': [
        {
          'id': 'CI-PH-29677883-0',
          'extension': [
            {
              'valueUrl': '(651) 321-3513',
              'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
            }
          ],
          'system': 'phone',
          'value': '6513213513',
          'use': 'home',
          'period': {
            'start': '2018-01-03T15:17:13.000Z'
          }
        }
      ],
      'gender': 'male',
      'birthDate': '1987-11-17',
      'address': [
        {
          'id': 'CI-24109485-0',
          'use': 'home',
          'text': '123 TEST STREET\nKANSAS CITY, MO 64089\nUSA',
          'line': [
            '123 TEST STREET'
          ],
          'city': 'KANSAS CITY',
          'state': 'MO',
          'postalCode': '64089',
          'country': 'USA',
          'period': {
            'start': '2017-02-28T17:22:36.000Z'
          }
        }
      ],
      'communication': [
        {
          'language': {
            'coding': [
              {
                'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
                'code': '151',
                'display': 'English',
                'userSelected': true
              },
              {
                'system': 'urn:ietf:bcp:47',
                'code': 'en',
                'display': 'English',
                'userSelected': false
              }
            ],
            'text': 'English'
          },
          'preferred': true
        }
      ]
    },
    'search': {
      'mode': 'match'
    }
  },
  {
    'fullUrl': 'https://fhir-
ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/REL-12541997-INT.ptntdemo-12510003',
    'resource': {
      'resourceType': 'Provenance',
      'id': 'REL-12541997-INT.ptntdemo-12510003',
      'meta': {
        'versionId': '12510003',
        'lastUpdated': '2019-08-28T21:37:08Z'
      },
      'text': {
        'status': 'generated',
        'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>Relat
edPerson/12510003-12541997</li></ul><p><b>Recorded</b>: Aug 28, 2019  4:37 P.M.
CDT</p><p><b>Agents</b>:</p><dl><dt>Cerner Test, Nurse Cerner</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent
Role</b>: Source</dd><dt>Default Prov Org</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>:
Source</dd></dl></div>'
      },
      'target': [
        {
```

```
            'reference': 'RelatedPerson/12510003-12541997'
          }
        ],
        'recorded': '2019-08-28T21:37:08Z',
        'agent': [
          {
            'type': {
              'coding': [
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/provenance-participant-type',
                  'code': 'author'
                }
              ],
              'text': 'Author'
            },
            'role': [
              {
                'coding': [
                  {
                    'system': 'http://terminology.hl7.org/CodeSystem/contractsignertypecodes',
                    'code': 'SOURCE'
                  }
                ],
                'text': 'Source'
              }
            ],
            'who': {
              'reference': 'Practitioner/607928',
              'display': 'Cerner Test, Nurse Cerner'
            },
            'onBehalfOf': {
              'reference': 'Organization/3398035',
              'display': 'Default Prov Org'
            }
          },
          {
            'type': {
              'coding': [
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/provenance-participant-type',
                  'code': 'author'
                }
              ],
              'text': 'Author'
            },
            'role': [
              {
                'coding': [
                  {
                    'system': 'http://terminology.hl7.org/CodeSystem/contractsignertypecodes',
                    'code': 'SOURCE'
                  }
                ],
                'text': 'Source'
              }
            ],
            'who': {
              'reference': 'Organization/3398035',
              'display': 'Default Prov Org'
            }
          }
        ]
      },
      'search': {
        'mode': 'include'
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a related person record by ID

File: r4-rest-api-v32/op-relatedperson-id-get.html

Duplicate relationships are only represented once. This operation supports the following authorization types:

Retrieves an individual related person record by the associated related person ID.

Note:

* Provider

* Patient

* System

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

: :

Example retrieval of a patient-level related person record.

Example Request

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/12724068-12724066
```

Example Response


```
{
    'resourceType': 'RelatedPerson',
    'id': '12724066-12724068',
    'meta': {
      'versionId': 145,
      'lastUpdated': '2021-03-22T14:12:37.000Z'
    },
    'text': {
      'status': 'extensions',
      'div': '<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Related Person</b></p><p><b>Name</b>: SMART,
    NANCY</p><p><b>DOB</b>: Aug 11, 1980</p><p><b>Gender</b>: female</p><p><b>Patient</b>: SMART,
    HAILEY</p><p><b>Status</b>: Active</p><p><b>Relationship</b>: Default Guarantor</p><p><b>Relationship</b>:
    Child</p><p><b>Relationship</b>: Family Member</p><p><b>Relationship Level</b>: Patient</p></div>'
    },
    'extension': [
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://hl7.org/fhir/resource-types',
              'code': 'Patient',
              'display': 'Patient'
            }
          ],
          'text': 'Patient'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
      }
    ],
    'identifier': [
      {
        'id': 'CI-490060520-1',
        'use': 'usual',
        'type': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
              'code': '22',
              'display': 'Military Id',
              'userSelected': true
            }
          ],
          'text': 'Military Id'
        },
        'system': 'urn:oid:2.16.840.1.113883.3.42.10001.100001.12',
        'value': '1000000105'
      },
      {
        'id': 'CI-490068325-2',
        'use': 'usual',
        'type': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
```

```
          'code': '2553236771',
          'display': 'Federated Person Principal',
          'userSelected': true
        },
        {
          'system': 'http://terminology.hl7.org/CodeSystem/v2-0203',
          'code': 'AN',
          'display': 'Account number',
          'userSelected': false
        }
      ],
      'text': 'Federated Person Principal'
    },
    'system': 'urn:oid:2.16.840.1.113883.3.13.6',
    'value': 'URN:CERNER:IDENTITY-
FEDERATION:REALM:EC2458F2-1E24-41C8-B71B-0E701AF7583D-CH:PRINCIPAL:RS7MH9WX3XB8VY63',
    'period': {
      'start': '2020-08-10T14:29:20.000Z'
    }
  },
  {
    'id': 'CI-490059563-3',
    'use': 'usual',
    'type': {
      'coding': [
        {
          'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
          'code': '2553236771',
          'display': 'Federated Person Principal',
          'userSelected': true
        },
        {
          'system': 'http://terminology.hl7.org/CodeSystem/v2-0203',
          'code': 'AN',
          'display': 'Account number',
          'userSelected': false
        }
      ],
      'text': 'Federated Person Principal'
    },
    'system': 'urn:oid:2.16.840.1.113883.3.13.6',
    'value': 'URN:CERNER:IDENTITY-
FEDERATION:REALM:E8A84236-C258-4952-98B7-A6FF8A9C587A-CH:PRINCIPAL:RS7MH9WX3XB8VY63',
    'period': {
      'start': '2020-06-30T20:06:42.000Z'
    }
  },
  {
    'id': 'CI-490153231-4',
    'use': 'usual',
    'type': {
      'coding': [
        {
          'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4',
          'code': '2553236785',
          'display': 'MIllennium Person Identifier',
          'userSelected': true
        }
      ],
      'text': 'MIllennium Person Identifier'
    },
    'value': '12868',
    'period': {
      'start': '2020-12-18T19:15:06.000Z'
    }
  }
],
'active': true,
'patient': {
  'reference': 'Patient/12724068',
  'display': 'SMART, HAILEY'
},
'relationship': [
  {
    'id': 'CI-490017041-0',
    'extension': [
      {
        'valuePeriod': {
          'start': '2019-12-26T16:06:26.000Z'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/2c400054-42d8-4e74-87b7-80b5bd5fde9f/codeSet/40',
              'code': '156',
              'display': 'Mother',
```

```
              'userSelected': true
            },
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
              'code': 'MTH',
              'display': 'mother',
              'userSelected': false
            }
          ],
          'text': 'Mother'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ],
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351',
        'code': '1150',
        'display': 'Default Guarantor',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
        'code': 'GT',
        'display': 'Guarantor',
        'userSelected': false
      }
    ],
    'text': 'Default Guarantor'
  },
  {
    'id': 'CI-490017055-1',
    'extension': [
      {
        'valuePeriod': {
          'start': '2019-12-26T16:06:26.000Z'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/2c400054-42d8-4e74-87b7-80b5bd5fde9f/codeSet/40',
              'code': '156',
              'display': 'Mother',
              'userSelected': true
            },
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
              'code': 'MTH',
              'display': 'mother',
              'userSelected': false
            },
            'text': 'Mother'
          ]
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ],
    'coding': [
      {
        'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351',
        'code': '1153',
        'display': 'Family Member',
        'userSelected': true
      },
      {
        'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
        'code': 'FAMMEMB',
        'display': 'family member',
        'userSelected': false
      }
    ],
    'text': 'Family Member'
  }
],
'name': [
  {
    'id': 'CI-12724066-0',
    'use': 'official',
    'text': 'SMART, NANCY',
    'family': 'SMART',
    'given': [
      'NANCY'
    ],
    'period': {
      'start': '2019-12-26T15:12:34.000Z'
    }
```

```
  }
],
'telecom': [
  {
    'id': 'CI-PH-29852801-0',
    'extension': [
      {
        'valueUrl': '13215555555',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '13215555555',
    'use': 'work',
    'period': {
      'start': '2021-03-22T14:12:37.000Z'
    }
  },
  {
    'id': 'CI-PH-29811918-1',
    'extension': [
      {
        'valueUrl': '(987)654-3210',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '9876543210',
    'use': 'home',
    'period': {
      'start': '2019-12-26T15:12:34.000Z'
    }
  },
  {
    'id': 'CI-PH-29844706-2',
    'extension': [
      {
        'valueUrl': '(503)400-8675',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '5034008675',
    'use': 'mobile',
    'period': {
      'start': '2021-01-25T05:00:00.000Z'
    }
  },
  {
    'id': 'CI-PH-29844710-3',
    'extension': [
      {
        'valueUrl': '5034008675',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '5034008675',
    'use': 'mobile',
    'period': {
      'start': '2021-01-25T05:00:00.000Z'
    }
  },
  {
    'id': 'CI-PH-29844690-4',
    'extension': [
      {
        'valueUrl': '5034008675',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '5034008675',
    'use': 'mobile',
    'period': {
      'start': '2021-01-25T05:00:00.000Z'
    }
  },
  {
    'id': 'CI-PH-29844708-5',
    'extension': [
      {
        'valueUrl': '5034008675',
        'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
      }
    ],
    'system': 'phone',
    'value': '5034008675',
    'use': 'mobile',
```

```
        'period': {
          'start': '2021-01-25T05:00:00.000Z'
        }
      },
      {
        'id': 'CI-PH-29844688-6',
        'extension': [
          {
            'valueUrl': '5034008675',
            'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
          }
        ],
        'system': 'phone',
        'value': '5034008675',
        'use': 'mobile',
        'period': {
          'start': '2021-01-25T00:33:18.000Z'
        }
      },
      {
        'id': 'CI-EM-29822894-0',
        'system': 'email',
        'value': 'workemail@gmail.com',
        'use': 'home',
        'period': {
          'start': '2020-06-30T19:47:10.000Z'
        }
      }
    ],
    'gender': 'female',
    'birthDate': '1980-08-11',
    'address': [
      {
        'id': 'CI-24313551-0',
        'use': 'home',
        'text': '3P\nKANSAS CITY, MO 641510001\nUS',
        'line': [
          '3P'
        ],
        'city': 'KANSAS CITY',
        'district': 'Jackson',
        'state': 'MO',
        'postalCode': '641510001',
        'country': 'US',
        'period': {
          'start': '2020-12-08T15:05:44.000Z'
        }
      }
    ],
    'communication': [
      {
        'language': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
              'code': '151',
              'display': 'English',
              'userSelected': true
            },
            {
              'system': 'urn:ietf:bcp:47',
              'code': 'en',
              'display': 'English',
              'userSelected': false
            }
          ],
          'text': 'English'
        },
        'preferred': 'true'
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-retrieve-encounter-level Response

: :

Example retrieval of an encounter-level related person record.

Example Request

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/E-12457994-97697434
```

Example Response


```
{
    'resourceType': 'RelatedPerson',
    'id': 'E-12457994-97697434',
    'meta': {
      'versionId': '21',
      'lastUpdated': '2018-10-30T14:17:47.000Z'
    },
    'text': {
      'status': 'extensions',
      'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Related Person</b></p><p><b>Name</b>: MATERNITY,
    TESTTHREE</p><p><b>DOB</b>: Jun 20, 1987</p><p><b>Gender</b>: female</p><p><b>Patient</b>: NEWBORN,
    TESTTHREE</p><p><b>Status</b>: Active</p><p><b>Relationship</b>: Default Guarantor</p><p><b>Relationship</b>:
    Child</p><p><b>Relationship Level</b>: Encounter</p><p><b>Encounter</b>: 97697434</p></div>'
    },
    'extension': [
      {
        'valueReference': {
          'reference': 'Encounter/97697434'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/related-person-encounter'
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://hl7.org/fhir/resource-types',
              'code': 'Encounter',
              'display': 'Encounter'
            }
          ],
          'text': 'Encounter'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relationship-level'
      }
    ],
    'active': true,
    'patient': {
      'reference': 'Patient/12457995',
      'display': 'NEWBORN, TESTTHREE'
    },
    'relationship': [
      {
        'id': 'CI-97982588-0',
        'extension': [
          {
            'valuePeriod': {
              'start': '2016-12-08T21:53:53.000Z'
            },
            'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
          },
          {
            'valueCodeableConcept': {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/2c400054-42d8-4e74-87b7-80b5bd5fde9f/codeSet/40',
                  'code': '156',
                  'display': 'Mother',
                  'userSelected': true
                },
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
                  'code': 'MTH',
                  'display': 'mother',
                  'userSelected': false
                }
              ],
              'text': 'Mother'
            },
            'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
          }
        ],
        'coding': [
          {
            'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/351',
            'code': '1150',
            'display': 'Default Guarantor',
            'userSelected': true
          },
          {
            'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
            'code': 'GT',
            'display': 'Guarantor',
            'userSelected': false
          }
        ],
        'text': 'Default Guarantor'
      }
```

```
    ],
    'name': [
      {
        'id': 'CI-12457994-0',
        'use': 'official',
        'text': 'MATERNITY, TESTTHREE',
        'family': 'MATERNITY',
        'given': [
          'TESTTHREE'
        ],
        'period': {
          'start': '2016-12-08T21:28:22.000Z'
        }
      }
    ],
    'telecom': [
      {
        'id': 'CI-PH-29583893-0',
        'extension': [
          {
            'valueUrl': '426-475-8753',
            'url': 'http://hl7.org/fhir/StructureDefinition/iso21090-TEL-address'
          }
        ],
        'system': 'phone',
        'value': '4264758753',
        'use': 'home',
        'period': {
          'start': '2016-12-08T21:28:22.000Z'
        }
      }
    ],
    'gender': 'female',
    'birthDate': '1987-06-20',
    'address': [
      {
        'id': 'CI-24079465-0',
        'use': 'home',
        'text': "3343 Genessee St\nKansas City, MO 64111\nUS",
        'line': [
          '3343 Genessee St'
        ],
        'city': 'Kansas City',
        'state': 'MO',
        'postalCode': '64111',
        'country': 'US',
        'period': {
          'start': '2016-12-08T21:05:59.000Z'
        }
      }
    ],
    'communication': [
      {
        'language': {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/36',
              'code': '151',
              'display': 'English',
              'userSelected': true
            },
            {
              'system': 'urn:ietf:bcp:47',
              'code': 'en',
              'display': 'English',
              'userSelected': false
            }
          ],
          'text': 'English'
        },
        'preferred': 'true'
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Patch a related person record

File: r4-rest-api-v32/op-relatedperson-id-patch.html

This operation supports the following authorization types:

Patches an existing encounter-level or patient-level related person record.

Note:

* This implementation follows the JSON Patch spec. See JavaScript Object Notation (JSON) Patch for more information.

* Only operations on the paths listed below are supported.

* Provider

* System

Path | Operation | Value Type

/identifier/- | add | Identifier

Description: Add a new identifier for the related person. Example: { "path": "/identifier/-", "op": "add", "value": { "type": { "coding": [{ "code": "MR", "system": "http://hl7.org/fhir/v2/0203" }] }, "system": "2.16.840.1.113883.6.1000", "value": "<unique identifier>", "period": { "start": "2016-01-02T00:00:00-05:00", "end": "2020-01-02T00:00:00-05:00" } } } Note: Must not contain the use field. Must contain the type, system, and value fields. When specifying a period, the fields must include a time component with a timezone.

/identifier/{index}/id | test | string

Description: Test the value of the Identifier.id of the identifier at the given {index} in the list of related person identifiers. Example: { "path": "/identifier/1/id", "op": "test", "value": "CI-490151266-1" }

/identifier/{index} | remove | N/A

Description: Remove the identifier at the given {index} in the list of related person identifiers. Example: { "path": "/identifier/1", "op": "remove" } Note: Requires a test operation to be provided for the Identifier.id of the identifier intended to be removed.

/relationship/- | add | CodeableConcept

Description The nature of the relationship between a patient and the related person. Example [{ "op": "add", "path": "/relationship/-", "value":{ "coding": [{ "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode", "code": "FAMMEMB" }], "extension": [{ "valuePeriod": { "start": "2016-12-19T16:44:25.000Z", "end": "2022-05-01T16:56:56.000Z" }, "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/period" }, { "valueCodeableConcept": { "coding": [{ "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode", "code": "SIS" }] }, "url": 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation' }] } }] Note: When specifying a period, the fields must include a time component with a timezone. Duplicate relationships may be added, but are only represented once on the resource. The coding field can contain codings from the standard or proprietary system, but not both. Only one coding may be provided.

/relationship/{index}/id | test | string

Description: Test the Relationship.id value of the relationship at the given {index} in the list of related person relationships. Example: { "op": "test", "path": "/relationship/0/id", "value": "CI-98278577-0" }

/relationship/{index} | remove | N/A

Description: Remove the relationship at the given {index} in the list of related person relationships. Example: { "op": "remove", "path": "/relationship/0" } Note: Requires a test operation to be provided for the Relationship.id field of the relationship to be removed.

/relationship/{index}/extension | replace | Extension

Description: Replace the relationship extension field at the given {index}. Example: { "op": "replace", "path": "/relationship/0/extension", "value": [{ "valuePeriod": { "start": "2016-12-30T22:26:08.000Z", "end": "2036-12-30T22:26:08.000Z" }, "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/period" }, { "valueCodeableConcept": { "coding": [{ "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode", "code": "SIS" }] }, "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/relation" }] } Note: Requires a test operation to be provided for the Relationship.id of the relationship whose extension value is intended to be replaced. When specifying a period, the fields must include a time component with a timezone. When either start or end are not provided the value is unset. The coding field can contain codings from the standard or proprietary system, but not both. Only one coding may be provided. If either the valuePeriod and valueCodeableConcept are not provided, any existing values for the absent period and relation extensions are removed.

/address/- | add | Address

Description: Add a new address for the related person. Example: { "op": "add", "path": "/address/-", "value": { "use": "home", "line": ["1212 Faircastle Drive", "Apartment 406"], "city": "KC", "district": "Jackson", "state": "KS", "postalCode": "64199", "country": "United States of America", "period": { "start": "2012-05-17T15:33:18.000Z" } } } Note: Must not contain the text field. Must contain the use field. When specifying a period, the fields must include a time component with a timezone. If line is provided, only the first four values are used. Additional values are ignored.

/address/{index}/id | test | string

Description: Test the Address.id value of the address at the given {index} in the list of related person addresses. Example: { "op": "test", "path": "/address/0/id", "value": "CI-24079465-0" }

/address/{index} | remove | N/A

Description: Remove the address at the given {index} in the list of related person addresses. Example: { "op": "remove", "path": "/address/0" } Note: Requires a test operation to be provided for the Address.id field of the address to be removed.

/telecom/- | add | ContactPoint

Description: Add a new contact detail for the related person. Example: { "op": "add", "path": "/telecom/-", "value": { "use": "home", "system": "email", "value": "test@test.com", "period": { "start": "2018-01-05T22:24:32.000Z" } } }, { "op": "add", "path": "/telecom/-", "value": { "use": "home", "system": "phone", "value": "913-444-8881", "period": { "start": "2018-01-05T22:24:32.000Z" } } } Note: Must contain the system, use, and value fields. When specifying a period, the fields must include a time component with a timezone.

/telecom/{index}/id | test | string

Description: Test the value of the ContactPoint.id of the telecom at the given {index} in the list of related person telecoms. Example: { "op": "test", "path": "/telecom/0/id", "value": "CI-PH-2807950-0" }

/telecom/{index} | remove | N/A

Description: Remove the telecom at the given {index} in the list of related person telecoms. Example: { "op": "remove" "path": "/telecom/0" } Note: Requires a test operation to be provided for the ContactPoint.id of the telecom intended to be removed.

/name/{index}/id | test | string

Description: Test the value of the HumanName.id of the name at the given {index} in the list of related person names. Example: { "op": "test", "path": "/name/0/id", "value": "CI-12467982-0" } Note: This operation is only supported for the first name in the list of related person names ({index} == 0).

/name/{index}/family | replace | string

Description: Replace the HumanName.family of the name at the given {index} in the list of related person names. Example: { "path": "/name/0/family", "op": "replace", "value": "Potter" } Note: Requires a test operation to be provided for the HumanName.id of the name whose family is intended to be replaced. This operation is only supported for the first name in the list of related person names ({index} == 0).

/name/{index}/prefix | replace | String

Description: Replace the HumanName.prefix of the name at the given {index} in the list of related person names. Example: { "op": "replace", "path": "/name/0/prefix", "value": ["Mr."] } Note: Requires a test operation to be provided for the HumanName.id of the name whose prefix is intended to be replaced. This operation is only supported for the first name in the list of related person names ({index} == 0). If no prefix value is provided, the prefix on the name is unset. Only one prefix value may be provided.

/name/{index}/given | replace | String

Description: Replace the HumanName.given of the name at the given {index} in the list of related person names. Example: { "path": "/name/0/given", "op": "replace", "value": ["Harry", "James"] } Note: Requires a test operation to be provided for the HumanName.id of the name whose given name is intended to be replaced. This operation is only supported for the first name in the list of related person names ({index} == 0). If there are fewer values provided than the HumanName.given list currently has, the additional existing values are removed. Up to two given names may be provided. If additional given names are required they must be provided in the second value as a single string separated by blank spaces.

/name/{index}/suffix | replace | String

Description: Replace the HumanName.suffix of the name at the given {index} in the list of related person names. Example: { "op": "replace", "path": "/name/0/suffix", "value": ["Jr."] } Note: Requires a test operation to be provided for the HumanName.id of the name whose suffix is intended to be replaced. This operation is only supported for the first name in the list of related person names ({index} == 0). If no suffix value is provided, the suffix on the name will be unset. Only one suffix value may be provided.

Request

Supported Media Types

* application/fhir+json

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* If-Match(required): string The current version of the RelatedPerson resource.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

content-type(required): string

Specifies the content type of the request entity body.

Example: application/json-patch+json

Body ()

The body of the patch operation.

Root Schema : RelatedPersonPatch

Type:

array

Minimum Number of Items:

A summary representation of the Patch (PATCH) operation for RelatedPerson.

Show Source

* Array of: object items


```
{
    "description":"A summary representation of the Patch (PATCH) operation for RelatedPerson.",
    "type":"array",
    "minItems":1,
    "items":{
        "type":"object",
        "required":[
            "op",
            "path"
        ],
        "properties":{
            "op":{
                "type":"string",
                "enum":[
                    "add",
                    "remove",
                    "replace",
                    "test"
                ]
            },
            "path":{
                "type":"string"
            },
            "value":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* op(required): string Allowed Values: ["add", "remove", "replace", "test"]

* path(required): string

* value: string

```
{
    "type":"object",
    "required":[
        "op",
        "path"
    ],
    "properties":{
        "op":{
            "type":"string",
            "enum":[
                "add",
                "remove",
                "replace",
                "test"
            ]
        },
        "path":{
            "type":"string"
        },
        "value":{
            "type":"string"
        }
    }
}
```

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Response

Supported Media Types

* application/json-patch+json

Default Response

: : : The response header indicates the current version to use on a subsequent update.

Example Request

```
PATCH https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/RelatedPerson/E-12467982-97731398
```

Example Request Body

```
[
    {
      'op': 'add',
      'path': '/relationship/-',
      'value': {
        'coding': [
          {
            'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
            'code': 'FAMMEMB'
          }
        ],
        'extension': [
          {
            'valuePeriod': {
              'start': '2016-12-19T16:44:25.000Z',
              'end': '2022-05-01T16:56:56.000Z'
            },
            'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
          },
          {
            'valueCodeableConcept': {
              'coding': [
                {
                  'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
                  'code': 'SIS'
                }
              ]
            },
            'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
          }
        ]
      }
    },
    {
      'op': 'add',
      'path': '/identifier/-',
      'value': {
        'type': {
          'coding': [
            {
```

```
          'code': 'MR',
          'system': 'http://hl7.org/fhir/v2/0203'
        }
      ]
    },
    'system': '2.16.840.1.113883.6.1000',
    'value': 'THIS:IS:A:UNIQUE:IDENTIFIER',
    'period': {
      'start': '2016-01-02T00:00:00-05:00',
      'end': '2020-01-02T00:00:00-05:00'
    }
  }
},
{
  'op': 'add',
  'path': '/address/-',
  'value': {
    'use': 'home',
    'line': [
      '1212 Faircastle Drive',
      'Apartment 406'
    ],
    'city': 'KC',
    'district': 'Jackson',
    'state': 'KS',
    'postalCode': '64199',
    'country': 'United States of America',
    'period': {
      'start': '2012-05-17T15:33:18.000Z'
    }
  }
},
{
  "op": 'add',
  "path": '/telecom/-',
  "value": {
    "use": 'home',
    "system": 'email',
    "value": 'test@test.com',
    "period": {
      "start": '2018-01-05T22:24:32.000Z'
    }
  }
},
{
  "op": 'add',
  "path": '/telecom/-',
  "value": {
    "use": 'home',
    "system": 'phone',
    "value": '913-444-8881',
    "period": {
      "start": '2018-01-05T22:24:32.000Z'
    }
  }
},
{
  'op': 'test',
  'path': '/telecom/0/id',
  'value': 'CI-PH-29811920-0'
},
{
  'op': 'remove',
  'path': '/telecom/0'
},
{
  'op': 'test',
  'path': '/address/0/id',
  'value': 'CI-24079465-0'
},
{
  'op': 'remove',
  'path': '/address/0'
},
{
  'op': 'test',
  'path': '/relationship/0/id',
  'value': 'CI-98278577-0'
},
{
  'op': 'replace',
  'path': '/relationship/0/extension',
  'value': [
    {
      'valuePeriod': {
        'start': '2016-12-30T22:26:08.000Z',
        'end': '2036-12-30T22:26:08.000Z'
      },
      'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/period'
```

```
      },
      {
        'valueCodeableConcept': {
          'coding': [
            {
              'system': 'http://terminology.hl7.org/CodeSystem/v3-RoleCode',
              'code': 'SIS'
            }
          ]
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/relation'
      }
    ]
  },
  {
    'op': 'test',
    'path': '/relationship/1/id',
    'value': 'CI-98909708-1'
  },
  {
    'op': 'remove',
    'path': '/relationship/1'
  },
  {
    'op': 'test',
    'path': '/name/0/id',
    'value': 'CI-12467982-0'
  },
  {
    'op': 'replace',
    'path': '/name/0/family',
    'value': 'Potter'
  },
  {
    'op': 'replace',
    'path': '/name/0/prefix',
    'value': ['Mr.']
  },
  {
    'op': 'replace',
    'path': '/name/0/given',
    'value': ['Harry', 'James']
  },
  {
    'op': 'replace',
    'path': '/name/0/suffix',
    'value': ['Jr.']
  }
]
```

Example Response

```
Cache-Control: no-cache
Content-Length: 0
Content-Type: application/fhir+json
Date: Tue, 26 Mar 2019 15:42:29 GMT
Etag: W/"7"
Last-Modified: Tue, 21 Jan 2020 15:57:25 GMT
Vary: Origin
X-Request-Id: 11111111-1111-1111-1111-111111111111
opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Schedule

File: r4-rest-api-v32/api-schedule.html

Description

Schedule

This topic describes information for the Schedule resource.

Overview

The Schedule resource provides a time period (planning horizon) where time slots are defined for booking an appointment. Consumers can query by schedule IDs. The schedule IDs can be obtained by querying for slots, which contain the references to associated schedules. A schedule belongs to only one service or resource (actor) and does not contain any information about actual appointments.

The following fields are returned if valued:

Schedule id

Active

Service Type

Actor

Terminology Bindings

Schedule.serviceType

Description: The specific service that is to be performed during this appointment.

Details: Practice Setting Code Value Set

System: http://snomed.info/sct

Details: For more information, see Code Set 14249 Scheduling Appointment Type Synonyms under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR_source_id>/codeSet/14249

Schedule

The operations from the Schedule category.

Get a list of schedules

Method: get Path: /Schedule/

Get a schedule by ID

Method: get Path: /Schedule/{id}

Get a list of schedules

File: r4-rest-api-v32/op-schedule-get.html

Retrieves a list of schedules that meet supplied query parameters.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Query Parameters

_id(required): array[string]

Minimum Number of Items:

The logical resource ID.

Example: _id=12345

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Bundle",
    "id":"69eeaac3-5def-4ffd-8e51-04150cbb5385",
    "type":"searchset",
    "link":[
        {
            "relation":"self",
            "url":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Schedule?_id=24477854-21304876-62852027-0"
        }
    ],
    "entry":[
        {
            "fullUrl":"https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Schedule/24477854-21304876-62852027-0",
            "resource":{
                "resourceType":"Schedule",
                "id":"24477854-21304876-62852027-0",
                "meta":{
                    "versionId":"34229204v992283",
                    "lastUpdated":"2020-07-06T18:46:08.000Z"
                },
                "text":{
                    "status":"generated",
                    "div":"<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Schedule</b></p><p><b>Status</b>:
    Active</p><p><b>Service Type</b>: Established Patient</p><p><b>Practitioner</b>: Cerner Test, Physician - Primary
    Care Cerner</p></div>"
                },
                "active":true,
                "serviceType":[
                    {
                        "coding":[
                            {
                                "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249",
                                "code":"24477854",
                                "display":"Established Patient",
                                "userSelected":true
                            }
                        ],
                        "text":"Established Patient"
                    }
                ],
                "actor":[
                    {
                        "reference":"Practitioner/593923",
                        "display":"Cerner Test, Physician - Primary Care Cerner"
                    }
                ]
            }
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
```

```
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
```

```
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string


```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```


Nested Schema : location

Type:

array

Show Source

* Array of: string


```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```


Nested Schema : coding

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code: string

* system: string


```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```


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Get a schedule by ID

File: r4-rest-api-v32/op-schedule-id-get.html

Retrieves a schedule by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Schedule",
    "id":"24477854-21304876-62852027-0",
    "meta":{
        "versionId":"34229204v992283",
        "lastUpdated":"2020-07-06T18:46:08.000Z"
    },
    "text":{
        "status":"generated",
        "div":"<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Schedule</b></p><p><b>Status</b>: Active</p><p><b>Service
     Type</b>: Established Patient</p><p><b>Practitioner</b>: Cerner Test, Physician - Primary Care Cerner</p></div>"
    },
    "active":true,
    "serviceType":[
        {
            "coding":[
                {
                    "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249",
                    "code":"24477854",
                    "display":"Established Patient",
                    "userSelected":true
                }
            ],
            "text":"Established Patient"
        }
    ],
    "actor":[
        {
            "reference":"Practitioner/593923",
            "display":"Cerner Test, Physician - Primary Care Cerner"
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : issue

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```


Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string


```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
```

```
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
```

```
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

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ServiceRequest

File: r4-rest-api-v32/api-servicerequest.html

Description

ServiceRequest

This topic describes information for the ServiceRequest resource.

Overview

The ServiceRequest resource returns the requests or orders for procedures, diagnostic tests or other services to be performed on a patient. This resource includes a wide range of requests including diagnostic procedures, therapies, patient care activities, referrals, and consults. Medication, dietary and appointment requests are not returned with the ServiceRequest resource. Referrals generated by Referral Management are also returned and return the associated referral order in the ServiceRequest.basedOn field.

This resource supports the US Core ServiceRequest Profile from HL7 FHIR US Core Implementation Guide STU 6.1.0.

The examples provided here are non-normative and replaying them in the public sandbox is not guaranteed to yield the results shown on the site. The field is present only for confidential laboratory resources.

Note:

meta.security

The following fields are returned if valued:

ServiceRequest ID

BasedOn

Status

Intent

Category

Priority

Code

Order detail

Quantity

Patient

Encounter

Occurrence

As needed

Authored on date

Requester

Performer

Location reference

Reason code

Body site

Note

Terminology Bindings

ServiceRequest.status

Description: The status of a service order.

Details: Request Status

System: http://hl7.org/fhir/request-status

ServiceRequest.intent

Description: The kind of service request.

Details: Request Intent

System: http://hl7.org/fhir/request-intent

ServiceRequest.category

Description: A code that classifies the request for service.

Details: For more information, see Code Set 6000 Catalog Type under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/6000

ServiceRequest.priority

Description: The level of importance to be assigned to actioning the request.

Details: Request Priority

System: http://hl7.org/fhir/request-priority

ServiceRequest.code

Description: A code that identifies a particular service that has been requested.

Details: SNOMED CT

System: http://snomed.info/sct

Details: LOINC

System: http://loinc.org

Details: For more information, see Code Set 200 Order Catalog under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/200

ServiceRequest.occurrence.timing

Description: The code for a known or defined timing pattern.

Details: TimingAbbreviation

System: https://hl7.org/fhir/r4/v3/GTSAbbreviation/cs.html

Details: SNOMED CT

System: http://snomed.info/sct

Details: For more information, see Code Set 4003 Frequency under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/4003

ServiceRequest.asNeededCodeableConcept

Description: A code representing the pre-condition for performing the service.

Details: SNOMED CT

System: http://snomed.info/sct

Details: For more information, see Code Set 4005 Reason for Medication under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/4005

ServiceRequest.reasonCode

Description: A reason code for why the service is being requested.

Details: SNOMED CT

System: http://snomed.info/sct

Details: ICD-9

System: http://hl7.org/fhir/sid/icd-9

Details: ICD-10

System: http://hl7.org/fhir/sid/icd-10

ServiceRequest.bodySite

Description: A code representing the anatomic location where the procedure should be performed.

Details: SNOMED CT

System: http://snomed.info/sct

Details: For more information, see Code Set 1028 Body Site under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/millennium/r4/proprietary-codes-and-systems/#code-set-1028-body-site

Details: For more information, see Code Set 14045 Surgical Procedure Modifiers under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/{EHR source ID}/codeSet/14045

ServiceRequest.meta.security

Description: These tags connect specific resources to the overall security policy and infrastructure.

Details: ValueSet

System: http://terminology.hl7.org/CodeSystem/v3-ActCode

ServiceRequest

The operations from the ServiceRequest category.

Get a list of service requests

Method: get Path: /ServiceRequest/

Get a service request by ID

Method: get Path: /ServiceRequest/{id}

Get a list of service requests

File: r4-rest-api-v32/op-servicerequest-get.html

Retrieves a list of service requests that meet supplied query parameters.

Note:

* When multiple pages of service request results are returned for a single query:

* All referrals (if any qualify for the query) are returned on the first page of results. The next item on this list does not apply to service request referrals.

* Querying for a service request with both code and category query parameters is not permitted.

* Querying for a service request with both _lastUpdated and authored query parameters is not permitted.

Authorization Types

This resource supports the following authorization types:

* Patient

* Provider

* System

Request

Query Parameters

_count: integer

Minimum Value:

The maximum number of results to return.

* The first page is unaffected because all referral data is displayed on the first page regardless of requested count.

* The default count is 50.

* From the second page onward, the returned item count may be less than requested.

Example: _count=15

_id: array[string]

Minimum Number of Items:

The logical resource ID. This parameter is required if the patient or subject parameter is not used.

Search by _id supports multiple IDs for both orders and referrals.

Example: _id=12345,R-23456

_lastUpdated: string

An explicit or implied date and time range within which the most recent clinically relevant update was made to the service request. Must include a time, and must be prefixed by ge or le.

May be provided:

* Once with a ge or le prefix representing the earliest date or latest date. Example: _lastUpdated=ge2015-01-01, _lastUpdated=le2016-01-01.

* Twice with the ge and le prefixes to indicate a specific range. Example: _lastUpdated=ge2015-01-01&_lastUpdated=le2016-01-01.

Example: _lastUpdated=ge2014-05-19T20:54:02.000Z

_revinclude: string

The Provenance resource entries to be returned as part of the bundle. Example:

* May be provided once with the Provenance:target value. Example: _revinclude=Provenance:target.

* May be provided with the _id or patient parameters. Example: _id=214938095&_revinclude=Provenance:target.

* When _revinclude is provided in a request to the closed endpoint, the OAuth2 token must include the scope corresponding with the authorization type, such as user/Provenance.read, patient/Provenance.read, or system/Provenance.read.

_revinclude=Provenance:target

authored: string

An explicit or implied date and time range within which the request was made.

Must include a time and can be prefixed by: eq (if prefix is not present, eq is assumed), ge, gt, le, or lt.

May be provided:

Once to indicate earliest date, latest date, or equal to a specific date and time (with or without prefix)

Examples:

* authored=2023-08-23T12:05:32Z (eq is assumed when no prefix is provided)

* authored=eq2023-08-23T12:05:32Z

* authored=ge2023-08-23T12:05:32Z

* authored=gt2023-08-23T12:05:32Z

* authored=le2023-08-23T12:05:32Z

* authored=lt2023-08-23T12:05:32Z

Twice to indicate a specific range (only or and or are supported)

Examples:

* authored=ge2023-08-23T12:05:32Z&authored=le2023-08-24T12:05:32Z

* authored=ge2023-08-23T12:05:32Z&authored=lt2023-08-24T12:05:32Z

* authored=gt2023-08-23T12:05:32Z&authored=le2023-08-24T12:05:32Z

* authored=gt2023-08-23T12:05:32Z&authored=lt2023-08-24T12:05:32Z

Example: authored=ge2023-09-24T07:36:13Z

category: array[string]

The category of service requests being requested. The system can be one of the following options and is not required:

* http://snomed.info/sct

* http://hl7.org/fhir/us/core/CodeSystem/us-core-category|sdoh

Example:

* category=http://snomed.info/sct|22337316

* category=22337316

Note: To retrieve results that contain referrals, you must use the following category: http://snomed.info/sct|386053000.

code: array[string]

What is being requested or ordered. The system can be one of the following options and is not required:

* https://fhir.cerner.com/{EHR source ID}/codeSet/200

* https://fhir.cerner.com/{EHR source ID}/codeSet/34

* http://snomed.info/sct

* http://loinc.org

Example:

* code=https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200|22337316

* code=22337316

patient: string

The specific patient to return service requests for. This parameter is required if the _id or subject parameter is not used.

Example: patient=12345

subject: string

The specific subject to return service requests for. Must represent a patient. May use the :Patient modifier. This parameter is required if the _id or patient parameter is not used.

Example:

* subject=Patient/12345

* subject:Patient=12345

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"Bundle",
    "id":"209b1587-251d-4ac8-9d6d-35660e9476fb",
    "type":"searchset",
    "link":[
        {
            "relation":"self",
            "url":"https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest?patient=12457977"
        }
    ],
    "entry":[
        {
            "fullUrl":"https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/294664205",
            "resource":{
                "resourceType":"ServiceRequest",
                "id":"294664205",
                "meta":{
                    "versionId":"2",
                    "lastUpdated":"2018-02-02T20:01:59.000Z"
                },
                "text":{
                    "status":"generated",
                    "div":"<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Service Request</b></p><p><b>Patient</b>:
    FRIAS,</p><p><b>Code</b>: Pain Assessment Adult</p><p><b>Occurrence</b>: every 4 hr</p><p><b>Priority</b>:
    Routine</p><p><b>Status</b>: Entered in Error</p><p><b>Intent</b>: Order</p><p><b>Requester</b>: Martin, MD, Mary
    Cerner</p><p><b>Order Detail</b>: 12/08/16 9:41:00 CST, every 4 hr</p></div>"
                },
                "status":"entered-in-error",
                "intent":"order",
                "category":[
                    {
                        "coding":[
                            {
                                "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                                "code":"2515",
                                "display":"Patient Care",
                                "userSelected":true
                            }
                        ],
                        "text":"Patient Care"
                    }
                ],
                "priority":"routine",
                "code":{
                    "coding":[
                        {
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                            "code":"2780687",
                            "userSelected":true
                        }
                    ],
```

```
                "text":"Pain Assessment Adult"
            },
            "orderDetail":[
                {
                    "text":"12/08/16 9:41:00 CST, every 4 hr"
                }
            ],
            "subject":{
                "reference":"Patient/12457977",
                "display":"FRIAS,"
            },
            "encounter":{
                "reference":"Encounter/97697398"
            },
            "occurrenceTiming":{
                "repeat":{
                    "boundsPeriod":{
                        "start":"2016-12-08T09:41:00.000-06:00",
                        "end":"2018-02-02T14:01:59.000-06:00"
                    }
                },
                "code":{
                    "coding":[
                        {
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4003",
                            "code":"696547",
                            "display":"every 4 hr",
                            "userSelected":true
                        },
                        {
                            "system":"http://terminology.hl7.org/CodeSystem/v3-GTSAbbreviation",
                            "code":"Q4H",
                            "display":"Q4H",
                            "userSelected":false
                        }
                    ],
                    "text":"every 4 hr"
                }
            },
            "authoredOn":"2016-12-08T09:41:35.000-06:00",
            "requester":{
                "reference":"Practitioner/11724002",
                "display":"Martin, MD, Mary Cerner"
            },
            "note":[
                {
                    "text":"Additional assessments may be required if pain level unacceptable"
                }
            ]
        }
    },
    {
        "fullUrl":"https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/294664221",
        "resource":{
            "resourceType":"ServiceRequest",
            "id":"294664221",
            "meta":{
                "versionId":"1",
                "lastUpdated":"2016-12-29T02:00:21.000Z"
            },
            "text":{
                "status":"generated",
                "div":"<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Service Request</b></p><p><b>Patient</b>:
FRIAS,</p><p><b>Code</b>: Consult to Case Management</p><p><b>Occurrence</b>: Dec  8, 2016  3:41 P.M.
UTC</p><p><b>Priority</b>: Routine</p><p><b>Status</b>: Revoked</p><p><b>Intent</b>: Order</p><p><b>Requester</b>:
Martin, MD, Mary Cerner</p><p><b>Order Detail</b>: 12/08/16 9:41:00 CST, Routine</p></div>"
            },
            "status":"revoked",
            "intent":"order",
            "category":[
                {
                    "coding":[
                        {
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                            "code":"636063",
                            "display":"Consults",
                            "userSelected":true
                        }
                    ],
                    "text":"Consults"
                }
            ],
            "priority":"routine",
            "code":{
                "coding":[
                    {
                        "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                        "code":"2551292823",
                        "userSelected":true
```

```
                    }
                ],
                "text":"Consult to Case Management"
            },
            "orderDetail":[
                {
                    "text":"12/08/16 9:41:00 CST, Routine"
                }
            ],
            "subject":{
                "reference":"Patient/12457977",
                "display":"FRIAS,"
            },
            "encounter":{
                "reference":"Encounter/97697398"
            },
            "occurrenceDateTime":"2016-12-08T09:41:00.000-06:00",
            "authoredOn":"2016-12-08T09:41:54.000-06:00",
            "requester":{
                "reference":"Practitioner/11724002",
                "display":"Martin, MD, Mary Cerner"
            },
            "note":[
                {
                    "text":"Appropriate discharge referrals and schedule follow-up appointment before discharge"
                }
            ]
        }
    },
    {
        "fullUrl":"https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/431633873",
        "resource":{
            "resourceType":"ServiceRequest",
            "id":"431633873",
            "meta":{
                "versionId":"0",
                "lastUpdated":"2024-08-08T14:19:14Z",
                "security":[
                    {
                        "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
                        "code":"PRS",
                        "display":"patient requested information sensitivity",
                        "userSelected":false
                    },
                    {
                        "system":"http://terminology.hl7.org/CodeSystem/v3-ActCode",
                        "code":"PHY",
                        "display":"physician requested information sensitivity",
                        "userSelected":false
                    }
                ]
            },
            "text":{
                "status":"generated",
                "div":"<div xmlns=\\"http://www.w3.org/1999/xhtml\\"><p><b>Service Request</b></p><p><b>Meta
Security</b>: patient requested information sensitivity, physician requested information
sensitivity</p><p><b>Patient</b>: FRIAS,</p><p><b>Code</b>: Basic Metabolic Panel</p><p><b>Occurrence</b>:
Once</p><p><b>Priority</b>: STAT</p><p><b>Reason</b>:</p><ul><li>Heart failure</li></ul><p><b>Status</b>:
Active</p><p><b>Intent</b>: Order</p><p><b>Requester</b>: PWSTORYAMBMA2, MD, CARDIO</p><p><b>Order Detail</b>:
Blood, Stat, 08/08/24 9:17:00 CDT, Once, Heart failure</p></div>"
            },
            "status":"active",
            "intent":"order",
            "category":[
                {
                    "coding":[
                        {
                            "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                            "code":"2513",
                            "display":"Laboratory",
                            "userSelected":true
                        },
                        {
                            "system":"http://snomed.info/sct",
                            "code":"108252007",
                            "display":"Laboratory procedure (procedure)",
                            "userSelected":false
                        }
                    ],
                    "text":"Laboratory"
                }
            ],
            "priority":"stat",
            "code":{
                "coding":[
                    {
                        "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                        "code":"2921288",
                        "userSelected":true
```

```
                        }
                    ],
                    "text":"Basic Metabolic Panel"
                },
                "orderDetail":[
                    {
                        "text":"Blood, Stat, 08/08/24 9:17:00 CDT, Once, Heart failure"
                    }
                ],
                "subject":{
                    "reference":"Patient/12457977",
                    "display":"FRIAS,"
                },
                "encounter":{
                    "reference":"Encounter/97697398"
                },
                "occurrenceTiming":{
                    "repeat":{
                        "boundsPeriod":{
                            "start":"2024-08-08T14:19:00Z",
                            "end":"2024-08-08T14:19:00Z"
                        }
                    },
                    "code":{
                        "coding":[
                            {
                                "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4003",
                                "code":"696531",
                                "display":"Once",
                                "userSelected":true
                            }
                        ],
                        "text":"Once"
                    }
                },
                "authoredOn":"2024-08-08T14:17:48Z",
                "requester":{
                    "reference":"Practitioner/12818101",
                    "display":"PWSTORYAMBMA2, MD, CARDIO"
                },
                "reasonCode":[
                    {
                        "coding":[
                            {
                                "system":"http://hl7.org/fhir/sid/icd-10-cm",
                                "code":"I50.9",
                                "display":"Heart failure, unspecified",
                                "userSelected":true
                            }
                        ],
                        "text":"Heart failure"
                    }
                ]
            },
            "search":{
                "mode":"match"
            }
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```


Nested Schema : issue

Type:

array

Show Source

* Array of: object items


```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
```

```
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
```

```
}
```

x-search-with-_revinclude Response

Example search by using .

_id

_revinclude

Example Request:

```
GET https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest?_id=445122305,R-
    1&_revinclude=Provenance:target
```

Example Response:


```
{
    {
      "resourceType": "Bundle",
      "id": "6c67r6de-b52d-43a0-bb52-68132c926s7c",
      "type": "searchset",
      "link": [
        {
          "relation": "self",
          "url": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest?_id=445122305%2CR-
    1&_revinclude=Provenance%3Atarget"
        }
      ],
      "entry": [
        {
          "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/445122305",
          "resource": {
            "resourceType": "ServiceRequest",
            "id": "445122305",
            "meta": {
              "versionId": "0",
              "lastUpdated": "2024-12-18T15:26:04Z"
            },
            "text": {
              "status": "generated",
              "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Service Request</b></p><p><b>Patient</b>: SMART,
    ANNA</p><p><b>Code</b>: Referral Ambulatory</p><p><b>Occurrence</b>: Dec 18, 2024  3:23 P.M.
    UTC</p><p><b>Priority</b>: Routine</p><p><b>Reason</b>:</p><ul><li>CHF - Congestive heart failure</li><li>Echo
    Stress Test</li></ul><p><b>Status</b>: Completed</p><p><b>Intent</b>: Order</p><p><b>Requester</b>: Cerner,
    Physician - Primary Care PCP One Cerner</p><p><b>Order Detail</b>: Medical Service: Cardiology, Reason: Echo Stress
    Test, Type: Procedure, Start: 12/18/24, Service By: 12/27/24</p></div>"
            },
            "status": "completed",
            "intent": "order",
            "category": [
              {
                "coding": [
                  {
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                    "code": "23276383",
                    "display": "Referral",
                    "userSelected": true
                  }
                ],
                "text": "Referral"
              }
            ],
            "priority": "routine",
            "code": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                  "code": "273989553",
                  "userSelected": true
                }
              ],
              "text": "Referral Ambulatory"
            },
            "orderDetail": [
              {
                "text": "Medical Service: Cardiology, Reason: Echo Stress Test, Type: Procedure, Start: 12/18/24,
    Service By: 12/27/24"
              }
            ],
            "subject": {
              "reference": "Patient/12746484",
              "display": "SMART, ANNA"
            },
            "encounter": {
              "reference": "Encounter/97973423"
            },
            "occurrenceDateTime": "2024-12-18T15:23:00Z",
            "authoredOn": "2024-12-18T15:23:54Z",
            "requester": {
              "reference": "Practitioner/11460104",
```

```
          "display": "Cerner, Physician - Primary Care PCP One Cerner"
        },
        "reasonCode": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/sid/icd-10-cm",
                "code": "I50.9",
                "display": "Heart failure, unspecified",
                "userSelected": true
              }
            ],
            "text": "CHF - Congestive heart failure"
          },
          {
            "text": "Echo Stress Test"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/R-1",
      "resource": {
        "resourceType": "ServiceRequest",
        "id": "R-1",
        "meta": {
          "versionId": "0",
          "lastUpdated": "2024-12-18T15:26:05Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Service Request</b></p><p><b>Patient</b>: SMART,
ANNA</p><p><b>Code</b>: Cardiology</p><p><b>Occurrence Start</b>: Dec 18, 2024  3:23 P.M. UTC</p><p><b>Occurrence
End</b>: Dec 28, 2024  5:59 A.M. UTC</p><p><b>Priority</b>: Routine</p><p><b>Authored On</b>: Dec 18, 2024  3:26
P.M. UTC</p><p><b>Reason</b>:</p><ul><li>CHF - Congestive heart failure</li></ul><p><b>Status</b>:
Active</p><p><b>Intent</b>: Order</p><p><b>Requester</b>: Cerner, Physician - Primary Care PCP One
Cerner</p><p><b>Categories</b>: Evaluation Procedure</p></div>"
        },
        "basedOn": [
          {
            "reference": "ServiceRequest/445122305"
          }
        ],
        "status": "active",
        "intent": "order",
        "category": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "386053000",
                "display": "Evaluation Procedure (procedure)"
              }
            ],
            "text": "Evaluation Procedure"
          }
        ],
        "priority": "routine",
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "183519002",
              "display": "Referral to cardiology service (procedure)"
            }
          ],
          "text": "Cardiology"
        },
        "subject": {
          "reference": "Patient/12746484",
          "display": "SMART, ANNA"
        },
        "encounter": {
          "reference": "Encounter/97973423"
        },
        "occurrencePeriod": {
          "start": "2024-12-18T15:23:00Z",
          "end": "2024-12-28T05:59:00Z"
        },
        "authoredOn": "2024-12-18T15:26:05Z",
        "requester": {
          "reference": "Practitioner/11460104",
          "display": "Cerner, Physician - Primary Care PCP One Cerner"
        },
        "reasonCode": [
          {
```

```
            "coding": [
              {
                "system": "http://hl7.org/fhir/sid/icd-10-cm",
                "code": "I50.9",
                "display": "Heart failure, unspecified"
              }
            ],
            "text": "CHF - Congestive heart failure"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/SER-
INT.med-445122305",
      "resource": {
        "resourceType": "Provenance",
        "id": "SER-INT.med-445122305",
        "meta": {
          "versionId": "445122305",
          "lastUpdated": "2024-12-18T15:26:04Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>Ser
viceRequest/445122305</li></ul><p><b>Recorded</b>: Dec 18, 2024  3:26 P.M.
UTC</p><p><b>Agents</b>:</p><dl><dt>Cerner, Physician - Primary Care PCP One Cerner</dt><dd><b>Agent Type</b>:
Author</dd><dd><b>Agent Role</b>: Source</dd><dt>Model Clinic 1</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent
Role</b>: Source</dd></dl></div>"
        },
        "target": [
          {
            "reference": "ServiceRequest/445122305"
          }
        ],
        "recorded": "2024-12-18T15:26:04Z",
        "agent": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
                  "code": "author"
                }
              ],
              "text": "Author"
            },
            "role": [
              {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                    "code": "SOURCE"
                  }
                ],
                "text": "Source"
              }
            ],
            "who": {
              "reference": "Practitioner/11460104",
              "display": "Cerner, Physician - Primary Care PCP One Cerner"
            },
            "onBehalfOf": {
              "reference": "Organization/685844",
              "display": "Model Clinic 1"
            }
          },
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
                  "code": "author"
                }
              ],
              "text": "Author"
            },
            "role": [
              {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                    "code": "SOURCE"
                  }
                ],
                "text": "Source"
```

```
                }
              ],
              "who": {
                "reference": "Organization/685844",
                "display": "Model Clinic 1"
              }
            }
          ]
        },
        "search": {
          "mode": "include"
        }
      },
      {
        "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Provenance/SER-INT.rfrl-1",
        "resource": {
          "resourceType": "Provenance",
          "id": "SER-INT.rfrl-1",
          "meta": {
            "versionId": "1",
            "lastUpdated": "2024-12-18T15:26:05Z"
          },
          "text": {
            "status": "generated",
            "div": "<div xmlns="http://www.w3.org/1999/xhtml"><p><b>Provenance</b></p><p><b>Target</b>:</p><ul><li>Ser
viceRequest/R-1</li></ul><p><b>Recorded</b>: Dec 18, 2024  3:26 P.M. UTC</p><p><b>Agents</b>:</p><dl><dt>Cerner,
Physician - Primary Care PCP One Cerner</dt><dd><b>Agent Type</b>: Author</dd><dd><b>Agent Role</b>:
Source</dd></dl></div>"
          },
          "target": [
            {
              "reference": "ServiceRequest/R-1"
            }
          ],
          "recorded": "2024-12-18T15:26:05Z",
          "agent": [
            {
              "type": {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
                    "code": "author"
                  }
                ],
                "text": "Author"
              },
              "role": [
                {
                  "coding": [
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/contractsignertypecodes",
                      "code": "SOURCE"
                    }
                  ],
                  "text": "Source"
                }
              ],
              "who": {
                "reference": "Practitioner/11460104",
                "display": "Cerner, Physician - Primary Care PCP One Cerner"
              },
              "onBehalfOf": {
                "extension": [
                  {
                    "valueCode": "unknown",
                    "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason"
                  }
                ],
                "display": "unknown"
              }
            }
          ]
        },
        "search": {
          "mode": "include"
        }
      }
    ]
  }
  }
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

x-search-with-confidential-laboratory Response

Example search by with confidential laboratory results.

patient

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest?patient=12457977&category=2513
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "209b1587-251d-4ac8-9d6d-35660e9476fb",
  "type": "searchset",
  "link":
  [
    {
      "relation": "self",
      "url": "https://fhir-
ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest?patient=12457977&category=2513"
    }
  ],
  "entry":
  [
    {
      "fullUrl": "https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/431633873",
      "resource":
        {
          "resourceType": "ServiceRequest",
          "id": "431633873",
          "meta":
            {
              "versionId": "0",
              "lastUpdated": "2024-08-08T14:19:14Z",
              "security":
                [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                    "code": "PRS",
                    "display": "patient requested information sensitivity",
                    "userSelected": false
                  },
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                    "code": "PHY",
                    "display": "physician requested information sensitivity",
                    "userSelected": false
                  }
                ]
            },
          "text":
            {
              "status": "generated",
              "div": "<div xmlns=\\"http://www.w3.org/1999/xhtml\\"><p><b>Service Request</b></p><p><b>Meta
Security</b>: patient requested information sensitivity, physician requested information
sensitivity</p><p><b>Patient</b>: FRIAS,</p><p><b>Code</b>: Basic Metabolic Panel</p><p><b>Occurrence</b>:
Once</p><p><b>Priority</b>: STAT</p><p><b>Reason</b>:</p><ul><li>Heart failure</li></ul><p><b>Status</b>:
Active</p><p><b>Intent</b>: Order</p><p><b>Requester</b>: PWSTORYAMBMA2, MD, CARDIO</p><p><b>Order Detail</b>:
Blood, Stat, 08/08/24 9:17:00 CDT, Once, Heart failure</p></div>"
            },
          "status": "active",
          "intent": "order",
          "category":
            [
              {
                "coding":
                  [
                    {
                      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                      "code": "2513",
                      "display": "Laboratory",
                      "userSelected": true
                    },
                    {
                      "system": "http://snomed.info/sct",
                      "code": "108252007",
                      "display": "Laboratory procedure (procedure)",
                      "userSelected": false
                    }
                  ],
                "text": "Laboratory"
              }
            ],
          "priority": "stat",
          "code":
            {
              "coding":
                [
                  {
```

```
                    "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                    "code": "2921288",
                    "userSelected": true
                  }
                ],
              "text": "Basic Metabolic Panel"
            },
          "orderDetail":
            [
              {
                "text": "Blood, Stat, 08/08/24 9:17:00 CDT, Once, Heart failure"
              }
            ],
          "subject":
            {
              "reference": "Patient/12457977",
              "display": "FRIAS,"
            },
          "encounter":
            {
              "reference": "Encounter/97697398"
            },
          "occurrenceTiming":
            {
              "repeat":
                {
                  "boundsPeriod":
                    {
                      "start": "2024-08-08T14:19:00Z",
                      "end": "2024-08-08T14:19:00Z"
                    }
                },
              "code":
                {
                  "coding":
                    [
                      {
                        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4003",
                        "code": "696531",
                        "display": "Once",
                        "userSelected": true

                      }
                    ],
                  "text": "Once"
                }
            },
          "authoredOn": "2024-08-08T14:17:48Z",
          "requester":
            {
              "reference": "Practitioner/12818101",
              "display": "PWSTORYAMBMA2, MD, CARDIO"
            },
          "reasonCode":
            [
              {
                "coding":
                  [
                    {
                      "system": "http://hl7.org/fhir/sid/icd-10-cm",
                      "code": "I50.9",
                      "display": "Heart failure, unspecified",
                      "userSelected": true
                    }
                  ],
                "text": "Heart failure"
              }
            ]
        },
      "search":
        {
          "mode": "match"
        }
    }
  ]
}
```


Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Back to Top

Get a service request by ID

File: r4-rest-api-v32/op-servicerequest-id-get.html

Retrieves a service request by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Patient

* Provider

* System

Request

Path Parameters

id(required): string

The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

200 Response

OK

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

Body

Example Response (application/fhir+json)


```
{
    "resourceType":"ServiceRequest",
    "id":"294672281",
    "meta":{
        "versionId":"1",
        "lastUpdated":"2018-02-05T16:01:45.000Z"
    },
    "text":{
        "status":"generated",
        "div":"<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Service Request</b></p><p><b>Patient</b>:
    FRIAS,</p><p><b>Code</b>: Pain Assessment Adult</p><p><b>Occurrence Starts</b>: Dec 12, 2016 12:00 P.M.
    UTC</p><p><b>Occurrence Ends</b>: Feb  5, 2018  4:01 P.M. UTC</p><p><b>Priority</b>: Routine</p><p><b>Status</b>:
    Revoked</p><p><b>Intent</b>: Instance Order</p><p><b>Requester</b>: Martin, MD, Mary Cerner</p><p><b>Order
    Detail</b>: 12/12/16 6:00:00 CST</p></div>"
    },
    "basedOn":[
        {
            "reference":"ServiceRequest/294664205"
        }
    ],
    "status":"revoked",
    "intent":"instance-order",
    "category":[
        {
            "coding":[
                {
                    "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
                    "code":"2515",
                    "display":"Patient Care",
                    "userSelected":true
                }
            ],
            "text":"Patient Care"
        }
    ],
    "priority":"routine",
    "code":{
        "coding":[
            {
                "system":"https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
                "code":"2780687",
```

```
                "userSelected":true
            }
        ],
        "text":"Pain Assessment Adult"
    },
    "orderDetail":[
        {
            "text":"12/12/16 6:00:00 CST"
        }
    ],
    "subject":{
        "reference":"Patient/12457977",
        "display":"FRIAS,"
    },
    "encounter":{
        "reference":"Encounter/97697398"
    },
    "occurrencePeriod":{
        "start":"2016-12-12T06:00:00.000-06:00",
        "end":"2018-02-05T10:01:45.000-06:00"
    },
    "authoredOn":"2016-12-08T09:41:35.000-06:00",
    "requester":{
        "reference":"Practitioner/11724002",
        "display":"Martin, MD, Mary Cerner"
    },
    "note":[
        {
            "text":"Additional assessments may be required if pain level unacceptable"
        }
    ]
}
```


Default Response

An describing any error that occurred.

OperationOutcome

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

Body ()

Root Schema : schema

Type:

object

Show Source

* issue(required): array issue

* resourceType(required): string Allowed Values: ["OperationOutcome"]


```
{
    "type":"object",
    "required":[
        "resourceType",
        "issue"
    ],
    "properties":{
        "resourceType":{
            "type":"string",
            "enum":[
                "OperationOutcome"
            ]
        },
        "issue":{
            "type":"array",
            "items":{
                "type":"object",
                "required":[
                    "severity",
                    "code"
                ],
                "properties":{
                    "severity":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
```

```
                    },
                    "details":{
                        "type":"object",
                        "properties":{
                            "text":{
                                "type":"string"
                            },
                            "coding":{
                                "type":"array",
                                "items":{
                                    "type":"object",
                                    "properties":{
                                        "system":{
                                            "type":"string"
                                        },
                                        "code":{
                                            "type":"string"
                                        }
                                    }
                                }
                            }
                        }
                    },
                    "location":{
                        "type":"array",
                        "items":{
                            "type":"string"
                        }
                    }
                }
            }
        }
    }
}
```

Nested Schema : issue

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "required":[
            "severity",
            "code"
        ],
        "properties":{
            "severity":{
                "type":"string"
            },
            "code":{
                "type":"string"
            },
            "details":{
                "type":"object",
                "properties":{
                    "text":{
                        "type":"string"
                    },
                    "coding":{
                        "type":"array",
                        "items":{
                            "type":"object",
                            "properties":{
                                "system":{
                                    "type":"string"
                                },
                                "code":{
                                    "type":"string"
                                }
                            }
                        }
                    }
                }
            },
            "location":{
                "type":"array",
                "items":{
                    "type":"string"
                }
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code(required): string

* details: object details

* location: array location

* severity(required): string

```
{
    "type":"object",
    "required":[
        "severity",
        "code"
    ],
    "properties":{
        "severity":{
            "type":"string"
        },
        "code":{
            "type":"string"
        },
        "details":{
            "type":"object",
            "properties":{
                "text":{
                    "type":"string"
                },
                "coding":{
                    "type":"array",
                    "items":{
                        "type":"object",
                        "properties":{
                            "system":{
                                "type":"string"
                            },
                            "code":{
                                "type":"string"
                            }
                        }
                    }
                }
            }
        },
        "location":{
            "type":"array",
            "items":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : details

Type:

object

Show Source

* coding: array coding

* text: string

```
{
    "type":"object",
    "properties":{
        "text":{
            "type":"string"
        },
        "coding":{
            "type":"array",
            "items":{
                "type":"object",
                "properties":{
                    "system":{
                        "type":"string"
                    },
                    "code":{
                        "type":"string"
                    }
                }
            }
        }
    }
}
```

Nested Schema : location

Type:

array

Show Source

* Array of: string

```
{
    "type":"array",
    "items":{
        "type":"string"
    }
}
```

Nested Schema : coding

Type:

array

Show Source

* Array of: object items

```
{
    "type":"array",
    "items":{
        "type":"object",
        "properties":{
            "system":{
                "type":"string"
            },
            "code":{
                "type":"string"
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* code: string

* system: string

```
{
    "type":"object",
    "properties":{
        "system":{
            "type":"string"
        },
        "code":{
            "type":"string"
        }
    }
}
```

x-confidential-laboratory Response

Example read confidential laboratory ServiceRequest.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/ServiceRequest/431633873
```

Example Response:

```
  {
    "resourceType": "ServiceRequest",
    "id": "431633873",
    "meta": {
      "versionId": "0",
      "lastUpdated": "2024-08-08T14:19:14Z",
      "security":
        [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "PRS",
            "display": "patient requested information sensitivity",
            "userSelected": false
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "PHY",
            "display": "physician requested information sensitivity",
            "userSelected": false
          }
        ]
```

```
},
"text": {
  "status": "generated",
  "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Service Request</b></p><p><b>Meta Security</b>: patient
requested information sensitivity, physician requested information sensitivity</p><p><b>Patient</b>: Smith,
BRIAN</p><p><b>Code</b>: Basic Metabolic Panel</p><p><b>Occurrence</b>: Once</p><p><b>Priority</b>:
STAT</p><p><b>Reason</b>:</p><ul><li>Heart failure</li></ul><p><b>Status</b>: Active</p><p><b>Intent</b>:
Order</p><p><b>Requester</b>: PWSTORYAMBMA2, MD, CARDIO</p><p><b>Order Detail</b>: Blood, Stat, 08/08/24 9:17:00
CDT, Once, Heart failure</p></div>"
},
"status": "active",
"intent": "order",
"category": [
  {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/6000",
        "code": "2513",
        "display": "Laboratory",
        "userSelected": true
      },
      {
        "system": "http://snomed.info/sct",
        "code": "108252007",
        "display": "Laboratory procedure (procedure)",
        "userSelected": false
      }
    ],
    "text": "Laboratory"
  }
],
"priority": "stat",
"code": {
  "coding": [
    {
      "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/200",
      "code": "2921288",
      "userSelected": true
    }
  ],
  "text": "Basic Metabolic Panel"
},
"orderDetail": [
  {
    "text": "Blood, Stat, 08/08/24 9:17:00 CDT, Once, Heart failure"
  }
],
"subject": {
  "reference": "Patient/12510003",
  "display": "Smith, BRIAN"
},
"encounter": {
  "reference": "Encounter/97905447"
},
"occurrenceTiming": {
  "repeat": {
    "boundsPeriod": {
      "start": "2024-08-08T14:19:00Z",
      "end": "2024-08-08T14:19:00Z"
    }
  },
  "code": {
    "coding": [
      {
        "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/4003",
        "code": "696531",
        "display": "Once",
        "userSelected": true
      }
    ],
    "text": "Once"
  }
},
"authoredOn": "2024-08-08T14:17:48Z",
"requester": {
  "reference": "Practitioner/12818101",
  "display": "PWSTORYAMBMA2, MD, CARDIO"
},
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "I50.9",
        "display": "Heart failure, unspecified",
        "userSelected": true
      }
    ],
    "text": "Heart failure"
```

```
      }
    ]
  }
```

Headers

X-Request-Id: string

Oracle troubleshooting identifier.

opc-request-id: string

Oracle troubleshooting identifier.

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Slot

File: r4-rest-api-v32/api-slot.html

Description

Slot

This topic describes information for the Slot resource.

Overview

The Slot resource returns time slots from a schedule which are available for booking an appointment. Slots contain no information about actual appointments, only availability and type.

Appointment types for use in FHIR scheduling workflows must be resource list based. The ability to retrieve slots and book appointments that are order role based are not supported in FHIR workflows.

When integrating your application with a client's production environment you will work with the client to determine the Practitioner and Location ids (Millennium personnel and location codes, respectively) which they want to make available to third-party applications for enabling scheduling functionality.

We understand this is a bit cumbersome, but we are always evaluating community feedback and look to improve the API in the future.

The following fields are returned if valued:

* Slot id

* ServiceType

* Schedule

* Status

* Start

* End

* Scheduling Location Extension (for more information, see the Extensions section of this topic).

* General Resource Extension (for more information, see the Extensions section of this topic).

Terminology Bindings

Slot.serviceType

Description: The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided, then this overrides the value provided on the availability resource.

Details: Practice Setting Code Value Set

System: http://snomed.info/sct

Details: Millennium Scheduling Appointment Synonyms or Types (for more information, see Code Set 14249 Scheduling Appointment Type Synonyms under the List of Code Sets section in Proprietary Codes and Systems.

System: https://fhir.cerner.com/<EHR source id>/codeSet/14249

Extensions

All URLs for custom extensions are defined as https://fhir-ehr.cerner.com/r4/StructureDefinition/{id}

Table - Custom Extensions

ID | Value[x] Type | Description
scheduling-location | Reference | Reference to the location corresponding to this schedule, where the appointment can be booked.
general-resource | string | A value string referencing a general string resource corresponding with the schedule.

Slot

The operations from the Slot category.

Get a list of slots

Method: get Path: /Slot

Get a slot by ID

Method: get Path: /Slot/{ID}

Patch a slot

Method: patch Path: /Slot/{ID}

Get a list of slots

File: r4-rest-api-v32/op-slot-get.html

Retrieves a list of slots that meet supplied query parameters.

Note:

* Valid IDs for the actor and -location search parameters are determined by the customer and provided when integrating your application with the customer's production environment.

* The retrieved slots are sorted first by start date and time in ascending order (earliest first), followed by service-type and Scheduling Location.

* The search operation only returns free slots that are available to be booked. However, you can still retrieve booked slots when using the _id parameter.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Query Parameters

-location: string

The location that the slot is associated with. May be included only once.

Example: -location=5678

_count: integer

The maximum number of results to return.

Example: _count=15

_id: string

This parameter is required if the , , and parameters are not used.

The logical resource ID.

-location

schedule.actor

service-type

Example: _id=12345

_include: string

Other resource entries to be returned as part of the bundle.

* May be provided once with the Slot:schedule or Schedule:actor:Practitioner value. Example: _include=Slot:schedule or _include=Schedule:actor:Practitioner.

* May be provided with the _id parameter. Example: _id=21265426-633867-3121665-0&_include=Slot:schedule.

* When _include=Slot:schedule is provided in a request to the closed endpoint, the OAuth2 token must include one of the user/Schedule.read, patient/Schedule.read, or system/Schedule.read scopes in addition to the Slot scope.

* When _include=Schedule:actor:Practitioner is provided in a request to the closed endpoint, the OAuth2 token must include one of the user/Practitioner.read, patient/Practitioner.read, or system/Practitioner.read scopes and one of the user/Schedule.read, patient/Schedule.read, or system/Schedule.read scopes in addition to the Slot scope.

Examples:

* _include=Slot:schedule

* _include=Schedule:actor:Practitioner

schedule.actor: string

The logical resource ID of the practitioner associated with the appointment.

* This parameter is required if the _id, service-type, and location parameters are not used.

* May be included only once.

Example: schedule.actor=Practitioner/593923

service-type: string

A single or comma-separated list of appointment types that can be booked into the slot.

* This parameter is required if the _id, -location, and schedule.actor parameters are not used.

* May be included only once.

* May contain standard codes or proprietary codes. If multiple service-type parameters are used, they must be all standard codes or all proprietary codes.

* When using standard codes, select codes bound to Slot.serviceType as noted in the Terminology Bindings table.

* When using proprietary codes, the system should use the following format where the code set is a Millennium code set and the code is a Millennium code value:

```
https://fhir.cerner.com/[your EHR source id>]/codeSet/[code set]
```

Example: service-type=https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611

start: string

The slot date and time.

* Must be provided when service-type, -location, or schedule.actor is provided.

* Both ge and lt prefixes must be used to indicate a specific range. The date and prefix pairs must define an upper and lower bound.

Example: start=ge2016-08-24T12:00:00.000Z&start=lt2017-01-24T12:00:00.000Z

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example search.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec245
    8f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-08T18:13:55Z&-
    location=Location/32216049
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': 'bf750997-bfdd-4ac3-bcc2-34776d05b46e',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhi
    r.cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&-location=32216049&start=ge2016-01-
    23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_count=50'
      },
      {
        'relation': 'next',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?-pageContext=5cafd537-be28-
    45ad-80ed-c5bc2a266678&-pageDirection=NEXT'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-0',
          'meta': {
            'versionId': '3423957v2380283',
            'lastUpdated': '2020-05-08T18:13:55Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
    Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017  7:00 A.M.
    UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Location/32216049'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
            }
          ],
```

```
        'serviceType': [
          {
            'coding': [
              {
                'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                'code': '4047611',
                'display': 'Surgery Rapid',
                'userSelected': true
              },
              {
                'system': 'http://snomed.info/sct',
                'code': '394576009',
                'display': 'Accident & emergency (qualifier value)',
                'userSelected': false
              }
            ],
            'text': 'Surgery Rapid'
          }
        ],
        'schedule': {
          'reference': 'Schedule/4047611-32216049-61518614-0'
        },
        'status': 'free',
        'start': '2017-01-23T06:00:00Z',
        'end': '2017-01-23T07:00:00Z'
      },
      'search': {
        'mode': 'match'
      }
    },
    {
      'fullUrl': 'https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-5',
      'resource': {
        'resourceType': 'Slot',
        'id': '4047611-32216049-61518614-5',
        'meta': {
          'versionId': '3423v2289979283',
          'lastUpdated': '2020-07-22T20:02:03Z'
        },
        'text': {
          'status': 'generated',
          'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
Rapid</p><p><b>Start</b>: May  8, 2020  2:00 P.M. UTC</p><p><b>End</b>: May  8, 2020  3:00 P.M.
UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-5</p><p><b>Status</b>: Free</p></div>'
        },
        'extension': [
          {
            'valueReference': {
              'reference': 'Location/39997807'
            },
            'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
          }
        ],
        'serviceType': [
          {
            'coding': [
              {
                'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                'code': '4047611',
                'display': 'Surgery Rapid',
                'userSelected': true
              },
              {
                'system': 'http://snomed.info/sct',
                'code': '394576009',
                'display': 'Accident & emergency (qualifier value)',
                'userSelected': false
              }
            ],
            'text': 'Surgery Rapid'
          }
        ],
        'schedule': {
          'reference': 'Schedule/4047611-32216049-61518614-5'
        },
        'status': 'free',
        'start': '2020-07-22T22:00:00Z',
        'end': '2020-07-22T23:00:00Z'
      },
      'search': {
        'mode': 'match'
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-using-include-of-schedule Response

Example search by _include=Schedule:acto:Practitioner.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec245
    8f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-
    08T18:13:55Z&_include=Schedule:acto:Practitioner&_count=5
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': 'f8dd806e-62bb-4657-8912-9c2705864d40',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhi
    r.cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&start=ge2016-01-
    23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_include=Schedule%3Aactor%3APractitionerQ&_count=50'
      },
      {
        'relation': 'next',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-
    0e701af7583d/Slot?_include=Schedule%3Aactor%3APractitioner&-pageContext=2d46d307-9a4c-4dc7-a1c2-88f1282d5aec&-
    pageDirection=NEXT'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-0',
          'meta': {
            'versionId': '3423957v2380283',
            'lastUpdated': '2020-05-08T18:13:55Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
    Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017  7:00 A.M.
    UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Location/32216049'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
            }
          ],
          'serviceType': [
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                  'code': '4047611',
                  'display': 'Surgery Rapid',
                  'userSelected': true
                },
                {
                  'system': 'http://snomed.info/sct',
                  'code': '394576009',
                  'display': 'Accident & emergency (qualifier value)',
                  'userSelected': false
                }
              ],
              'text': 'Surgery Rapid'
            }
          ],
          'schedule': {
            'reference': 'Schedule/4047611-32216049-61518614-0'
          },
          'status': 'free',
          'start': '2017-01-23T06:00:00Z',
          'end': '2017-01-23T07:00:00Z'
        },
        'search': {
          'mode': 'match'
        }
```

```
  },
  {
    "fullUrl": 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Practitioner/593923',
    "resource": {
      "resourceType": 'Practitioner',
      "id": '593923',
      "meta": {
        "versionId": '21',
        "lastUpdated": '2020-07-23T16:37:26.000Z'
      },
      "text": {
        "status": 'generated',
        "div": 'PractitionerName:Applegate MD, ChristinaIdentifiers: EXTERNALID: 1221221111,
EXTERNALID:CERNERPHYSPCP, MESSAGING: 593923Status: Active'
      },
      "extension": [
        {
          "valueBoolean": true,
          "url": 'https://fhir-ehr.cerner.com/r4/StructureDefinition/is-physician'
        }
      ],
      "identifier": [
        {
          "use": 'usual',
          "type": {
            "text": 'External Identifier'
          },
          "system": 'urn:oid:2.16.840.1.113883.3.42.10001.100001.12',
          "value": '1221221111',
          "period": {
            "start": '2021-07-13T20:12:34.000Z'
          }
        },
        {
          "use": 'usual',
          "type": {
            "text": 'External Identifier'
          },
          "_system": {
            "extension": [
              {
                "valueCode": 'unknown',
                "url": 'http://hl7.org/fhir/StructureDefinition/data-absent-reason'
              }
            ]
          },
          "value": 'CERNERPHYSPCP',
          "period": {
            "start": '2016-04-28T15:02:02.000Z'
          }
        },
        {
          "use": 'usual',
          "type": {
            "coding": [
              {
                "system": 'http://terminology.hl7.org/CodeSystem/v2-0203',
                "code": 'U',
                "display": 'Unspecified identifier'
              }
            ],
            "text": 'Messaging'
          },
          "_system": {
            "extension": [
              {
                "valueCode": 'unknown',
                "url": 'http://hl7.org/fhir/StructureDefinition/data-absent-reason'
              }
            ]
          },
          "value": '593923',
          "period": {
            "start": '2015-08-18T05:00:00.000Z'
          }
        }
      ],
      "active": true,
      "name": [
        {
          "use": 'usual',
          "text": 'Applegate MD, Christina',
          "family": 'Applegate',
          "given": [
            'Christina'
          ],
          "suffix": [
            'MD'
          ],
```

```
            "period": {
              "start": '2020-07-23T16:37:25.000Z'
            }
          }
        ],
        "telecom": [
          {
            "system": 'email',
            "value": 'test@testcerner.direct.com',
            "use": 'work'
          }
        ],
        "qualification": [
          {
            "code": {
              "coding": [
                {
                  "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600',
                  "code": '677091',
                  "display": 'MD',
                  "userSelected": true
                },
                {
                  "system": 'http://terminology.hl7.org/CodeSystem/v2-0360%7C2.7',
                  "code": 'MD',
                  "display": 'Doctor of Medicine'
                }
              ],
              "text": 'MD'
            },
            "period": {
              "start": '2020-05-31T05:00:00.000Z'
            }
          }
        ]
      },
      "search": {
        "mode": 'include'
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-using-include-of-slot Response

Example search by _include=Slot:schedule.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec245
    8f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-
    08T18:13:55Z&_include=Slot:schedule&_count=5
```

Example Response:

```
{
    'resourceType': 'Bundle',
    'id': 'f8dd806e-62bb-4657-8912-9c2705864d40',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhi
    r.cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&start=ge2016-01-
    23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_include=Slot%3Aschedule&_count=50'
      },
      {
        'relation': 'next',
        'url': 'https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?_include=Slot%3Aschedule&-
    pageContext=2d46d307-9a4c-4dc7-a1c2-88f1282d5aec&-pageDirection=NEXT'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-0',
          'meta': {
            'versionId': '3423957v2380283',
            'lastUpdated': '2020-05-08T18:13:55Z'
          },
```

```
      'text': {
        'status': 'generated',
        'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017  7:00 A.M.
UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
      },
      'extension': [
        {
          'valueReference': {
            'reference': 'Location/32216049'
          },
          'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
        }
      ],
      'serviceType': [
        {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
              'code': '4047611',
              'display': 'Surgery Rapid',
              'userSelected': true
            },
            {
              'system': 'http://snomed.info/sct',
              'code': '394576009',
              'display': 'Accident & emergency (qualifier value)',
              'userSelected': false
            }
          ],
          'text': 'Surgery Rapid'
        }
      ],
      'schedule': {
        'reference': 'Schedule/4047611-32216049-61518614-0'
      },
      'status': 'free',
      'start': '2017-01-23T06:00:00Z',
      'end': '2017-01-23T07:00:00Z'
    },
    'search': {
      'mode': 'match'
    }
  },
  {
    'fullUrl': 'https://fhir-
open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Schedule/4047611-32216049-61518614-0',
    'resource': {
      'resourceType': 'Schedule',
      'id': '4047611-32216049-61518614-0',
      'meta': {
        'versionId': '3423v4vv168v283',
        'lastUpdated': '2020-08-11T15:48:54Z'
      },
      'text': {
        'status': 'generated',
        'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
Rapid</p><p><b>Start</b>: May  8, 2020  2:00 P.M. UTC</p><p><b>End</b>: May  8, 2020  3:00 P.M.
UTC</p><p><b>Status</b>: Free</p></div>'
      },
      'active': true,
      'serviceType': [
        {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
              'code': '4047611',
              'display': 'Surgery Rapid',
              'userSelected': true
            },
            {
              'system': 'http://snomed.info/sct',
              'code': '394576009',
              'display': 'Accident & emergency (qualifier value)',
              'userSelected': false
            }
          ],
          'text': 'Surgery Rapid'
        }
      ],
      'actor': [
        {
          'display': 'OR Add-On 02'
        }
      ]
    },
    'search': {
      'mode': 'include'
    }
```

```
      }
    ]
  }
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth Response

Example search using patient authorization.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec2458
    f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-08T18:13:55Z&-
    location=Location/32216049
```

Example Response:

```
{
    'resourceType': 'Bundle',
    'id': 'bf750997-bfdd-4ac3-bcc2-34776d05b46e',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhir
    .cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&-location=32216049&start=ge2016-01-
    23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_count=50'
      },
      {
        'relation': 'next',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?-pageContext=5cafd537-be28-
    45ad-80ed-c5bc2a266678&-pageDirection=NEXT'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-0',
          'meta': {
            'versionId': '3423957v2380283',
            'lastUpdated': '2020-05-08T18:13:55Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient
    Friendly Display - Surgery Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017
    7:00 A.M. UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Location/32216049'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
            }
          ],
          'serviceType': [
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                  'code': '4047611',
                  'display': 'Patient Friendly Display - Surgery Rapid',
                  'userSelected': true
                },
                {
                  'system': 'http://snomed.info/sct',
                  'code': '394576009',
                  'display': 'Accident & emergency (qualifier value)',
                  'userSelected': false
                }
              ],
              'text': 'Patient Friendly Display - Surgery Rapid'
            }
          ],
          'schedule': {
            'reference': 'Schedule/4047611-32216049-61518614-0'
          },
          'status': 'free',
          'start': '2017-01-23T06:00:00Z',
          'end': '2017-01-23T07:00:00Z'
```

```
        },
        'search': {
          'mode': 'match'
        }
      },
      {
        'fullUrl': 'https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-5',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-5',
          'meta': {
            'versionId': '3423v2289979283',
            'lastUpdated': '2020-07-22T20:02:03Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient
    Friendly Display - Surgery Rapid</p><p><b>Start</b>: May  8, 2020  2:00 P.M. UTC</p><p><b>End</b>: May  8, 2020
    3:00 P.M. UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-5</p><p><b>Status</b>: Free</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Location/39997807'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
            }
          ],
          'serviceType': [
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                  'code': '4047611',
                  'display': 'Patient Friendly Display - Surgery Rapid',
                  'userSelected': true
                },
                {
                  'system': 'http://snomed.info/sct',
                  'code': '394576009',
                  'display': 'Accident & emergency (qualifier value)',
                  'userSelected': false
                }
              ],
              'text': 'Patient Friendly Display - Surgery Rapid'
            }
          ],
          'schedule': {
            'reference': 'Schedule/4047611-32216049-61518614-5'
          },
          'status': 'free',
          'start': '2020-07-22T22:00:00Z',
          'end': '2020-07-22T23:00:00Z'
        },
        'search': {
          'mode': 'match'
        }
      }
    ]
  }
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth-using-include-of-schedule Response

Example search by _include=Schedule:acto:Practitioner using patient authorization.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec2458
    f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-
    08T18:13:55Z&_include=Schedule:acto:Practitionere&_count=5
```

Example Response:


```
{
    'resourceType': 'Bundle',
    'id': 'f8dd806e-62bb-4657-8912-9c2705864d40',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhir
    .cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&start=ge2016-01-
```

```
23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_include=Schedule%3Aactor%3APractitioner&_count=50'
  },
  {
    'relation': 'next',
    'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-
0e701af7583d/Slot?_include=Schedule%3Aactor%3APractitioner&-pageContext=2d46d307-9a4c-4dc7-a1c2-88f1282d5aec&-
pageDirection=NEXT'
  }
],
'entry': [
  {
    'fullUrl': 'https://fhir-
ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
    'resource': {
      'resourceType': 'Slot',
      'id': '4047611-32216049-61518614-0',
      'meta': {
        'versionId': '3423957v2380283',
        'lastUpdated': '2020-05-08T18:13:55Z'
      },
      'text': {
        'status': 'generated',
        'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient
Friendly Display - Surgery Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017
7:00 A.M. UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
      },
      'extension': [
        {
          'valueReference': {
            'reference': 'Location/32216049'
          },
          'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
        }
      ],
      'serviceType': [
        {
          'coding': [
            {
              'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
              'code': '4047611',
              'display': 'Patient Friendly Display - Surgery Rapid',
              'userSelected': true
            },
            {
              'system': 'http://snomed.info/sct',
              'code': '394576009',
              'display': 'Accident & emergency (qualifier value)',
              'userSelected': false
            }
          ],
          'text': 'Patient Friendly Display - Surgery Rapid'
        }
      ],
      'schedule': {
        'reference': 'Schedule/4047611-32216049-61518614-0'
      },
      'status': 'free',
      'start': '2017-01-23T06:00:00Z',
      'end': '2017-01-23T07:00:00Z'
    },
    'search': {
      'mode': 'match'
    }
  },
  {
    "fullUrl": 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Practitioner/593923',
    "resource": {
      "resourceType": 'Practitioner',
      "id": '593923',
      "meta": {
        "versionId": '21',
        "lastUpdated": '2020-07-23T16:37:26.000Z'
      },
      "text": {
        "status": 'generated',
        "div": 'PractitionerName: Applegate MD, ChristinaIdentifiers: EXTERNALID: 1221221111,
EXTERNALID:CERNERPHYSPCP, MESSAGING: 593923Status: Active'
      },
      "extension": [
        {
          "valueBoolean": true,
          "url": 'https://fhir-ehr.cerner.com/r4/StructureDefinition/is-physician'
        }
      ],
      "identifier": [
        {
          "use": 'usual',
          "type": {
            "text": 'External Identifier'
```

```
    },
    "system": 'urn:oid:2.16.840.1.113883.3.42.10001.100001.12',
    "value": '1221221111',
    "period": {
      "start": '2021-07-13T20:12:34.000Z'
    }
  },
  {
    "use": 'usual',
    "type": {
      "text": 'External Identifier'
    },
    "_system": {
      "extension": [
        {
          "valueCode": 'unknown',
          "url": 'http://hl7.org/fhir/StructureDefinition/data-absent-reason'
        }
      ]
    },
    "value": 'CERNERPHYSPCP',
    "period": {
      "start": '2016-04-28T15:02:02.000Z'
    }
  },
  {
    "use": 'usual',
    "type": {
      "coding": [
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v2-0203',
          "code": 'U',
          "display": 'Unspecified identifier'
        }
      ],
      "text": 'Messaging'
    },
    "_system": {
      "extension": [
        {
          "valueCode": 'unknown',
          "url": 'http://hl7.org/fhir/StructureDefinition/data-absent-reason'
        }
      ]
    },
    "value": '593923',
    "period": {
      "start": '2015-08-18T05:00:00.000Z'
    }
  }
],
"active": true,
"name": [
  {
    "use": 'usual',
    "text": 'Applegate MD, Christina',
    "family": 'Applegate',
    "given": [
      'Christina'
    ],
    "suffix": [
      'MD'
    ],
    "period": {
      "start": '2020-07-23T16:37:25.000Z'
    }
  }
],
"telecom": [
  {
    "system": 'email',
    "value": 'test@testcerner.direct.com',
    "use": 'work'
  }
],
"qualification": [
  {
    "code": {
      "coding": [
        {
          "system": 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/29600',
          "code": '677091',
          "display": 'MD',
          "userSelected": true
        },
        {
          "system": 'http://terminology.hl7.org/CodeSystem/v2-0360%7C2.7',
          "code": 'MD',
          "display": 'Doctor of Medicine'
```

```
                }
              ],
              "text": 'MD'
            },
            "period": {
              "start": '2020-05-31T05:00:00.000Z'
            }
          }
        ]
      },
      "search": {
        "mode": 'include'
      }
    }
  ]
}
```

Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth-using-include-of-slot Response

Example search by _include=Slot:schedule using patient authorization.

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https://fhir.cerner.com/ec2458
    f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249|4047611&start=ge2016-01-23T22:05:47Z&start=lt2020-06-
    08T18:13:55Z&_include=Slot:schedule&_count=5
```

Example Response:

```
{
    'resourceType': 'Bundle',
    'id': 'f8dd806e-62bb-4657-8912-9c2705864d40',
    'type': 'searchset',
    'link': [
      {
        'relation': 'self',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?service-type=https%3A%2F%2Ffhir
    .cerner.com%2Fec2458f2-1e24-41c8-b71b-0e701af7583d%2FcodeSet%2F14249%7C4047611&start=ge2016-01-
    23T22%3A05%3A47Z&start=lt2020-06-08T18%3A13%3A55Z&_include=Slot%3Aschedule&_count=50'
      },
      {
        'relation': 'next',
        'url': 'https://fhir-ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot?_include=Slot%3Aschedule&-
    pageContext=2d46d307-9a4c-4dc7-a1c2-88f1282d5aec&-pageDirection=NEXT'
      }
    ],
    'entry': [
      {
        'fullUrl': 'https://fhir-
    ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0',
        'resource': {
          'resourceType': 'Slot',
          'id': '4047611-32216049-61518614-0',
          'meta': {
            'versionId': '3423957v2380283',
            'lastUpdated': '2020-05-08T18:13:55Z'
          },
          'text': {
            'status': 'generated',
            'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient
    Friendly Display - Surgery Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017
    7:00 A.M. UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
          },
          'extension': [
            {
              'valueReference': {
                'reference': 'Location/32216049'
              },
              'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
            }
          ],
          'serviceType': [
            {
              'coding': [
                {
                  'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                  'code': '4047611',
                  'display': 'Patient Friendly Display - Surgery Rapid',
                  'userSelected': true
                },
                {
                  'system': 'http://snomed.info/sct',
                  'code': '394576009',
```

```
                'display': 'Accident & emergency (qualifier value)',
                'userSelected': false
              }
            ],
            'text': 'Patient Friendly Display - Surgery Rapid'
          }
        ],
        'schedule': {
          'reference': 'Schedule/4047611-32216049-61518614-0'
        },
        'status': 'free',
        'start': '2017-01-23T06:00:00Z',
        'end': '2017-01-23T07:00:00Z'
      },
      'search': {
        'mode': 'match'
      }
    },
    {
      'fullUrl': 'https://fhir-
ehr.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Schedule/4047611-32216049-61518614-0',
      'resource': {
        'resourceType': 'Schedule',
        'id': '4047611-32216049-61518614-0',
        'meta': {
          'versionId': '3423v4vv168v283',
          'lastUpdated': '2020-08-11T15:48:54Z'
        },
        'text': {
          'status': 'generated',
          'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient
Friendly Display - Surgery Rapid</p><p><b>Start</b>: May  8, 2020  2:00 P.M. UTC</p><p><b>End</b>: May  8, 2020
3:00 P.M. UTC</p><p><b>Status</b>: Free</p></div>'
        },
        'active': true,
        'serviceType': [
          {
            'coding': [
              {
                'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
                'code': '4047611',
                'display': 'Patient Friendly Display - Surgery Rapid',
                'userSelected': true
              },
              {
                'system': 'http://snomed.info/sct',
                'code': '394576009',
                'display': 'Accident & emergency (qualifier value)',
                'userSelected': false
              }
            ],
            'text': 'Patient Friendly Display - Surgery Rapid'
          }
        ],
        'actor': [
          {
            'display': 'OR Add-On 02'
          }
        ]
      },
      'search': {
        'mode': 'include'
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a slot by ID

File: r4-rest-api-v32/op-slot-id-get.html

Retrieves an individual slot by the associated ID.

Authorization Types

This operation supports the following authorization types:

* Provider

* Patient

* System

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example retrieval.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0
```

Example Response:


```
{
    'resourceType': 'Slot',
    'id': '4047611-32216049-61518614-0',
    'meta': {
      'versionId': '3423957v2380283',
      'lastUpdated': '2020-05-08T18:13:55Z'
    },
    'text': {
      'status': 'generated',
      'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Surgery
    Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017  7:00 A.M.
    UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
    },
    'extension': [
      {
        'valueReference': {
          'reference': 'Location/32216049'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
      }
    ],
    'serviceType': [
      {
        'coding': [
          {
            'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
            'code': '4047611',
            'display': 'Surgery Rapid',
            'userSelected': true
          },
          {
            'system': 'http://snomed.info/sct',
            'code': '394576009',
            'display': 'Accident & emergency (qualifier value)',
            'userSelected': false
          }
        ],
        'text': 'Surgery Rapid'
      }
    ],
    'schedule': {
      'reference': 'Schedule/4047611-32216049-61518614-0'
    },
    'status': 'free',
    'start': '2017-01-23T06:00:00Z',
    'end': '2017-01-23T07:00:00Z'
  }
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

x-with-patient-auth Response

Example retrieval using patient authorization.

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0
```

Example Response:


```
{
    'resourceType': 'Slot',
    'id': '4047611-32216049-61518614-0',
    'meta': {
      'versionId': '3423957v2380283',
      'lastUpdated': '2020-05-08T18:13:55Z'
    },
    'text': {
      'status': 'generated',
      'div': '<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Slot</b></p><p><b>Service Type</b>: Patient Friendly
    Display - Surgery Rapid</p><p><b>Start</b>: Jan 23, 2017  6:00 A.M. UTC</p><p><b>End</b>: Jan 23, 2017  7:00 A.M.
    UTC</p><p><b>Schedule Reference</b>: 4047611-32216049-61518614-0</p><p><b>Status</b>: Free</p></div>'
    },
    'extension': [
      {
        'valueReference': {
          'reference': 'Location/32216049'
        },
        'url': 'https://fhir-ehr.cerner.com/r4/StructureDefinition/scheduling-location'
      }
    ],
    'serviceType': [
      {
        'coding': [
          {
            'system': 'https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/14249',
            'code': '4047611',
            'display': 'Patient Friendly Display - Surgery Rapid',
            'userSelected': true
          },
          {
            'system': 'http://snomed.info/sct',
            'code': '394576009',
            'display': 'Accident & emergency (qualifier value)',
            'userSelected': false
          }
        ],
        'text': 'Patient Friendly Display - Surgery Rapid'
      }
    ],
    'schedule': {
      'reference': 'Schedule/4047611-32216049-61518614-0'
    },
    'status': 'free',
    'start': '2017-01-23T06:00:00Z',
    'end': '2017-01-23T07:00:00Z'
  }
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Patch a slot

File: r4-rest-api-v32/op-slot-id-patch.html

Patches an existing slot.

Note:

* This implementation follows the JSON PATCH specification.

* Only operations on the paths listed below are supported.

Authorization Types

This operation supports the following authorization types:

* Provider

* System

Patch Operations

Path | Operation | Value Type
/status | replace | code
Description: The status of the slot. Example: { "op": "replace", "path": "/status" "status": "free" } Note: Only free and busy-tentative are supported.

Request

Supported Media Types

* application/fhir+json

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* Authorization(required): string Contains the credentials to authenticate a consumer to the service. This should be the OAuth2 Bearer Token.

* If-Match(required): string The current version of the Slot resource.

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

* content-type(required): string Specifies the content type of the request entity body.

Body ()

The body of the patch operation

Root Schema : SlotPatch

Type:

array

Minimum Number of Items:

Show Source

* Array of: object items

```
{
    "type":"array",
    "minItems":1,
    "items":{
        "type":"object",
        "required":[
            "op",
            "path"
        ],
        "properties":{
            "op":{
                "type":"string",
                "enum":[
                    "replace"
                ]
            },
            "path":{
                "type":"string",
                "enum":[
                    "/status"
                ]
            },
            "value":{
                "type":"string",
                "enum":[
                    "free",
                    "busy-tentative"
                ]
            }
        }
    }
}
```

Nested Schema : items

Type:

object

Show Source

* op(required): string Allowed Values: ["replace"]

* path(required): string Allowed Values: ["/status"]

* value: string Allowed Values: ["free", "busy-tentative"]

```
{
    "type":"object",
    "required":[
        "op",
        "path"
    ],
    "properties":{
        "op":{
            "type":"string",
            "enum":[
                "replace"
            ]
        },
        "path":{
```

```
            "type":"string",
            "enum":[
                "/status"
            ]
        },
        "value":{
            "type":"string",
            "enum":[
                "free",
                "busy-tentative"
            ]
        }
    }
}
```

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Response

Supported Media Types

* application/json-patch+json

Default Response

The response header indicates the current version to use on a subsequent update.

Example Request:

```
PATCH https://fhir-ehr-code.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Slot/4047611-32216049-61518614-0
```

Example Request Body:

```
[
    {
      'op': 'replace',
      'path': '/status',
      'value': 'busy-tentative'
    }
  ]
```

Example Response:

```
  Cache-Control: no-cache
  Content-Length: 20
  Content-Type: application/fhir+json
  Date: Tue, 06 Apr 2021 15:36:47 GMT
  Etag: W/"410v795030969834"
  Last-Modified: Tue, 06 Apr 2021 15:36:47 GMT
  Vary: Origin
  X-Request-Id: 11111111-1111-1111-1111-111111111111
  opc-request-id: /11111111111111111111111111111111/11111111111111111111111111111111
```

ETag

If-Match

Headers

* Etag: string For Update or Patch versioning controls. Related to If-Match. When a resource performs an operation that creates or updates a record, an Etag value is returned as a header. This same value should be included in request headers as an If-Match for any subsequent update to that record.

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Specimen

File: r4-rest-api-v32/api-specimen.html

Description

Specimen

This topic describes information for the Specimen resource.

Overview

The Specimen resource provides information related to any sample to be used for diagnostic and environmental testing.

The Specimen resource is used for gathering, maintaining, and processing the specimen, and determining where the specimen originated.

This resource supports the US Core Specimen Profile from HL7 FHIR US Core Implementation Guide STU 6.1.

The following fields are returned if valued:

* ID

* Accession Identifier
* Type
Subject
* Reference (Patient)
* Received Time
Collection
Collector
* Reference (Practitioner)
* Collected Date and Time
* Body Site
Container
* Identifier
* Type
Terminology Bindings
Specimen.type
Description: A code to identify the type of the specimen.
Details:
SNOMED CT
System: https://snomed.info/sct
* For more information, see under the section in .
Details:
Code Set 1306 Anatomic Pathology Specimen
List of Code Sets
Proprietary Codes and Systems
System: https://fhir.cerner.com/{EHR source ID}/codeSet/1306
* For more information, see under the section in .
Details:
Code Set 2052 General Laboratory Specimen Type
List of Code Sets
Proprietary Codes and Systems
System: https://fhir.cerner.com/{EHR source ID}/codeSet/2052
Specimen.collection.bodySite
Description: A code representing the anatomic location from which the specimen was collected.
Details:
SNOMED CT
System: https://snomed.info/sct
* For more information, see under the section in .
Details:
Code Set 1028 Body Site
List of Code Sets
Proprietary Codes and Systems
System: https://fhir.cerner.com/{EHR source ID}/codeSet/1028
Specimen.container.type
Description: A code representing the type of container associated with the specimen.
Details:
SNOMED CT
System: https://snomed.info/sct
* For more information, see under the section in .
Details:
Code Set 2051 Container Type
List of Code Sets
Proprietary Codes and Systems

System: https://fhir.cerner.com/{EHR source ID}/codeSet/2051

Specimen

The operations from the Specimen category.

Get a list of specimens

Method: get Path: /Specimen

Get a specimen by ID

Method: get Path: /Specimen/{ID}

Get a list of specimens

File: r4-rest-api-v32/op-specimen-get.html

Retrieves a list of specimens that meet supplied query parameters.

Request

Query Parameters

* _id: string The logical resource ID associated with the resource. Example: _id=7891

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Specimen_id=275745679,275667708
```

Example Response:


```
{
  "resourceType": "Bundle",
  "id": "08642ec2-4613-4d42-966b-cbc949db9d6f",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Specimen?_id=275745679%2C275667708"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Specimen/275667708",
      "resource": {
        "resourceType": "Specimen",
        "id": "275667708",
        "meta": {
          "versionId": "2",
          "lastUpdated": "2020-08-04T21:56:18Z"
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Specimen</b></p><p><b>Accession Identifier</b>:
    0000120202160000087</p><p><b>Type</b>: Nasopharyngeal swab (specimen)</p><p><b>Container</b>: Swab (physical
    object)</p></div>"
        },
        "accessionIdentifier": {
          "use": "official",
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                "code": "ACSN",
                "display": "Accession ID"
              }
            ],
            "text": "Accession ID"
          },
          "value": "000012020216000087"
        },
```

```
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2052",
          "code": "312624",
          "userSelected": false
        },
        {
          "system": "http://snomed.info/sct",
          "code": "258500001",
          "display": "Nasopharyngeal swab (specimen)",
          "userSelected": false
        }
      ]
    },
    "subject": {
      "reference": "Patient/12742837"
    },
    "receivedTime": "2020-08-03T16:27:00Z",
    "collection": {
      "collector": {
        "reference": "Practitioner/12736052"
      },
      "collectedDateTime": "2020-08-03T16:00:00Z"
    },
    "container": [
      {
        "id": "275667707",
        "identifier": [
          {
            "use": "official",
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                  "code": "ACSN",
                  "display": "Accession ID"
                }
              ],
              "text": "Accession ID"
            },
            "value": "000012020216000087"
          }
        ],
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2051",
              "code": "309510",
              "userSelected": false
            },
            {
              "system": "http://snomed.info/sct",
              "code": "408098004",
              "display": "Swab (physical object)",
              "userSelected": false
            }
          ]
        }
      }
    ]
  }
},
{
  "fullUrl": "https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Specimen/275745679",
  "resource": {
    "resourceType": "Specimen",
    "id": "275745679",
    "meta": {
      "versionId": "3",
      "lastUpdated": "2023-02-21T20:49:00Z"
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Specimen</b></p><p><b>Accession Identifier</b>:
00000202305200000 4</p><p><b>Type</b>: Blood specimen (specimen)</p><p><b>Container</b>: Aerobic blood culture bottle
 (physical object)</p></div>"
    },
    "accessionIdentifier": {
      "use": "official",
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "ACSN",
            "display": "Accession ID"
          }
        ],
        "text": "Accession ID"
```

```
          },
          "value": "000002023052000004"
        },
        "type": {
          "coding": [
            {
              "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2052",
              "code": "1765",
              "userSelected": false
            },
            {
              "system": "http://snomed.info/sct",
              "code": "119297000",
              "display": "Blood specimen (specimen)",
              "userSelected": false
            }
          ]
        },
        "subject": {
          "reference": "Patient/12783130"
        },
        "receivedTime": "2023-02-21T20:25:00Z",
        "collection": {
          "collector": {
            "reference": "Practitioner/12742563"
          },
          "collectedDateTime": "2023-02-21T13:50:00Z"
        },
        "container": [
          {
            "id": "275745680",
            "identifier": [
              {
                "use": "official",
                "type": {
                  "coding": [
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                      "code": "ACSN",
                      "display": "Accession ID"
                    }
                  ],
                  "text": "Accession ID"
                },
                "value": "000002023052000004"
              }
            ],
            "type": {
              "coding": [
                {
                  "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2051",
                  "code": "2569819877",
                  "userSelected": false
                },
                {
                  "system": "http://snomed.info/sct",
                  "code": "767385003",
                  "display": "Aerobic blood culture bottle (physical object)",
                  "userSelected": false
                }
              ]
            }
          }
        ]
      }
    }
  ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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Get a specimen by ID

File: r4-rest-api-v32/op-specimen-id-get.html

Retrieves an individual specimen by the associated specimen ID.

Request

Path Parameters

* ID(required): string The logical resource ID associated with the resource.

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

This operation supports the following authorization types:

* Provider

* Patient

* System

Example Request:

```
GET https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/Specimen/275745679
```

Example Response:


```
{
 "resourceType": "Specimen",
 "id": "275745679",
 "meta": {
   "versionId": "3",
   "lastUpdated": "2023-02-21T20:49:00Z"
 },
 "text": {
   "status": "generated",
   "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p><b>Specimen</b></p><p><b>Accession Identifier</b>:
   00000202305200004</p><p><b>Type</b>: Blood specimen (specimen)</p><p><b>Container</b>: Aerobic blood culture bottle
    (physical object)</p></div>"
 },
 "accessionIdentifier": {
   "use": "official",
   "type": {
     "coding": [
       {
         "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
         "code": "ACSN",
         "display": "Accession ID"
       }
     ],
     "text": "Accession ID"
   },
   "value": "000002023052000004"
 },
 "type": {
   "coding": [
     {
       "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2052",
       "code": "1765",
       "userSelected": false
     },
     {
       "system": "http://snomed.info/sct",
       "code": "119297000",
       "display": "Blood specimen (specimen)",
       "userSelected": false
     }
   ]
 },
 "subject": {
   "reference": "Patient/12783130"
 },
 "receivedTime": "2023-02-21T20:25:00Z",
 "collection": {
   "collector": {
     "reference": "Practitioner/12742563"
   },
   "collectedDateTime": "2023-02-21T13:50:00Z"
 },
 "container": [
   {
     "id": "275745680",
     "identifier": [
       {
         "use": "official",
         "type": {
           "coding": [
             {
               "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
               "code": "ACSN",
```

```
              "display": "Accession ID"
            }
          ],
          "text": "Accession ID"
        },
        "value": "000002023052000004"
      }
    ],
    "type": {
      "coding": [
        {
          "system": "https://fhir.cerner.com/ec2458f2-1e24-41c8-b71b-0e701af7583d/codeSet/2051",
          "code": "2569819877",
          "userSelected": false
        },
        {
          "system": "http://snomed.info/sct",
          "code": "767385003",
          "display": "Aerobic blood culture bottle (physical object)",
          "userSelected": false
        }
      ]
    }
  }
 ]
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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StructureDefinition

File: r4-rest-api-v32/api-structuredefinition.html

Description

StructureDefinition

This topic describes information for the StructureDefinition resource.

Overview

The StructureDefinition resource describes a FHIR structure including data elements and their usage. Our current implementation uses this resource to define custom extensions.

The following fields are returned if valued:

* Id

* URL

* Name

* Title

* Status

* Date

* Publisher

* Description

* Fhir version

* Kind

* Abstract

* Type

* Context

* Base definition

* Snapshot

* Differential

List of Structure Definitions

ID | Description
account-balance | A representation of the account balance.
account-presented-form | A link to the PDF statement image related to the statement account.
account-related-parts | A reference to other related accounts.
account-state | The status of the account in the billing or correspondence workflow.
action-comment | The comment associated with the action performed on the resource.

apply-first-in-first-out | An indication of whether to apply FIFO logic for recurring encounters.
associated-encounter | A reference to the encounter associated with the resource.
bill-code-schedule | A defined group of bill codes that drives billing behavior.
choice-answer | An indication that answers come from a list of options.
client-instruction | A representation of instructions for an order that are intended for healthcare providers.
client-organization | The financially responsible organization.
clinical-instruction | The clinical instructions for an order that are intended for healthcare providers.
condition-course | An indication of a condition's progress since diagnosis.
condition-lifecycle-status | An indication of whether a condition is active, inactive, resolved, and so on.
condition-result | An indication of the presence (positive) or absence (negative) of a given condition.
converted-measurement | A converted measurement of a different measurement system than the original quantity.
coverage-encounter | A reference to the encounter associated with the encounter-level coverage.
communication-preference | The defined communication methods preferred by a patient.
custom-attribute | A customer-defined custom attribute for the resource.
description | A description providing additional details of the resource.
diagnosis-type | A representation of the type of diagnosis.
email-status | The status of the electronic communication.
estimated-financial-responsibility-amount | The estimated amount to be collected for the encounter.
estimated-financial-responsibility-not-collected-reason | The reason that no estimated amount is collected for the encounter.
familymemberhistory-severity | A qualification of the seriousness or impact on health of the family member condition.
financial-transaction-account-number | A value associated with the specific payment method usually represented as the last four digits of a credit card, the check number, the EFT number, or Lockbox number.
financial-transaction-alias | A customer-defined value to represent the combination of the type, subtype, and reason describing the financial transaction.
financial-transaction-allocation | A definition of how the payment or adjustment is to be allocated across other resources.
financial-transaction-amount | The total amount of the financial transaction.
financial_transaction_authorization_code | An alphanumeric string representing the approval of a credit card transaction.
financial-transaction-card-brand | The brand of credit card when a credit card is used as a payment method.
financial-transaction-date | A representation of the expiration date if the method is card, check date if the method is check, EFT date if the method is EFT, or Lockbox date if the method is Lockbox.
financial-transaction-location | A customer-configured value representing the location or workflow that the payment was received in.
financial-transaction-method | The method of payment for the financial transaction.
financial-transaction-tendered-amount | The amount of cash originally tendered for payment. This value should be greater than or equal to the amount of the cash payment.
financial-transaction-type | The classification of the transaction.
formatted-text | A link to the formatted text for a note. This link is used in cases when the note is not in plain text.
general-resource | A representation of information for appointments scheduled with a resource classified as general.
group-appointment-id | The ID of the appointment group that this appointment is part of.
infuse-over-time | The length of time in minutes it took to infuse a medication.
is-modifiable | An indication of whether data is modifiable.
is-physician | An indication of whether the provider is a physician.
is-reschedulable | An indication of whether the resource can be rescheduled.
military-service-connected-indicator | An indication of whether an encounter is connected to military service.
modifier-code | A code providing additional detail about a product or service.
national-drug-product | The national drug product used in care.
net-price | The quantity times the unit price for a resource total price.
note | Additional details related to an element in the resource with author and date and time information.
note-allowed | An indication of whether an additional comment is permitted.
note-type | The type of note represented. This information can be useful when multiple notes exist.
offset-by | An indication of a resource that this resource is offset by. This resource is no longer active when offset.
patient-adopted | An indication of whether a patient is adopted.
patient-friendly-display | An expression of the original string in terminology that patients should be able to understand.
payment-collection-status | The status of the payment collection for the encounter.
performing-location | A location where the resource was performed.
period | A time period defined by a start and end date and time.
pharmacy-verification-status | A representation of whether a medication request is verified by a pharmacist.
precision | An indication of the precision of a given value.
priority | The priority of the element in a list.
procedure-code | A code providing information about the procedure performed on the patient associated with the resource.
quantity-conversion-factor | The conversion factor used to calculate the quantity for billing.
related-person-encounter | A reference to the encounter associated with the encounter-level related person.
relation | The related person's familial relationship to the patient.
relationship-level | The resource's relationship to either the patient or encounter level.
replaced-by | The resource containing this link must no longer be used. The link points forward to another resource that must be used in lieu of the resource that contains this link.
replacing | A reference to a resource that this resource is replacing.
reply-to | A link to a resource that the reply should be directed to.
reschedule_reason | The reason associated with a rescheduled appointment.
revenue-code | The type of revenue or cost center providing the product, service, or both.
scheduling-location | A reference to the location corresponding with this schedule where the appointment can be booked.
transmitting-organization | An organization that transmitted or participated in the creation of a resource, but not the author.
unit-price | The price of a single unit for the resource.

StructureDefinition

The operations from the StructureDefinition category.

Get a structure definition by ID

Method: get Path: /StructureDefinition/{ID}

Get a structure definition by ID

File: r4-rest-api-v32/op-structuredefinition-id-get.html

Retrieves an individual structure definition by the associated ID.

Note:

* Authentication is not required to access the StructureDefinition resource.

* You can retrieve this resource by its defining URL or from the StructureDefinition resource located at the service root URL. For example, both of the following URLs work:

* https://fhir-ehr.cerner.com/r4/StructureDefinition/account-balance

* https://fhir-open.cerner.com/r4/ec2458f2-1e24-41c8-b71b-0e701af7583d/StructureDefinition/account-balance

Authorization Types

Authorization is not required.

Request

Path Parameters

* ID(required): string The logical resource ID.

Header Parameters

* accept(required): string The media type to be requested. See what the resource's operation produces for what is supported.

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Response

Supported Media Types

* application/fhir+json

Default Response

Example Request:

```
GET https://fhir-ehr.cerner.com/r4/StructureDefinition/account-balance
```

Example Response:


```
{
  "resourceType": "StructureDefinition",
  "id": "account-balance",
  "url": "https://fhir-ehr.cerner.com/r4/StructureDefinition/account-balance",
  "name": "AccountBalance",
  "title": "Account Balance",
  "status": "active",
  "date": "2019-08-13",
  "publisher": "Oracle Health",
  "description": "Represents the sum of all credits and all debits associated with the account. May be positive, zero or
     negative.",
  "fhirVersion": "4.0.0",
  "kind": "complex-type",
  "abstract": false,
  "type": "Extension",
  "context": [
    {
      "type": "element",
      "expression": "Account"
    }
  ],
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Extension",
  "snapshot": {
    "element": [
      {
        "id": "Extension",
        "path": "Extension",
        "short": "Represents the account balance",
        "definition": "Represents the sum of all credits and all debits associated with the account. May be positive,
    zero or negative.",
        "min": 0,
        "max": "1",
        "base": {
          "path": "Extension",
          "min": 0,
          "max": "*"
        },
        "type": [
          {
```

```
        "code": "Extension"
      }
    ]
  },
  {
    "id": "Extension.id",
    "path": "Extension.id",
    "representation": [
      "xmlAttr"
    ],
    "short": "xml:id (or equivalent in JSON)",
    "definition": "unique id for the element within a resource (for internal references).",
    "min": 0,
    "max": "1",
    "base": {
      "path": "Extension.id",
      "min": 0,
      "max": "1"
    },
    "type": [
      {
        "code": "id"
      }
    ]
  },
  {
    "id": "Extension.extension",
    "path": "Extension.extension",
    "sliceName": "extension",
    "short": "Extension",
    "definition": "An Extension",
    "min": 0,
    "max": "0",
    "base": {
      "path": "Extension.extension",
      "min": 0,
      "max": "*"
    },
    "type": [
      {
        "code": "Extension"
      }
    ]
  },
  {
    "id": "Extension.url",
    "path": "Extension.url",
    "representation": [
      "xmlAttr"
    ],
    "short": "identifies the meaning of the extension",
    "definition": "Source of the definition for the extension code - a logical name or a URL.",
    "comment": "The definition may point directly to a computable or human-readable definition of the extensibility codes, or it may be a logical URI as declared in some other specification. The definition should be version specific.  This will ideally be the URI for the Resource Profile defining the extension, with the code for the extension after a #.",
    "min": 1,
    "max": "1",
    "base": {
      "path": "Extension.url",
      "min": 1,
      "max": "1"
    },
    "type": [
      {
        "code": "uri"
      }
    ],
    "fixedUri": "https://fhir-ehr.cerner.com/r4/StructureDefinition/account-balance"
  },
  {
    "id": "Extension.valueMoney",
    "path": "Extension.valueMoney",
    "short": "Value of extension",
    "definition": "Value of extension - may be a resource or one of a constrained set of the data types (see Extensibility in the spec for list).",
    "min": 1,
    "max": "1",
    "base": {
      "path": "Extension.value[x]",
      "min": 0,
      "max": "1"
    },
    "type": [
      {
        "code": "Money"
      }
    ]
  }
```

```
    ]
  },
  "differential": {
    "element": [
      {
        "id": "Extension",
        "path": "Extension",
        "short": "Represents the account balance",
        "definition": "Represents the sum of all credits and all debits associated with the account. May be positive,
    zero or negative.",
        "min": 0,
        "max": "1",
        "type": [
          {
            "code": "Extension"
          }
        ]
      },
      {
        "id": "Extension.extension",
        "path": "Extension.extension",
        "sliceName": "extension",
        "max": "0"
      },
      {
        "id": "Extension.url",
        "path": "Extension.url",
        "type": [
          {
            "code": "uri"
          }
        ],
        "fixedUri": "https://fhir-ehr.cerner.com/r4/StructureDefinition/account-balance"
      },
      {
        "id": "Extension.value[x]",
        "path": "Extension.value[x]",
        "min": 1,
        "type": [
          {
            "code": "Money"
          }
        ]
      }
    ]
  }
}
```


Headers

* X-Request-Id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the X-Request-Id, if present.

* opc-request-id: string Unique Oracle-assigned identifier for the request. If you need to contact Oracle about a particular request, provide the opc-Request-Id, if present.

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